

THE SUREKEY

INTEGRATED SCIENCE

Pupil's **BOOK**

6

THE HUMAN EAR



DISTILLATION METHOD



VEGETATIVE PROPAGATION

Planting the sucker



SIMPLE WATER PURIFICATION

Filtration



For Predominance in Integrated Science

AUTHOR: JAMIE HUMPHREY

Based on the Current Standard Curriculum

THE SUREKEY INTEGRATED SCIENCE

Pupil's Book 6

First Edition



"Don't speak for quality, let quality speak for itself"

MESSAGE FROM THE AUTHOR



Dear learner, once again, I'm overwhelmed to unveil this academic material to you. For a learner to shine in any subject excellently, he/she must have a better understanding and knowledge of that subject in line with the current curriculum designed for that particular level.

Given this fact Surekey Integrated Science Pupil's Book 6 is the first edition published by Surekey Examinations Board (SKEB) to help you achieve this target. The book mainly targets primary six learners who are preparing for primary seven. However, the P.6 teachers and researchers of integrated science will as well find this reference book helpful to them.

The approach used in this book is content based with simple and clear language that a learner can read and understand on his/her own. The content given on each topic is sufficient as per learner's level. A number of carefully selected evaluation activities on UNEB standard have been provided to help in revision of the learnt concepts. Do not read the book just for the purpose of answering these questions but use the questions to help you understand and retain what you have learnt.

It is my sincere hope that you will find this book a reliable key reference during the teaching and learning of Primary Six Integrated Science as a subject. Therefore, may this reference book be your guide and companion, fostering a love for learning. Embrace curiosity, question boldly and let knowledge be your constant companion.

Lastly, I regret and apologise for any typing errors that you may come across in the course of using this handbook. **Remember!** Errors are to human.

Note: In case of any questions, concerns and inquiries about the book, please contact me using the telephone numbers or e-mail address on the back cover.

With humble regards,

Jamie Humphrey

ACKNOWLEDGEMENT



I take this golden opportunity to cast a vote of thanks to the following people for their great contribution towards the compilation and production of this reference book.

Your guidance and consultations have played a fundamental role during the course of writing this academic material.

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"Your support in crafting this reference material in the world of science is truly commendable"

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TABLE OF CONTENTS

TERM ONE

THEME : THE WORLD OF LIVING THINGS.....1

TOPIC 1: CLASSIFICATION OF ANIMALS.....1

Vertebrates.....1

Mammals.....3

Birds.....8

Reptiles.....17

Fish.....21

Amphibians.....24

Non-vertebrates (Invertebrates).....28

Insects.....28

Arachnids.....32

Myriapods.....33

Crustaceans.....33

Molluscs.....35

Worms.....36

Caring for and protecting vertebrates and invertebrates.....39

THEME :MATTER AND ENERGY.....40

TOPIC 2 : SOUND ENERGY.....40

Sources of sounds.....40

Musical Instruments.....41

Pitch, Frequency and Volume of sound.....44

Echoes.....52

How sound is produced, stored and reproduced.....54

The Human Ear.....59

Diseases and disorders of the Human Ear.....61

Ways of caring for the Human Ear.....62

THEME : HUMAN BODY.....64

TOPIC 3 : CIRCULATORY SYSTEM.....64

Blood Circulation.....64

Structure and function of the Human Heart.....65

Composition and functions of blood.....71

Diseases and Disorders of the Circulatory System.....75

HIV/AIDS and blood.....77

Ways of increasing the volume of blood in the Human Body.....79

THEME : HUMAN HEALTH.....	82
TOPIC 4: ALCOHOL, SMOKING AND DRUGS.....	82
Alcohol.....	82
Alcoholism.....	84
Smoking.....	86
Drugs in the society.....	90
Narcotic Drugs/Drugs of Dependency.....	94
TERM TWO	
THEME : THE WORLD OF LIVING THINGS.....	98
TOPIC 5 : CLASSIFICATION OF PLANTS.....	98
Flowering plants.....	98
Non-flowering plants.....	103
Seed Dispersal.....	105
Plant Propagation.....	108
THEME : SCIENCE IN HUMAN ACTIVITIES AND OCCUPATIONS.....	115
TOPIC 6 : KEEPING CATTLE	115
Types of cattle.....	115
Breeds of cattle.....	117
Breeding in cattle.....	119
External features of a cow.....	120
Caring for cattle.....	120
Grazing cattle.....	120
Housing cattle.....	125
Mating, Gestation period and Calving in cattle.....	126
Cattle management practices.....	129
Milking and milk products.....	133
Cattle pests (parasites) and diseases.....	137
Practices that harm cattle and domestic animals.....	142
Starting a livestock farm.....	142
Record keeping on a livestock farm.....	143
THEME : THE ENVIRONMENT.....	146
TOPIC 7 : RESOURCES IN THE ENVIRONMENT.....	146
Non-living things as resources.....	146
Living things as resources.....	155
Types of resources.....	159
Environmental degradation.....	160
Managing wastes in our environment.....	164
THEME : HUMAN BODY.....	168
TOPIC 8 : RESPIRATORY SYSTEM.....	168

Respiratory organs.....	168
Lungs and breathing.....	170
Respiration in relation to food and oxygen.....	173
Diseases That Affect The Respiratory System.....	174
Disorders Of The Respiratory System.....	175

TERM THREE

THEME : SCIENCE IN HUMAN ACTIVITIES OCCUPATIONS...177

TOPIC 9 : SCIENCE AT HOME AND IN OUR COMMUNITY.....177

Preparation of Clean and Safe Water for Drinking and Washing.....177

Preparing local salt from plant ash using filtration.....179

Cleaning Clothes in a Home.....183

THEME : HUMAN HEALTH.....187

TOPIC 10 : ACCIDENTS AND FIRST AID.....187

Burns and Scalds.....187

Fever and convulsions.....188

Near Drowning.....190

Fainting.....191

Foreign Bodies in Passages.....192

Poisoning.....195

THEME : HUMAN HEALTH.....197

TOPIC 11 : SANITATION.....197

Ways of promoting sanitation.....197

Toilets and latrines.....200

Ventilated Improved Pit latrine (VIP latrine).....200

Ecosan latrines.....201

Conventional/Ordinary Pit Latrine and Potty.....202

The water closet/water borne toilets.....205

THEME : HUMAN BODY.....209

TOPIC 12 : THE REPRODUCTIVE SYSTEM.....209

Puberty and adolescence.....209

Primary and Secondary sex Changes.....209

Reproductive Organs and Cells.....212

Fertilization, Conception and Pregnancy.....214

Reproductive Health.....217

Teenage pregnancy.....217

Common Diseases and Disorders of the Reproductive

System.....218

Family planning.....223

IRRELEVANT CONCEPTS IN PRIMARY SIX INTEGRATED SCIENCE

TOPIC	IRRELEVANT CONCEPTS
Classification of Animals	Placental and non-placental mammals, structure of animal and plant cells, terms like oviparous, viviparous, ovoviviparous, ruminants, non ruminants, animates, inanimate, hibernation and aestivation, homoeothermic and poikilothermic, classification of fish as bony, cartilaginous and lung fish, structure of the gills, their parts, function of parts of the gills and adaptations of the gills filaments/gills to their function, classification of hoofed mammals as odd toed and even toed mammals, classification of carnivorous mammals as dog and cat family, lagomorphs as a group of mammals, Details on feathers i.e. structure and types of feathers, parts of an egg, sponges, echinoderms, coelenterates, concepts on single celled organisms (protocista kingdom) i.e. amoeba, euglena, algae, paramecia and viruses
Sound Energy	Types of sound, units of sound
Circulatory System	Blood groups, the whole illustration of blood circulation in The human body , specific names for valves i.e. tricuspid Valve, bicuspid valve etc...and types blood circulation
Classification of Plants	Classification of fruits as drupes, berries, pomes , dehiscent Indehiscent, false fruits, true fruits, multiple, aggregate and Compound fruits, tropisms, testing for starch in the leaf, Definition and experiment on osmosis, terms like translocation And guttation, types of transpiration. Note: even teaching leaves, flowers, pollination, roots, stems Transpiration, seeds, germination, photosynthesis are not Necessary in primary six as they were already covered in p.4 Therefore, in p.6, concentrate on, 1. Flowering plants (legumes and cereals only) 2. Non-flowering plants 3. Seed dispersal 4. Plant propagation as indicated in the curriculum.

Keeping Cattle	Digestive system of a bull/cow, ruminants and non-ruminants Reproductive system of a cow and a bull, developing foetus In a cow's womb, pasture and fencing concept.
Respiratory System	Types of respiration i.e. aerobic and anaerobic respiration, Details on each type of respiration, structure of alveolus.
Science at Home and in our Community	Concepts on hard and soft water, properties of water, diseases associated with water i.e. water borne diseases, water cleaned diseases etc.. (should be taught in P.7 under population and Health (poor water supply)
The Reproductive System	Concepts on birth and labour, twins, sex determination/Chromosomes i.e. X and Y chromosomes.

SPECIAL ADVICE TO TEACHERS

1. Teach following the limits of the curriculum.
2. Observe and internalise the competencies in the curriculum.
3. Read the suggested activities and apply them when developing test items.
4. Use practical approaches when handling certain topics in Primary Six.
5. Introduce each lesson with at least six vocabularies to enable learners spell, pronounce and read words correctly. This promotes language competence.
6. Make your lesson objectives **SMART** i.e. Specific, Measurable, Achievable, Realistic and Time bound.
7. A good teacher of Integrated Science should be **CAR** i.e. Current, Accurate/Appropriate and Relevant in teaching his/her lessons

Theme:

Topic

1: THE WORLD OF LIVING THINGS CLASSIFICATION OF ANIMALS

Biological features/characteristics used to classify animals.

- ❖ The mode of feeding
- ❖ The body structure/shape
- ❖ The mode of reproduction
- ❖ Mode of fertilisation

Reasons why animals are classified.

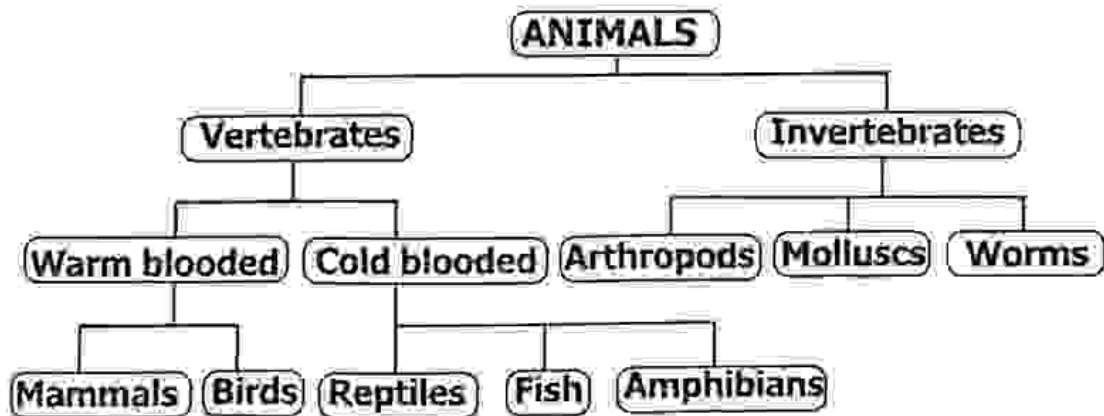
- ❖ For easy identification of animals.
- ❖ For easy study of animals.
- ❖ For easy naming of animals.
- ❖ For easy understanding of animals.

Characteristics of animals

- ❖ Animals are made up of many cells.
- ❖ Animals feed on ready-made food.
- ❖ Animals reproduce sexually.
- ❖ Animals have natural defensive mechanisms

Classification table for animals

Classification of vertebrates and invertebrates



Groups of animals

- ❖ vertebrates
- ❖ invertebrates

VERTEBRATES

Vertebrates are animals that have backbone.

Characteristics of vertebrates

- ❖ They have backbone.
- ❖ They have endoskeleton.
- ❖ They undergo sexual reproduction.

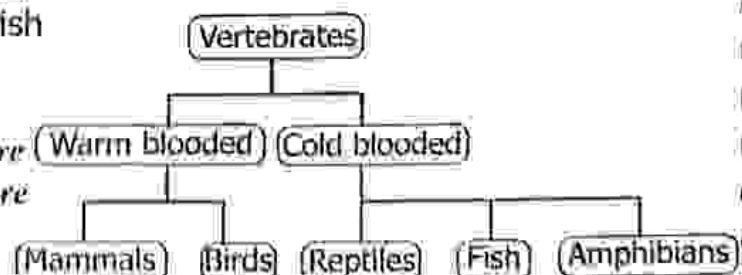
Classes / groups of vertebrates

- ❖ mammals
- ❖ birds
- ❖ fish
- ❖ reptiles
- ❖ amphibians

Mnemonic used: MR FAB

Some groups of vertebrates are warm blooded while others are cold blooded.

Warm blooded vertebrates



Don't speak for Quality, let Quality speak for itself

Warm blooded vertebrates are animals that have constant body temperature.

Mammals and **birds** are warm blooded vertebrates.

Reason. Their body temperature is constant.

Cold blooded vertebrates

Cold blooded vertebrates are animals whose body temperature changes according to that of the surrounding.

Reptiles, fish and **amphibians** are cold blooded vertebrates.

Reason. Their body temperature changes according to that of the surrounding.

Evaluation activity 1.1

1. Mention any **two** biological features used to classify animals.
2. Give any **one** characteristic common to all animals.
3. Why is a bat referred to as a warm-blooded vertebrate?
4. Write down any **two** reasons why animals are called living things.
5. List any **two** ways in which animals are different from plants.
6. What do we call the smallest unit of living things?

The diagram below shows an animal. Use it to answer questions 7 and 8.



7. To which group of animals does the animal above belong?

8. Why is the above animal grouped under vertebrates?

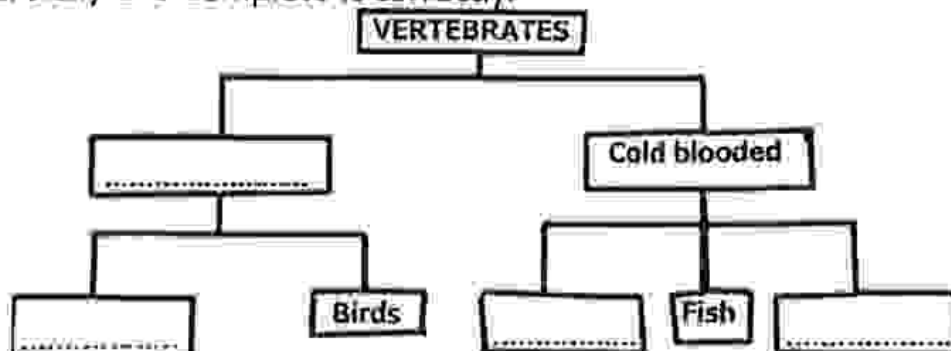
9. Name any **one** class of;

- i) Warm blooded vertebrates
- ii) Cold blooded vertebrates

10. Study the list of vertebrates below and use it to answer the questions that follow.

Rat, toad, cat.

- a) Which **two** vertebrates can be classified together?
 - b) Give a reason for your answer in a) above.
 - c) State any **one** way in which the toad is different from the rat and cat on the list above.
11. The table below shows a simple classification of vertebrates. Study carefully and complete it correctly.



MAMMALS

Mammals are warm blooded vertebrates with mammary glands which produce milk.

Characteristics of mammals

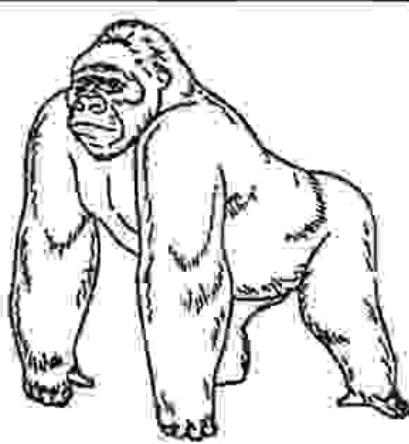
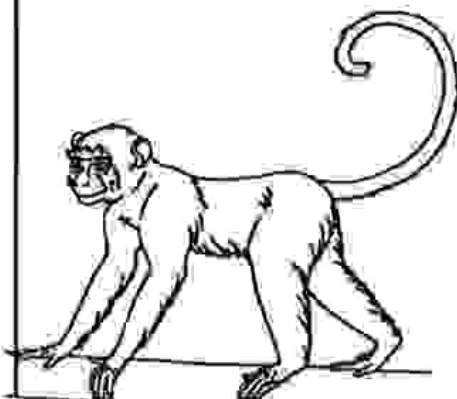
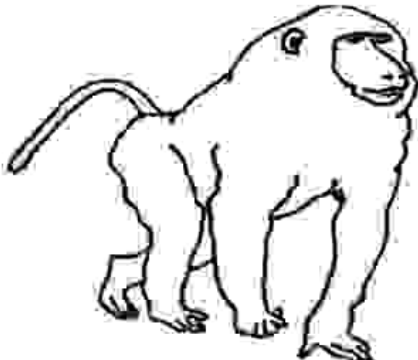
- ❖ They have mammary glands.
Mammary glands produce milk on which their young ones are fed.
- ❖ They have fur on their bodies.
Hair / fur insulates their body against heat loss.
- ❖ Most mammals produce live young ones.

Groups of mammals

Primates

These are the most advanced mammals with well-developed brain.
They are omnivores (feed on both flesh and plant materials)

Examples of primates

Humans	Mountain gorilla	Bush Baby
		
Chimpanzee	Monkey	Baboon
		

Hoofed mammals

These are mammals which have hooves on their feet. They are also called ungulates.

Uses of hooves to ungulates

- ❖ They support the weight of the animal on the ground.
- ❖ They protect the tissues and foot bones.
- ❖ They provide traction to the animal.

Structure of hooves



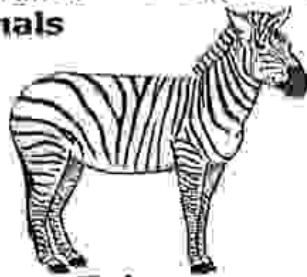
Examples of hoofed mammals

❖ Cows ❖ Zebras ❖ Giraffes ❖ Goats ❖ Sheep

Examples of hoofed mammals



Cow



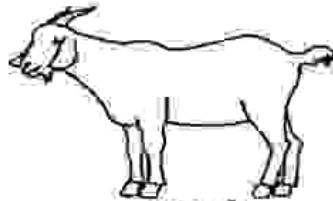
Zebra



Sheep



Giraffe



goat

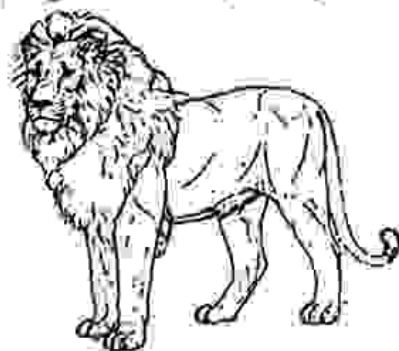
Flesh eating mammals.

These are mammals which feed on meat. They are also called carnivorous mammals.

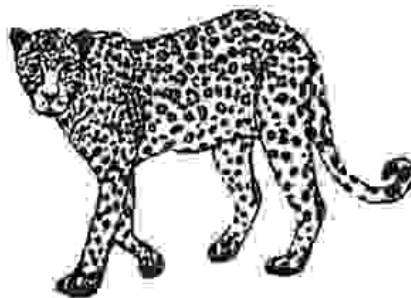
Characteristics of flesh-eating mammals

- ❖ They have well developed canine teeth for tearing the flesh.
- ❖ They have strong sharp claws for holding the prey firmly.
- ❖ They have a keen eyesight to spot their prey.
- ❖ They have soft pads on their feet to attack their prey without being noticed.
- ❖ They have a strong sense of smell for tracing their prey.
- ❖ They run very fast to be able to chase their prey.

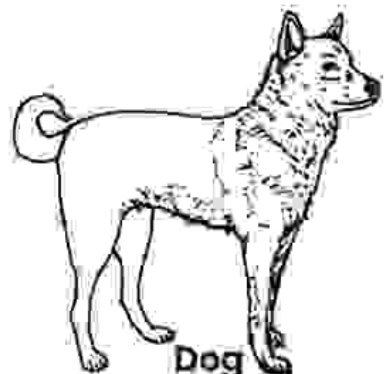
Examples of flesh-eating mammals are cat, lion, cheetah, leopard, tiger, dog, skunk, hyena, jackal, jaguar and fox



Lion



Leopard



Dog

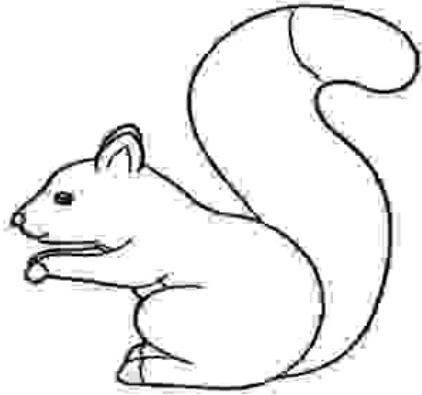
Rodents

These are gnawing mammals which have long tails and well-developed incisor teeth for biting plant materials rapidly. These plant materials include; grass, cereals, root tubers and stem tubers. Rodent are crop pests.

Characteristics of rodents.

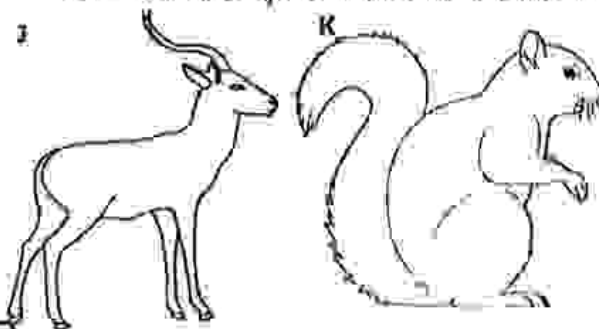
- ❖ They have well developed incisor teeth.
- ❖ They feed on plant materials.
- ❖ They have sharp claws for digging up root crops.
- ❖ They have long tails

Examples of rodents

Rat	Squirrel
	
Gopher	Porcupine
	

Evaluation activity 1.2

- Name the group of mammals which have well developed;
 - Canine teeth
 - Incisor teeth
- What type of fertilization do mammals undergo?
- Why is a lion able to run after its prey without being noticed?
- To which group of mammals does a gopher belong?
- Give any **one** example of flesh-eating mammals.
 - Why do flesh eating mammals have well developed canine teeth?
 - State any **two** other characteristics of flesh-eating mammals.
- The diagrams below show groups of mammals. Study and use them to answer the questions that follow.



- Identify the group of mammals marked **J** and **K**
 - State any one danger of mammal marked **K** to crop farmers.
 - Why is the mammal marked **J** referred to as herbivore?
- Why is a monkey grouped under primates?

Sea/marine mammals

These are mammals that live in the water of seas and oceans.

They are also called **cetaceans**.

Characteristics of sea mammals

- ❖ They have a layer of fats called blubber. Blubber keeps sea mammals warm during cold weather.
- ❖ They live in the cold water of the sea or ocean.
- ❖ Their forelimbs are modified into flippers for swimming in water.
- ❖ They breathe through lungs.
- ❖ They have a streamlined body.

Examples of sea mammals are; whales, otters, seals, dolphins, porpoises, dugongs, polar bears, sea lion, manatees and walrus



Seal



Dolphin



Whale

Adaptations of sea mammals to living in water

- ❖ They have blubber which keeps them warm during cold weather.
- ❖ They use echoes to dodge obstacles in water.
- ❖ Their forelimbs are modified into flippers for swimming in water.

Insect eating mammals

These are mammals which feed on insects.

Characteristics of insect eating mammals

- ❖ They have strong claws for digging the ground to look for insects.
- ❖ They have long sticky tongue for catching insects.
- ❖ They are mostly nocturnal.
- ❖ They have a sensitive snout.
- ❖ They have spines for protection.

Examples of insectivores are hedgehog, moles, pangolin, armadillo and elephant shrew.



hedgehog



shrew



Pangolin

Flying mammals

These are mammals whose limbs are modified into wings.

They are also called chiroptera.

An example is a **bat**.

Reasons why a bat is called a mammal.

- ❖ It has mammary glands.
- ❖ It has a hairy body.
- ❖ It produces live young ones.

 *Note well: A bat is a nocturnal animal because it is more active at night. A bat uses echoes to locate food and dodge obstacles.*

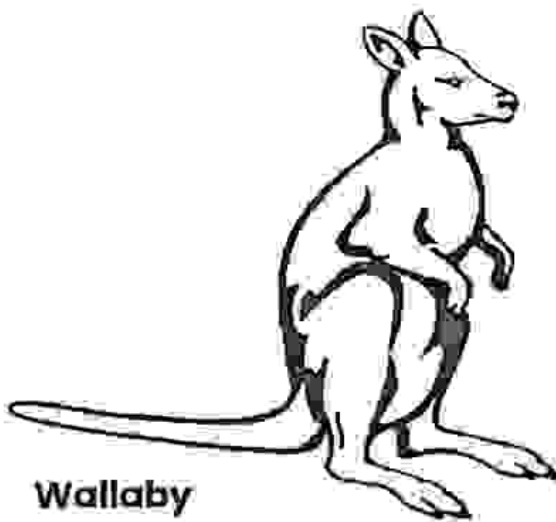


Pouched mammals

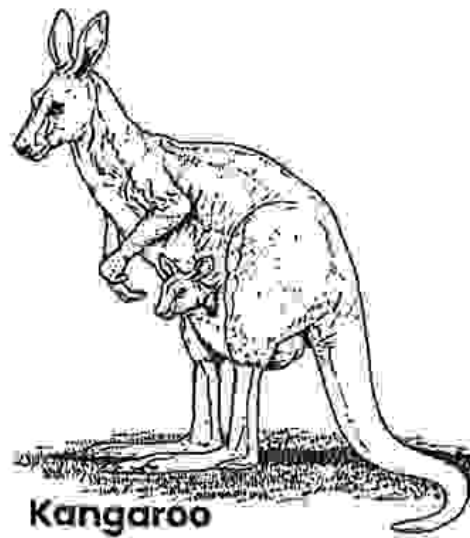
These are mammals which have a pouch for carrying their joeys. They are also called **marsupials**.

Examples of pouched mammals

➤ kangaroo ➤ wombat ➤ koala bear ➤ wallaby ➤ opossum



Wallaby



Kangaroo

-  *Points to note:*
- *Fur keeps the kangaroo warm.*
 - *The pouch keeps the joey for further growth.*
 - *The tail provides stability during hopping.*

Egg laying mammals

These are mammals which reproduce by laying eggs.

They are also called **monotremes**.

They are mammals because they feed their young ones on milk produced from mammary glands after hatching.

Monotremes are most primitive mammals because;

- They have characteristics of reptiles, birds and mammals.
- They lay eggs and have a beak similar to that of birds.



Duck billed platypus



Spiny anteater

Importance of mammals in the environment

- ✓ Some mammals are source of food.
- ✓ Mammals provide hides and skins used in leather industries.
- ✓ Their hooves and horns are used for making buttons and glue.
- ✓ Their dung, droppings and urine are used to produce biogas.
- ✓ Some mammals provide labour on farms.
- ✓ Some mammals help in pollination e.g. bats.

Evaluation activity 1.3

1. Name the layer of fats that keeps sea mammals warm when in cold water of an ocean.
 2. Write down any **two** examples of sea mammals.
 3. How is blubber useful to a whale?
 4. State the importance of strong claws to a mole.
 5. To which group of mammals does each of the following animals belong?
 - i) pangolin
 - ii) Wallaby
 - iii) Bat
 - iv) platypus
 6. Give any **two** ways in which mammals are important in the environment.
- The diagram below shows a mammal. Use it to answer questions 7 and 8.



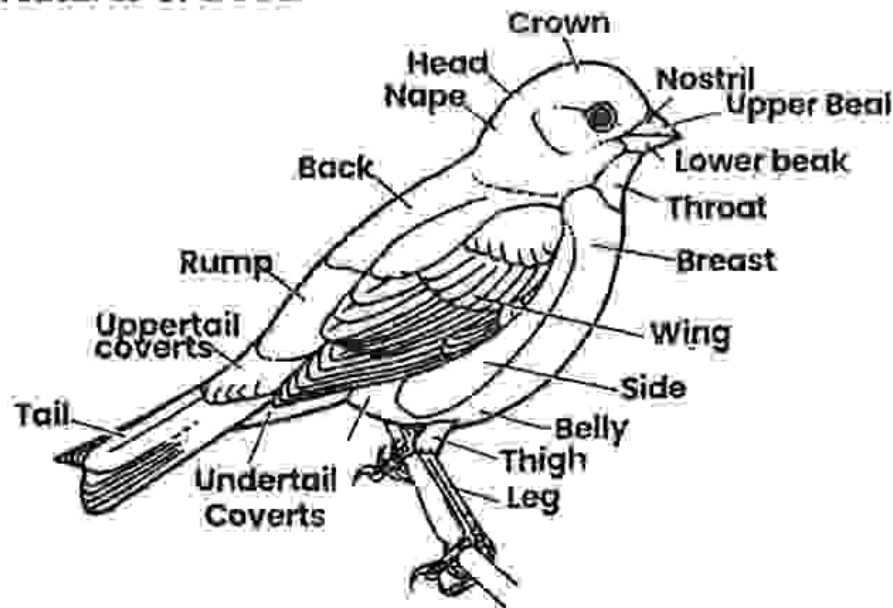
7. Identify the group of mammals to which the animal above belongs.
 8. How is part marked K useful to the mammal above?
 10. The list below shows groups of mammals. Study and use it to answer the questions that follow.
seal, shrew, spiny anteater, porpoise
- a) Identify **two** mammals that can be grouped together.
 - b) Which mammal from the list above feeds on termites?
 - c) How is the spiny anteater different from all other mammals on the list above in the way they reproduce?

BIRDS

Characteristics of birds

- ❖ They have feathers covering their bodies.
- ❖ They have scales on their legs.
- ❖ They have streamlined bodies. This helps them to reduce friction in air during flight.
- ❖ They have hollow bones. This helps to reduce weight in air during flight.
- ❖ They have a nictitating membrane on their eyes. This protects their eyes against strong wind during flight.
- ❖ They are warm blooded vertebrates.
- ❖ They undergo internal fertilization.
- ❖ They reproduce by laying eggs.
- ❖ They breathe through lungs.

External features of a bird.



Functions of some features of the bird.

Beak

- ❖ For picking and breaking food.
- ❖ For protection against enemies.
- ❖ For pecking the tree to look for insects.
- ❖ For turning eggs during incubation.

Feathers

- ❖ For flight
 - ❖ They protect the skin from mechanical injuries.
 - ❖ They keep the bird warm.
- Birds often raise their feathers during cold weather.

Reason. To maintain their constant body temperature.

Nostril

- ❖ For smelling
- ❖ Lets in air during breathing.

Wings: For flight

Claws

- ❖ For holding the prey firmly.
- ❖ For perching on trees.
- ❖ For climbing trees.
- ❖ For scratching the ground to get food.
- ❖ For protection.

Tail

- ❖ For changing direction during flight.

Eyes.

- ❖ For seeing.
- ❖ They have a nictitating membrane. This protects the eyes against strong wind during flight.

Earlobes

- ❖ For protecting the ears of the bird.

GROUPS OF BIRDS

Factors to consider when classifying birds

- Mode of feeding,
- The shape of the beak.
- The nature of the feet,
- The mode of movement.

Birds of prey

They feed on the prey hunted and killed by themselves.

A prey is an animal hunted and killed by another animal for food.

A predator is an animal which hunts and kills others for food.

Characteristics of birds of prey

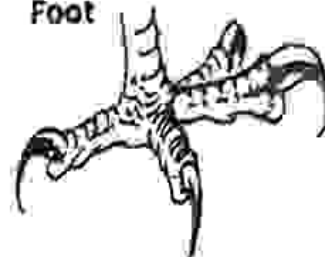
- They have strong sharp hooked beak.
The hooked beak enables them to tear flesh of the prey.
- They have strong sharp curved talons.
For gripping or holding the prey firmly.
- They have strong eyesight.
To spot their prey even at a distance.

Examples of birds of prey are; eagles, kites, falcons, hawks, owls, buzzards and secretary birds

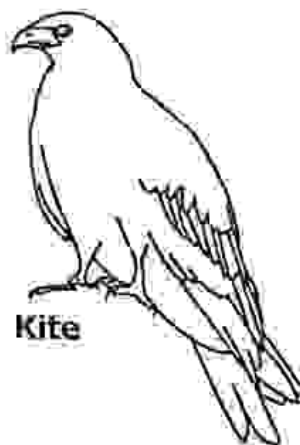
Beak



Foot



Eagle



Kite



owl

i **Learn, unlearn and relearn:** it is wrong to say that birds of prey feed on fresh flesh or flesh/meat because even scavenger birds feed on flesh/meat.

Birds of prey help to feed on crop pests like rats and squirrels. However, they hunt and kill chicks. This affects a poultry farmer.

Scavenger birds

They feed on leftover flesh/abandoned flesh.

They have a strong hooked beak for tearing carcasses.

Scavenger birds help to clean the environment by feeding on abandoned flesh of dead animals. They reduce bad smell from carrion in an area.

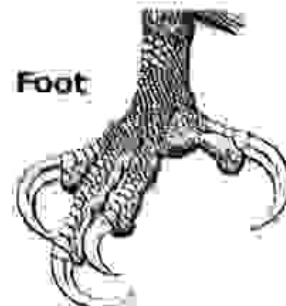
Examples of scavenger birds are; marabou storks, crows and vultures.

Jackals, hyena and foxes are scavenger mammals.

Beak



Foot

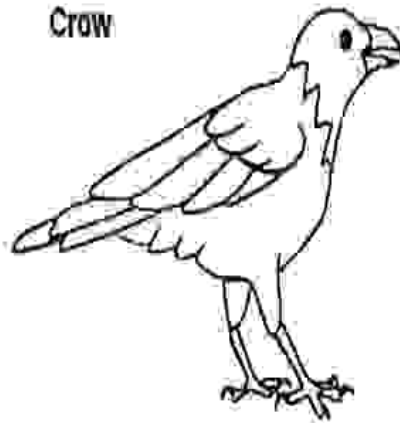


Examples of scavenger birds

Vulture



Crow



Marabou Stork



i **Learn, unlearn and relearn:** it is wrong to say that scavenger birds feed on carrion as their food because they can also feed on meat/flesh slaughtered at that very time. Note that carrion is the decaying flesh of a dead animal.

Vultures lack feathers on their heads to easily keep themselves clean when eating leftover flesh.

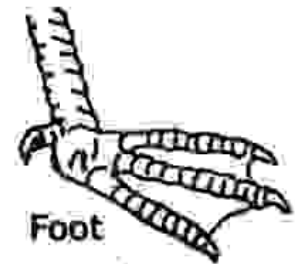
A marabou stork has a throat pouch for catching and holding food.

Swimming birds

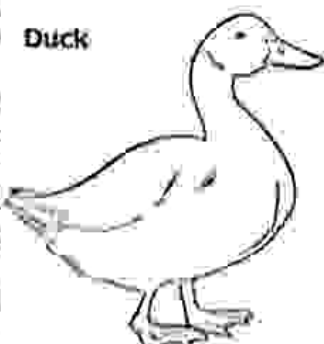
Characteristics of swimming birds.

- ✓ They have webbed feet that enable them to swim in water.
The webs are used for paddling in water during swimming.
- ✓ They have spoon shaped beak for easy sieving of food from mud/water.
- ✓ They feed on small fish, tadpoles and aquatic plants.
- ✓ They have glands that produce oil to keep their skin waterproof.

Examples of swimming birds are; duck, swans, pelican, goose, cormorant, penguin, seagull and gadwall.



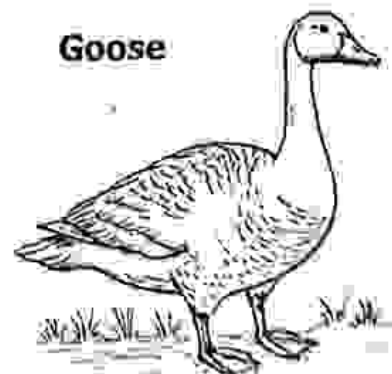
Duck



Swan



Goose



Don't speak for Quality, let Quality speak for itself



Wading birds.

These are birds that walk through water or mud.

Characteristics of wading birds

- ❖ They have thin long legs for walking through water or mud.
- ❖ They have thin beaks for catching fish, frogs and worms.
- ❖ They have half webbed toes widely spread out to prevent them from sinking in water.

Examples of wading birds are; flamingo, crested crane, heron, ibis, egret, jacana and kingfisher.

Note: egrets are commonly seen on cattle to eat up the insects that fall on cattle.

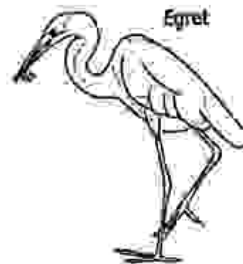
Examples of wading birds



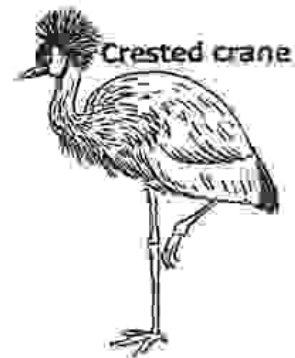
Heron



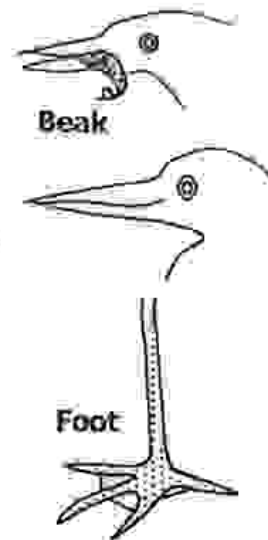
Flamingo



Egret



Crested crane



Beak

Foot

Evaluation activity 1.4

1. State any **two** characteristics of birds of prey.
2. a) Name the feature on an eagle that enables it to;
i) tear flesh ii) hold the prey
b) Apart from an eagle, name any **two** other examples of birds of prey.
3. State any **two** features used to classify birds.
4. a) Write down any **two** examples of scavenger birds.
5. b) How is the feeding of scavenger birds different from that of birds of prey?
c) In which way are scavenger birds useful in the environment?
6. a) Give any **two** uses of feathers to birds.
b) Apart from having feathers, state any **two** characteristics common to all birds.
7. The diagram below shows the foot and beak of the bird. Study and use it to answer the questions that follow.



Evaluation activity 1.4Continued

7. a) Which group of birds have the foot and beak shown above?
- b) Give any **two** examples of birds which have the foot and beak shown above.
- c) What kind of food do birds with the foot and beak shown above eat?
8. The diagrams below are of a foot and a beak of a bird. Study and use them to answer the questions that follow.



- a) Identify the group of birds with the type of foot and beak shown.
- b) How do the parts of the bird shown above help it live well in its environment?
 - i) Foot.....
 - ii) Beak.....
- c) Give any **one** example of a domestic bird with such type of beak and foot.

Scratching birds

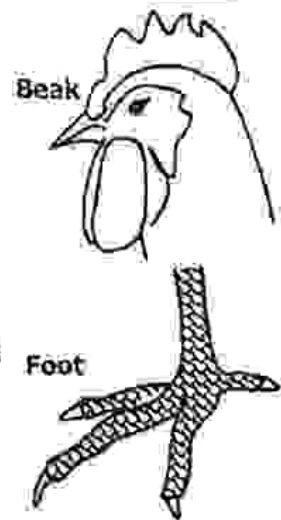
These are birds which scratch the surface of the ground to get food.

They eat small insects, worms and seeds from soil.

Characteristics of scratching birds

- ✓ They have strong blunt claws for scratching the ground.
- ✓ They have strong and short pointed beaks for picking food from the ground.

Examples of scratching birds are; chickens, turkeys and guinea fowls

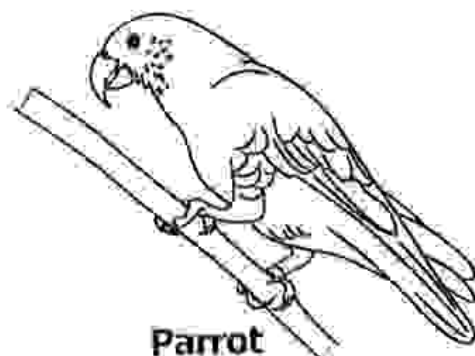


Climbing birds

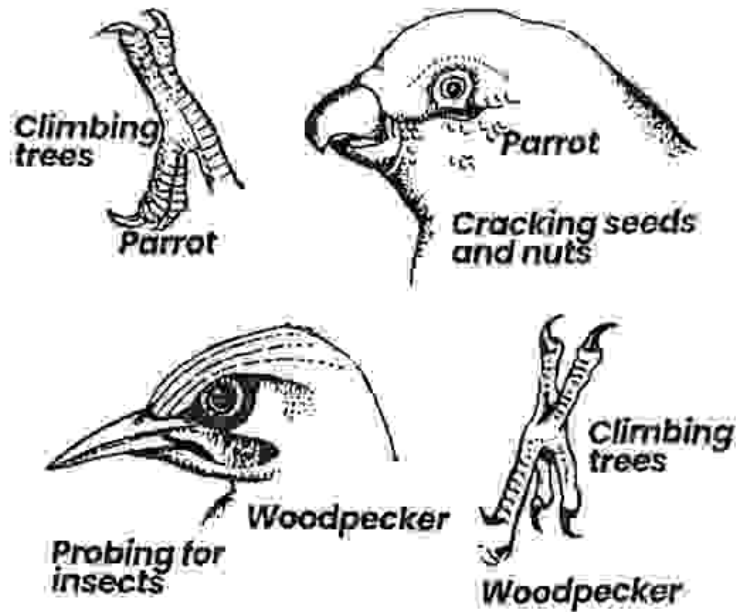
These are birds that have the ability to climb trees to look for insects to eat.

Characteristics of climbing birds

- ❖ They have two toes pointing forward and two toes pointing backwards. This helps them to climb trees to look for insects.
- ❖ They have strong pointed and chisel shaped beaks. This enables them to dig holes in tree trunks and break seeds.



Feet and Beaks of climbing birds



Perching birds

These are birds which rest and stay on tree branches.

They have three toes pointing forward and one toe pointing backwards.

This enables them to hold firmly onto the tree branch.

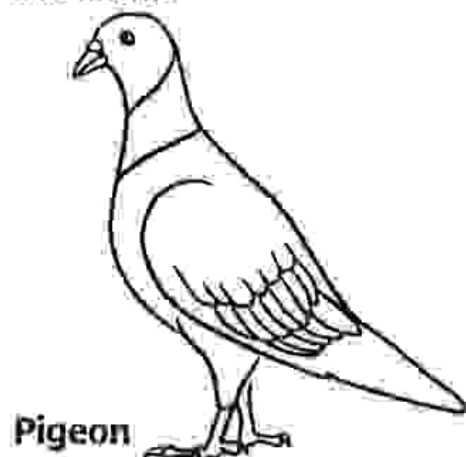
Groups of perching birds

Seed eaters

They eat seeds and grains.

They have short and strong conical beaks for breaking seeds.

Examples are; pigeons, doves, weaver birds and finches



Insect eaters

They have short narrow beaks for catching insects during flying.

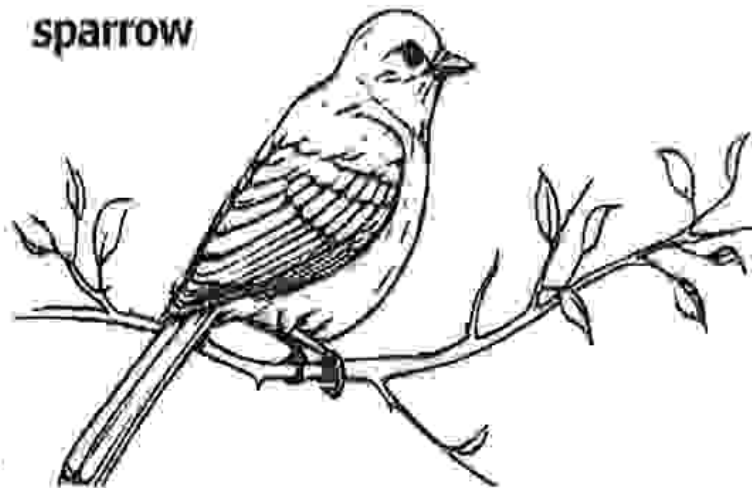
They feed on harmful insects to humans.

Examples of insect eaters are; sparrows, robins, swallows, swifts and bee eaters.



robin

sparrow



Sparrow



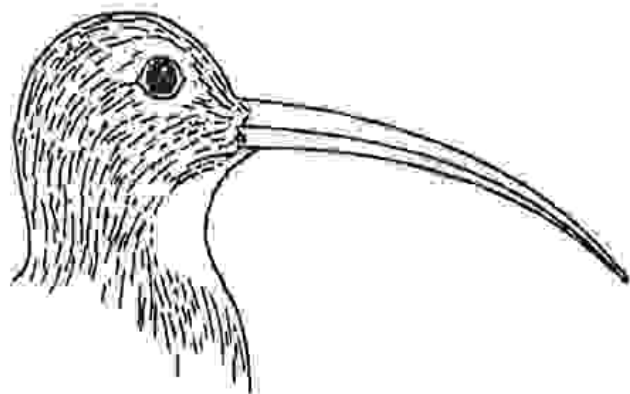
Swift

Nectar suckers

They have long slender beaks for sucking nectar from the bottom of the flowers. They help in pollination.

Examples of nectar suckers are; sunbird and humming bird

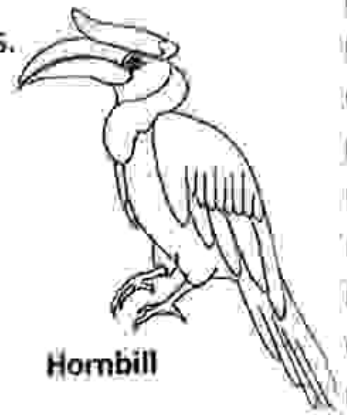
A humming bird is able to fly both forward and backwards while other birds only fly forward.



Fruit eaters

They have a long stout beak for collecting fruits from trees.
They help in seed dispersal.

An example is a hornbill.



Hornbill

Flightless birds

These are birds that are unable to fly.

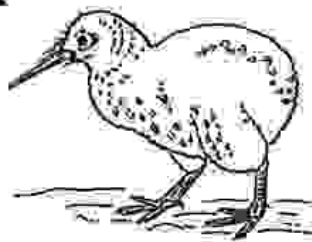
Reason They have small and weak wings compared to their body size.

They feed on plant materials, insects and worms.

Examples of flightless birds are; ostrich, emu, kiwi, rhea, penguin, cassowary and kakapo



Penguin



Kiwi



Cassowary



Ostrich

Reasons why birds fly.

- ✓ To look for food.
- ✓ To look for mates.
- ✓ To look for protection.
- ✓ To look for shelter.
- ✓ To look for their young ones.

Reasons why birds are able to fly.

- ↓ Their forelimbs are modified to form wings.
- ↓ They have flight feathers on their wings.
- ↓ They have strong pectoral muscles for moving wings during flight.

Importance of birds

- ❖ Birds like chickens provide meat and eggs to people.
- ❖ Birds like sunbirds help in pollination of flowers.
- ❖ Birds like chickens are sold for money.
- ❖ Birds help to feed on insect pests.
- ❖ Birds like vultures clean an area by feeding on abandoned flesh.

Dangers of birds in the environment.

- Birds like weaver birds spoil crops.
- Birds keep vectors like fleas.
- Birds cause accidents on runways at airports.
- Birds of prey kill and eat chicks.

Similarities between birds and mammals.

- Both are warm blooded vertebrates.
- Both undergo internal fertilization.
- Both breathe through lungs.
- Both have streamlined bodies.

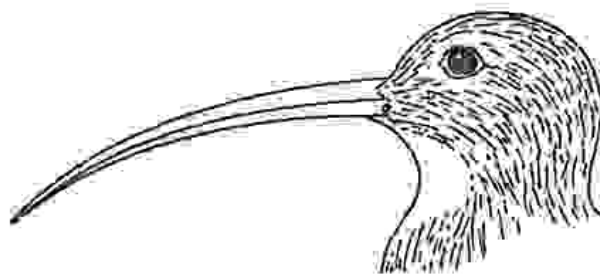
Differences between birds and mammals.

- ↓ Birds have feathers on their bodies while mammals have fur/hair on their bodies.
- ↓ Birds lay eggs while mammals produce live young ones except egg laying mammals.
- ↓ Birds have scales on their legs while mammals have hair on their legs.
- ↓ Birds move by flying except flightless birds while mammals move by walking, hopping or flying.

Evaluation activity 1.5

1. State any **two** ways in which birds are similar to mammals.
2. Why are sunbirds able to suck nectar from the bottom of flowers?
3. Give any **one** reason why birds are able to fly in air.
4. How is a humming bird different other birds in the way it flies?
5. Why is an ostrich unable to fly?
6. How is the feeding of a pigeon different from that of a robin?
7. In which way is the feeding of a hornbill useful in the environment?
8. Identify the group of birds to which a turkey belongs.
9. How is reproduction in a dove similar to that in a spiny anteater?

The diagram below shows a beak of a bird. Use it to answer questions 10 and 11.



10. What kind of food does the bird with beak shown above eat?
11. How is the feeding of the bird with beak above helpful to a crop farmer?

REPTILES

These are cold blooded vertebrates whose body is covered with scales and move by crawling.

Characteristics of reptiles.

- ✓ Their body is covered with scales.
- ✓ They are cold blooded.
- ✓ They reproduce by laying eggs.
- ✓ They breathe through lungs.
- ✓ They undergo moulting.
- ✓ They undergo internal fertilization

Examples of reptiles are; snakes, lizards, crocodiles, turtles, chameleon, tortoise, terrapin

Snakes

Characteristics of snakes

- ↓ They are limbless.
- ↓ They move by gliding/slithering
- ↓ They have a forked tongue.

The forked tongue is used for smelling and touch.

↑ They moult to grow the new skin and increase in size.

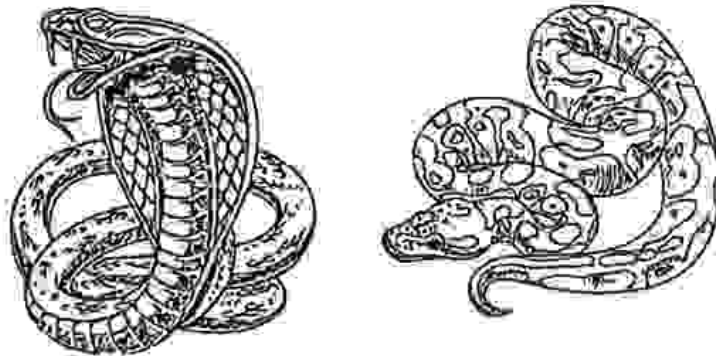
↑ They eat frogs, toads, rats, lizards and rats

Groups of snakes

Group	Characteristics	Examples
Poisonous/ venomous snakes	-They have fangs. -They have a triangular head. -They have poison glands. -They leave two fang marks at the bitten part of the body.	• Cobra • puff adder • black mamba • gaboon viper • boom slung
Non poisonous/ non venomous snakes	-They have solid teeth. -They have a round head. -They have no venom glands. -They leave four marks in the bitten part of the body.	• Green grass snakes • brown snakes • house snakes • garter snakes • bull snakes
Constrictors	- They kill their prey by coiling around them to squeeze them to death. This helps them to break the bones for easy digestion after swallowing. - They are non-poisonous. - They lick their killed prey to make it slippery for easy swallowing.	• Python • anaconda • boa • milk snake • king snake

Diagrams of different groups of snakes

Poisonous snakes



The structure of the head of a poisonous snake



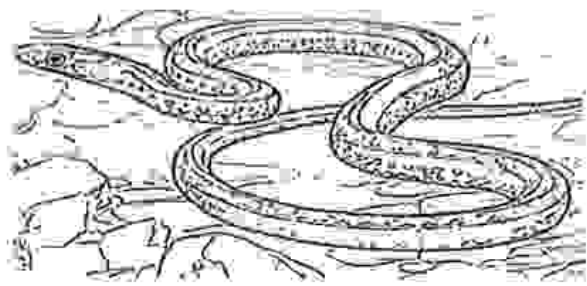
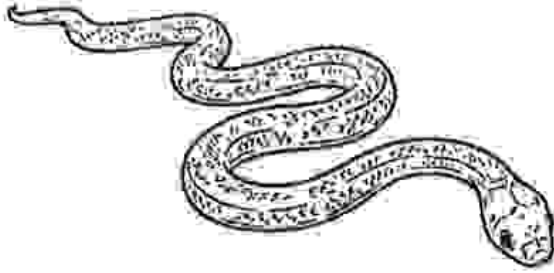
Fangs. They are used to inject venom into the body of an enemy.

-They are hollow to inject venom easily into an enemy.

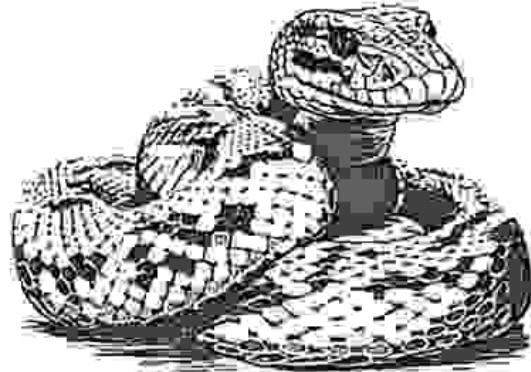
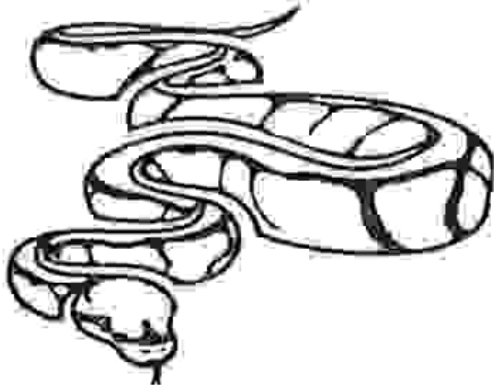
-They point backwards to prevent the prey from escaping from the mouth.

Forked tongue. It is used for smelling and touch.

Non poisonous snakes



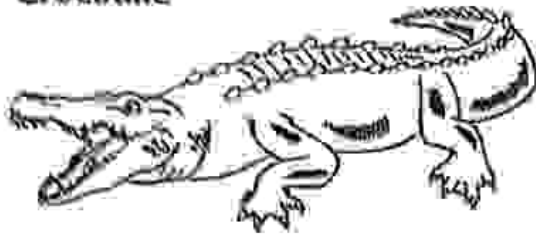
Constrictors



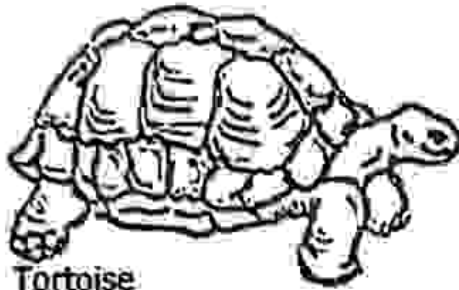
Reptile	Features/Characteristics
Chameleon	- It has bulging eyes which enable it to see all directions at the same time. - It changes its colour for protection and to trap its prey. - It has a long sticky tongue to trap its prey. - It has well developed feet and tail for catching and gripping small twigs branches of trees. - Chameleons feed on harmful insects like houseflies and mosquitoes.
Gecko	- It has sticky tiny hair under its feet that grip to the wall/ceiling. - It feeds on harmful insects like houseflies and mosquitoes.
Tortoises Turtles	- They protect themselves by hiding into their hard shells. - Turtles have flippers for swimming in water. - They feed on plants, mushrooms, insects, sea snails and jellyfish.
Crocodiles	- They have long strong tails and teeth for protection against enemies. - They use their tails and teeth for attacking their prey. - They live both in water and on land. They can hold the breath for long time under water. - They lay eggs in sand. Sand provides warmth for eggs to hatch. - Sand protects their eggs from being eaten by predators.

Diagrams of reptiles

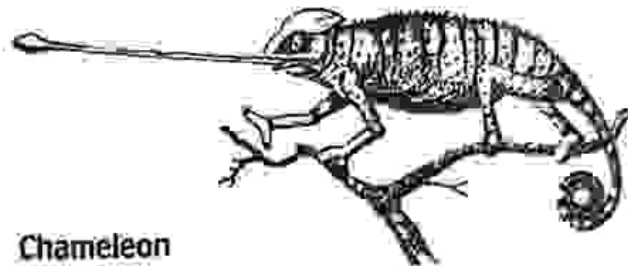
Crocodile



Gecko



Tortoise



Chameleon

Importance of reptiles

- ✓ Reptiles are source of food to some people.
- ✓ Reptiles provide skins for leather industries.
- ✓ Reptiles like snakes eat crop pests.
- ✓ Reptiles control some vectors like houseflies.
- ✓ Poisonous snakes provide venom used to make serum (antidote)
Serum (antidote) is used for treating snake bites.

Similarities between reptiles and birds.

- ❖ Both reproduce by laying eggs.
- ❖ Both undergo internal fertilization.
- ❖ Both breathe through lungs.
- ❖ Both undergo moulting.

Differences between reptiles and birds.

- Reptiles are cold blooded while birds are warm blooded.
- Reptiles have scales on their bodies while birds have feathers on their bodies except on the legs.
- Reptiles move by crawling while birds move by flying or walking.

Evaluation activity 1.6

1. Which group of cold-blooded vertebrates undergo internal fertilization?
2. State any **two** features used to identify poisonous snakes.
3. What name is given to the long teeth of a cobra?
4. Apart from cobra, give any **two** other examples of venomous snakes.
5. State the difference in structure between the teeth of a puff adder and those of a brown snake.
6. Why are fangs of a cobra positioned pointing backwards?
7. How are fangs of a gaboon viper able to inject poison into an enemy?
8. a) Give the reason why a python;
i) Squeezes its prey..... ii) Licks its prey.....
b) Apart from a python, name any **two** other examples of constrictors.

Evaluation activity 1.6 Continued

The diagram below shows a head of a black mamba. Study use it to answer questions 9 and 10.



9. Name the part marked X.
10. Give any **one** way in which part marked Y is useful to a black mamba.
11. How is a snake useful to a crop farmer?
12. In which way do snakes help in the control of insect borne diseases in humans?

FISH

Characteristics of fish

- ✓ They are cold blooded vertebrates.
- ✓ They reproduce by laying eggs.
- ✓ They have a slippery body to escape from predators.
- ✓ They have streamlined bodies to reduce friction during swimming.
- ✓ They breathe through gills.
- ✓ They entirely live in water.
- ✓ They undergo external fertilisation.
- ✓ They have fins for swimming in water.

A fish dies when removed from water.

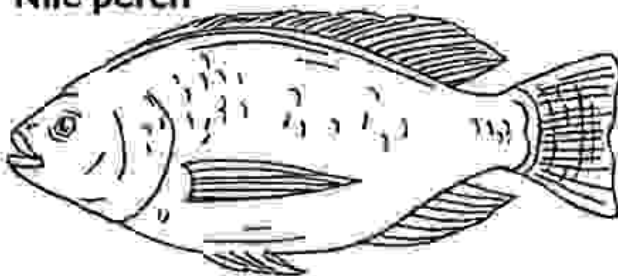
Reasons.

A fish lacks dissolved oxygen when removed from water.

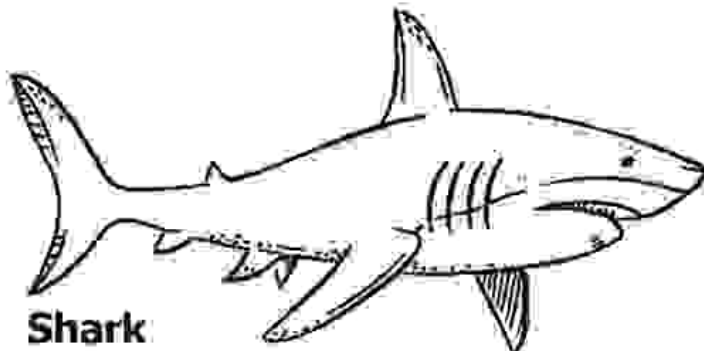
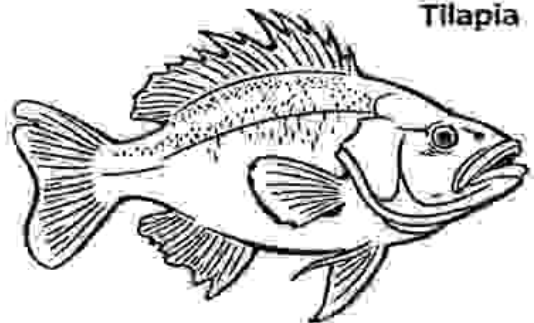
The muscular system of the mouth and gills cannot work in air.

Examples of fish are; tilapia, Nile perch, mudfish, catfish, sharks, dogfish, silver cyprinid (mukene) and salmon

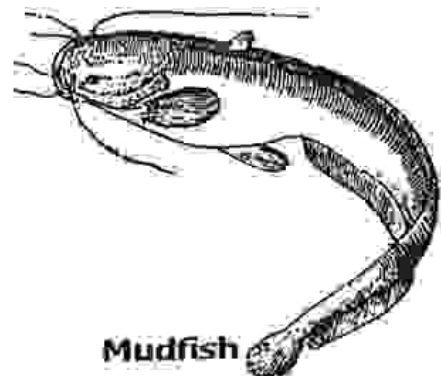
Nile perch



Tilapia

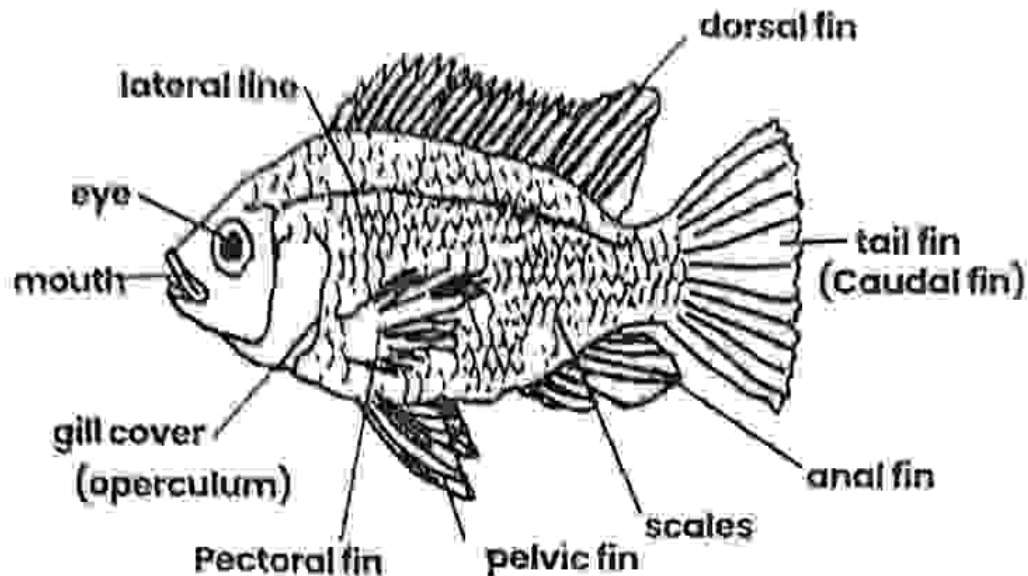


Shark



Mudfish

External parts of a fish



Function of each part the fish

Mouth.

- It used for feeding.
- It used to take in water to extract dissolved oxygen.

Nostril/nare

It is used for smelling food.

NB: the nostril works like the feelers of an insect, nose of humans and tentacles of a snail.

Lateral line

- It is used to detect sound vibrations in water.
- It used to sense danger in water.

NB: the lateral line works like the human ear and feelers of an insect.

Dorsal fin

- It is used for protection.
- The dorsal fin and anal fin (Ventral fin) are used for balancing the fish in water during swimming.
- They prevent the fish from rolling when swimming.
- They enable the fish to swim sideways and stabilize the fish in water.
- The dorsal fin has sharp and pointed spines for protection.

- They are short and large to increase the stability of the fish.

NB: Dorsal and anal fin work like semicircular canals of the human ear, brain and halteres of an insect.

Pelvic and pectoral fins

- They enable the fish to swim upwards and downwards.
- They increase or decrease speed when the fish is swimming.

NB: the pelvic and pectoral fins work like gears of the vehicle.

Caudal/tail fin

- It enables the fish to change direction and move forward.
- It has strong muscles to control the turning of the fish during swimming.

NB: the tail fin works like the steering wheel of a car.

Gills:

- They are used for breathing.
- They work like lungs, spiracles, lung books and stomata in plant leaves.

Gill cover/operculum

- It protects gills from external injuries.
- NB:** the gill cover works like the rib cages.

Scales

- They protect the body of the fish from external injuries and predators.
- The scales are slippery to enable the fish slip off the predator.
- The scales are arranged in the way that they overlap.

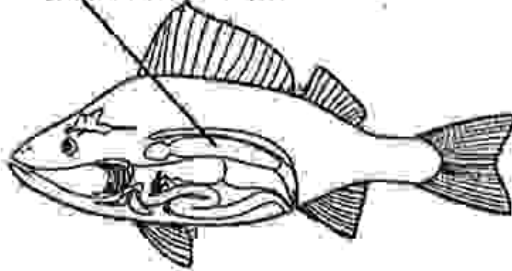
Reason. In order to reduce friction in water during swimming.

Importance of fish.

- They are source of food to people.
- They are source of income when sold.
- Fish bones are used to make animal feeds and glue.
- Fish farming is a source of employment to people.
- They feed on harmful insects like mosquitoes.

Note: A fish has a swim bladder to keep it buoyant in water

swim bladder



A swim bladder maintains fish buoyancy by adjusting the gas volume within it to match the surrounding water pressure, allowing the fish to remain neutrally buoyant

Reasons why fish are able to live in water.

- Fish have gills for breathing in water.
- Fish take in dissolved oxygen.

Similarities between fish and reptiles.

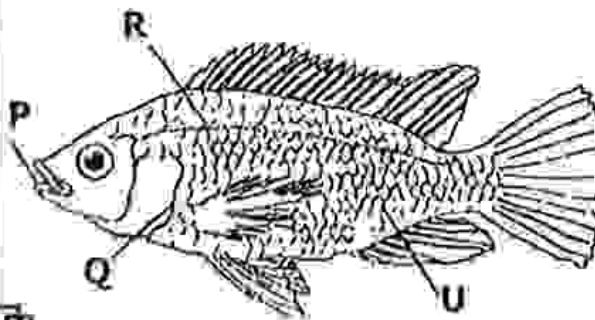
- Both are cold blooded vertebrates.
- Both reproduce by laying eggs.
- Both have scales on their bodies.
- Both have endoskeleton.

Differences between fish and reptiles.

- Fish undergo external fertilization while reptiles undergo internal fertilization.
- Fish move by swimming while reptiles move by crawling.
- Fish live entirely in water while some reptiles live both in water and on land.
- Fish breathe through gills while reptiles breathe through lungs.

Evaluation activity 1.7

1. To which group of vertebrates does a shark belong?
2. State any **two** characteristics of fish.
3. How are the following features useful to a fish?
 - i) Streamlined body.....
 - ii) Slippery body.....
4. Give any **one** reason why fish are able to live in water.
5. Why does a fish die soon after it has been removed from water?
6. The diagram below is of a fish. Use it to answer questions that follow.



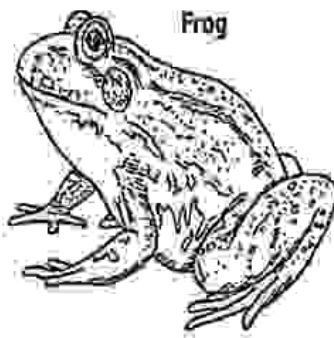
- a) Name the part marked P and U
- b) Which structures are protected by part marked Q?
- c) Identify the part on the fish above that has the same function as the human ear.

AMPHIBIANS

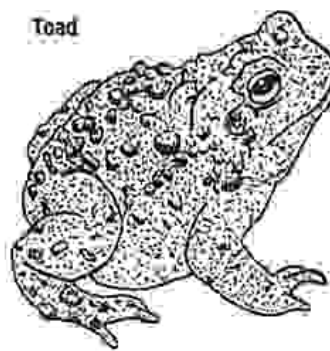
Characteristics of amphibians.

- ✓ They live both on land and in water.
 - ✓ They lay eggs covered in jelly.
 - ✓ They move by hopping or leaping.
 - ✓ They use gills to breathe at their early stages.
 - ✓ They undergo external fertilization except caecilians.
 - ✓ They are cold blooded.
 - ✓ They have webbed toes.
 - ✓ They have external ears.
- Caecilians undergo Internal fertilization.

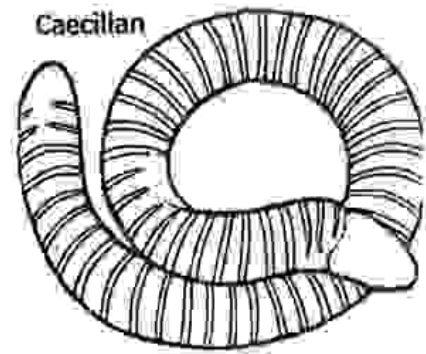
Examples of amphibians



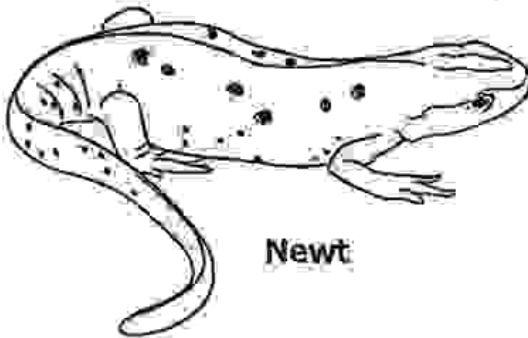
Frog



Toad



Caecilian



Newt

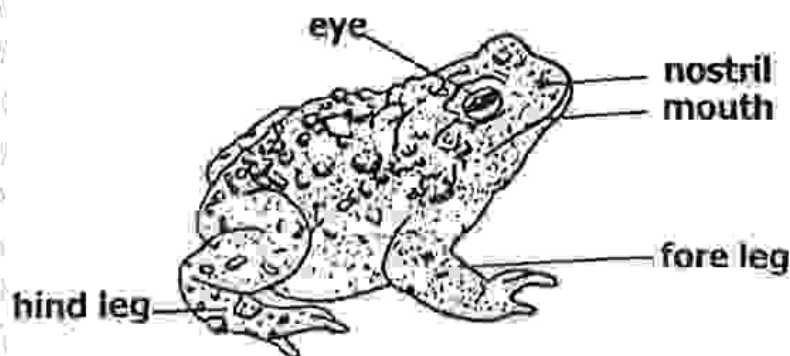


Salamander

Points to note:

- ✍ Caecilians are limbless amphibians while other amphibians have limbs.
- A frog breathes through its moist skin when in water and through its lungs when on land. This makes it to live both in water and on land.

External parts of a toad



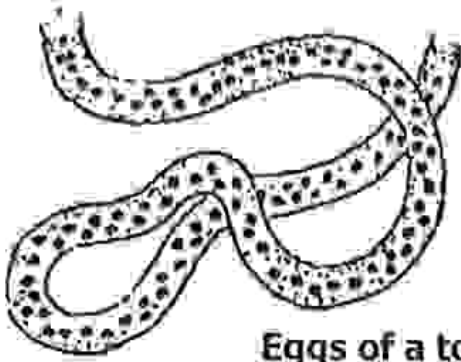
Functions of each part

- Mouth:** for feeding
- Nostril:** For smelling.
- Eyes:** for seeing,
- Legs:** for movement.

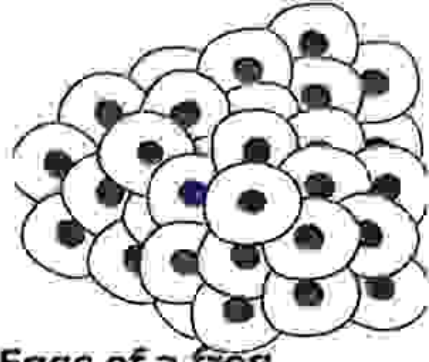
Differences between a frog and a toad

- A frog has brown tadpoles while a toad has black tadpoles.
- A frog moves by leaping while a toad moves by hopping.
- A frog has a smooth slippery skin while a toad has a dry rough skin.

- A frog breathes through lungs and moist skin while a toad breathes only through lungs.
- A frog lives mostly in water while a toad lives mostly on land.
- A frog has fully webbed feet while a toad has partly webbed feet.
- A frog has long hind legs while a toad has short hind legs.
- A frog lays eggs in mass rafts while a toad lays eggs in ribbon like strings.



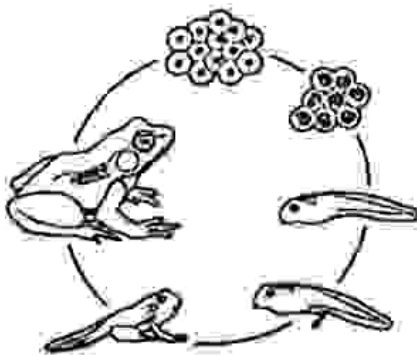
Eggs of a toad



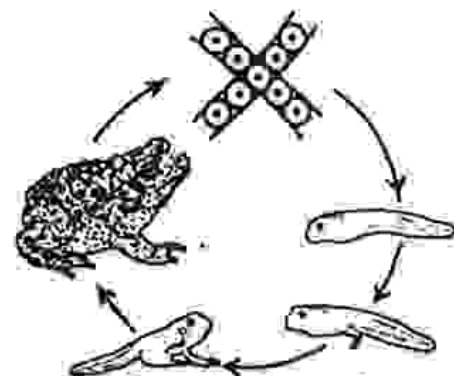
Eggs of a frog

Reproduction in amphibians

- They reproduce by laying eggs which are fertilized externally.
- Eggs laid by amphibians are called spawn.
- Sun heat enables the eggs to hatch.
- The eggs are laid covered with jelly.
- Young amphibians are called tadpoles.
- Tadpoles breathe through gills like fish.
- Tadpoles feed on decaying vegetation in water.



Life cycle of a frog



Life cycle of a toad

Reasons why amphibians lay many eggs

- ❖ To increase chances for fertilization.
- ❖ Some eggs are eaten by predators.

Ways in which jelly is important to the eggs of amphibians.

- o It protects the eggs from being eaten by predators.
- o It prevents eggs from being swept away by water waves.
- o It holds eggs together for easy fertilization.
- o It holds eggs to float on water.
- o It prevents eggs from drying up.

Feeding in amphibians

- Adult frogs and toads are carnivores. They feed on worms, beetles, cockroaches, mosquitoes and houseflies.
- They have long sticky tongue to trap their prey.

Breathing in amphibians

- A frog breathes through lungs and moist skin.
- A toad breathes through lungs.
- Tadpoles breathe through gills.
- *Note: a frog has fully webbed feet for swimming in water.*

How amphibians protect themselves

Frogs have poison glands on their bodies.
Frogs have a slippery body to easily slide off the enemy.

Importance of amphibians

- ❖ They are eaten by some people.
- ❖ They feed on insect vectors like mosquitoes.
- ❖ Meat from amphibians is used as baits for catching fish.
- ❖ Caecilians help to aerate the soil thus improve soil fertility.
- ❖ Caecilians release wastes which add humus into the soil.

Similarities between amphibians and fish

- Both are cold blooded vertebrates.
- Both undergo external fertilization.
- Both lay eggs for reproduction.
- Both have endoskeleton.

Differences between fish and amphibians.

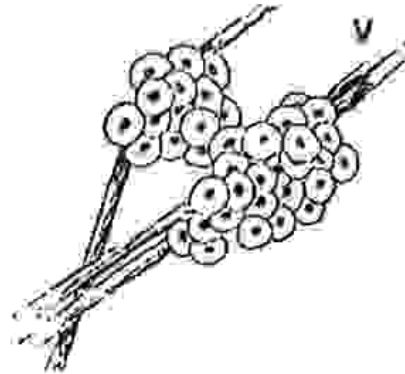
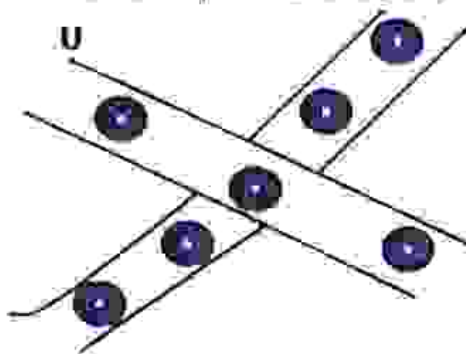
- Fish have scales while amphibians have moist or rough skin.
- Fish have lateral line to detect sound while amphibians use external ears to detect sound.
- Fish use fins for movement while amphibians use limbs for movement.
- Fish live their entire life in water while amphibians live both in water and on land.

Evaluation activity 1.8

- 1.a) Write down any **two** reasons why toads are grouped under amphibians.
b) In which **two** ways are fish different from amphibians?
2. Give a reason why amphibians are referred to as cold blooded vertebrates.
3. Write down any **two** examples of vertebrates that help to control malaria at home.
4. Give any **two** reasons why amphibians lay eggs covered in jelly.
5. Why is a frog able to live both in water and on land?
6. Give any **one** reason why amphibians lay many eggs.
7. How are caecilians different from other amphibians in their body structure?
8. Which class of vertebrates use external gills for breathing in their early growth stage?
9. Give **one** reason why a frog is able to live in water.

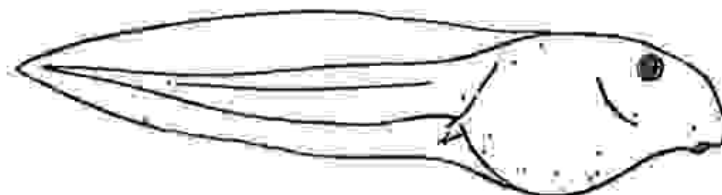
Evaluation activity 1.8 ... Continued

10. State any **one** difference between a frog and a toad.
11. The diagram below shows eggs laid by amphibians. Study and use it to answer the questions that follow.

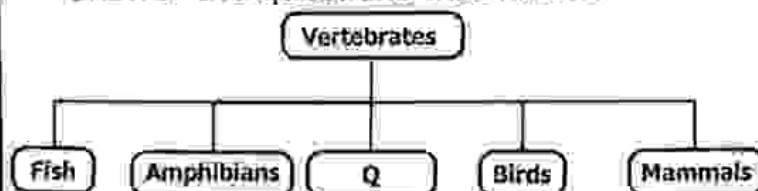


- a) Name the amphibian that lays eggs marked **U** and **V**.
- b) How are the above eggs protected from predators?
- c) What type of fertilization do eggs marked **U** and **V** undergo?

The diagram below shows the stage in the growth of vertebrates. Use it to answer questions **12** and **13**.



12. Which class of vertebrates develop from the stage shown below?
13. In which way is the breathing of the above stage similar to that of a tilapia?
14. Give any **two** ways in which amphibians are important to people.
15. a) To which class of vertebrates do toads belong?
- b) Give **one** way in which the breathing of a toad is different from that of a tadpole.
- c) State any **two** ways in which the jelly round the eggs of toads is important.
16. The diagram below shows classes of vertebrates. Study and use it to answer the questions that follow.



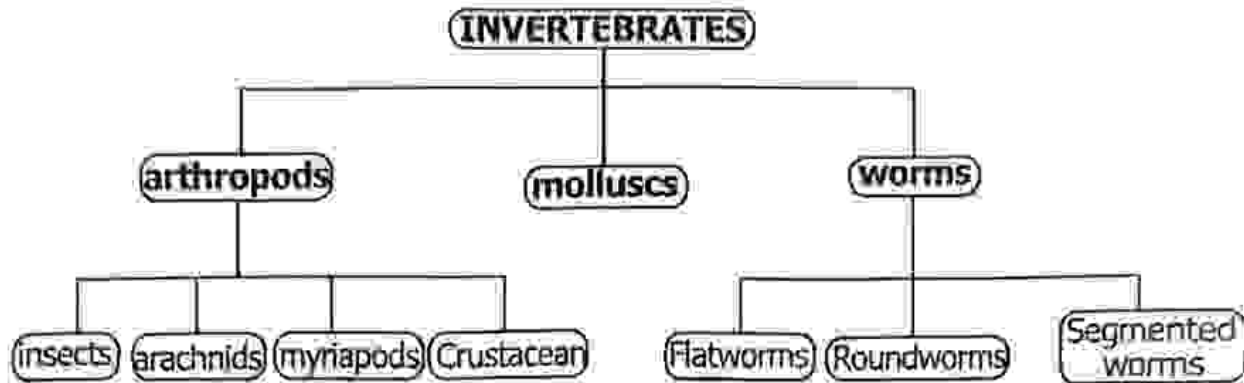
- (a) Name the class of vertebrate represented by **Q**.

- (b) Apart from having a backbone, give **one** characteristic common to;
- (i) All fish, amphibians and vertebrates in class **Q**.
- (ii) Birds and mammals.
- (c) State the difference in reproduction between vertebrates in class **Q** and amphibians.

NON-VERTEBRATES (INVERTEBRATES)

Invertebrates are animals that have no backbone.

Classification table for invertebrates.



Groups/classes of invertebrates

➤ arthropods

➤ molluscs

➤ worms

ARTHROPODS

Arthropods are invertebrates that have segmented bodies and jointed legs.

This is the largest group of invertebrates

➤ They have jointed legs.

➤ They have segmented bodies.

➤ They have exoskeleton.

➤ They moult to grow.

Mnemonic used: **JESM**

Groups of arthropods

▪ insects

▪ arachnids

▪ crustaceans

▪ myriapods

INSECTS

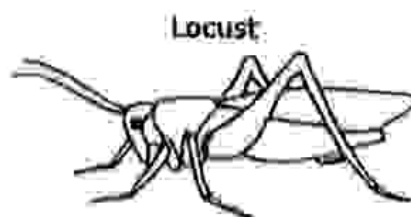
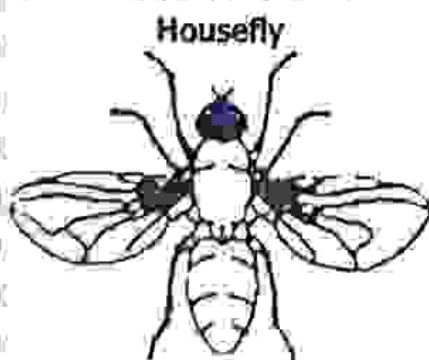
Characteristics of insects

❖ They have three main body divisions.

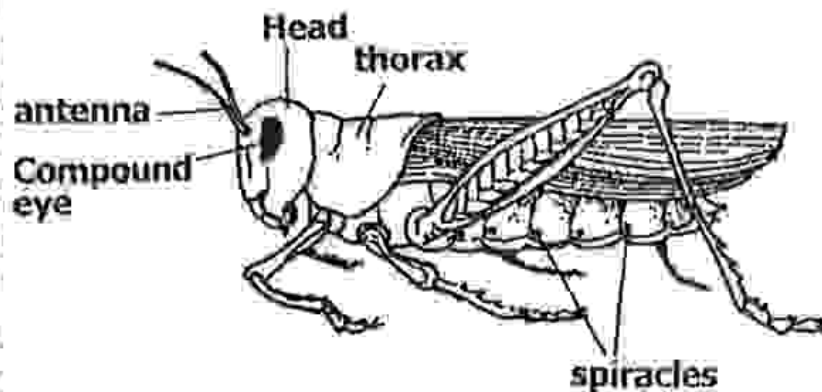
❖ They breathe through spiracles.

❖ They have three pairs of jointed legs.

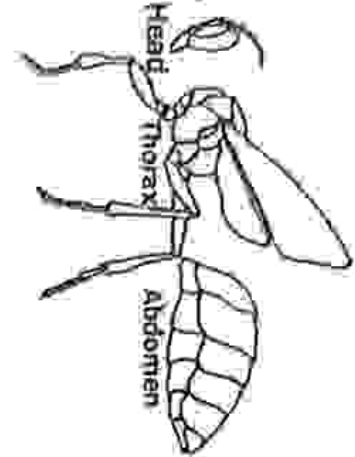
Examples of insects are: houseflies, bees, grasshoppers, locusts, cockroaches, mosquitoes...



The structure of an insect



3 Main body divisions



Functions of each part of an insect

Antennae/feelers

- They are used for feeling.
- They are used for detecting sound vibrations.
- They are used for detecting change in temperature.
- They are used for sensing danger.
- They are used for smelling and tasting.

NB: The feelers work like human ears, lateral line and nostril of the fish, human skin, the tongue, human nose and tentacles of a snail.

Thorax: provides attachment for wings, halteres and legs.
Halteres balance the body of an insect while flying.

Spiracles

They are used for breathing.

Note: when an insect passes through water or flour, it dies.

Reason. Water or flour closes the spiracles of an insect.

Spiracles work like stomata, lungs, lung books and gills.

Compound eyes. For seeing.

Wings. They are used for flying.

Ovipositor. It is used for laying eggs in female insects.

Proboscis/mandibles . For feeding.

Some insects have proboscis for sucking fluids

e.g. houseflies, mosquitoes, bees and tsetse flies.

While others have mandibles for chewing e.g. grasshoppers, cockroaches, locusts and termites.

Reproduction in insects

Insects reproduce by laying eggs except tsetse flies.

They undergo internal fertilization.

Metamorphosis

Metamorphosis is the change in body structure as an insect passes through different stages of its life cycle.

It is an example of a biological change that causes growth in animals.

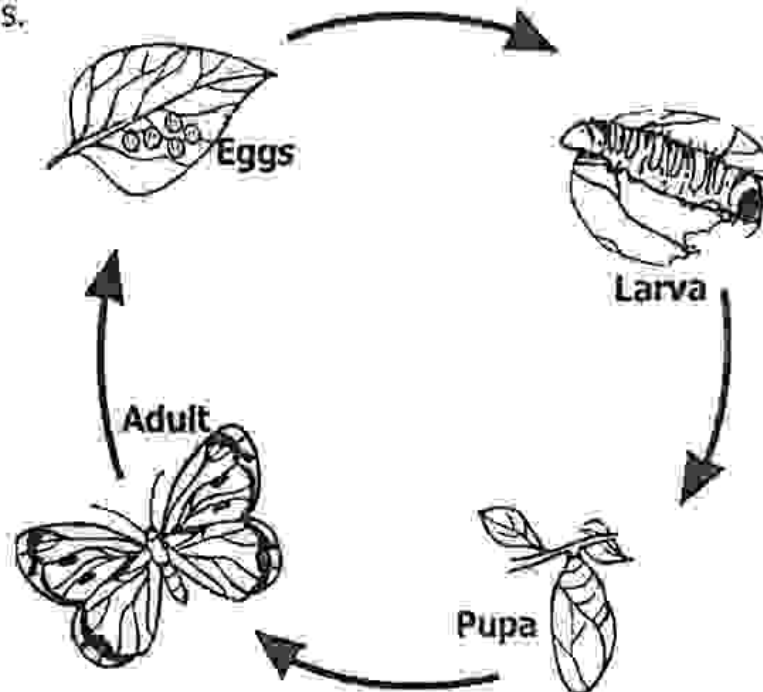
Types of metamorphosis

Complete metamorphosis

In this type of life cycle, the larva stage is totally different from the adult in structure and function.

These stages are eggs, larva, pupa and adult.

Most sucking insects (insects with proboscis) undergo complete metamorphosis.



Examples of insects that undergo complete metamorphosis

Houseflies, bees, butterflies, moths, tsetse flies, fleas, mosquitoes, wasps and beetles. Note. (Lice undergo incomplete metamorphosis though they have proboscis).

Points to note:



- Butterflies lay eggs under plant leaves.

Reasons. To protect them from direct sunshine//To protect them from predators

- They lay eggs on leaves to enable caterpillars get food after being hatched.

- The caterpillars feed on a lot of food so that food can be used in pupa stage.

- The pupa stage is dormant.

Reasons. It neither feeds nor moves

- Also eggs are dormant.

- The pupa stage of a butterfly is called chrysalis.

- The larva stage of a butterfly is called caterpillar. They eat up crop leaves.

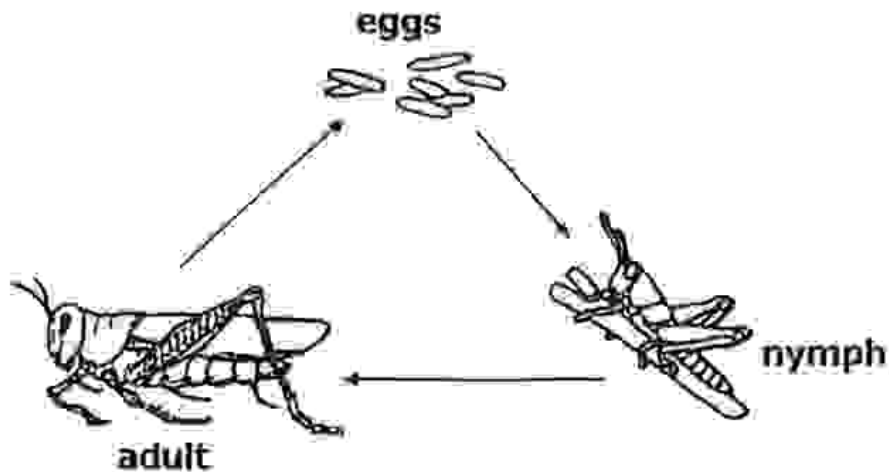
- The adult stage pollinates farmer's crops.

Incomplete metamorphosis

In this type of lifecycle, the nymph resembles the adult stage. i.e. the nymph in grasshopper and cockroach resembles the adult in structure.

Their difference is that the nymph does not have wings while the adult has wings.

All insects which have mandibles (chewing insects) undergo incomplete lifecycle.



Examples of insects that undergo incomplete metamorphosis.

- grasshoppers ➤ locusts ➤ cockroaches ➤ crickets
- dragon flies ➤ praying mantis

Importance of insects.

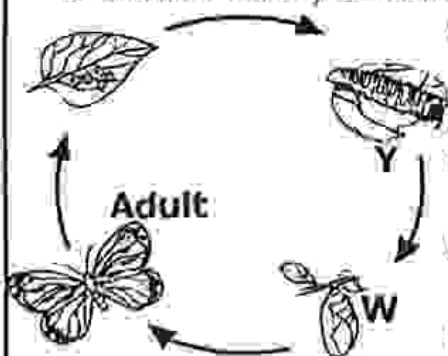
- ❖ Some insects are source of food.
- ❖ Bees provide honey and wax to people.
- ❖ Some insects pollinate flowers.
- ❖ Some insects are sold for money.

Dangers of insects.

- ✓ Some insects are crop pests.
- ✓ Some insects sting people.
- ✓ Insects like cockroaches destroy clothes, books
- ✓ Some insects spread germs.
- ✓ Some insects are ectoparasites.

Evaluation activity 1.9

1. Name the biological change that causes growth in insects.
2. What type of life cycle does a tsetse fly undergo?
3. Why do butterflies lay their eggs under plant leaves?
4. The diagram below shows a type of life cycle in insects. Study and use it to answer the questions that follow.



- a) What type of life cycle is shown in the diagram above?
- b) Name the stages marked **X** and **W**.
- c) How is the stage marked **Y** destructive to a crop farmer?
5. Write down any **two** ways in which insects are important to people.
6. How is the life cycle of a locust different from that of a flea?
7. Why do butterflies lay their eggs on leaves of the plant?
8. How are tsetse flies different from other insects in the way they reproduce?
9. Give any **two** ways in which some insects are dangerous to people.

Evaluation activity 1.9... Continued

10. The diagram below shows an arthropod. Study and use it to answer the questions that follow.



- a) In which group of arthropods does the animal above belong?
- b) How is the above arthropod useful to a crop farmer?
- c) What name is given to the larva stage of the above arthropod?
- d) What type of metamorphosis does the arthropod above undergo?

ARACHNIDS

Characteristics of arachnids

- ✓ They have two main body divisions i.e. cephalothorax and abdomen
- ✓ They have four pairs of jointed legs.

Examples of arachnids



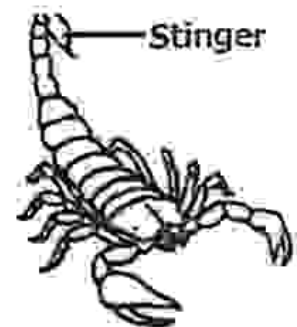
Spider



mite



tick



Stinger

scorpion

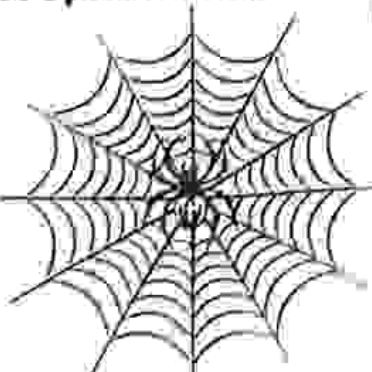
 **Note the following:**

Spiders and scorpions use book lungs for breathing while ticks and mites use spiracles for breathing.

Scorpions produce live young ones while other arachnids lay eggs.

Scorpions have a poisonous stinger for protection.

A spider web



Uses of spider web to a spider

- ✓ For trapping its prey.
- ✓ For shelter
- ✓ For defending itself.

 **Note:** spiders feed on insect vectors like houseflies, cockroaches and mosquitoes.

Importance of arachnids.

- ✓ Scorpions are eaten by some people.
- ✓ Spiders control disease vectors like mosquitoes.

Dangers of arachnids

- ❖ Ticks and mites spread diseases to other organisms.
- ❖ Scorpions sting people causing a lot of pain.

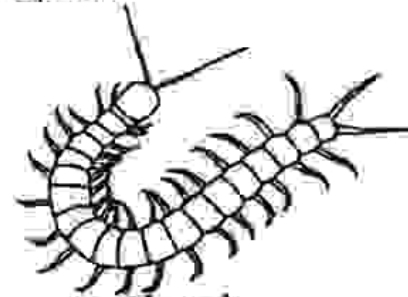
MYRIAPODS

Myriapods are arthropods with many jointed legs and segmented bodies.

Examples of myriapods are; millipede and centipede



millipede



centipede

Points to note:

- *A centipede feeds on insects and worms while a millipede feeds on plant materials. Thus a centipede is a carnivore while a millipede is a herbivore.*
- *A millipede protects itself by curling/coiling while centipede uses its poisonous claws to sting an enemy.*
- *A millipede also produce bad smell for protection.*
- *Both a millipede and a centipede breathe through spiracles located along the sides of their bodies.*
- *Millipedes and centipedes aerate the soil by burrowing into the soil.*
- *Their wastes add humus into the soil.*

CRUSTACEANS

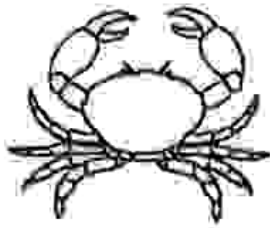
Crustaceans are arthropods with a hard crusty skin.

Characteristics of crustaceans

- ↓ They have a hard crusty skin.
 - ↓ They breathe through gills.
 - ↓ They have many jointed legs.
 - ↓ They have two main body divisions i.e. cephalothorax and abdomen
 - ↓ They have open circulatory system.
- Blood does not flow through a closed loop(no blood vessels) Blood flow to the heart through small holes called ostia.

Examples of crustaceans are; crabs, lobster, woodlice, shrimps, Cyclops, barnacles, crayfish, sand hopper, and prawn

Examples of crustaceans



Crab



woodlouse



lobster

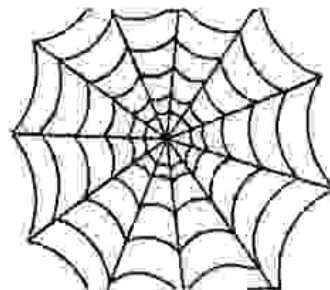
Importance of crustaceans

- They are source of food to some people.
- They are source of income when sold.

Evaluation activity 1.10

1. Which group of invertebrates have segmented bodies and jointed legs?
2. Under which group of arthropods is;
 - i) Crab?
 - ii) Locust?
 - iii) Tick?
 - iv) Centipede?

The diagram below shows a structure built by a spider. Use it to answer questions 3 and 4.



3. Name the structure shown above.
4. Give any **one** way in which the structure shown above is useful to a spider.

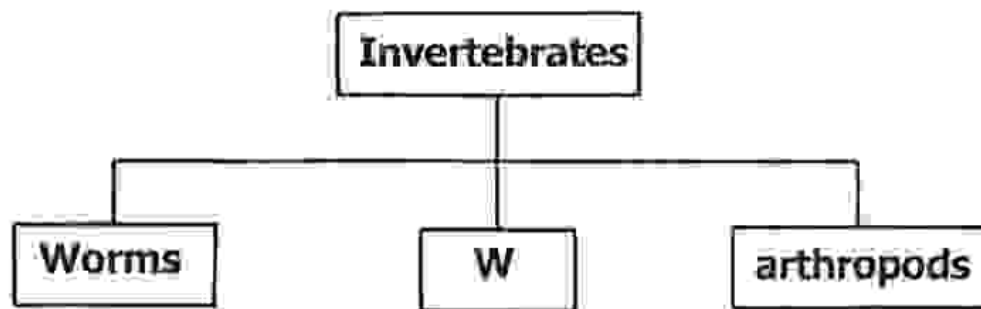
The list below shows animals belonging to different classes of arthropods. Study and use it to answer the questions that follow.

Shrimp, lady bird, scorpion, lobster

- a) To which class of arthropods does a lady bird from the list above belong?
- b) How is the scorpion from the list above different from other arthropods in the way they reproduce?
- c) Name any **one** animal from the list above which is a crustacean.
- d) Which class of arthropods is not represented on the list above?

Evaluation activity 1.10 ...Continued

6. The table below shows groups of invertebrates. Study and use it to answer the questions that follow.



- Name the group of invertebrates represented by **W**.
 - State any **one** similarity between invertebrates at **W** and worms.
 - Mention any **one** invertebrate that can be put at **W**.
 - Identify the group of invertebrates from the table above to which a termite belongs.
7. The diagrams below show animals in arthropod group. Study and use them to answer the questions that follow.



- Name the group of arthropods to which animals **X** and **Y** belong.
- Give any **one** reason why animal **Y** does not belong to the same group as animal **Z**.
- How are animals **X**, **Y** and **Z** similar in the way they reproduce?

MOLLUSCS

Molluscs are soft bodied invertebrates.

Characteristics of molluscs

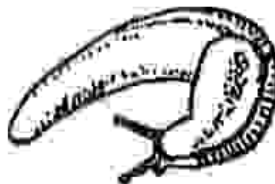
- ✓ They have soft unsegmented bodies.
- ✓ They have a hard shell or mantle for protection.
- ✓ Some live in water e.g. oysters, octopus and cuttlefish while others like snail and slugs live both in water and on land.
- ✓ They have hydrostatic skeleton.
- ✓ They have tentacles.
- ✓ Molluscs feed on plant materials and dead water animals.

Don't speak for Quality, let Quality speak for itself

Examples of molluscs are; snail, slug, octopus, oyster, clam, mussel, cuttlefish, winkle, squid and cowrie.



Snail

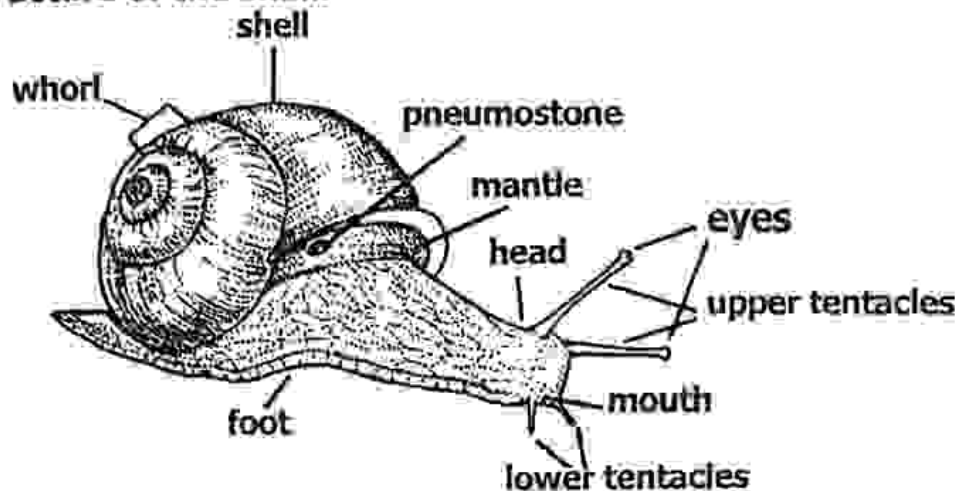


slug



Octopus

The structure of the snail.



Functions of parts of a snail.

Shell: It is used for protection against its enemies.

Eyes: They are used for sight.

Tentacles: They are used for feeling, sensing danger, sucking food and smelling.

Foot: It is used for movement.

A snail secretes slime which helps it to reduce friction during movement and protects its body against scratches on the ground.

A snail moves by crawling.

Importance of molluscs.

↑ They are eaten by some people.

↑ They provide shells for making ornaments.

↑ They provide shells which are used to enrich mash with calcium.

Dangers of molluscs

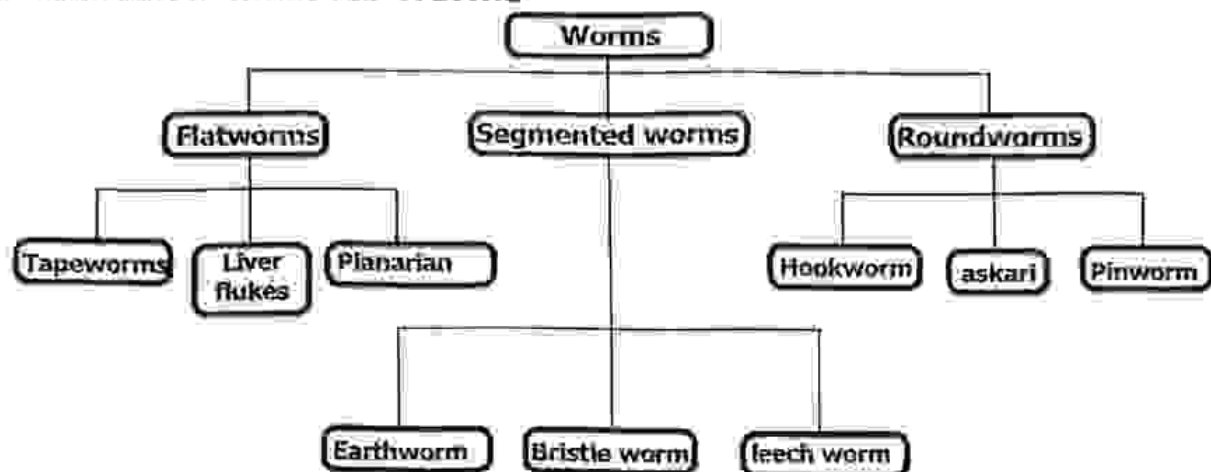
- Fresh water snails spread bilharziasis.

- Land snails are crop pests.

WORMS

Worms are long, thin and soft bodied invertebrates.

Classification chart for worms



Characteristics of worms.

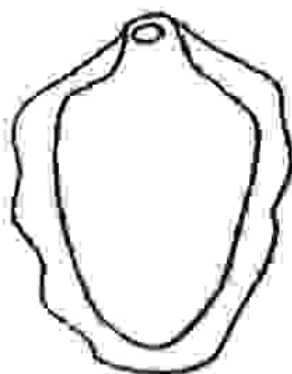
- ✓ They use their moist skin for breathing.
- ✓ They have hydrostatic skeleton.
- ✓ They reproduce by laying eggs.
- ✓ They have no legs.

Groups of worms

Group	Characteristics	Examples
Flatworms (Platyhelminthes)	• They have flat and segmented bodies. • They are parasites to animals.	Tapeworm , liver flukes and planarian
Segmented worms (annelids)	• Their body is made up of rings. • They live in soil and water.	Earthworm, leech worm, Lugworm and bristle worm
Roundworms (nematodes)	• They have a round shaped body. • They are not segmented	Hookworm, askari, Pinworm, filarial worm, eelworm

Diagrams showing different types/groups of worms

Flatworms



Liver fluke



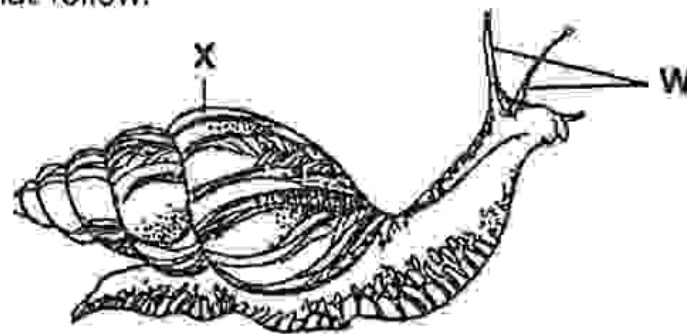
Tapeworm

CARING FOR AND PROTECTING VERTEBRATES AND INVERTEBRATES

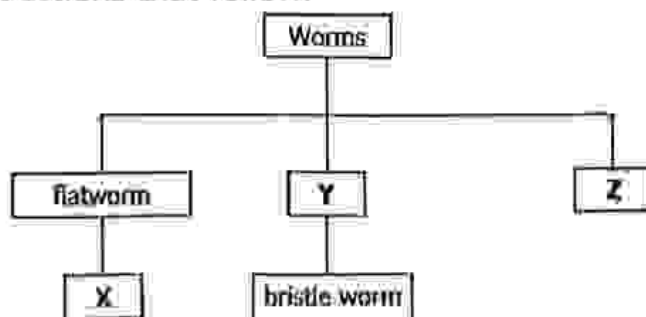
- ❖ Giving enough food and water to animals.
- ❖ Treating animals when they are sick.
- ❖ Vaccinating and deworming animals.
- ❖ Building houses for domestic animals.
- ❖ Protecting and conserving habitats for wild animals.
- ❖ Enforcing laws against poaching.

Evaluation activity 1.11

1. Name the group of invertebrates to which an octopus belongs.
2. (a) Name the group of worms to which the following belong.
 - i) Liver flukes..... ii) Hookworms
 (b) Apart from improving soil aeration, identify any other **two** ways in which earthworms are important in the soil.
3. How are molluscs different from all other invertebrates?
4. Why does an earthworm die when oil is poured on its body?
5. The diagram below is of an invertebrate. Study and use it to answer the questions that follow.



- a) Name the part labeled **X**.
- b) Give any **two** functions of part labeled **W** to the invertebrate above.
- c) Identify the group of invertebrates to which the animal above belongs.
6. Give an example of a mollusc which is a vector to man.
7. State the mode of reproduction in molluscs.
8. Why do earthworms come out of the soil after it has rained?
9. The table below shows groups of worms. Study and use it to answer the questions that follow:



- (a) Identify the group of worms put at **Y** and **Z**.
- (b) Name any **one** example of worm that can be put at **X**.
- (c) State the difference in structure between the bristle worm and the group of worms at **Z**.

Theme: **MATTER AND ENERGY**
Topic **2: SOUND ENERGY**

Sound is a form of energy produced by vibration of an object.
The ear uses sound energy in order for it to function.

Sound is called a form of energy because;

- Sound has the ability to do work.
- Sound can be transformed into other forms of energy.

Ways in which sound is important.

- ✓ For communication.
- ✓ For locating objects.
- ✓ For education through songs, debates, talk shows and drama.
- ✓ For entertainment.
- ✓ For income when music is sold.
- ✓ For evidence in courts of law.

Dangers of sound to people.

- It causes noise pollution.
- It can lead to partial deafness.
- It can lead to loss of concentration at work.

SOURCES OF SOUND.

Natural sources

- thunder
- waterfall
- wind
- earthquake
- rain
- animals

Artificial sources of sound.

- radios
- TV sets
- pianos
- drums
- bells
- flutes
- keyboards
- phones
- whistles
- guitars

Reasons why some animals make sound.

- To call their young ones.
- To communicate to each other.
- To express their feelings.
- Due to pain.
- To meet their mates.
- To threaten their enemies.
- Due to fear.

Evaluation activity 2.1

1. Name the form of energy that stimulates the sense of hearing.
2. Which body organ uses sound energy in order to function?
3. Give any **one** reason why sound is referred to as a form of energy.
4. Write down any **two** natural sources of sound.
5. In which **two** ways is sound important to people?
6. State any **one** danger of listening to loud sound on human health.

Evaluation activity 2.1Continued

The diagram below shows a person playing a guitar. Use it to answer questions 7 and 8.



7. Identify the form of energy produced by guitar being played by the person above.
8. Give any **one** way in which the person above benefits from playing the guitar,
9. State any **two** reasons why animals make sound.
10. What form of energy is produced when it is raining?

MUSICAL INSTRUMENTS

These are devices that are played to produce musical sound.

Groups of musical instruments.

- Percussion musical instruments
- Wind musical instruments
- String musical instruments.

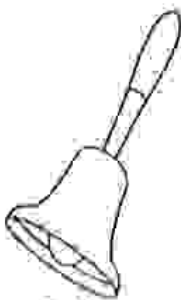
Percussion musical instruments.

These are musical devices that produce sound by vibration of their surfaces when hit or shaken.

The methods used to produce sound are; hitting/beating/striking and shaking.

Examples of percussion musical instruments.

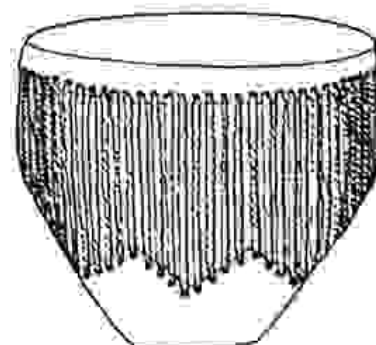
- | | | | |
|--------------------------------|----------------|--------------|------------|
| ➤ drum | ➤ xylophone | ➤ shakers | ➤ bell |
| ➤ rattles | ➤ thumb pianos | ➤ brass band | ➤ maracas |
| ➤ castanets | ➤ cymbals | ➤ lambourine | ➤ triangle |
| ➤ hand drum (This uses a lens) | | | |



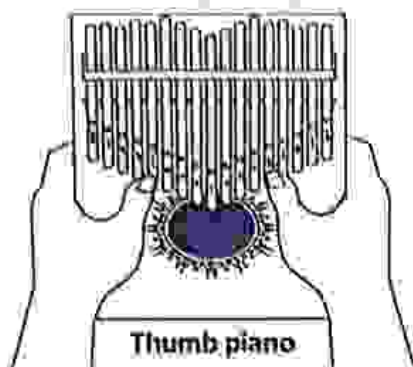
Bell



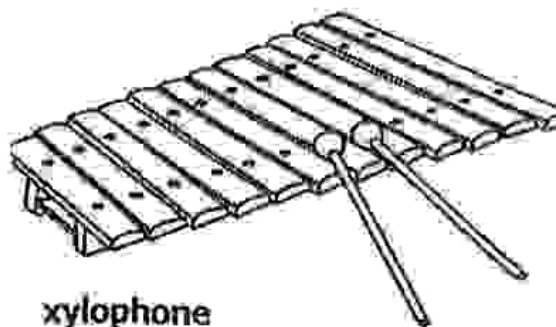
Maracas



Drum



Thumb piano



xylophone

Wind musical instruments

These are musical devices that produce sound by vibration of air molecules blown in them.

The method used to produce sound is blowing.

Examples of wind musical instruments.

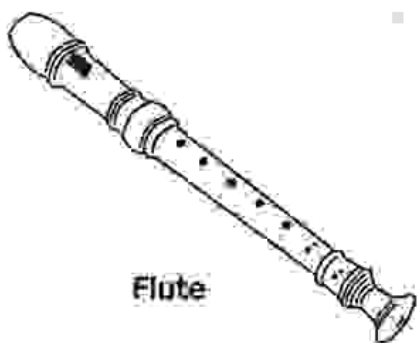
- flute
- bottle
- trombone
- panpipes
- horn
- piccolo
- trumpet
- saxophone
- tuba
- whistle
- bugle
- oboe
- clarinet



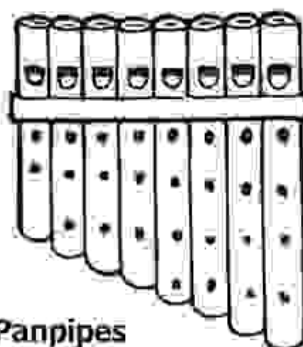
Trumpet



Horn



Flute



Panpipes



whistle

String musical instruments

These are musical devices that produce sound by vibration of strings when plucked or rubbed.

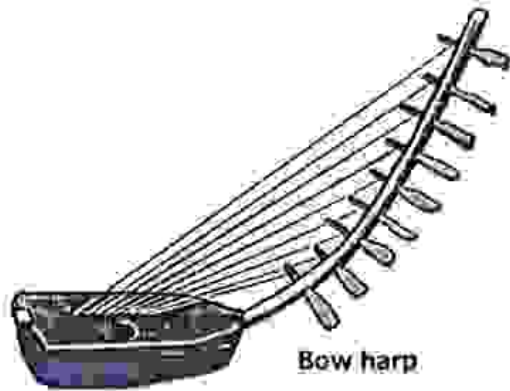
The method used to produce is plucking.

Examples of string musical instruments.

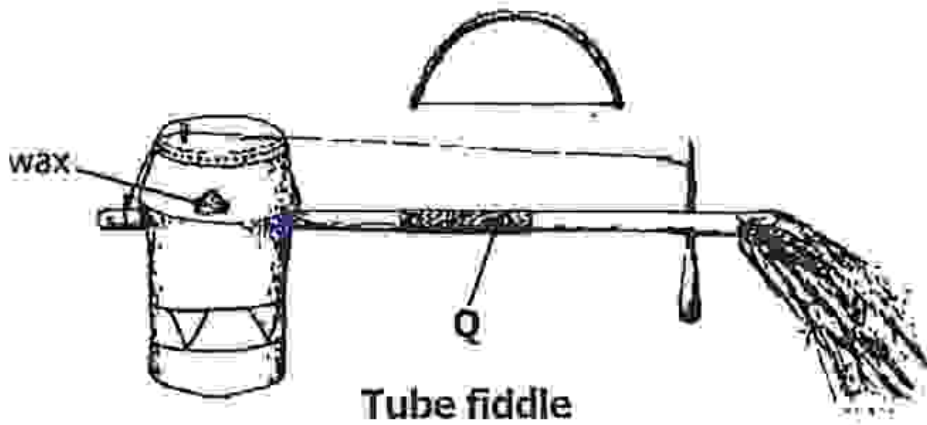
- ❖ guitar
- ❖ lyre
- ❖ bow harp
- ❖ violin
- ❖ banjo
- ❖ tube fiddle
- ❖ cello
- ❖ viola
- ❖ sitar



Guitar



Bow harp



Tube fiddle



Violin



Harp

Evaluation activity 2.2

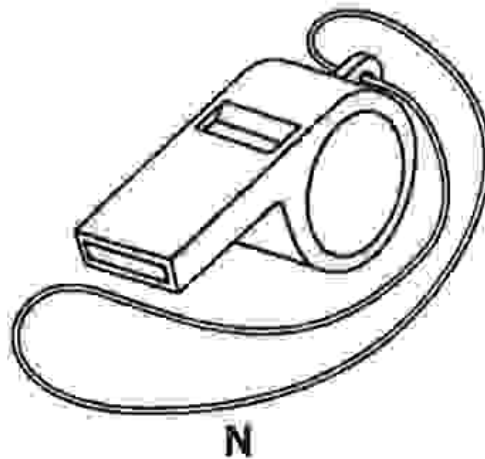
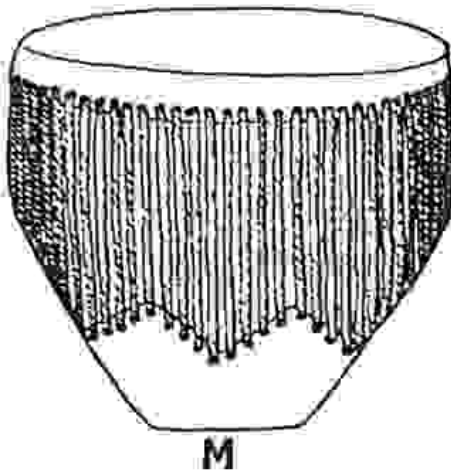
1. Mention any **two** groups of musical instruments.
2. To which group of musical instruments does a saxophone belong?
3. How does a drum produce sound?
4. Mention any **one** method used to produce sound in percussion musical instruments.
5. The list below shows musical instruments. Study and use them to answer the questions that follow.

cymbal, violin, shakers, trumpet

- a) Which **two** musical instruments from the list above can be grouped together?
- b) Give a reason for your answers in a) above.

Evaluation activity 2.2 ... Continued

- c) How does a violin in the list above produce sound?
4. Mention any **two** examples of;
- a) Wind musical instruments
 - b) String musical instruments
6. Which method is used to produce sound in wind musical instruments?
7. The diagrams below show two musical instruments. Study and use them to answer the question that follow.



- a) Identify the group of musical instruments to which device N belongs.
- b) How is sound production in musical instrument M different from that in N?
- c) Write down any **two** musical instruments that can be grouped together with M.

PITCH, FREQUENCY AND VOLUME OF SOUND.

Pitch of sound

Pitch of sound is the highness or lowness of sound.

Factors that determine the pitch of sound.

❖ Frequency.

Frequency is the number of vibrations produced by an object per second.

Frequency is measured in units called **Hertz**.

The higher the frequency, the higher the pitch and the lower the frequency, the lower the pitch of sound.



High pitched note Low pitched note

❖ **Length of the vibrating space or surface.**

The shorter the vibrating surface, the higher the pitch of sound and the longer the vibrating surface, the lower the pitch of sound.

Refer to the bow harp below.



Explanation:

String **A** produces sound of low pitch.

Reason. It is longer thus producing low frequency.

String **J** produces sound of high pitch.

Reason. It is shorter thus produces high frequency.



Points to note. The sound hole allows out sound when plucking the strings.

The pegs are used to tune the strings.

The sound box (resonator) amplifies sound produced by strings.

❖ **Wavelength.**

Pitch of sound increases with the decrease in wavelength.

❖ **Nature of the medium in which vibration occurs.**

Molecules in solids vibrate faster than those in liquids and gases.

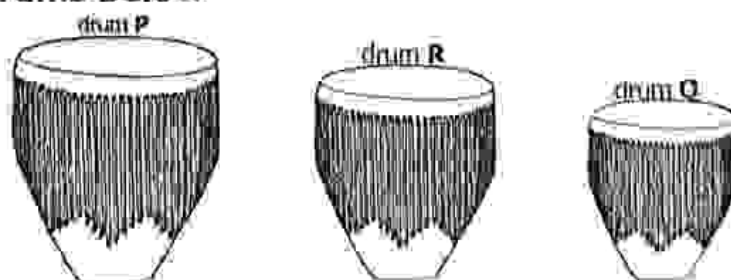
Reason. Molecules in solids are closely packed together.

❖ **Size of the object. / Surface area for vibration.**

Smaller objects produce higher pitch of sound while bigger objects produce lower pitch of sound.

Reason. Smaller objects have a smaller vibrating surface/space compared to the bigger objects.

Refer to the drums below



Explanation:

In the diagrams above, drum **P** will produce low pitched sound.

Reason. Drum **P** has a bigger vibrating surface.

Drum **Q** will produce a high pitched sound.

Reason. Drum **Q** has a smaller vibrating surface.

Point to note: Drums are usually put in sunshine before playing them.

Reason. To increase their pitch/To tighten the skin in order to increase tension.

❖ Tension of the vibrating object.

- Tension is how tight or loose a material is.
- Tight materials vibrate faster thus produce higher pitch of sound.
- Loose materials vibrate slowly thus produce sound of low pitch.

❖ Nature of the vibrating object.

- Metallic objects produce higher pitch of sound than plastic and wooden materials.
- The strings of guitars are made up of metals.

Reason. Metallic strings produce higher pitch of sound.

- Additionally, thick strings produce sound of low pitch while thin strings produce sound of high pitch.

Using bottles to determine pitch of sound

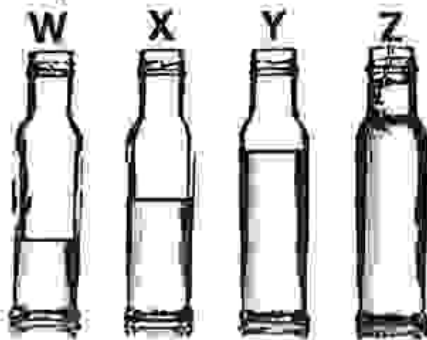
Materials needed:

Water and 4 empty bottles

What to do.

- Get four empty bottles of the same brand and same size.
- Pour different amounts of water in each bottle.
- Blow air into each bottle, one at a time.

Set up:



Findings:

- Bottle **Z** produces the highest pitch.

Reason. It has a small vibrating surface area.

- Bottle **W** produces the lowest pitch.

Reason. It has a bigger space for vibration.

Note: In case water was replaced with milk, the pitch of sound would remain the same.

Reason. Both water and milk are liquids.

Bottles produce sound by vibration of air molecules when blown inside.

VOLUME OF SOUND.

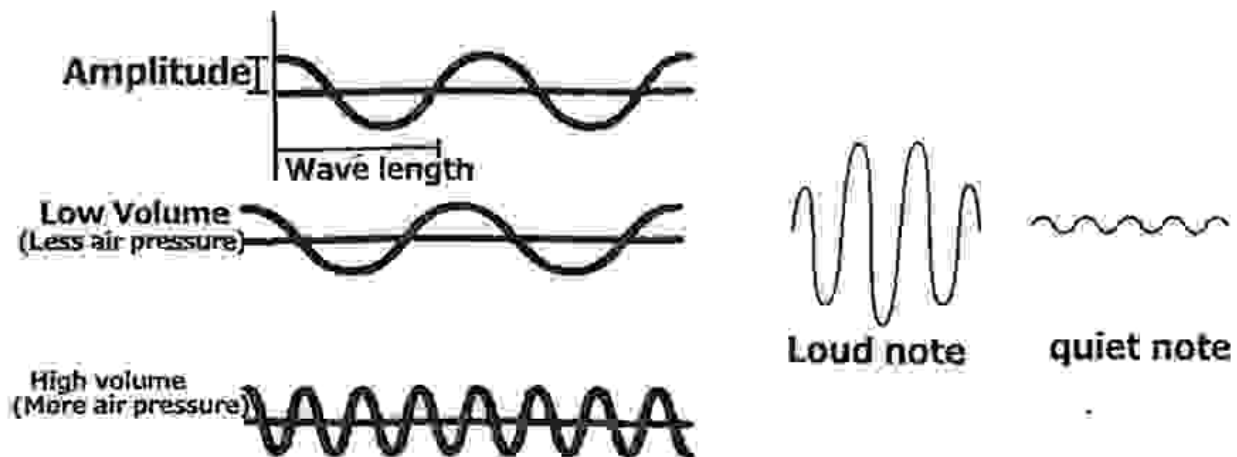
Volume of sound is the loudness or softness of sound.

The force used to produce sound determines the loudness or softness of sound produced by the vibrating object. Loudness of sound is measured in **decibels**.

Examples.

- If a person hits a drum softly, soft sound is produced and if a person hits a drum hard, a loud sound is produced.
 - In hospitals, doctors use a stethoscope to raise the volume of the patient's heartbeat.
 - Some people put ear buds or earphones in their ears to be able to listen to music from their phones loudly without interfering other people.
 - Volume of sound is determined by amplitude or intensity of sound.
- Amplitude is the width of vibration from the point of rest.

Volume of sound



Factors that determine the pitch of sound produced by musical instruments.

Percussion musical instruments	Wind instruments	String instruments
• Nature of the instrument. • Size of the instrument.	❖ Length of the tube. ❖ Space left in the instrument.	• Tension of the strings. • Length of the strings. • Nature of the strings. • Thickness of the strings.

How pitch of some musical instruments be varied/changed.

Musical instrument	How its pitch can be varied.
drum	- By tightening the drum's skin to produce higher pitch. - By putting a thicker skin on the drum to lower the pitch. - By striking the drum closer to the centre to produce lower pitch.
Xylophone	- By altering the length of the slab. - Cutting off part of the arch to lower the pitch. - Chopping off the sides to increase the pitch.

Thumb piano	Adjusting the length of the metal tines. Longer tines produce lower pitch while shorter tines produce higher pitch.
Flute	- By closing and opening the holes at different intervals. - The pitch of the flute can be increased by shortening the air column and lowered by lengthening the air column
trumpet	- By changing the position of the lips on the mouth piece when blowing. - By pushing down one or more valves to increase pitch. - By pulling valves up to lower the pitch.
Guitar and bow harp	- By tightening or loosening the strings.
Panpipes	- By moving the panpipes side by side on lips while blowing. - By reducing their length to increase the pitch of sound
Whistle	By altering the force applied when blowing.

Evaluation activity 2.3

1. What is meant by the term pitch of sound?
2. State any **two** factors that determine the pitch of sound produced by an object.
3. How can the pitch produced by the musical device below be varied?



4. The diagram below shows a musical instrument. Study and use it to answer the questions that follow.



- a) To which group of musical instruments does the device above belong?
- b) Which of the strings on the diagram above will produce sound of high pitch?
- c) Why would the string given in b) above produce sound of the high pitch?
- d) State any **one** factor that determines the pitch of sound produced by the musical device above?

Evaluation activity 2.3.... Continued

5. The diagrams below show two drums of different sizes. Study and use it to answer the questions that follow.

drum W



drum X



- Identify the drum that will produce sound of the highest pitch.
 - Why would the drum you have identified in a) above produce sound of the highest pitch?
 - State any **one** factor that determines the pitch of sound produced by the above drums.
- d) Name the group of musical instruments in which you would put the above drums.
6. Give the meaning of the term frequency in sound.
7. How does frequency affect the pitch of sound produced by an object?
8. In which units are the following measured?
- Frequency of sound
 - Volume of sound
9. In the table below, part **A** shows terms used to describe sound and part **B** has the meanings of the terms

Part A	Part B
Volume	Number of vibrations per second.
Pitch	Reflected sound.
Frequency	Loudness or softness of sound.
Echo	Highness or lowness of sound.

Use the meanings in part B of the table to match the terms below.

- Volume
- Pitch
- Frequency
- Echo

HOW SOUND TRAVELS.

Sound travels through sound waves.

Sound waves travel in all directions from the source and they require a medium for transmission.

The medium can be liquid, gas or solid due to the presence of molecules/particles that vibrate.

A person receives a phone call from another person through electromagnetic sound waves.

 **Point to note:**
Sound does not travel through the vacuum.

Reason. The vacuum has no medium to transmit sound waves.

A vacuum is a space without matter.

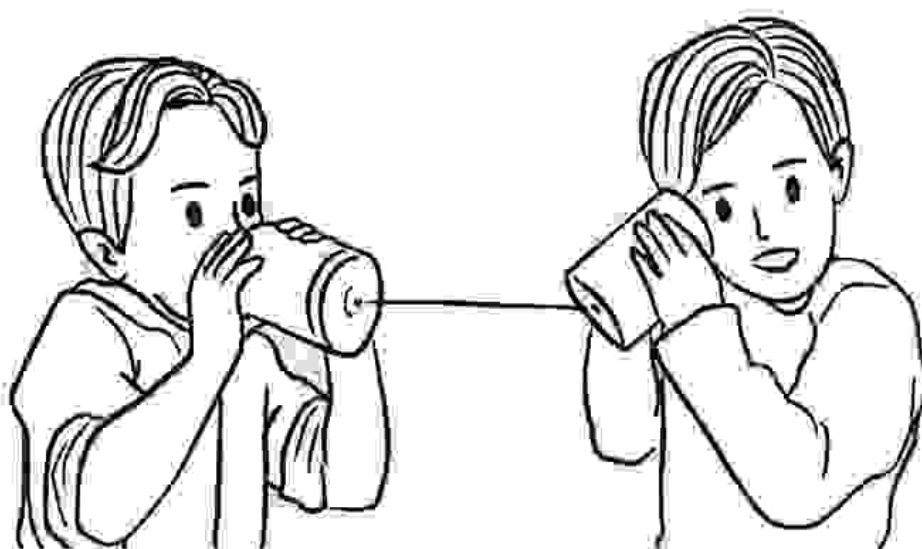
Experiment to find out if sound travels through solids.

Materials needed: 2 empty tins, string

Procedure:

- i. Get two tins and make a hole through each tin at the centre of its bottom.
- ii. Fix a string and tie it through both holes to join both tins with a knot at each end of the string.
- iii. Let one of you talk through the tin while another one listens from the other tin.

Set up:



Observation:

Sound travels through the string (solid) and the person listening is able to hear what the other person is talking. Therefore, sound travels through solids as a medium of sound waves.

THE SPEED OF SOUND.

Sound travels fastest in solids and slowest in gases.

Sound travels slowest in gases because particles in gases are far apart from each other.

Sound travels faster in solids than in liquids and gases.

Reason: Solids have closely packed molecules.

Simple calculations on the speed of sound.

Examples.

1. A man fired a pistol towards a cliff and he heard the echo from the cliff after 8 seconds. If the speed of sound in air is 330m/sec, how far is the man from the cliff?

Solution

$$8 \div 2 = 4 \text{ sec}$$

$$\text{Distance} = \text{speed} \times \text{time}$$

$$\text{Distance} = 330\text{m/sec} \times 4 \text{ sec}$$

$$\text{Distance} = 1320\text{m}$$

Note well:

 Sound travelled from the gun to the cliff and back (2 times)

Remember original sound and an echo move the same speed in the medium.

2. Coccidiosis and Pneumonia are standing across the valley which is 990m away from the hill. If they shout, how long will it take to hear their echoes?

$$\text{Time} = \frac{\text{distance}}{\text{Speed}}$$

$$\text{Time} = \frac{990\text{m} \times 2}{330\text{m/sec}}$$

$$\begin{aligned} \text{Time} &= 3 \times 2 \\ \text{Time} &= 6 \text{ sec} \end{aligned}$$

Note well:

 The distance is multiplied by 2 because the sound covers the same distance when moving to and fro.

Factors that affect the speed of sound.

Temperature:

High temperature increases the speed of sound while low temperature reduces the speed of sound.

People are able to hear clearly at night because the temperature is low which makes sound waves to travel near the ground level.

It is the same reason why radio signals are clearer at night than during day time.

The heat during day time makes the sound waves to rise high. This makes it difficult to hear sound clearly during day time.

Altitude:

The speed of sound reduces on higher altitude and increases in places with lower altitude.

Direction of wind.

Wind can carry sound waves if it is blowing in the same direction.

Wind obstructs sound and it becomes fainter and obstructed.

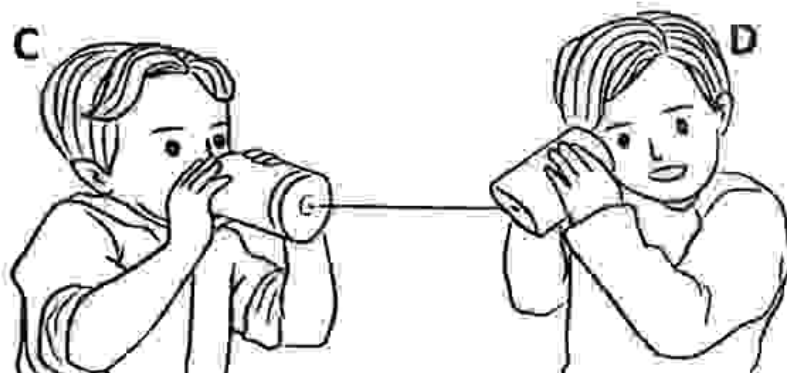
❖ Mnemonic used is **WHAT** i.e wind, humidity, altitude, temperature.

Evaluation activity 2.4

1. Name the form of energy that requires a medium of transmission.
2. How does sound travel?
3. Why is sound unable to travel through the vacuum?
4. In which state of matter does sound travel fastest?
5. Give a reason why radio waves are clearer at night than during day time.

Evaluation activity 2.4... Continued

The diagram below shows two boys demonstrating using a string telephone. Study and use it to answer the questions that follow.



- Which medium of sound transmission is being demonstrated by the boys above?
 - How is the boy at point D able to hear sound from the boy at point C?
 - Why did the boys use the string to connect the two tins?
 - What is the purpose of the experiment demonstrated by the boys above?
- Write down any **two** factors that affect the speed of sound.
 - Why does sound travel fastest in solids compared to other states of matter?
 - Identify the state of matter in which sound travels slowest.

ECHOES

An echo is a reflected sound.

An echo is formed by obstruction of sound waves.

An echo is formed when sound waves hit a barrier or a solid surface.

These include; hard wall, cliff, hill or mountain.

Sound produced as echoes may not be heard clearly.

Reason. Echoes have low waves and their pitch is low.

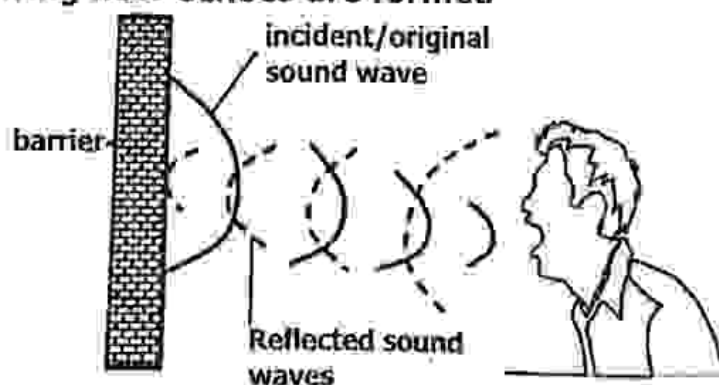
Echoes cannot be heard in a room full of people.

Reason. People's bodies and clothes absorb sound waves.

Point to note.

 Sound energy, heat energy and light energy are forms of energy that can be reflected. Both an echo and an image are results of reflection.

Diagram showing how echoes are formed



Importance of echoes in the environment.

- ✓ Sailors use echoes to determine the depth of water bodies. Sailors use a fathometer (echosounder) to measure the depth of water bodies. It can also be done using the SONAR (Sound Navigation and Ranging)
- ✓ Echoes help sailors to dodge obstacles on their way.
- ✓ Pilots use echoes to avoid accidents in mountainous areas
- ✓ Echoes help bats to locate their prey.
- ✓ Echoes help bats to find their way. This is called echolocation.
- ✓ Echoes help whales, dolphin and seals to dodge obstacles when swimming in water.
- ✓ Echoes help blind people to dodge obstacles on their way using an echostick.
- ✓ Echoes help fishermen to locate the shoals of fish in water bodies.

Disadvantages of echoes in the environment.

- Echoes can distort message.
- Echoes can cause accident.
- Echoes interfere with music.
- Echoes cause noise pollution.

Ways in which echoes can be reduced in the cinema halls.

- ❖ Putting softboards on ceilings.
- ❖ Covering walls with thick and soft clothes.
- ❖ Use of thick curtains in windows and doors.
- ❖ Covering the floor with thick carpets.
- ❖ Covering seats with cushions.

Materials used to reduce echoes in cinema halls.

Soft boards, thick curtains, thick carpets, thick clothes, blankets, bamboo and cushions

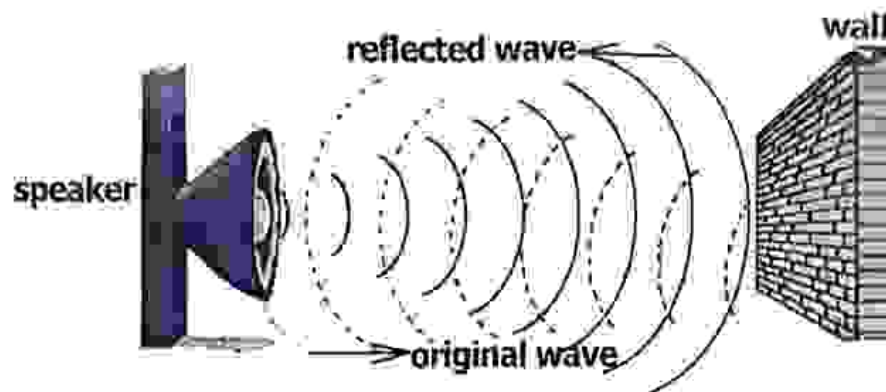
The above materials absorb sound waves.

Evaluation activity 2.4

1. Name the sound formed when a person shouts in an empty hall.
2. Explain how sound is formed. (02 marks)
3. Why are echoes not heard in a classroom when learners are reading a passage?
4. Name any **one** mammal that makes use of echoes.
5. In which **one** way are echoes useful to sailors?
6. Write down any **two** places where echoes can be heard.
7. Why are cinema hall covered with thick porous materials?
8. State any **two** ways in which echoes can be reduced in cinema halls.
9. How are echoes useful to pilots while flying an aeroplane?
10. Give any **two** materials that can be used to reduce echoes in cinema halls.

Evaluation activity 2.4... Continued

The diagram below shows a property of sound. Use it to answer questions 11 and 12.



11. Identify the property of sound shown above.
12. In which **one** way is the above property of sound useful to a bat?
13. How does covering the music recording studios with thick carpets and curtain help to control echoes when recording music?

HOW SOUND IS PRODUCED, STORED AND REPRODUCED

How sound is produced

Sound is produced by vibration of an object.

Vibration is to and fro movement of an object or a particle.

Examples of vibrating objects include; ruler, pen, knife, string, stick, panga and spring.

Music is an organised sound produced by regular vibrations while noise is a disorganised sound produced by irregular vibrations.

How different animals produce sound.

Humans. By vibration of vocal cords.

Bees. By rapid flapping of their wings.

Grasshoppers. By rubbing their hind legs or wings together.

Toads. By inflating their vocal sacs.

Birds. By vibration of the rings of cartilage in the trachea.

EXPERIMENT

Producing sound using a ruler.

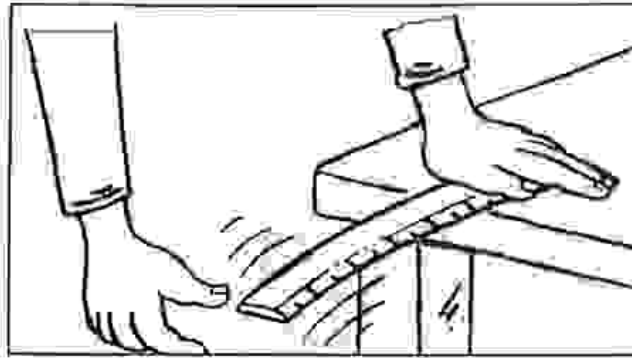
Materials needed:

A ruler, a table/desk.

Procedure:

- i. Put the ruler at the edge of a table.
- ii. Let a bigger part of the ruler stick beyond the edge of the table/desk.
- iii. Place one hand on the part of the ruler that is on the table/desk and press hard.
- iv. Use the other hand to pluck the free end part of the ruler.
- v. The ruler vibrates and produces sound. Thus sound is produced by vibration of the ruler when plucked.

Set up:



Note. The pitch of sound produced by the ruler can be changed by increasing or decreasing the length of the ruler.

Increasing the length of the ruler reduces the pitch of sound produced by the ruler.

Decreasing the length of the ruler increases the pitch of sound produced by the ruler.

Methods of storing sound

Mechanical method

Sound is stored on plastic discs in form of grooves made on the side by physical means.

This can be done using a gramophone, vinyl record, wood record, paper record and wax cylinder.

Sound can be reproduced by the movement of the needle along the grooves as the disc turns.

Magnetic method

Sound is stored on plastic objects coated with magnetic materials. Such plastic objects include radio cassette tapes and magnetic recording tapes. Sound can be reproduced by the turning of the magnetized tape over the play head (coil).

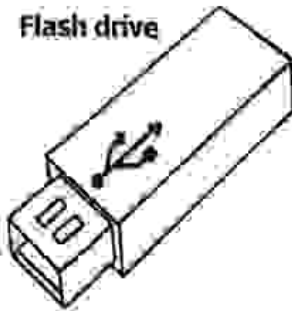
Electromagnetic method

Sound is stored on plastic discs with the use of laser light. Such plastic discs include Compact Discs (CDs) and hard disc drives.

Sound can be reproduced by directing a weak laser beam on the disc.

Devices used to store sound

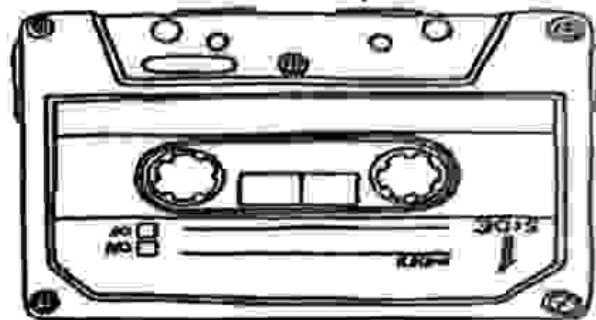
Flash drive



Memory card



Cassette Tape



Compact disc



Other devices used to store sound are;
mobile phones,
magnetic tapes
and mp3

Notation

This is a method of writing music using symbols to indicate pitch and rhythm.

Forms of writing pieces of music.

Solfa notation

This is the form of notation using solfa notes.

Solfa notes are musical notes or symbols used to write music.

m:d:s | m:d:s | m:- :- | d:- :-d |

r:- :r | r :m :r | d:- :- | - : - |

Sol-fa notation

C	D	E	F	G	A	B	C
1	2	3	4	5	6	7	8/1

Doh Ray Me Fah Soh Lah Te Doh

Staff notation.

This is a method of writing music using staff notes or symbols written on a stave.



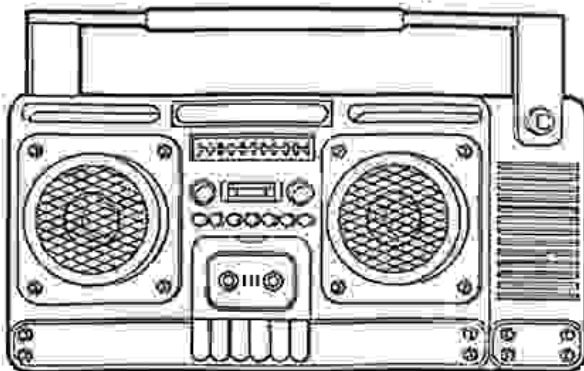
Reproducing stored sound.

- Using a piano, keyboards and guitars.
- Playing the recorded sound using DVDs.

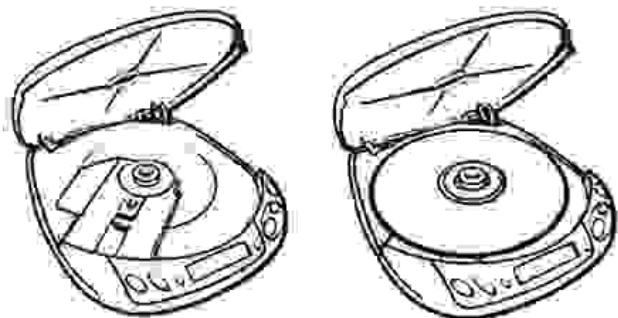
Devices used to reproduce stored sound

- radio cassette player
- CD players
- video player
- gramophone
- mobile phone
- woofers
- digital versatile disc
- computer disc player

Radio Cassette Player



Compact Disc player



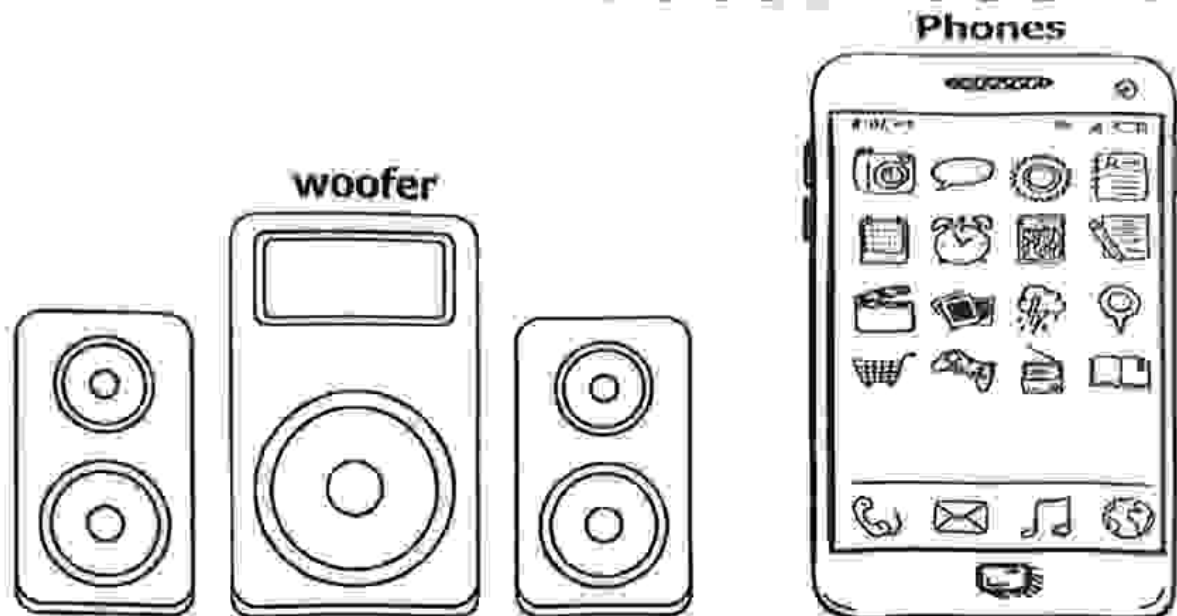
DVD Player



Gramophone



Don't speak for Quality, let Quality speak for itself



Instruments used to reproduce sound stored.

- > keyboard > flute > piano > xylophone > thumb piano.
- > horn > recorder > bowharp > panpipes > lyre
- > violin > trumpet > guitar > tuba

Similarities between sound and light energy

- Both are forms of energy.
- Both travel in all directions.
- Both can be reflected and absorbed.

Differences between sound and light energy

- Sound travels slowly while light travels faster.
- Sound does not travel through the vacuum while light travels through the vacuum.
- Sound requires a medium of transmission while light does not.

Evaluation activity 2.5

1. Which form of energy is produced by vibration of an object?
2. Explain how sound is produced. (2 marks)
3. Apart from being forms of energy, state any **one** way in which sound energy is similar to light energy

The diagram below shows a method of storing music. Use it to answer questions 4 and 5.

C	D	E	F	G	A	B	C
1	2	3	4	5	6	7	8/1
Doh	Ray	Me	Fah	Soh	Lah	Te	Doh

4. Name the method of storing music shown above.
5. Give any **one** way in which music stored using the above method can be reproduced

Evaluation activity 2.5... Continued

The diagram below shows a symbol used in writing musical notes. Use it to answer questions 6 and 5.



6. Identify the method of writing music in which the above musical symbol is used.
7. Apart from the method above, give **one** other method used to write musical notes.
8. Write down any **two** devices used to store sound.
9. Apart from mechanical method, mention any **two** other ways of storing sound.
10. Name any **two** instruments used to reproduce the sound stored by magnetic method.
11. Give any **two** devices that can be used to reproduce stored music.

The diagram below is of a device used to store sound. Use it to answer questions 12 and 13.



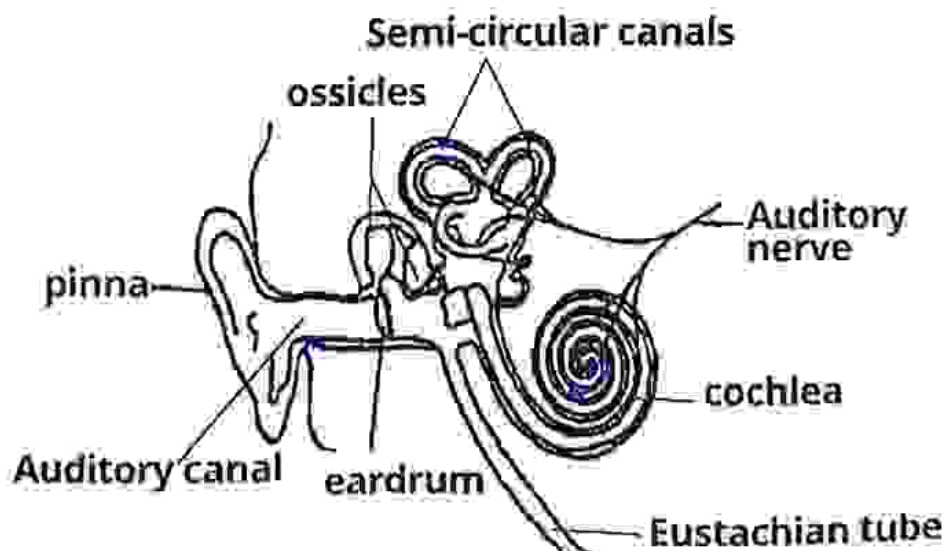
12. Name the method of storing music that requires the use of the device shown above.
13. Give any **one** disadvantage of using the device above when storing music.

THE HUMAN EAR

Functions of the human ear.

- For hearing.
- For body balance.

The structure of the human ear



Regions of the human ear with their parts and functions of parts.

The outer region

Pinna/earflap

- It collects sound waves from the surroundings.
- It directs sound waves to the auditory canal.
- It judges the direction from which sound waves come.

Its adaptations.

- It is made up of cartilage which keeps it open all the time.
- It is broad shaped to trap sound waves effectively and accurately.

Auditory canal.

It directs sound waves from the pinna to the eardrum.

It contains wax which;

- Traps foreign bodies.
- Lubricates the ear canal.
- Kills bacteria to control ear infections.

However, too much wax in the ear causes partial deafness.

Eardrum

- It changes sound waves into vibrations.
- It transmits vibrations to the ossicles.
- It is also called tympanum

The middle ear

Ossicles

They amplify sound vibrations.

The word ossicles is a collective term for the three small bones found in the human ear

These small bones are;

- The hammer (malleus)
- The anvil (incus)
- The stirrup (stapes)

Mnemonic used: **HAS/MIS**

- The ossicles also link the eardrum to the oval window.

Eustachian tube

It equalises air pressure on both sides of the eardrum.

- The eustachian tube is not involved in the process of hearing.

The inner ear

Cochlea

It converts sound vibrations into nerve impulses.

Its adaptations

- It is coiled to increase the surface area for converting vibrations into nerve impulses.
- It contains perilymph which transmits vibrations and causes the cochlea membrane to move up and down.

- It has sensory hairs which produce nerve impulses which are sent to the brain through auditory nerve.
- The cochlea is considered as the centre of hearing.

Semi-circular canals

They are used for body balance.

Their adaptation.

- They have endolymph which transmits signals to the hair cells for body balance.
- The semicircular canal works like the halteres of an insect and pelvic and pectoral fins of the fish.
- The semi-circular canals are not involved in hearing.

Auditory nerve

It transmits sound impulses to the brain for interpretation.

The process of hearing

- The pinna picks up the sound waves from air, the auditory canal sends these sound waves to the eardrum.
- The eardrum changes the sound waves into sound vibrations and transmits them to ossicles where they are amplified.
- The cochlea converts the vibrations into nerve impulses.
- The nerve impulses are later sent to the brain for interpretation through the auditory nerve and the brain tells us what we hear.

Diseases that affect the human ear

- > otitis
- > otalgia
- > ear cancer
- > meniere's disease

Signs and symptoms of a sick ear

- Rise in temperature.
- Pus in the ear.
- Inflammation of the ear.
- Painful ear
- Itching inside the ear

Disorders / defects of the human ear

- ear boils
- deafness
- tinnitus (ringing in the ear)
- deformed pinna
- impacted earwax

Deafness

Deafness is the inability to hear.

Forms of deafness

Permanent deafness

Here a person's ears are unable to detect sound.
Rubella can make unborn baby to become deaf.
Permanent deafness cannot be corrected.

Partial deafness

Partial deafness is a condition in which a person is unable to hear properly. It is caused by too much wax in the ear canal.

Partial deafness can be corrected in two ways namely;

- Removing excess wax from the ear using a soft material.
- Put oil ear drops in the ear overnight. This helps to soften the wax.

These oil ear drops can be olive oil drops, mineral oil drops or baby oil drops.

- Syringing the ear using warm water.

Warm water helps to remove accumulated wax from the ear.

Sensory deafness

This is a condition in which a person is unable to differentiate between sounds.

Causes of sensory deafness

- ✓ Diseases or injury of the auditory canal
- ✓ Serious fracture on the skull.
- ✓ Old age
- ✓ Living in places with too much noise for a long time.

How to control sensory deafness

- ❖ Good feeding to keep healthy even at old age.
- ❖ Travelling in vehicles in good mechanical conditions.
- ❖ Early treatment of ear infections.

Ways of caring for the human ears.

- ✓ Going for regular medical check up.
- ✓ Clean the ears using soft materials like ear cotton swabs.
Sharp objects damage the eardrum.
- ✓ Avoid playing rough games like slide kicks.
- ✓ Listen to music at a low volume.



Learn, unlearn and relearn: Earbud is a small earphone inserted into the ear so they are not used to clean the ears.

Comparing human ears with organs of hearing in other animals.

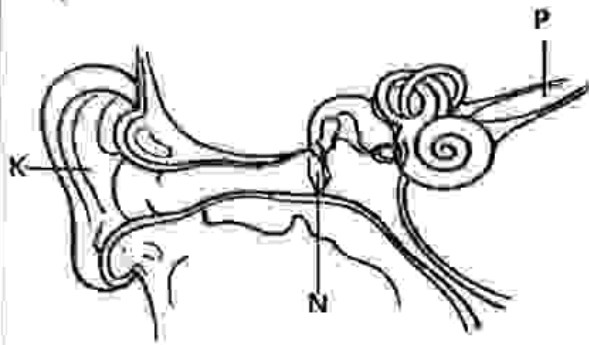
Different animals have organs that help them to detect sound vibrations just like the human ears.

The following are some of the animals and body organs they use for hearing.

Animal	Hearing organ/part
fish	lateral line
snakes	jawbone
insects	feelers
amphibians	external eardrum
birds and chameleon	ears

Evaluation activity 2.6

1. Name the human body organ used to detect sound vibrations.
2. How is the eardrum useful in the process of hearing?
3. Which part of the human ear equalizes pressure in and outside the ear?
4. Name the part of the fish that has the same function as the human ears.
5. Why is it not advisable to clean our ears using sharp objects?
6. Study the diagram of the human ear below and use it to answer the questions that follow.



- a) Name the parts marked **P** and **N**.
 - b) Give the function of part marked **P**.
 - c) Use letter **B** to show the part of the ear responsible for balancing the body.
7. What is the danger of having too much wax in the human ear?
 8. Apart from hearing, give **one** other function of the human ear.
 9. Which part of the human ear amplifies sound vibrations?
 10. How are ears useful to a deaf person?
 11. In which way are the antennae of a grasshopper similar to the ears of a person?
 12. Which part of the human ear has the same function as the halteres of an insect?
 13. The diagram below shows parts of the human ear. Study and use it to answer the questions that follow.



- a) Name the parts marked **Q** and **R**.
 - b) Which part of an insect has the same function as part marked **Q**?
 - c) How is part marked **R** useful in the process of hearing?
14. Which defect of the human ear is caused due to accumulation of wax in the auditory canal?
 15. Give any **two** ways of caring for the human ears.

Theme:

HUMAN BODY

Topic

3: CIRCULATORY SYSTEM

The circulatory system is the group of human body organs that transport materials in the human body.

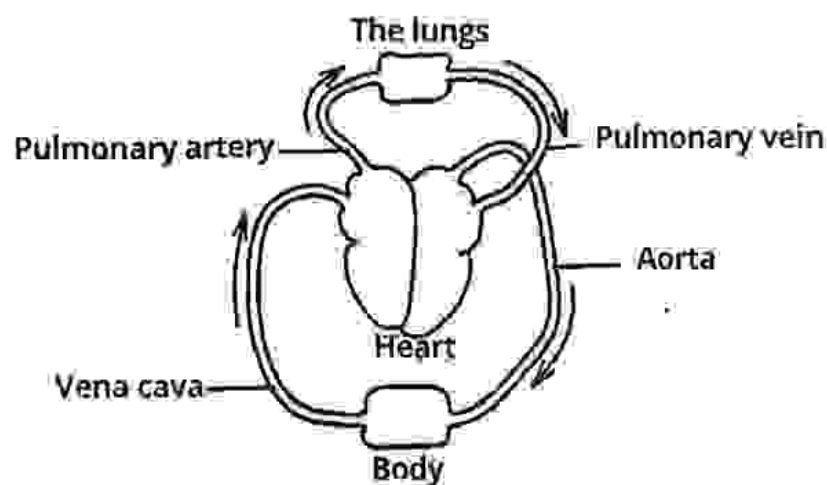
These materials include; food materials, body wastes, heat, antibodies and hormones

BLOOD CIRCULATION

Blood circulation is the continuous movement of blood in the body.

In the human body, blood is pumped by the heart and it is transported by blood vessels.

Diagram showing blood circulation in the human body.



Why blood goes to some body organs.

Lungs:

To drop carbon dioxide and pick oxygen.
Supplied by pulmonary artery.

Kidney:

To be filtered/purified.
Supplied by renal artery.

Intestines:

To facilitate the absorption of nutrients from digested food.
Supplied by the mesenteric artery

Brain:

To deliver oxygen and glucose.
Supplied by carotid artery.

Liver

To remove harmful materials from food.
To dilute sugar in food.
Supplied by hepatic portal vein.

Components of the circulatory system

➤ heart ➤ blood vessels ➤ blood

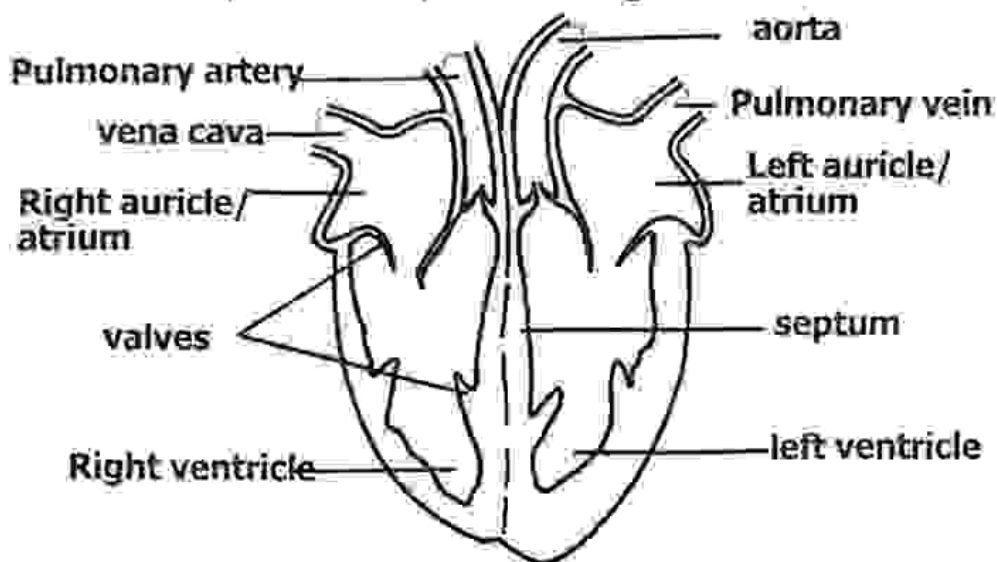
Organs of the circulatory system are; the heart and blood vessels.



Learn, unlearn and relearn: lungs, kidney, liver, brain and intestines are not organs of the circulatory system. These organs just help in blood circulation in the human body.

STRUCTURE AND FUNCTION OF THE HUMAN HEART.

- The heart is the major organ of the circulatory system.
- The heart pumps blood to all parts of the body.
- The heart is located in the chest cavity surrounded by pericardium membrane.
- The heart is protected by the **rib cage**.



Functions of each part of the human heart.

Pulmonary artery

It transports deoxygenated blood from the heart to the lungs.
Blood goes to the lungs to drop carbon dioxide and pick oxygen.

Pulmonary vein

It transports oxygenated blood from the lungs to the heart.
Blood returns to the heart to be pumped to all parts of the body.

Aorta

It transports oxygenated blood from the heart to all parts of the body.
Aorta is the biggest/main artery in the human body.

Vena cava

It transports deoxygenated blood from all parts of the body to the heart.

Septum.

It prevents oxygenated blood from mixing with deoxygenated blood.
The septum separates the left side of the heart from the right side.

Valves

They prevent the backflow of blood.

The chambers of the human heart

The right atrium/auricle.

It receives deoxygenated blood from the body.

The left atrium/auricle

It receives oxygenated blood from the lungs by the pulmonary vein.

The right ventricle

It pumps deoxygenated blood to the lungs.

The left ventricle

It pumps oxygenated blood to all body parts.

It has thicker and stronger cardiac muscles to pump blood at a higher pressure.

Note well:

-  - The atria receive blood to the heart while the ventricles pump blood out of the heart. This is the reason why atria do not have thick muscular walls.
- The normal heartbeat of a person is 72 times per minute. It is listened using a stethoscope.
- Fear, excitement, fright and vigorous exercises can lead to increased heartbeat.

Ways in which the heart is able to pump blood in the human body

- The heart is made up of cardiac muscles to withstand high blood pressure.
- The heart has valves to prevent the backflow of blood.
- The heart has septum to prevent oxygenated blood from mixing with deoxygenated blood.
- The cardiac muscles have the ability to contract and relax when pumping blood.
- The left side of the heart is thick walled to pump blood at a higher pressure.

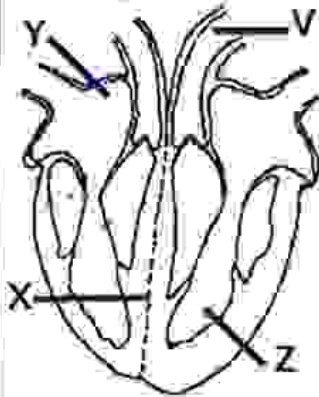
Conditions that affect the heart

- coronary heart disease
- heart valve disease
- cardiac arrest
- heart failure
- heart attack

Evaluation activity 3.1

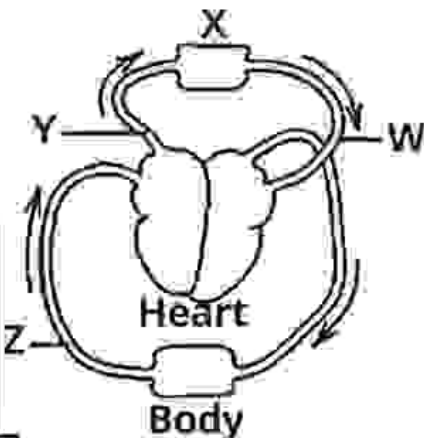
1. Name the body system that transports materials in the human body.
2. Mention any **two** components of the circulatory system.
3. State the main function of the human heart.
4. Why does blood go to the following organs during circulation?
i) Lungs.....ii) Kidney
5. How are the following important in the human heart?
i) Septum.....ii) Valves

6. The diagram below shows the human heart. Study and use it to answer the questions that follow.



- Name the parts marked V and Y.
- State the function of part marked X in the human heart.
- How is part marked Z able to pump blood at a higher pressure?

- How is the rib cage useful to the circulatory system?
- State the features in the human heart that prevent the backflow of blood.
- Give any **two** ways in which the heart is able to pump blood to all body parts.
- Why is the left side of the heart able to pump blood to most parts of the body?
- The diagram below shows circulation of blood in the human body. Study and use it to answer the questions that follow:



- Name the organ marked with letter X.
- Which blood vessel is marked with letter W?
- State the similarity between blood vessel Y and blood vessel Z.
- Why does blood vessel W bring back blood to the heart?

BLOOD VESSELS

Blood vessels are muscular tubes that transport blood in the human body.

Types of blood vessels are; arteries, veins and blood capillaries

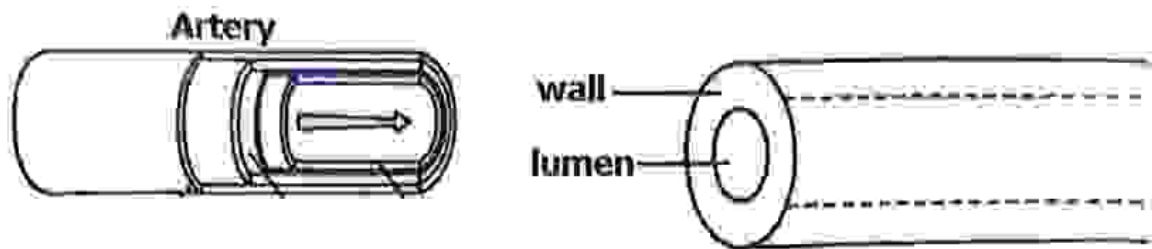
Mnemonic: **VAC**

Arteries

Arteries are blood vessels that transport blood away from the heart.

Arteries transport oxygenated blood except pulmonary artery.

Pulmonary artery transports deoxygenated blood.



Examples of arteries are; aorta, pulmonary artery, renal artery and carotid artery

Characteristics of arteries.

- They are thick walled.
The thick walls help arteries to withstand the high blood pressure.
- They have a narrow lumen.
The narrow lumen helps to maintain blood pressure flowing through arteries.
- They do not have valves.

Reason. Blood that flows through them is under high pressure.

They are able to prevent the backflow of blood by transporting blood at a higher pressure.

Adaptation of arteries to their function.

They have thick walls to withstand the high pressure at which blood flows through them.

Points to note:

- The aorta is the largest artery and blood vessel in the human body.
- The arteriole is the smallest artery in the human body.
- Pulse is the rhythmic flow of blood from the heart through an artery as it contracts and relaxes.
- It is important to know one's pulse rate

Reason. To know whether the person is alive or dead.

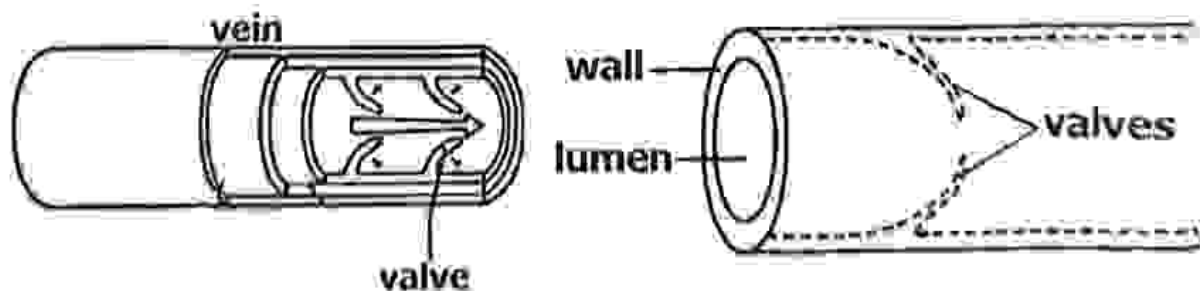
- When an artery is cut, blood oozes in spurts and fast.

Veins

Veins are blood vessels that transport blood towards the heart.

Veins transport deoxygenated blood except pulmonary vein.

Pulmonary vein transports oxygenated blood.



Examples of veins in the human body are; vena cava, pulmonary vein and renal vein

Characteristics of veins.

❖ They have valves.

The valves in veins prevent the backflow of blood.

Valves allow blood to flow in one direction.

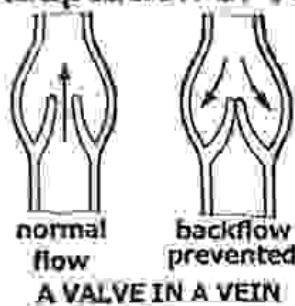
❖ They have wider lumens.

The wider lumen allows more blood to flow with less resistance.

❖ They have thin walls.

Reason. The blood that flows through veins has low pressure and speed.

Adaptation of veins to their function.



Veins have valves to prevent the backflow of blood



Points to note:

- The vena cava is the largest vein in the human body.

- The venule is the smallest vein in the human body.

- When a vein is cut, blood oozes/seeps slowly.

Similarities between an artery and a vein.

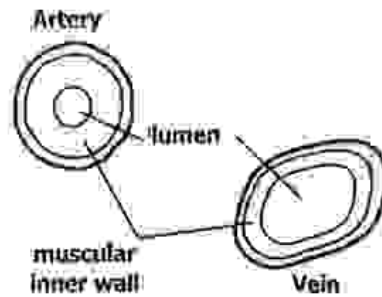
- ✓ Both transport blood in the body.
- ✓ Both have a lumen.
- ✓ Both have walls.

Structural differences between an artery and a vein.

- An artery has thick walls while a vein has thin walls.
- An artery has a narrow lumen while a vein has a wider lumen.
- An artery has no valves while a vein has valves.

Functional differences between an artery and a vein.

- An artery transports blood away from the heart while a vein transports blood towards the heart.
- An artery transports oxygenated blood except pulmonary artery while a vein transports deoxygenated blood except pulmonary vein.
- An artery transports blood with high pressure while a vein transports blood with low pressure.



Blood capillaries

These are the smallest blood vessels in the human body.

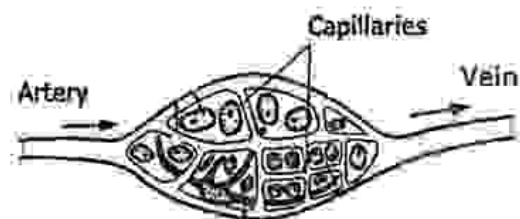
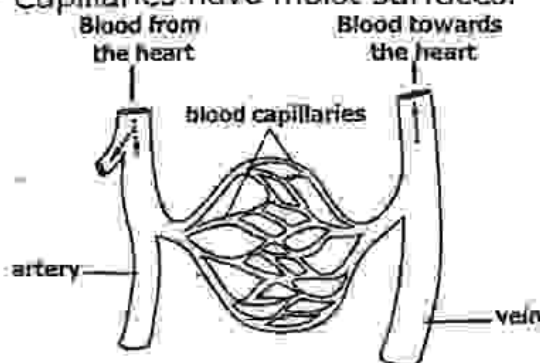
Capillaries join veins to arteries

Their function

- They allow the exchange of materials in the body.
- These body materials are; food nutrients, oxygen and body wastes.

Their adaptations

- Capillaries have thin walls.
- Capillaries have moist surfaces.

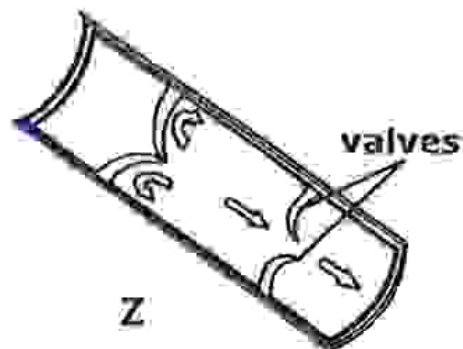
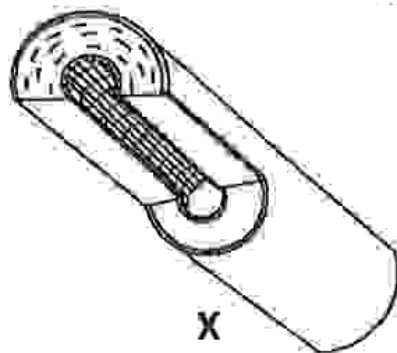


Evaluation activity 3.2

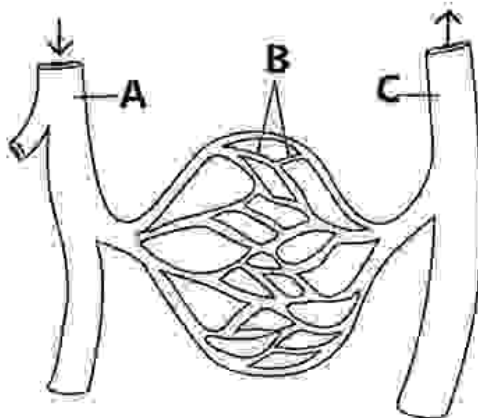
- 1.a) Name the largest vein in the human body.
b) Name the type of blood that is transported by the pulmonary artery.
c) State **one** characteristic of arteries that enable them to carry blood at high pressure.
d) Give **one** way in which the structure of capillaries enables them to exchange body materials.
2. How is the function of the pulmonary vein different from that of other veins?
- 3.a) Which type of blood vessels return blood to the heart?
b) What is the function of valves in the blood vessels during blood circulation?
c) What type of blood is carried by most blood vessels with valves?
d) Give any **one** material transported by blood capillaries.

Evaluation activity 3.2 ...Continued

4. The diagram below shows blood vessels in the human circulatory system. Study and use it to answer the questions that follow.



- Name the blood vessels marked **X** and **Z**.
 - State any **one** structural difference between blood vessels marked **X** and **Z**.
 - How are valves useful to the blood vessel marked **Z**?
5. Name the blood vessel which takes blood from the heart to the lungs.
6. State any **one** difference in function between an artery and a vein.
7. The diagram below is of two blood vessels. Study and use it to answer the questions that follow.



- Name the blood vessels marked **B** and **C**.
- State any **one** way in which blood vessel marked **A** is able to transport blood with high pressure.
- Give any **one** function of blood vessel marked **B** during blood circulation.

COMPOSITION AND FUNCTIONS OF BLOOD.

Blood is a red fluid that continuously flows in the human body.

Blood is a living tissue because it has living cells.

Types of blood

oxygenated blood

This is blood rich in oxygen.

deoxygenated blood

This is blood rich in carbon dioxide.

Components of blood.

➤ blood plasma ➤ red blood cells ➤ white blood cells ➤ blood platelets
Red blood cells, white blood cells, and blood platelets are solid components.

Note well: Red blood cells and white blood cells are the only examples of blood cells in the body.

Blood plasma

Blood plasma is the liquid part of blood.

Functions of blood plasma.

- It transports digested food nutrients, antibodies and hormones in the body.
- It distributes heat in the body.
- It carries red blood cells, white blood cells and platelets.

Substances carried by blood plasma.

- | | | |
|--------------|---------------------------|------------|
| ➤ antibodies | ➤ digested food nutrients | ➤ hormones |
| ➤ water | ➤ carbon dioxide | ➤ urea |

Red blood cells(erythrocytes)

Red blood cells are disc shaped and thin in size.

This maximises the surface area for oxygen absorption.

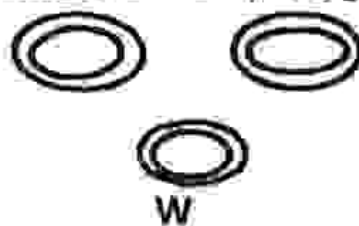
Red blood cells do not have a nucleus.

Reasons

- To have more haemoglobin so as to increase the surface area for carrying oxygen.
- To have a bi-concave shape that helps in diffusion of oxygen.

Red blood cells contain haemoglobin. Haemoglobin is a red pigment that enables red blood cells to carry oxygen in the body. When oxygen combines with haemoglobin they form oxyhaemoglobin.

The function of red blood cells is to carry oxygen round the body



Characteristics of red blood cells.

- They have haemoglobin.
- They have no nucleus.
- They are disc shaped.
- They are many in number.

i *Learn, unlearn and relearn:* Red blood cells do not transport oxygen but oxygen. Transporting blood is done by blood vessels not blood cells. Secondly, red blood cells do not carry oxygen around the body but round the body.

 *Point to note: Red blood cells are produced in the red bone marrows of bones like sternum, scapula, pelvis and ribs. The circulatory system supplies oxygen to the body cells for respiration.*

Ways in which red blood cells are able to carry oxygen in the body.

- ✓ They have haemoglobin which absorbs more oxygen to be carried.
- ✓ They have a disc shape which creates more space for oxygen.
- ✓ They are small and flexible to fit in narrow blood vessels.

Diseases that affect red blood cells

- malaria. It leads to the destruction of red blood cells.
- anaemia. Lack of blood in the body.
- sickle cells. It leads to abnormal shape of red blood cells
- sleeping sickness

White blood cells(leucocytes)

White blood cells have irregular shape.

They have a nucleus which enables them to make control the activities of the cells.



Function. To protect the body against disease germs.

How white blood cells fight against germs in the body

- By engulfing and digesting germs.
- By producing antibodies against germs.

Characteristics of white blood cells.

- They have a nucleus.
- They have an irregular shape.

How white blood cells are adapted to their function.

- They have a nucleus which controls the activities of the cells.
- They have an irregular shape to easily engulf and capture germs.

Diseases that affect white blood cells.

AIDS. Leads to the destruction of white blood cells.

leukaemia. Leads to increased production of immature white blood cells



Learn, unlearn and relearn: White blood cells do not fight against diseases in the body. They only fight against germs.

 **Note:** They are made in the red bone marrow of long bones, lymph nodes and spleen.

Blood platelets. These are the smallest blood cells in the body.



- Platelets help in blood clotting when there is bleeding.

They clot blood by producing a fibrin that forms a

fibrous network around the bleeding wound.

When a blood vessel is injured, the platelets move to

the damaged site and form a mesh like structure that helps to clot blood.

- Clotting of blood helps to stop bleeding.
Platelets are also called thrombocytes.

Diseases that affect blood platelets

- haemophilia
- thrombosis

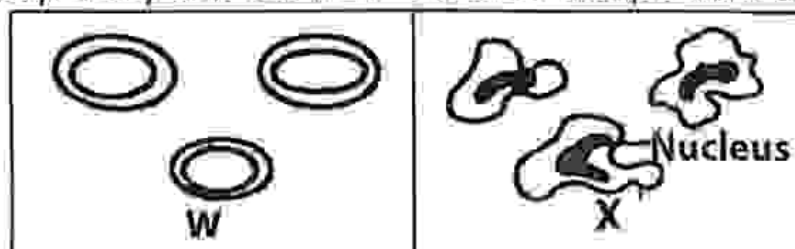
Functions of blood in the human body

- ❖ Blood carries substances in the body.
- ❖ Blood distributes heat in the body.
- ❖ Blood defends the body against germs.

Substances transported by blood in the body are; hormones, oxygen, carbon dioxide, antibodies, digested food nutrients, waste materials and medicines

Evaluation activity 3.3

1. Name the component of blood affected by sickle cell disease.
2. Why are red blood cells able to carry oxygen round the body?
3. Write down any **two** components of blood.
4. Why is blood referred to as a living tissue?
5. Name the component of blood destroyed by plasmodia parasites.
6. The diagrams below show the structure of different blood cells in the human body. Study and use them to answer the questions that follow.



- a) Name the blood cell marked **X**.
- b) Give the function of the blood cells marked **W** and **X**.
- c) Identify any **one** disease that destroys blood cell **W**.

Evaluation activity 3.3... Continued

7. Give **two** ways in which white blood cells fight against germs in the human body.
- 8.a) Apart from respiratory gases and body wastes, name any **two** other materials carried by blood.
b) Name the respiratory gas carried by red blood cells in the human body.
c) Which disease germ attacks the white blood cells in humans?
9. Name the human organ where each of the following takes place
 - i) Filtration of blood
 - ii) Blood gets oxygen while carbon dioxide is removed.b) Give the function of each of the following components of blood in the human body.
 - i) White blood cells
 - ii) Blood platelets
10. How is the circulatory system useful to the respiratory system?

DISEASES OF THE CIRCULATORY SYSTEM

sickle cells, haemophilia, anaemia, leukaemia, AIDS and malaria

Mnemonic used: **SHALAM**

Sickle cell disease

- ↑ It is a hereditary disease that affects the red blood cells.
- ↑ Sickle cell disease causes deformation of red blood cells.
- ↑ It reduces the ability of red blood cells to carry oxygen in the body.
- ↑ It makes red blood cells to become crescent-shaped
- ↑ It blocks the blood vessels and prevents easy flow of blood.

Sickle cells



Signs and symptoms of sickle cells

- | | | |
|-----------------------------------|-----------------------|------------------|
| ✓ Pale skin | ✓ Shortness of breath | ✓ Dizziness |
| ✓ Passing out dark | ✓ coloured urine. | ✓ Yellowish eyes |
| ✓ Swollen legs, fingers and arms. | | ✓ Painful joint |

Leukaemia

This is also called blood cancer.

It is a condition in which a person has a large number of immature white blood cells.

The immature white blood cells prevent the production of red blood cells

and platelets.

Leukaemia causes severe anaemia which may lead to death of an individual.

Haemophilia

This is a hereditary disease. It is a condition in which one's blood is not able to clot easily.

It can be caused due to lack of vitamin K in the body.

However, haemophilia is not a food deficiency disease.

A person who has haemophilia is given vitamin K by injection.

The drug contains **phyloquinone** which helps in blood clotting.

Malaria

- It is caused by protozoa.

- Plasmodia parasites destroy the red blood cells. This leads to anaemia.

Malaria is spread through the bites of a female anopheles mosquito.

Male anopheles mosquitoes do not transmit malaria to people. **Reason.**

They do not feed on human blood.

How a female anopheles mosquito is able to transmit malaria.

- It feeds on human blood.

- It has proboscis to suck blood.

- Plasmodia germs survive in its abdomen.

Signs and symptoms of malaria

↑ Fever ↓ Headache ↓ Muscle pain ↓ Nausea and vomiting

↓ Shivering

Ways of preventing malaria.

❖ Sleeping under a treated mosquito net.

It helps to prevent bites from female anopheles mosquitoes.

❖ Draining stagnant water.

It destroys the breeding places for mosquitoes.

❖ Pouring oil on stagnant water.

Oil suffocates mosquito larvae.

❖ Spraying using insecticides.

Insecticides kill adult mosquitoes.

❖ Clearing bushes around the home.

It destroys hiding places for mosquitoes.



Learn, unlearn and relearn: Sleeping under a treated mosquito net does not control mosquitoes but controls the spread of malaria. Therefore, sleeping under a treated mosquito net does not prevent mosquitoes from breeding.

Anaemia

Anaemia is a deficiency disease that attacks the circulatory system. It is a condition in which the person's body lacks blood. It is caused by lack of iron in a diet.

Other causes of anaemia are; Excessive bleeding, malaria parasites and hookworm infestation.

Signs of anaemia

- ✓ Pale skin, eyelids and tongue.
- ✓ White fingers
- ✓ Dizziness
- ✓ General body weakness

Ways of preventing anaemia.

⚡ Feeding on food rich in iron.

Such food include; sea fish, kidney, liver and dark green leafy vegetables.

⚡ Taking iron supplement tablets.

HIV/AIDS and blood

- AIDS is the viral disease of the circulatory system.
- It stands for acquired immune deficiency syndrome
- Acquired means got from, immune means protected against, **deficiency** means lack of and **syndrome** means many signs and symptoms.
- AIDS is caused by a virus called HIV
- HIV stands for human immunodeficiency virus.
- HIV destroys the white blood cells making them unable to fight against germs in the body.

Ways in which HIV/AIDS spreads.

- ✓ Through having unprotected sexual intercourse with an infected person.
- ✓ Through sharing unsterilized sharp objects like syringes with an infected person.
- ✓ Mother to child transmission.
 - Through placenta during pregnancy.
 - Through blood during breastfeeding
 - Through blood during birth.
- ✓ Through cultural circumcision and genital mutilation.

 **Learn, unlearn and relearn:** blood is screened before transfusion. Therefore, reject through blood transfusion.

Signs and symptoms of HIV/AIDS

- ✓ Prolonged dry cough
- ✓ Widespread itching and skin rash.
- ✓ Loss of body weight.
- ✓ Blood-stained sputum.
- ✓ Prolonged diarrhoea
- ✓ Persistent fever
- ✓ Excessive sweating at night.

Diseases mistaken as AIDS are; Tuberculosis, cholera, measles and typhoid. Such diseases are called opportunistic diseases/secondary diseases. They attack the body when its immunity becomes weak.

A person suffering from HIV/AIDS is easily attacked by other diseases.

Reasons. HIV/AIDS weakens the immunity of a person.

- HIV/AIDS destroys white blood cells which fight against germs in the body.

Effects of HIV/AIDS to an individual.

- ❖ It weakens the immunity of a person.
- ❖ It increases the chances of getting other diseases.
- ❖ It leads to stigmatization.
- ❖ It leads to loss of one's job.
- ❖ It leads to death of an individual.

Effects of HIV/AIDS to the family

- ↑ It leads to poverty due to high treatment costs.
- ↑ It leads to loss of important people in a family.
- ↑ It causes loss of trust and respect among family members.

Effects of HIV/AIDS to the community.

- It leads to loss of skilled labour in the family.
- It leads to loss of friends in the community.
- It leads to shortage of productive population.
- It causes child headed families.
- It increases the number of street children.

Ways of preventing the spread of HIV/AIDS among people.

- ✓ Abstaining from pre-marital sex. This is the best way for school children.
- ✓ Being faithful to one's sexual partner. This is best for married couples.
- ✓ Avoid sharing sharp objects like nail clippers and razorblades.
- ✓ Using condoms when having sex with untrusted sex partner.

Ways in which HIV positive mother can protect unborn baby from getting HIV/AIDS.

- Taking ARVs.
- Giving birth in hospitals under the supervision of health workers.



Learn, unlearn and relearn: AIDS does not affect the reproductive system. Therefore, it only attacks the circulatory system.

Note well.



-It is easier to prevent AIDS than malaria.

Reason. It is easier to abstain from sex than avoiding mosquito bites.

A person suffering from malaria dies faster than one suffering from AIDS.

Reason: Malaria destroys red blood cells while AIDS makes the white blood cells weak.

- A person who has HIV is one who is infected with the virus that causes AIDS while an AIDS patient is one who suffers from AIDS and shows signs like chronic cough, herpes zoster and skin rash.

- Female anopheles mosquito cannot spread HIV/AIDS.

Reasons.-HIV is destroyed in its abdomen.

Disorders of the circulatory system

a. heart failure; The heart can't pump blood effectively, leading to fatigue shortness of breath, and fluid buildup.

b. heart attack; Blood flow to part of the heart is blocked, causing damage to the heart muscle and chest pain.

c. high blood pressure (hypertension); The heart works harder to pump blood, potentially causing heart damage, artery damage, and increasing stroke risk.

d. low blood pressure (hypotension); Inadequate blood flow can cause dizziness, fainting and potential organ damage due to insufficient oxygen delivery.

Ways of increasing the volume of blood in the human body

- ✓ Through blood transfusion.
- ✓ Feeding on foods rich in iron.
- ✓ Taking iron supplement tablets.
- ✓ Drinking plenty of fruit juice and water regularly.

Practices that can affect the circulatory system.

- ↓ Poor feeding ↓ Alcoholism ↓ Regular smoking
 - It hardens the arteries.
 - It raises blood pressure.
- ↓ Eating oil/fatty food.
 - It Leads to clogged arteries due to too much fats.
 - It can lead to increased blood pressure.

Ways of maintaining the proper functioning of the circulatory system

- ❖ Feeding on a balanced diet.
- ❖ Doing regular physical exercises.
 - It helps to strengthen the cardiac muscles.
 - It improves blood circulation in the body.
- ❖ Going for regular medical checkups.
- ❖ Avoid smoking.
- ❖ Limit eating oily/fatty food.

Evaluation activity 3.4

1. Name the disease that leads to abnormal production of immature white blood cells.
 2. Write down any **two** other diseases that affect the circulatory system.
 3. Which body system is affected by HIV/AIDS?
 4. Give any **two** ways of increasing the volume of blood in the human body.
 5. Name the component of blood that is attacked by sickle cell disease.
 6. How does malaria lead to anaemia in a person's body?
 7. In which way does HIV/AIDS affect the white blood cells?
- The diagram below shows the shape of red blood cells when attacked by a certain disease. Use it to answer question 8.



8. Which disease makes the red blood cells to have the shape shown above?
9. When it is said that a person is anaemic. What is lacking in his/her blood?
10. Which body system is affected by malaria parasites?
11. a) Write down any **two** signs of HIV/AIDS.
b) Give any **one** effect of HIV/AIDS to an individual.
c) How best can a primary school pupil protect him/herself from contracting HIV/AIDS?
12. Give any **one** way in which doing regular physical exercises is important to the circulatory system.
13. a) Identify any **two** characteristics of the female anopheles mosquito that make it a good vector for malaria.
b) How do the following methods help in controlling the spread of malaria?
i) Pouring oil on stagnant water.
ii) Sleeping under a mosquito net.
14. State **one** reason why a person suffering from HIV/AIDS is easily attacked by many other diseases.
15. Give a reason why male anopheles mosquitoes do not transmit malaria parasites to humans.
16. Why is it easier to prevent HIV/AIDS than malaria?
17. a) Give any **two** possible ways through which baby can get HIV/AIDS from a mother.
b) State any **two** ways in which an HIV positive pregnant woman can protect her unborn baby from getting HIV/AIDS.
18. Write down any **three** PIASCY messages that protect children from getting HIV/AIDS

Theme:

HUMAN HEALTH

Topic

4: ALCOHOL, SMOKING AND DRUGS

ALCOHOL

Alcohol is a substance that makes people drunk when taken in excess.

Examples of alcoholic drinks; wine, spirit and beer

Types of alcohol

Ethanol alcohol

This is a toxic addictive drug contained mostly in all alcoholic drinks like spirits and beers.

Uses of ethanol alcohol

- It is drunk by some people.
- It is used as fuel.

Methanol alcohol

This is the most dangerous type of alcohol.

It is highly acidic and can lead to blindness when it gets into contact with the eyes.

It can also lead to death when drunk.

Uses of methanol alcohol

- It is used to mix some medical drugs.
- It is used to sterilize medical equipment.
- It is used as fuel in some engines.
- It is used to mix paint.

How alcohol is made

The methods of making alcohol are; fermentation and distillation.

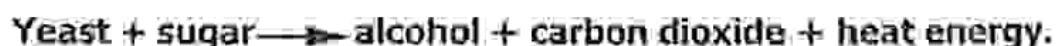
Fermentation

Fermentation is a chemical process by which fruit juice is turned into alcohol.

The fruit juice can be extracted from bananas, pineapples, oranges, mangoes and grapes.

The fungus used in fermentation is yeast. It is obtained from maize, wheat, millet or sorghum.

Yeast produces zymase that speeds up fermentation.



Examples of plant materials used to produce alcohol

- ripe bananas
- cassava flour
- maize flour
- millet malt
- sorghum flour

Alcoholic drinks obtained by fermentation are; banana brew, millet brew and maize brew

 **Points to note:**

Alcohol made by fermentation is of low alcoholic content.

Reason: It contains a lot of impurities.

Fermentation is a chemical change because;

-It forms new permanent substances.

-It is irreversible.

How to obtain alcohol locally at home by fermentation.

Materials needed:

Pineapple, clean water, cup and sieve

What to do:

- i) Make juice from pineapples and add some little water and sugar to it.
- ii) Sieve the juice and store it in a covered container. Keep it in a safe place.
- iii) Check on the contents after two or three days.
- iv) Bubbles of carbon dioxide will be seen in a cup and the content in the cup tastes sugary.

 **Points to note:**

- Bubbles from the cup show carbon dioxide produced during fermentation.

- The sugary taste is the taste of alcohol produced from fruit juice.

- The sugar in juice has been broken down to form alcohol and carbon dioxide.

Other ways in which fermentation is useful to people:

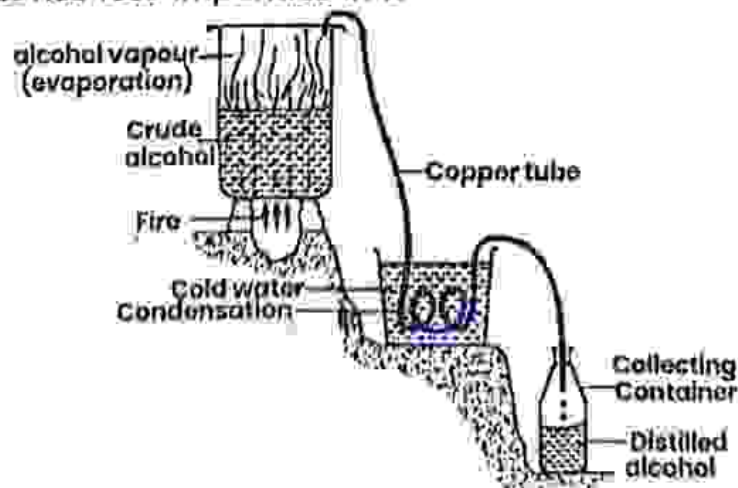
- It helps in the production of biogas.
- It helps in the making of dairy products e.g. yoghurt
- It helps in baking.

Distillation

Distillation is the method of obtaining pure alcohol from fermented alcohol by evaporation and condensation.

This method produces the highest concentration of alcohol. **Reasons.**
It removes most of the water from alcohol.

It has less impurities in it



Roles of some features during distillation.

Cold water:

-It cools and condenses the vapour into alcohol.

The delivery tube is coiled to increase the surface area for condensation.

Sealed drum.

-To prevent alcohol vapour from escaping.

-To enable alcohol vapour flow to the cold water.

Delivery tube.

-It transports alcohol vapour to the cold water.

-It transports condensed alcohol to the collecting container.

The delivery tube is made of copper wire.

Reasons.

- Copper does not rust once exposed to moisture.
- Copper cools faster.



Alcoholic drinks prepared through distillation.

- wine
- whisky
- beer

Ways in which alcohol is useful to people.

- ↑ It is sold for money.
- ↑ It is used as a drink.
- ↑ It is used as fuel in some engines.
- ↑ It is used in thermometers like six's thermometer.
- ↑ It is used to dissolve dye and paint.

Uses of alcohol in hospitals.

- ✓ It is used as a disinfectant to clean wounds.
- ✓ It is used to mix some medical drugs.
- ✓ It is used as surgical spirit.
- ✓ It is used to sterilize medical equipment.
- ✓ It is used to clean an area on the skin where an injection is to be taken.

Evaluation Activity 4.1

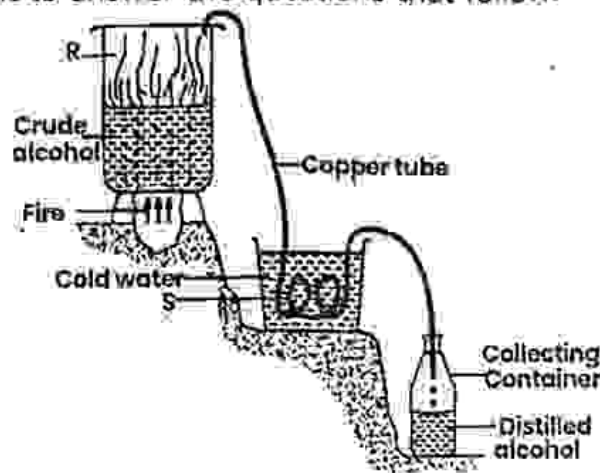
1. By what process is alcohol made from fruit juice?
2. Which type of alcohol causes blindness when taken?
3. Name the fungus used during fermentation of alcohol.
4. Which method of making alcohol does not require heat?
5. Apart from brewing alcohol, give any two other ways in which fermentation is useful to people.
6. Give any **two** ways in which alcohol is useful in hospitals.
7. Identify the method of making alcohol that involves the production of carbon dioxide.

Evaluation Activity 4.1...continued

8. Study the equation below about a way in which alcohol is produced and answer the questions that follow.



- Name the process of making alcohol shown in the equation above.
 - State the role of yeast in the process above.
 - Identify the substance obtained at Y.
 - Mention any **one** alcoholic drink made from the above process.
9. The diagram below shows the distillation method of making alcohol. Study and use it to answer the questions that follow.



- Identify the physical changes taking place;
 - At part marked R
 - In part S
 - Give one reason why the container with crude alcohol should be properly sealed.
 - Give the function of the copper tube in the method shown above.
10. Cannibalism wanted to make alcohol locally at home. He had the following materials;
- ripe banana, clean water, cup and sieve**
- Which method of making alcohol did Cannibalism use?
 - Why did cannibalism include ripe banana on the list of his materials?
 - Describe how Cannibalism will make alcohol locally using the materials shown on the list above? **(02 marks)**

ALCOHOLISM

Alcoholism is the condition in which one is addicted to alcohol.

An alcoholic is a person who is addicted to alcohol.

Factors that lead to alcoholism.

- Peer pressure
- Boredom
- Idleness

- Frustration
- Family background

Reasons why some people drink alcohol

- ✓ To pastime
- ✓ To feel mature
- ✓ To celebrate events.
- ✓ To socialize with others.
- ✓ To relax their minds.
- ✓ Due to peer pressure.

Effects of alcoholism to an individual.

- It can lead to self-neglect.
- It can lead to poverty.
- It leads to peptic ulcers.
- It can lead to lack of self-control.
- It damages body organs like the liver.

Alcohol irritates the lining of the stomach which causes peptic ulcer



Points to note:



- The liver, brain, pancreas and the kidney are body organs that are damaged due to alcoholism.
- The liver is the body organ that is greatly damaged by excessive drinking of alcohol.
- Absorption of alcohol takes place in the ileum and stomach.



Learn, unlearn and relearn: The biggest absorption of alcohol takes place in the small intestines not in the stomach.

Effects of alcoholism to the family.

- ↑ It can lead to poverty.
- ↑ It may lead to child and spouse abuse.
- ↑ It sets bad habits to children.
- ↑ It may lead to family neglect.
- ↑ It may lead to domestic violence.

Effects of alcoholism to the community

- It leads to increased crime rate.
- It leads to increased rate of accidents.
- It leads to poor job performance.
- It leads to public nuisance

Don't speak for Quality, let Quality speak for itself

Ways of avoiding alcohol.

- Desist bad peer groups.
- Join good social clubs.
- Joining health clubs.
- Ignore adverts that praise alcoholic drinks like beer.

Evaluation Activity 4.2

1. Give the meaning of the following term:
 - i) Alcoholism
 - ii) alcoholic
2. State any **two** factors that lead to alcoholism.
3. Name the body organ that is greatly damaged by excessive drinking of alcohol.
4. Apart from damaging organs, state any two other effects of alcoholism to an individual.
5. Give any **two** reasons why some people drink alcohol.
6. Name the disease that affects the stomach because of alcoholism.
7. Mention any **one** part of the body where the absorption of alcohol takes place.
8. How does excessive drinking of alcohol lead to peptic ulcers?
9. State any **two** effects of alcoholism to the family.
10. Give any **two** life skills that children need to develop in order to safeguard themselves against alcoholism.

SMOKING

Smoking is the inhaling of tobacco smoke from the burning material through the mouth.

Ways in which people smoke

- ✓ Through burning pipes.
- ✓ Through burning cigarette.

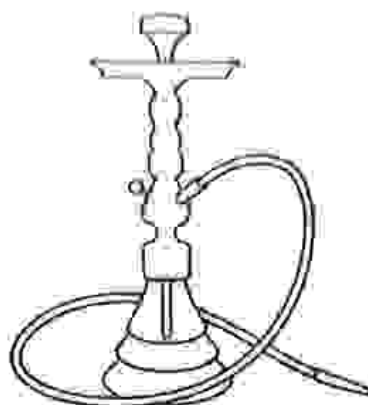


Examples of smoking devices

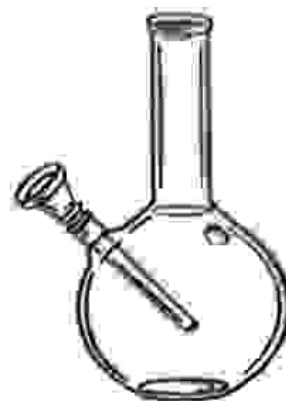
- hand pipes
- bubblers
- bong
- dab rigs
- hookahs



Dab rig



Hookah



Bong

Forms/types of smoking

Active smoking

This is the inhaling of tobacco smoke directly from a burning cigarette or smoking pipe.



Passive smoking

This is the inhaling of tobacco smoke from an active smoker. A person who inhales smoke from an active smoker is called a passive smoker.



Note well:

Passive smoking is more harmful than active smoking

Reasons

1. *Passive smokers are exposed to unfiltered tobacco smoke from the cigarette*
2. *Tobacco smoke from passive smoking has higher levels of toxic substances like tar and carbon monoxide.*



Read more

Passive smoking, or secondhand smoke, can be more harmful than active smoking for several reasons:

- a. *Unfiltered exposure: When a person smokes actively, they inhale the smoke through the filter of the cigarette, which reduces some harmful substances. Passive smokers, however, inhale the smoke that comes from the burning end of a cigarette (sidestream smoke) and the smoke exhaled by the smoker (mainstream smoke). Sidestream smoke is not filtered and contains higher concentrations of harmful substances.*

Don't speak for Quality, let Quality speak for itself

- b. *Chemical composition: sidestream smoke contains higher levels of certain toxic substances compared to the smoke inhaled by the smoker. These include carcinogens like benzene, formaldehyde, and arsenic. studies show that sidestream smoke can have up to three times more nicotine and tar, five times more carbon monoxide, and many more carcinogens*

Factors that lead people to smoke

- Peer pressure
- Boredom
- Idleness

Reasons why some people smoke

- ↓ To fit in groups
- ↓ To feel warm.
- ↓ To concentrate at work.
- ↓ To pastime
- ↓ Due to peer group influence.

Harmful or dangerous substances found in tobacco smoke

Tar

- It makes blood vessels narrow.
- It causes lung cancer.
- It stains teeth.

Nicotine

- It is the addictive drug found in tobacco which causes addiction.
- Nicotine in tobacco smoke induces hormonal changes in the body.
- Nicotine leads to clotting of blood.

Carbon monoxide

- It is the poisonous gas found in tobacco smoke.
- It lowers oxygen content in blood and body cells.

Effects of smoking to an individual

- ↓ It causes lung cancer and emphysema.

Tobacco smoke contains carcinogens that damage the cells lining the lungs.

- ↓ It may lead to self-neglect.
- ↓ It leads to reduced life expectancy.
- ↓ Tobacco smoke stains teeth.
- ↓ It causes bad breath from the mouth.
- ↓ It worsens diseases like tuberculosis, bronchitis, pneumonia and asthma.
- ↓ It causes loss of appetite. Nicotine causes addiction thus reducing appetite for food.



Learn, unlearn and relearn: bronchitis is not caused due to tobacco smoking. It is caused by bacteria.

Dangers of tobacco smoking to a pregnant woman.

- ✓ It leads to birth of underweight baby.
- ✓ It leads to miscarriage.
- ✓ It leads to still birth. ✓ It leads to premature birth.

Effects of smoking to the family.

- Smoking leads to poverty.
- It may lead to family neglect.
- Children may copy smoking habits.
- It can cause child and spouse abuse.
- Smoking increases risks of fire outbreak.

Effects of smoking to the community.

- ❖ Tobacco smoke causes air pollution.
- ❖ Smoking leads to loss of community labour.
- ❖ Children may copy smoking habits.
- ❖ It can lead to fire outbreak in an area.

Ways of avoiding smoking in the society.

- ✓ Joining good social groups
- ✓ Learning dangers of smoking.
- ✓ Engaging in sports during leisure time.
- ✓ Keep away from active smokers.



Evaluation Activity 4.3

1. Which substance in tobacco causes coating on teeth?
2. How is active smoking different from passive smoking?
3. Give any **two** reasons why some people smoke.
4. Name any **two** dangerous substances found in tobacco smoke.
5. Write down any **one** respiratory disease caused due to smoking.
6. State any **one** way in which you can safeguard yourself against smoking.
7. Identify the dangerous gas found in tobacco smoke.
8. Which substance in tobacco smoke leads to addiction?
9. The diagram below shows substances commonly used by some people in the community. Study and use it to answer the questions that follow.



- a) Name the habit that involves the use of the substances shown above.
 - b) Mention any **two** body organs that are damaged by the habit named in a) above.
 - c) Why should the substances shown above be kept away from the reach of children?
10. Give any **two** diseases that are worsened due to smoking.
 11. How does tobacco smoking lead to loss of appetite for food?
 12. Why does tobacco smoking cause addiction when used for a long time?
 13. In which way does tobacco smoking cause lung cancer?

DRUGS IN THE SOCIETY

A drug is a chemical substance which changes the normal functioning of the body when taken in.

Types/classes/groups of drugs

➤ essential drugs ➤ narcotic drugs

ESSENTIAL DRUGS

Essential drugs are drugs which meet people's health needs when used correctly

Ways in which essential drugs meet people's health needs.

- They cure diseases.
- They prevent unwanted pregnancies.
- They treat diseases.
- They reduce pain.
- They prevent immunisable diseases.

Characteristics of essential drugs.

- ✓ They are affordable.
- ✓ They are available/accessible.
- ✓ They are common.
- ✓ They cure diseases

Mnemonic used: **CACE**

Qualities of essential drugs

- ↓ They should be affordable.
- ↓ They should be accessible.
- ↓ They should have curative value.
- ↓ They should be effective.
- ↓ They should be common.

Groups/types/categories/forms of essential drugs.

Curative drugs

These are essential drugs used in curing diseases.

Examples are; chloroquine, coartem, quinine, cough syrup and fansider.

Preventive drugs.

These are essential drugs used to prevent diseases.

Examples are vaccines.

Pain killers

These are essential drugs used to reduce or relieve pain when taken.

Examples are Panadol, hedex, curamol, paracentamol, dichlofenac and ibuprofen

Contraceptive drugs

These are essential drugs used in family planning.

Examples are birth control pills and birth control injections.

Drug prescription

Drug prescription is an instruction written by a medical worker to a patient on how to use given drugs.

Pieces of information found on prescribed drugs.

- ❖ Number of drugs to be taken.
- ❖ Time at which to take the drug.
- ❖ Side effects of the drug to the patient.
- ❖ Name of the given drug.

Factors considered when prescribing drugs.

- Age of the patient
- Weight of the patient.
- Strength of the drug.
- Condition of the illness.
- Side effects of the drug to the patient.
- Disease the patient is suffering from.

Reject: gender (sex) of the patient

Note: The above factors enable the health worker to give the right drugs and dosage so as to prevent body poisoning.

How to interpret drug prescription

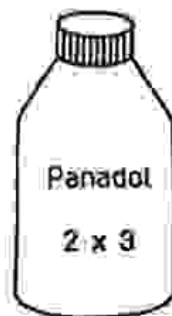


Diagram 1



Diagram 2

Explanation.

In diagram 1, the patient is supposed to take 2 tablets at once 3 times a day. When you divide the 24 hours of a day by 3, you get 8 hours. This means that the patient has to wait for 8 hours before taking the drug again.

In diagram 2, the patient is supposed to take 1 tablet 2 times a day. When you divide the 24 hours by 2, you get 12 hours. Therefore, the patient has to wait for 12 hours after taking the drug before taking another drug.

A medical form



MEDICAL FORM

PATIENT INFORMATION

Full Name
Date of Birth Gender ☐ Male ☐ Female ☐ Other
Address Zip Code
Phone Number E-mail

MEDICAL HISTORY

Please check any of the following conditions that apply to you

- ☐ Allergies
- ☐ Previous surgeries
- ☐ Family medical history
- ☐ Chronic illness
- ☐ Medication currently taking
- ☐ other Medical conditions

REASON FOR VISIT

Describe your symptoms or reason for this visit:

Importance of labels on medical drugs

- ✓ They inform the patient the name of the drug.
- ✓ They inform the patient the composition of the drug.
- ✓ They inform the patient the safety precaution of the drug.
- ✓ They inform the patient about expiry date of the drugs.

Importance of drug prescription.

- ❖ It prevents the patient from taking wrong drugs.
- ❖ It enables the patient to take the right dose of drugs.
- ❖ It ensures correct treatment of the disease.
- ❖ It enables a patient to know when to take the drugs.
- ❖ It enables the patient to take the right drugs.

Dangers of buying essential drugs from local shops

- | | |
|---|-----------------------------------|
| ↑ One may buy expired drugs. | ↑ One may buy contaminated drugs. |
| ↑ The drugs may not be properly prescribed. | ↑ One may buy sub-standard drugs. |
| | ↑ The drugs might be expired. |

STORAGE OF ESSENTIAL DRUGS

Conditions under which essential drugs should be kept.

A place away from direct sunlight.

- To prevent drugs from getting spoilt.
- To prevent drugs from losing curative value.

A place away from moisture.

- To prevent drugs from getting spoilt.

A dust free place.

- To prevent contamination.

Out of reach of children.

- To prevent child poisoning.
- To prevent drug misuse.
- To prevent children from contaminating drugs.

Well-sealed.

- To maintain the strength of the drugs.

Learn, unlearn and relearn:

Traditional drugs and laboratory manufactured drugs are not the types of essential drugs.

Evaluation Activity 4.4

1. State any **two** characteristics of essential drugs.
2. Give any **two** factors to consider when prescribing drugs to patients.
3. Give **one** reason why drugs should be kept;
 - i) Away from moisture and sunlight
 - ii) Out of reach of children.

4. Name any **two** groups of essential drugs.

5. A patient was given a packet of drugs labelled with information as shown in the diagram below. Use it to answer the questions that follow.



- What name is given to such information written on a packet of drugs?
- Give any **one** reason why such information is important to patients.
- How many times a day is the patient supposed to take the drug?
- The patient took the drug in the morning. For how many hours should the patient wait before taking the drug again?

6. Name the type of essential drugs that;

- Relieves the body of pain.
- Cures diseases.
- Prevents diseases.
- Controls unwanted pregnancy.

NARCOTIC DRUGS/DRUGS OF DEPENDENCY

Narcotic drugs are drugs that cause addiction when used for a long time.

They are also called **drugs of dependency**.

Examples of narcotic drugs

- | | | | |
|-------------|-----------|--------------|------------|
| ▪ alcohol | ▪ tobacco | ▪ cocaine | ▪ heroin |
| ▪ marijuana | ▪ opium | ▪ miraa/khat | ▪ cannabis |

Drug abuse, drug misuse and drug dependency

Drug abuse is the use of drugs in the way that is harmful to one's health.

Drug dependency is the condition in which a person is addicted to drugs.

Reasons why some people abuse drugs.

- | | |
|-------------------------|--------------------------------------|
| ✓ To relax their minds. | ✓ To get sleep. |
| ✓ To pastime. | ✓ To boost their mental performance. |
| ✓ To stay awake. | ✓ To fit in peer groups. |

Commonly abused drugs and how they are abused.

Drug	How it is abused
opium	By smoking
heroin	By sniffing By swallowing By injection
cocaine	By sniffing powder

Drug	How it is abused
miraa	By chewing leaves
petrol	By sniffing
alcohol	By drinking
cannabis	By smoking
marijuana	By smoking
caffeine	By drinking
nicotine	By smoking , By chewing tobacco leaves



Points to note:

- Caffeine is got from coffee
- Cocaine is got from coca plant.
- Nicotine is got from tobacco.



Tobacco



Marijuana



Opium



Beer

Effects of drug abuse/dependency to an individual

- Leads to loss of appetite.
- Leads to loss of sleep.
- Leads to staining of teeth.
- Leads to brain and liver damage.
- Leads to self-neglect.
- Leads to coronary heart disease.
- Leads to reduced life expectancy.

Effects of drug abuse/dependency to a family.

- Leads to poverty.
- Leads to child and spouse abuse.
- Leads to broken homes.
- Leads to family neglect.
- Leads to loss of family labour.

Effects of drug abuse/dependency to a community

- ❖ Leads to increased crime rate.
- ❖ Leads to increased road accidents.
- ❖ May lead to loss of community labour.

Ways of avoiding drug dependency

- ✓ Desist bad peer groups.
- ✓ Keep busy with books during leisure time.
- ✓ Join good social clubs.
- ✓ Engage in sports activities.

Life Skills to Safeguard against Alcoholism, Smoking and Drug Dependency

A life skill is a positive behaviour that enables a person to effectively deal with challenges in life.

Life skills prevent temptations of taking narcotic drugs/drugs of dependency.

Life skill	Explanation and use
Self-esteem	This is the confidence and respect one has for oneself. - It empowers an individual to make healthier choices and believe in their ability to overcome challenges without relying on drugs.
Self-awareness	This is the ability to recognize and understand one's own emotions, thoughts and behaviour. -It allows individuals to identify triggers for drug use and understand their own vulnerabilities.
Critical thinking	This is the ability to analyse information objectively, evaluate evidence and form a reasoned judgement. -It enables an individual to assess the risks and consequences of drug use more accurately.
Self-control	This is the ability to regulate one's emotions, thoughts and behaviour in the face of temptations and impulses. -It enables an individual to resist the immediate use of drugs even in challenging situations.
Decision making	This is the process of making choices by identifying options, gathering information and assessing alternatives. -It enables individuals to make conscious, informed choices about their behaviour and lifestyle. -It enables individuals to avoid situations where drug use is likely to occur
Assertiveness	This is the ability to express one's opinions, needs and feelings openly respectfully. -It enables individuals to communicate their refusal to use drugs clearly and confidently. -It enables them to stand up for themselves in social situations and seek support when needed rather than passively accepting drug use.

Peer resistance	This is the ability to withstand pressure from friends or peers to engage in behaviour that one does not agree with or that are harmful. -It allows individuals to say no to drug offers and avoid situations where drug use is prevalent. -It enables them to maintain their personal boundaries and make independent decisions despite social pressures.
Creative thinking	This involves looking at problems or situations from fresh perspective and generating innovative solutions. -It can help an individual to find new and healthier ways to cope with stress, boredom or emotional pain that might lead to drug use.
Coping with emotions	This involves managing and responding to emotional experiences in a healthy way. -It reduces the likelihood of using drugs as a way to escape or numb negative emotions such as sadness, anger or anxiety. healthy coping mechanisms might include talking to friends, exercising or engaging in hobbies.
Coping with stress	This involves handling stressful situations and pressures without resorting to unhealthy behaviour. -It prevents individuals from turning to drugs as a way to relieve stress.
Effective communication	This is the ability to express oneself clearly and listen to others actively and empathetically. -It helps in setting boundaries and refusing offers of drugs assertively.
Problem solving	It involves identifying a problem, generating potential solutions, evaluating options and implementing a plan. -It helps an individual to tackle the underlying issues that might drive drug use such as stress, anxiety or social problems. -It aids in creating action plans for recovery and dealing with relapses constructively.

Drug misuse

Drug misuse is the wrong use of essential drugs.

Ways in which drugs are misused.

- Sharing the same drugs.
- Taking wrong dose.
- Taking drugs for a wrong disease.
- Taking drugs at a wrong time.
- Taking drugs when one is not sick.

Overdose

Overdose is the taking in of more drugs than what is required.

Dangers of overdose to the body.

- It can lead to body poisoning.
- It can lead to death.

- It can damage body organs.

Under dose

Under dose is the taking in of fewer drugs than what is required.

Causes of under dose

- ↓ Fear of taking medicine.
- ↓ Bad smell of the drug.
- ↓ Bitterness of the drug.
- ↓ Bad side effects in one's body.

Dangers of under dose to an individual

- ❖ It reduces drug resistance.
- ❖ It leads to germ resistance in the body.

Evaluation Activity 4.5

- (a) State any **two** effects of chewing miraa (khat) to human body.
(b) Mention any **two** things teenager can do to avoid the habit of chewing miraa (khat)
- Give any **two** life skills that children can develop to safeguard themselves against drug dependency.
- How is taking under dose dangerous to a patient?
- Mention any **two** causes of taking under dose to a patient.
- State any **two** ways in which people misuse drugs.
- In which way is drug abuse different from drug misuse?
- Name narcotic drug obtained from;
i) Coffee..... ii) cocoa.....
- Give any **two** reasons why some people abuse drugs.
- How are the following drugs abused?
i) Opium..... ii) Marijuana
- The diagram below shows a leaf of a certain narcotic drug. Study and use it to answer the questions that follow.



- Identify the narcotic drug whose leaf is shown above.
 - State any **one** way in which people abuse the narcotic drug shown above.
 - Give any **one** way in which the drug whose leaf is shown above affects the human health.
 - Apart from the narcotic shown above, mention any **one** other example of a narcotic drug.
- A doctor prescribed Panadol as 2x3 to a patient. The patient instead took 1x3.
(a) What term is used to describe the habit practised by the patient?
(b) Give any **two** reasons why the patient practised the habit above.
(c) State **one** effect of the habit practised by the patient.

Theme:
Topic

THE WORLD OF LIVING THINGS

5: CLASSIFICATION OF PLANTS

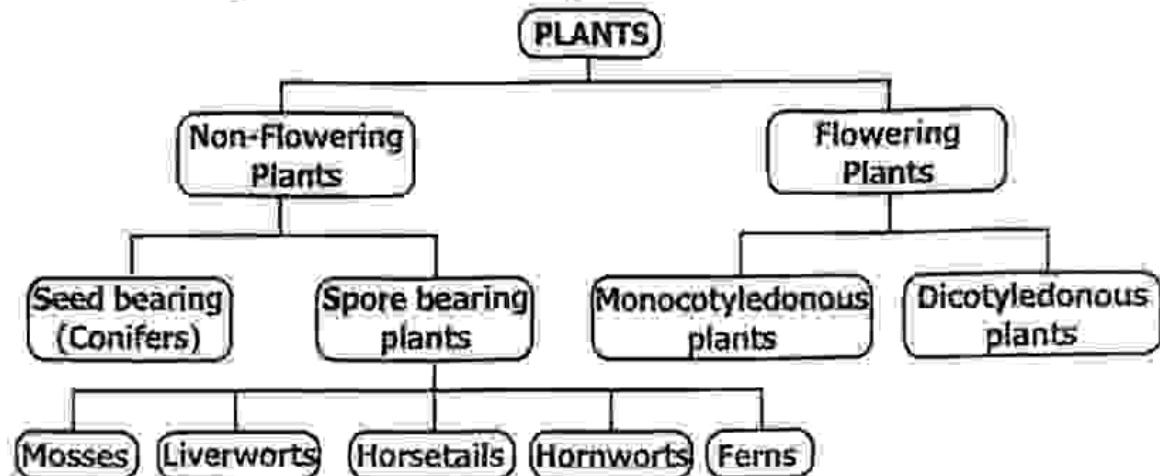
Characteristics of plants

- ❖ They make their own food.
- ❖ They have chlorophyll.
- ❖ They have many cells.

Biological factors/features/characteristics used to classify plants

- Mode of reproduction
- The structure of the plant.

Table showing classification of plants.



Groups of plants

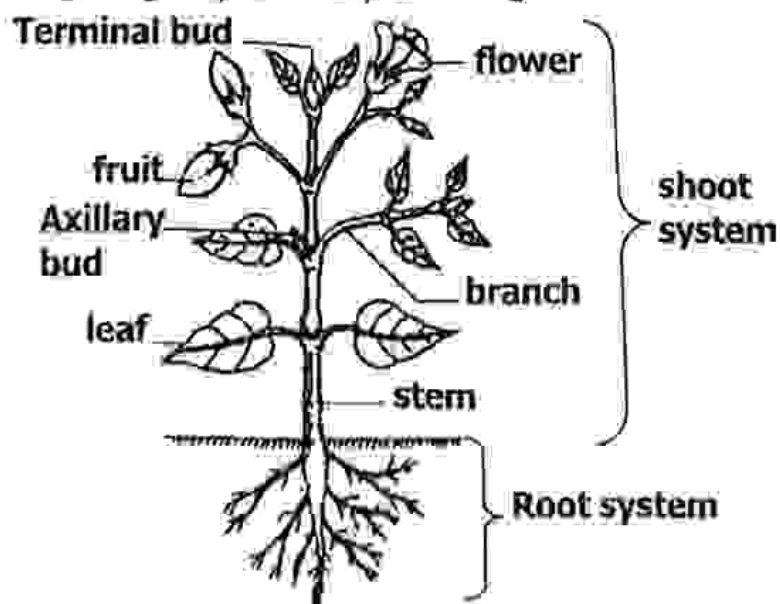
➤ flowering plants

➤ non flowering plants

FLOWERING PLANTS

Flowering plants are plants that bear flowers.

They are the largest group of the plant kingdom.



Functions of the parts of a flowering plant.

Terminal bud. It grows to increase the height of the plant.

Axillary bud/lateral bud. It develops into a side branch.

Stem.

- It holds the leaves, flowers and fruits of a plant.
- It transports water and mineral salts from roots to other parts of a plant.
- It transports manufactured food from leaves to other parts of a plant.

Leaves

- ❖ They make food for the plant.
- ❖ They carry out transpiration.
- ❖ Some leaves store food for the plant.
- ❖ They have stomata for breathing.

Roots

- ✓ Roots absorb water and mineral salts for the plant.
- ✓ Roots hold the plant firmly into the soil.
- ✓ Some roots store food for the plants.

These include; carrots, sweet potatoes, turnips, cassava, beetroots and radishes

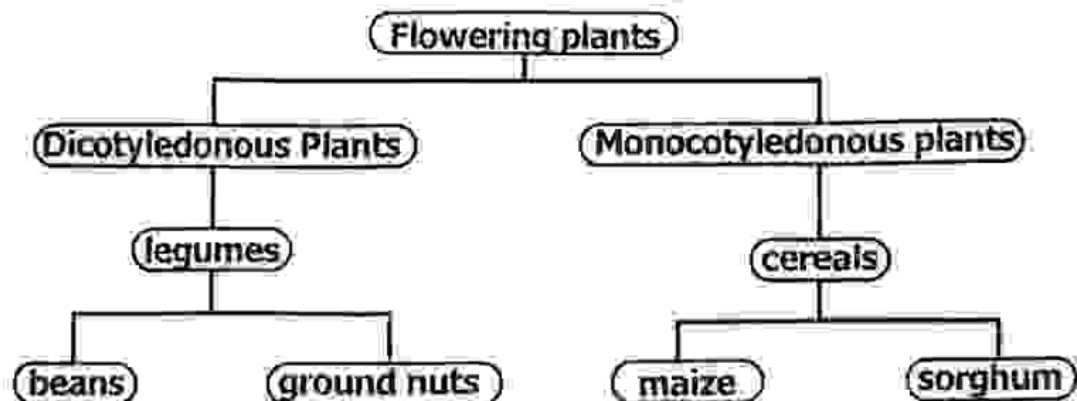
- ✓ Breathing roots are used for breathing.

Flowers.

For reproduction. Fruits and seeds develop from flowers.

Fruits. They store seeds.

Table showing classification of flowering plants.



Groups of flowering plants

- dicotyledonous plants
- monocotyledonous plants.

Dicotyledonous plants

These are flowering plants that bear seeds with two cotyledons.

Characteristics of dicotyledonous plants

- ✓ They bear seeds with two cotyledons.
- ✓ They have network leaf venation.

- ✓ Their seeds store food in the cotyledon.
- ✓ They have taproot system.
- ✓ Most dicot seeds undergo epigeal germination.

Examples of dicot plants are;

bean plants, ground nut plants, pea plants, mango plants, orange plants, cabbage plants, carrot plants, pawpaw plants.



Soya Bean



Pea



Bean Plant



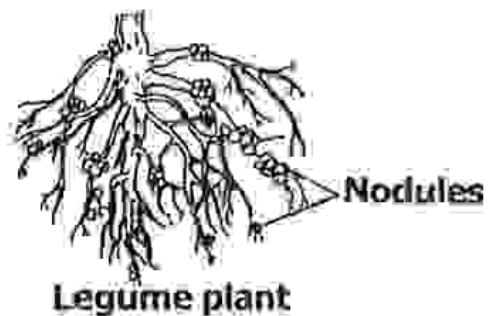
Ground nut

LEGUMES

Legumes are dicotyledonous plants that have nodules on their roots.
 Root nodules store nitrogen fixing bacteria.
 Nitrogen fixing bacteria fix nitrogen into the soil. This helps to improve soil fertility.

Root hairs absorb water and mineral salts from the soil.

Examples of legumes are ; bean plants, pea plants, ground nut plants, soya bean plants, French bean plants and castor plants



Points to note:



- Legumes help to add nitrates into the soil.
- Legumes can be identified by checking for root nodules on their roots
- Cowpeas, avocados, green peas and ground nuts are dicotyledons that undergo hypogeal germination.
- Simsim plants are not legumes because they lack root nodules.

How legumes differ from other dicotyledonous plants.

- Legumes have root nodules.
- Legumes store their seeds in pods

Importance of legumes

- ✓ They are source of proteins.
- ✓ They are sold for money.
- ✓ They improve soil fertility.

Monocotyledonous plants

These are flowering plants that bear seeds with one cotyledon.

Characteristics of monocotyledonous plants

- ↓ They bear seeds with one cotyledon.
- ↓ They have parallel leaf venation.
- ↓ They store their food in the endosperm.
- ↓ Their seeds undergo hypogeal germination.
- ↓ They have fibrous root system.

Examples of monocotyledonous plants are

maize plants, millet plants, sorghum plants, onion plants....

CEREALS

Cereals are monocotyledonous plants that produce grains.

Characteristics of cereals.

- ✓ They are annual plants.
- ✓ They produce large quantities of seeds.
- ✓ They have fibrous root system
- ✓ They are rich in carbohydrates.
- ✓ They have parallel veined leaves.

Examples of cereal plants are

maize, millet, sorghum, oats, barley, wheat and rice.



Sorghum



Millet



Maize

Importance of cereal plants

- ❖ They are eaten by many people.
- ❖ Cereals like wheat are used to make baking powder.
- ❖ Oats are used to make oat cakes.
- ❖ Cereals are used to make flour for porridge and posho.
- ❖ Cereals are used to make animal feeds.
- ❖ Maize grains can be used to make corn oil.
- ❖ Maize cobs are used to make bioethanol.



Wheat

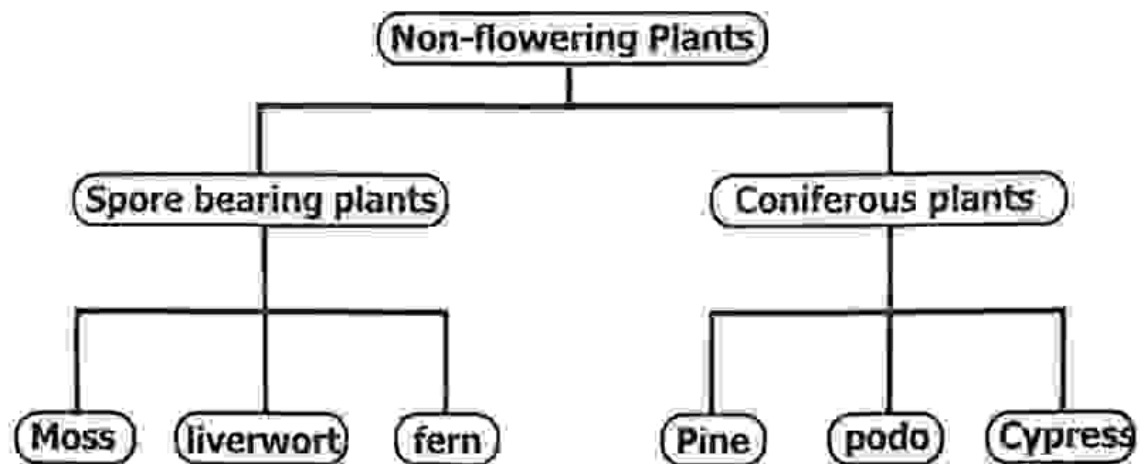
Evaluation Activity 5.1

1. State any **one** biological feature used to classify plants.
2. Mention **two** groups of plants.
3. State any **two** characteristics of dicotyledonous plants.
4. How are legumes different from other dicotyledonous plants?
5. Name the group of flowering plants to which;
 - i) Cowpeas belong
 - ii) Millet belong
6. In which way do legumes make use of nitrogen gas?
7. Mention any **one** way in which cereals are important to people.
8. How can **one** identify a legume from other dicotyledonous plants?
9. What is the role of nitrogen fixing bacteria in the soil?
10. Where do legumes store their seeds?

NON-FLOWERING PLANTS

Non flowering plants are plants that do not bear flowers.
They make their own food because they contain chlorophyll.

Classification table for non-flowering plants



Groups of non-flowering plants

- Seed bearing non flowering plants.
- Spore bearing plants

Spore bearing plants

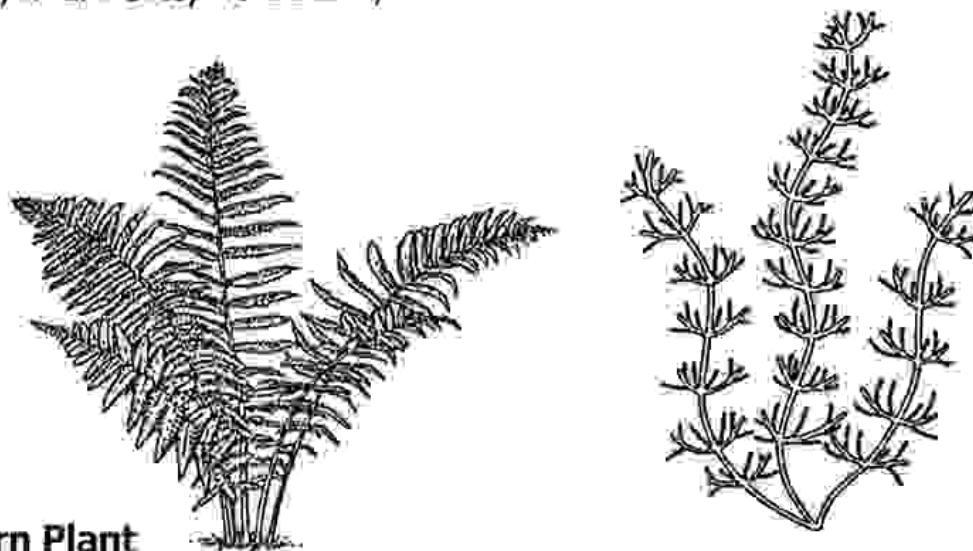
Spore bearing plants are non-flowering plants that reproduce by means of spores.

A spore is a single cell that can develop into new organisms.

Fungi like mushrooms, puff balls and toad stools also reproduce by means of spores.

Examples of spore bearing plants are;

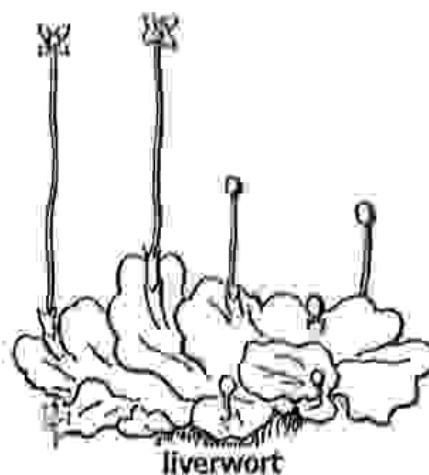
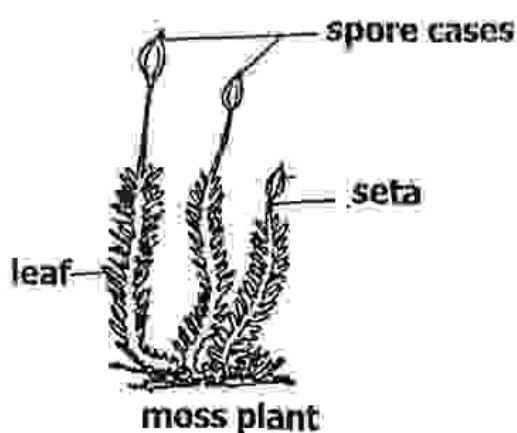
mosses, liverworts, horsetails, hornworts and ferns.



Fern Plant

Leaves of a fern plant are called **fronds**

They make food for the fern plant



The spore case produces spores, seta holds the spore case and leaves make food for the plant.

Spore bearing plants mainly grow in damp places like verandas, logs, on rocks.

Seed bearing non-flowering plants. (Conifers)

Conifers are groups of non-flowering plants that reproduce by means of seeds.

Conifers store their seeds in cones.

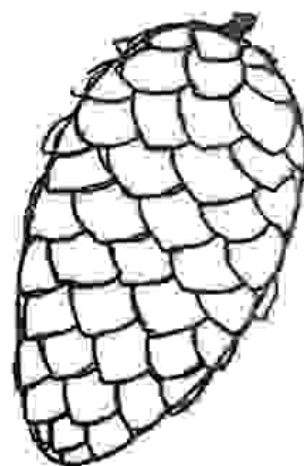
The cone stores seeds used for reproduction.

The staminate cone produces pollen grains while pistillate cone produces ovules.

Examples of conifers are; pine, fir, cedar, cypress, cycad, podocarp and ginkgo.

Importance of conifers

- ✓ They produce soft wood for timber
- ✓ They are source of income to people.
- ✓ They provide shade to people.
- ✓ They are used to make soft board ceilings.
- ✓ They are used for making match boxes, papers, match sticks, book covers and wooden boards.
- ✓ They are used as windbreaks.

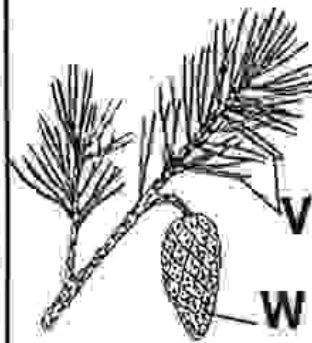


Evaluation Activity 5.2

1. Name the group of seed bearing non flowering plants.
2. How is a mushroom similar to a fern in the way they reproduce?
3. Where are seeds of conifers stored?
4. Give any **one** way in which conifers are useful to people.
5. Which group of non-flowering plants is used for making books?

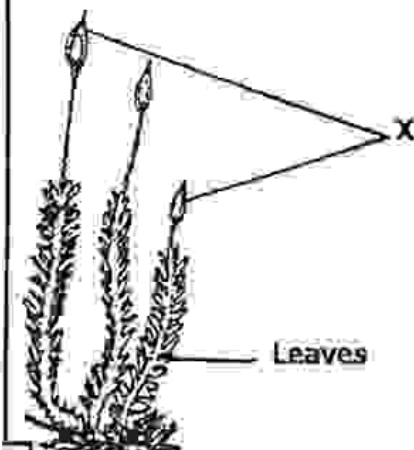
Evaluation Activity 5.2... continued

The diagram below shows a structure found on conifers. Use it to answer questions 6 and 7.



6. Name the part of the conifer marked **V**.
7. State the function of part marked **W** of the conifer?
8. Write down any **two** examples of;
 - i) Spore bearing plants
 - ii) Seed bearing non flowering plants
9. Why are non-flowering plants able to make their own food?
10. Which group of non-flowering plants reproduce by means of seeds?

11. How is a pine plant similar to a bean plant in the way they reproduce?
12. The diagram below shows a non-flowering plant. Study and use it to answer the questions that follow.



- a) Name the non-flowering plants shown above.
- b) How is part labelled **X** useful to the above non-flowering plant?
- c) Name any **two** other non-flowering plants that reproduce in the same as the above plant.

SEED DISPERSAL

Seed dispersal is the scattering of seeds from the parent plant to other areas.

Mechanisms/types/modes/methods of seed dispersal.

wind dispersal, water dispersal, animal dispersal and self-dispersal

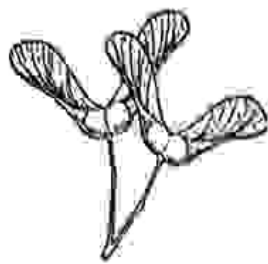
Wind dispersal

Characteristics of seeds dispersed by wind.

- ✓ They are small and light. (Reject: they are "small" alone)
- ✓ Seeds are enclosed in fruits which have parachute-like structures.
- ✓ They have feathery structures.

Examples of seeds dispersed by wind are;

tridax, dandelion, jacaranda, silk cotton, maple, orchid, Poppy and phyllon.



Maple seed



Poppy seeds



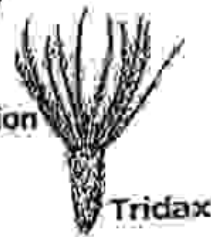
Orchid



Jacaranda



Dandelion



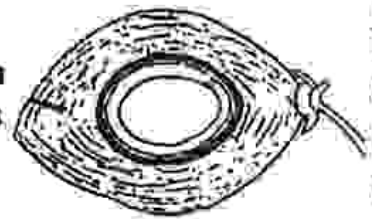
Tridax

Water dispersal

Characteristics of seeds dispersed by water

- Their mesocarp has air spaces.
- They are light to float on water.
- They have spongy mesocarp.
- They have hollow mesocarp.
- They have hard testa to prevent the entry of water.

Husk with air spaces



Examples of seeds dispersed by water are; coconut, water lily, lotus and cattail

Animal dispersal

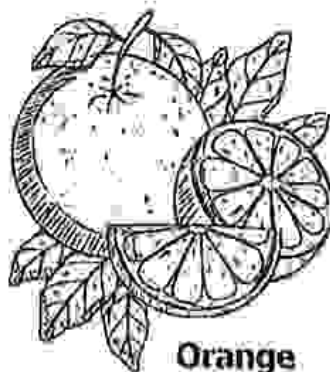
Characteristics of seeds dispersed by animals.

- ❖ They have hooks to attach on the animal's body e.g. black jack
- ❖ They are sweet and juicy.
- ❖ They have hard testa to resist digestion. e.g. tomato seeds, passion fruit seeds, red pepper seeds guava seeds.
- ❖ Their fruits produce scent when mature e.g. jackfruit. This attracts animals for seed dispersal.
- ❖ Their fruits are edible.
- ❖ Their fruits have bright epicarps e.g. mangoes, guavas, oranges.

Examples of seeds dispersed by animals are; mango seeds, orange seeds, black jack, beggar ticks, sandbur, mimosa, love grass and tangerines



Guava



Orange



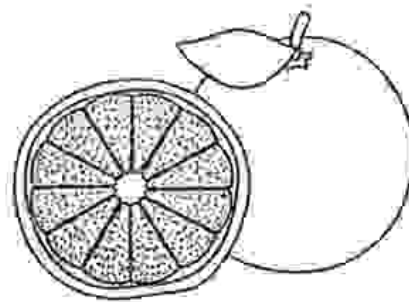
Black jack



Mango



Beggar ticks



Lemon

Self-dispersal (explosive mechanism)

Seeds are born in pods. The pods burst when dry to release seeds.

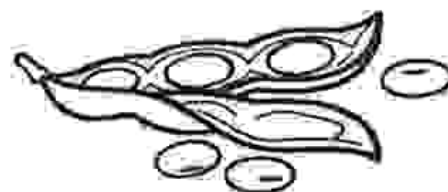
Factors that lead to self-dispersal

- ↑ Long exposure to sun heat
- ↑ Maturity of the pods.
- ↑ Presence of the lines of weakness on the wall of the pod.

Examples of seeds dispersed by self-dispersal are; castor oil seeds, bean seeds, pea seeds



Before dispersal



After dispersal

Agents of seed dispersal are water, animals and wind.

Importance/advantages of seed dispersal to plants

- ✓ It enables plants to colonise new areas.
- ✓ It prevents overcrowding of plants.
- ✓ It reduces competition for sunlight, nutrients and space.
- ✓ It prevents plant extinction.

Disadvantages of seed dispersal

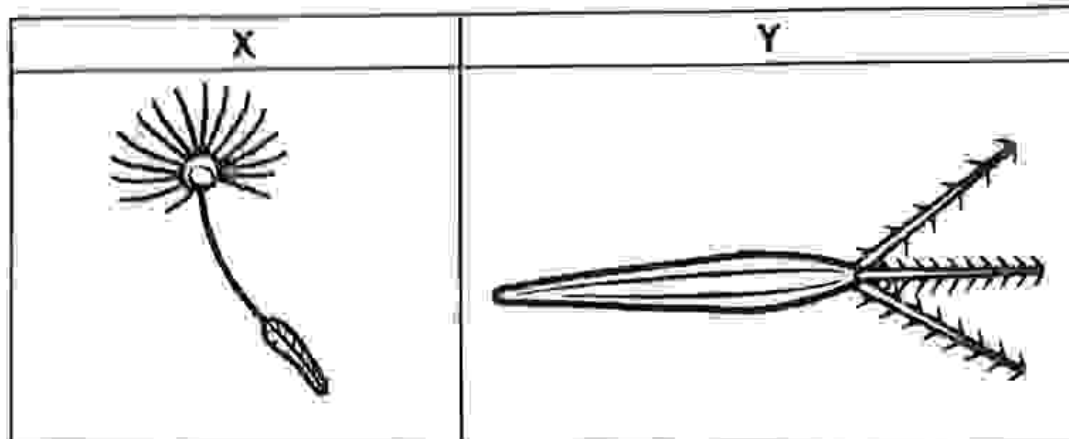
- Increases the growth of weeds.
- It may increase pests and diseases in an area.

Evaluation Activity 5.3

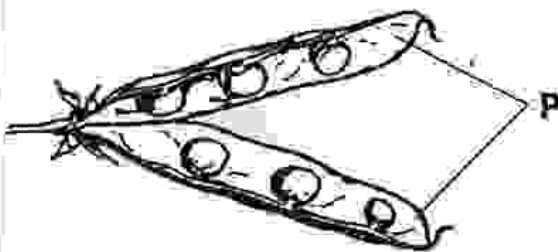
1. Give the meaning of the term seed dispersal.
2. a) Apart from animals, name any **two** other agents of seed dispersal.
b) State any **two** characteristics of seeds dispersed by animals.
3. Why does a jack fruit send off a smell when mature?
4. How is a coconut seed adapted to water dispersal?

Evaluation Activity 5.3... continued

5. The diagram below show seeds. Study and use them to answer the questions that follow.



- How is the seed marked **X** dispersed?
 - In which way is the seed marked **Y** adapted to its mechanism of dispersal?
 - Name any **two** other seeds that can be dispersed in the same way as seed marked **X**.
6. State any **one** characteristic of seeds dispersed by wind.
7. Which agent of seed dispersal scatters lotus seeds?
8. a) Name the method of seed dispersal in;
- Black jack
 - Coconut
- b) Give any **two** ways in which seed dispersal is important to plants.
9. The diagram below shows a method of seed dispersal. Study it to answer the questions that follow.



- Identify the method of seed dispersal shown in the diagram above.
- State any **one** factor that makes the structure marked **P** to split during seed dispersal.

- Name any **two** other plants whose seeds are dispersed through the diagram shown above.

PLANT PROPAGATION

Plant propagation is the way of obtaining new plants from the parent plants.

Ways of propagating plants

➤ Seed propagation

➤ Vegetative propagation

Some plants are propagated by means of seeds. Plants propagated by seeds include; Millet, maize, beans, tomatoes, passion fruits, mangoes,

ground nuts, oranges...

Vegetative propagation

This is a way of obtaining new plants from parent plants using other parts other than seeds.

Parts of the plant used for vegetative propagation are; stems, leaves and axillary buds.

Vegetative propagation is a form of asexual reproduction in plants.

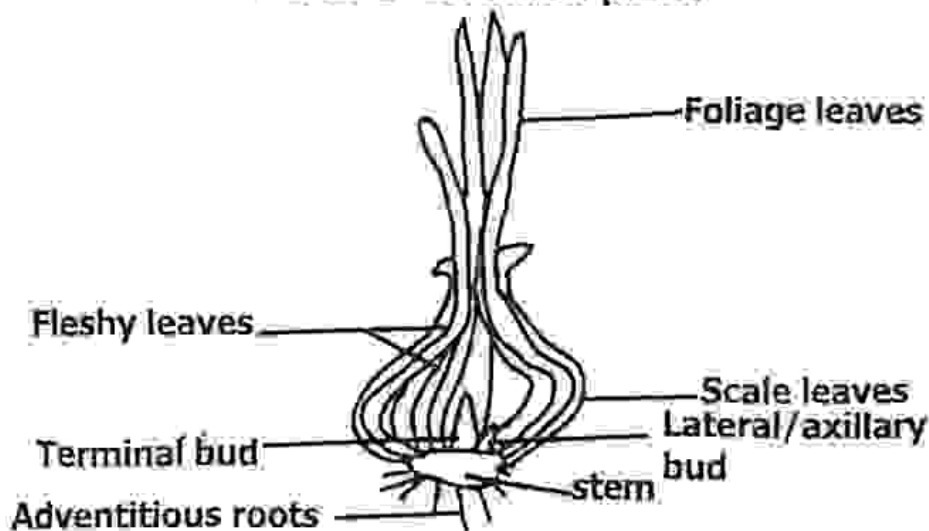
Natural methods of vegetative propagation

Planting the bulbs

This is used in onion, garlic, shallots and spider lily

Presence of axillary buds that develops into the new shoot.

Parts of an onion plant



Functions of each part of an onion plant

Foliage leaves. They make food for the onion.

Fleshy leaves. They store food for the onion.

Scale leaves. They protect the fleshy leaves.

Adventitious leaves. They hold the onion plant firmly into the soil.

Axillary bud. It develops into the new shoot.

Points to note: An onion can also be propagated by planting seeds.

An onion has fibrous root system and adventitious type of roots.

It contains mineral salts (potassium) and vitamins.

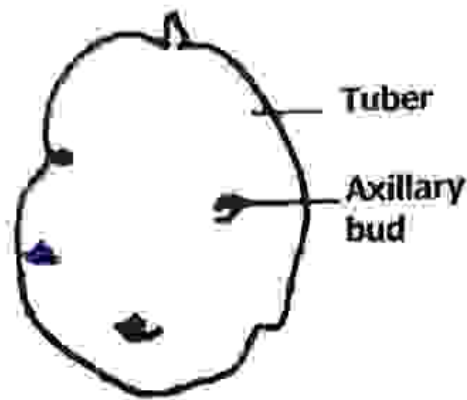
It undergoes hypogeal germination. It has parallel leaf venation and it is a monocot plant.

Planting the stem tuber

Such plants are Irish potatoes and cocoyams.

Presence of the axillary bud that develop into the new shoot.

Don't speak for Quality, let Quality speak for itself.



Tuber

It stores food for the plant
It supplies food to the growing shoot during propagation of an Irish potato.

Lateral bud/axillary bud

It develops into the new shoot.

Scale leaf

It protects the axillary bud from physical damage.

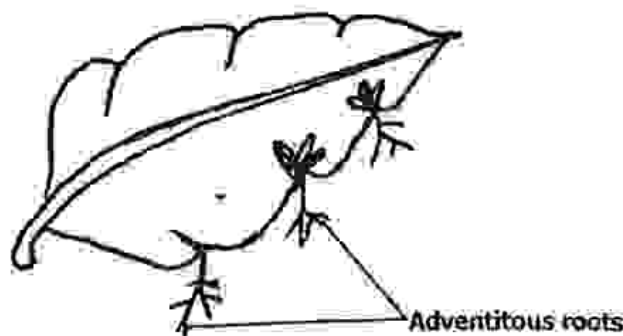
Adventitious roots

They absorb water and mineral salts from the soil.

They hold the plant firmly into the soil.

Planting leaves

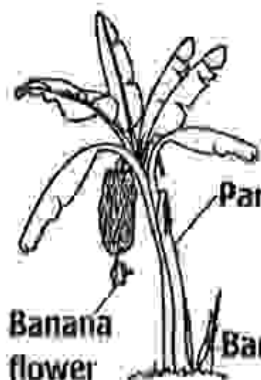
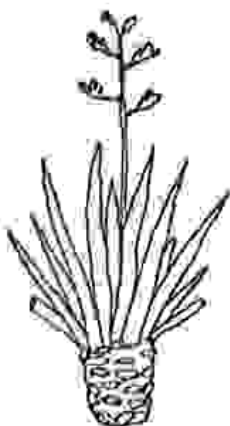
An example is a Bryophyllum plant. Adventitious roots and axillary buds develop at the edges of the leaf margin when leaves are buried into fertile soil.



Planting the suckers

Such plants are; bananas, sisal, pineapples and aloe vera.

Sisal

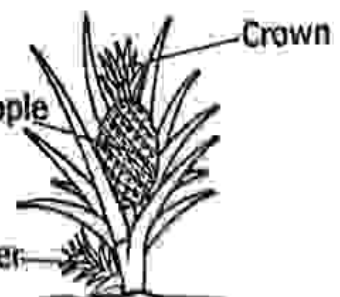


Banana flower

Parent plant

Banana sucker

Pineapple



Crown

Sucker

Points to note:

 Pineapples can also be propagated by planting the crown and slips.
Banana suckers are planted in wider pits

Reasons

- ✓ To prevent overcrowding of the sucker.
- ✓ To create large surface area for trapping water for proper growth of suckers.
- ✓ To ease expansion and penetration of roots.

Before transplanting the banana sucker, leaves and roots are cut off.

- ✓ Leaves are cut off to reduce the rate of transpiration.
- ✓ Roots are cut off;

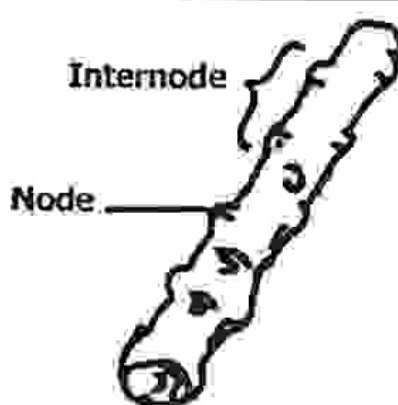
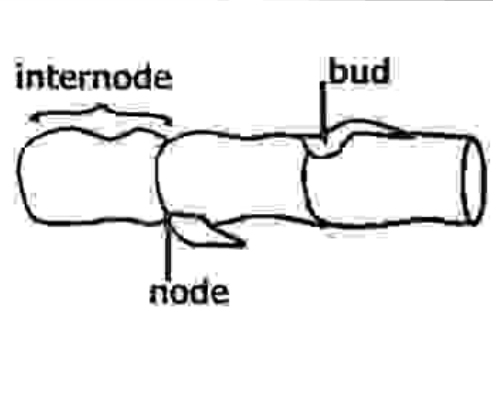
1. To remove hiding places for pests. e.g. nematodes
2. To promote the growth of new healthy roots.
3. To ensure that a clean sucker is used.

Artificial methods of vegetative propagation

Planting stem cuttings

Such plants are; cassava, sugar cane and sweet potatoes

They have nodes that develop into the new shoot.

Presence of the nodes with axillary buds. Axillary buds develop into the new shoot.	Presence of the nodes with axillary buds. Axillary buds develop into the new shoot.
	

Shoot grafting

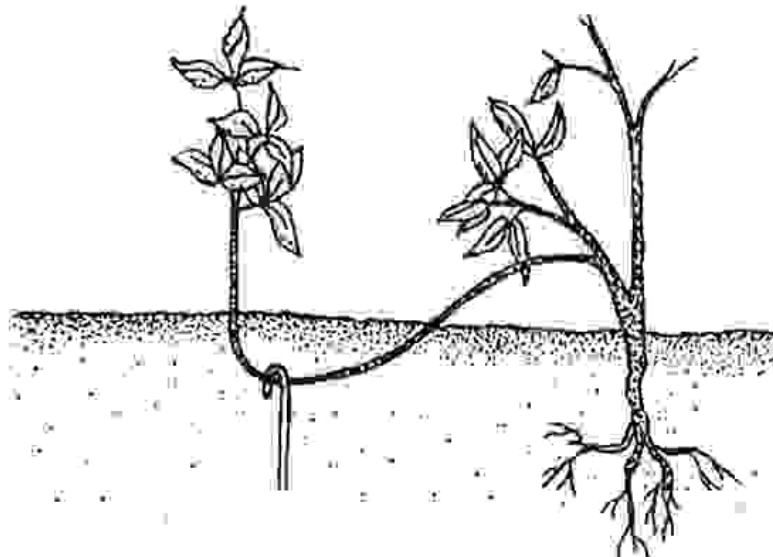
In this method, a shoot (scion) from one plant is joined to the stock from another plant. Mangoes, oranges, lemons, guavas, tangerines and grapes are propagated by grafting.



Layering

In this method of propagation, a branch from a plant is wounded at a point and pegged into the fertile soil.

Adventitious roots grow from the wounded part. After sometime, the pegged branch is cut off from the parent plant and left to grow into an individual plant.

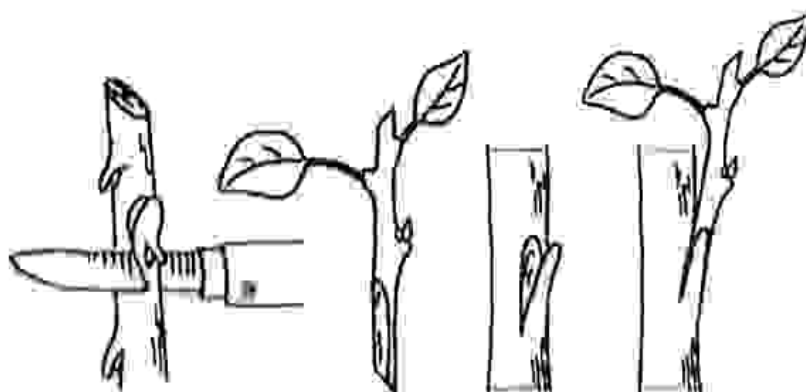


Budding

Budding is the placing of a bud of one plant on the root system of another plant.

This is a method used in nursey beds to propagate fruit trees.

A bud from any good quality tree is taken. A small T-shaped cut is made on another tree. The bud brought is then inserted in the cut and tied to keep in position. It is used in plants like oranges and lemons.

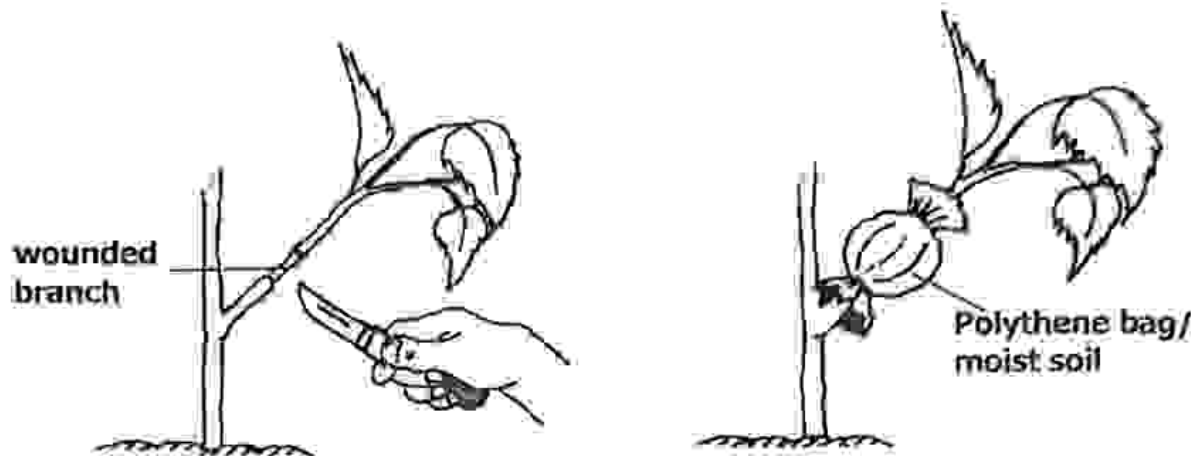


Plants propagated by budding are lemons, roses, avocados and oranges.

Marcotting

In marcotting, a branch of a plant is wounded and a polythene bag or sack containing moist fertile soil is tied around the wounded part.

Fertile soil in the sack encourages the growth of adventitious roots.



Pawpaws and roses can be propagated by marcotting.

Advantages of vegetative propagation

- ✓ Plants grow and mature faster.
- ✓ The farmer gets high yields.
- ✓ Desired varieties are obtained.
- ✓ Young plants get food from the parent plant.
- ✓ Fruit trees which take long time to bear fruits take short time to bear fruits.

Economic values of plants to people

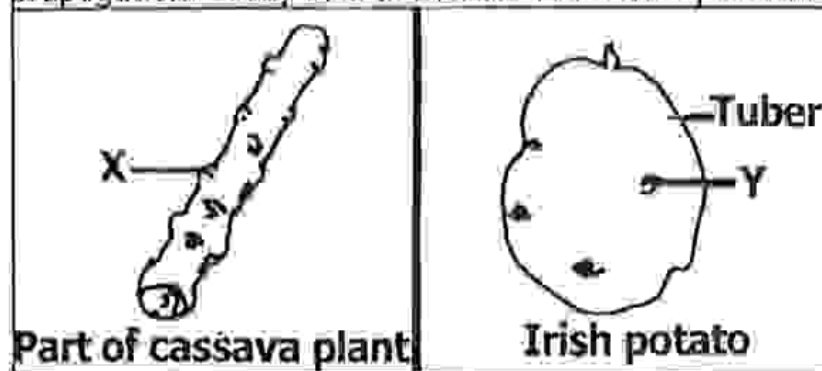
- Natural forests are tourist attractions.
- Plants are source of income when sold.
- Plants are used as building materials.
- Some plants are source of timber for making furniture.
- Plants provide raw materials for craft industries.

Evaluation Activity 5.4

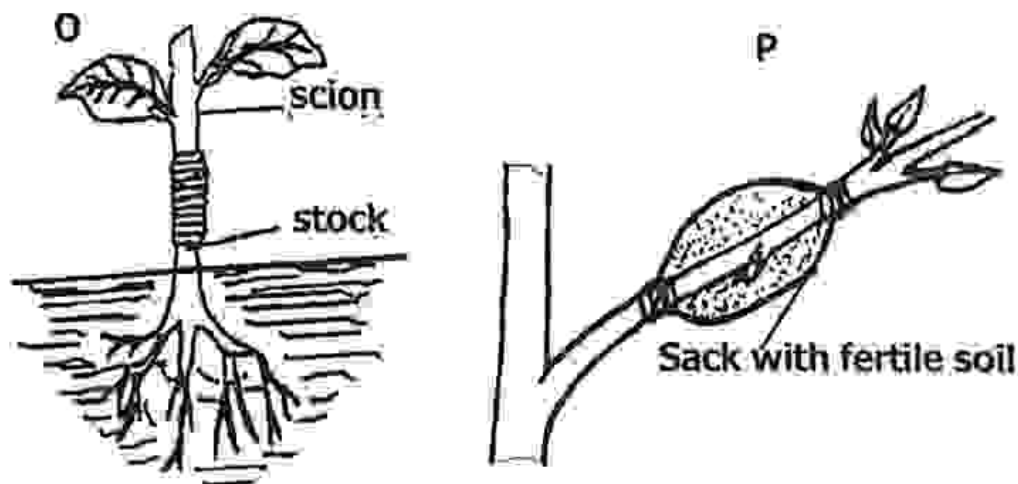
1. (a) Some fruit trees take long to bear fruits. What is the best method of making such trees produce fruits within a year?
(b) How are the following plants propagated?
(i) sisal hemp..... (ii) cocoyam..... (iii) sugar cane.....
2. Name the method of plant propagation where a scion and a stock are joined together..
3. State any **one** advantage of using vegetative propagation.
4. Apart from planting sucker, give any **one** other method used to propagate pineapples.
5. How is the propagation of sweet potatoes different from that of an Irish potato?
6. Mention any **two** parts of the plant that can be used for vegetative propagation.

Evaluation Activity 5.4 ... continued

7. The diagrams below show part of a cassava plant and an irish potato that are used in propagation. Study and use it to answer the questions that follow.



- Name the part of the cassava plant marked X.
 - State the method of vegetative propagation where the part of a cassava plant shown above is used.
 - In which **one** way is the function of part X of the cassava plant similar to that of part Y of the irish potato?
 - Give the importance of the tuber in the propagation of the irish potato.
8. The diagram belows shows artificial methods of vegetative plant propagation. Study and use it to answer the questions that follow.



- Name the method of vegetative plant propagation marked O and P
 - O
 - P
- State the importance of the fertile soil found in the sack on the method marked P.
- Apart from the above methods, mention any **one** other artificial method of vegetative plant propagation.

Theme: SCIENCE IN HUMAN ACTIVITIES AND OCCUPATIONS

Topic 6: KEEPING CATTLE

Examples of cattle are; bulls, cows, calves, bullocks and heifers

Reasons why people keep cattle

- ↑ To get milk
- ↑ To get meat
- ↑ To sell and get money.
- ↑ To provide labour for ploughing.

Importance of keeping cattle

- Cattle provide us with meat and milk.
- Their hides are used in leather industry.
- Oxen are used for transport and ploughing.
- Their dung is used for making farmyard manure.
- Cattle are sold for money.

TYPES OF CATTLE

Type of cattle refers to group of cattle kept for a specific purpose.

The types of cattle are;

- dairy cattle
- beef cattle
- dual purpose cattle
- draught cattle

Dairy cattle

Dairy cattle are cattle mainly kept for the purpose of milk production.

Characteristics of dairy cattle

- ❖ They have large udder.
- ❖ They have large hind quarters
- ❖ They have triangular body shape.
- ❖ They have long legs.

Examples of dairy cattle are;

Friesian, Guernsey, Jersey, Brown swiss, Jamaican hope and Ayrshire

Beef cattle

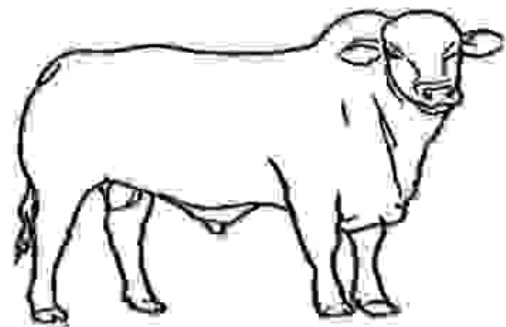
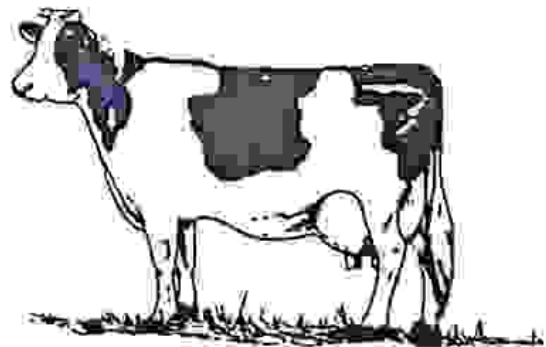
Beef cattle are cattle kept for the purpose of meat production.

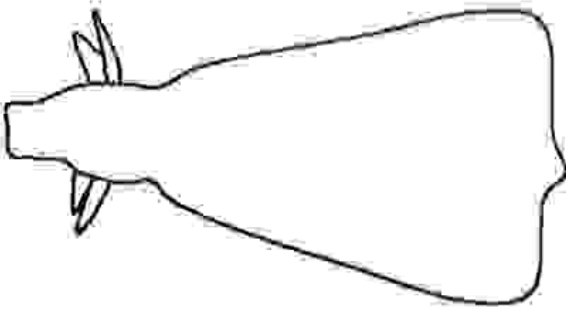
Characteristics of beef cattle

- ✓ They have small hind legs.
- ✓ They have short legs.
- ✓ They produce plenty of meat.
- ✓ They have rectangular shape.

Examples of beef cattle

Borān, Hereford, Galloway, Charolais, Aberdeen Angus, American beef master



Beef cattle	Dairy Cattle
	

Differences between dairy cattle and beef cattle.

- Dairy cattle have big udder while beef cattle have small udder.
- Dairy cattle have triangular body shape while beef cattle have rectangular body shape.
- Dairy cattle have wide hind quarters while beef cattle have small hindquarters.
- Dairy cattle have long legs while beef cattle have short legs.

Dual purpose cattle

Dual purpose cattle are cattle kept for both meat and milk.

Examples of dual-purpose cattle are Redpoll, Sahiwal and Zebu cattle.

Draught cattle/work type cattle

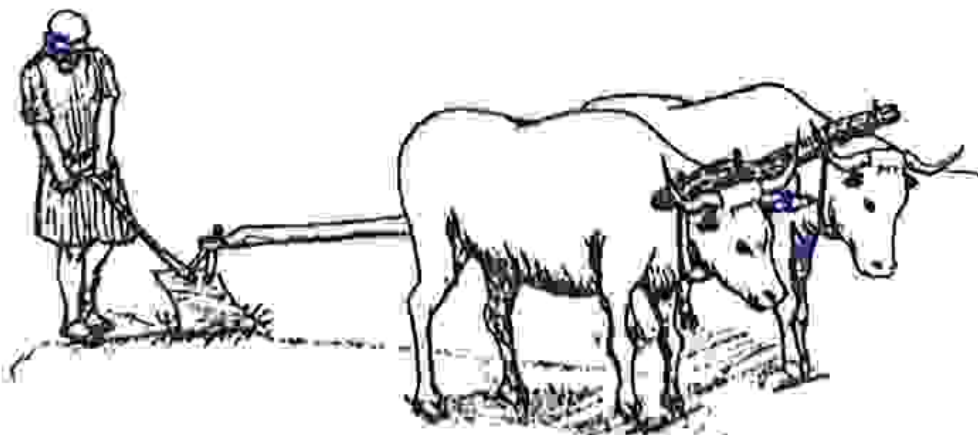
Draught cattle are cattle kept to provide labour on a farm.

Examples of draught cattle are oxen.

They provide labour for ploughing and transport.

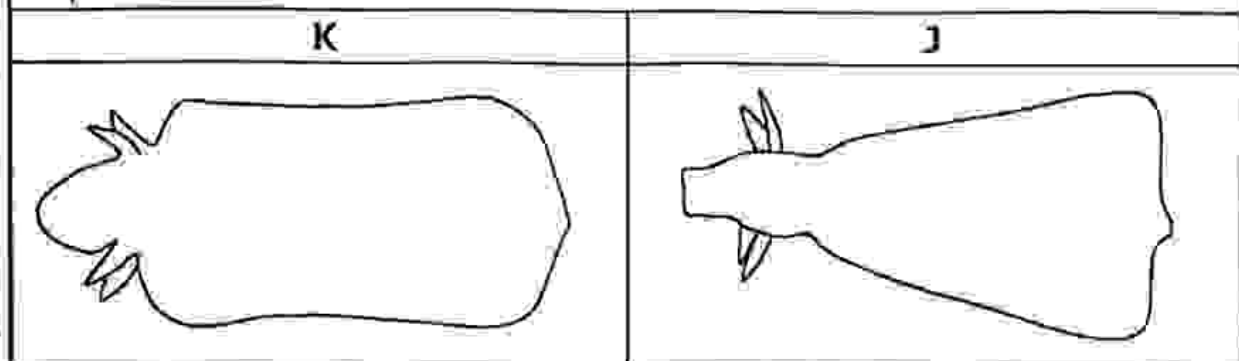
Their adaptations are;

- They have low water demand.
- They are disease resistant.
- They are docile since they are castrated.
- They are drought resistant.



Evaluation Activity 6.1

1. State the type of cattle kept to provide labour on the farm.
2. Give any **two** reasons why people keep cattle.
3. Name any **two** exotic breeds of cattle for;
 - i) Milk production
 - ii) Meat production
4. State any **two** characteristics of dairy cattle.
5. How is keeping cattle helpful to leather industries?
6. Which type of cattle helps in the production of yoghurt?
7. The diagrams below show the shapes of cattle. Use them to answer the questions below.



- a) Name the type of cattle with the shapes marked K and J.
- b) Identify the main product obtained from cattle with shape K.
- c) State any **one** characteristic of cattle with shape J.

BREEDS OF CATTLE

Breeds of cattle are groups of cattle that have similar characteristics.

These characteristics are; colour, size, body shape, productivity and growth rate.

Heredity, resistance to diseases, body weight, growth rate, production rate are considered

when selecting the breed of cattle to keep.

Breeds of cattle include; local breeds, exotic breeds and crossbreeds

Local breeds of cattle

They are also called indigenous cattle

Characteristics of local breeds of cattle

- ✓ They have different colours.
- ✓ They are dual purpose in nature.
- ✓ The bulls have big humps.
- ✓ They have big dewlaps.

Examples of local breeds of cattle

- Boran cattle
- Ankole cattle
- Zebu cattle
- Sanga cattle

Advantages of local breeds of cattle.

- They can survive on poor pasture.
- They are resistant to tropical diseases.
- They are resistant to harsh weather conditions.
- They produce good flavoured beef.
- They have ability to walk long distances.

Disadvantages of local breeds of cattle.

- They grow and mature slowly.
- They produce less milk and beef.

Ways in which local breeds of cattle can be improved.

- Crossbreeding
- Castration
- Proper housing
- Proper feeding
- Regular vaccination

Exotic breeds of cattle

These are breeds of cattle that are imported from overseas.

Characteristics of exotic breeds of cattle.

- ↓ The bulls have very short or no humps.
- ↓ They have very short or no horns.
- ↓ They have small dewlaps.
- ↓ They are kept for a specific product.

Examples of exotic breeds of cattle

Friesians, Ayrshire, Guernsey and jersey

Advantages of exotic breeds of cattle

- They grow and mature quickly.
- They produce high quantities of milk and beef.

Disadvantages of exotic breeds of cattle.

- They are not tolerant to tropical diseases.
- They need a lot of care and attention.
- Their meat is very soft and not tasty.
- They are not resistant to harsh weather conditions.

Crossbreeds

A crossbreed is a breed of cattle got by mating a pure local breed and an exotic breed.

They are also called hybrids.

Crossbreeds are more resistant to diseases than exotic breeds.

Crossbreeds produce more meat and milk than exotic breeds.

BREEDING IN CATTLE

Breeding in cattle refers to improving and maintaining the desired characteristics in livestock.

Such characteristics are; fast growth, resistance to diseases and ability to produce more meat and milk.

Methods of breeding in cattle

Inbreeding.

This is the mating of closely related animals.

Inbreeding helps to maintain the desired characteristics in livestock.

However, inbreeding can result into production of poor quality animals.

Inbreeding can be controlled by; paddocking, castration and artificial insemination.

Outbreeding.

This is the mating of livestock of the same breed but from different herds/stocks.

It helps to restore the qualities in cattle that might be disappearing.

Line breeding

This is the mating of distantly related animals.

Crossbreeding

This is the mating of a pure local breed with an exotic breed.

Crossbreeding helps to improve on the quality of local breeds.

Upgrading

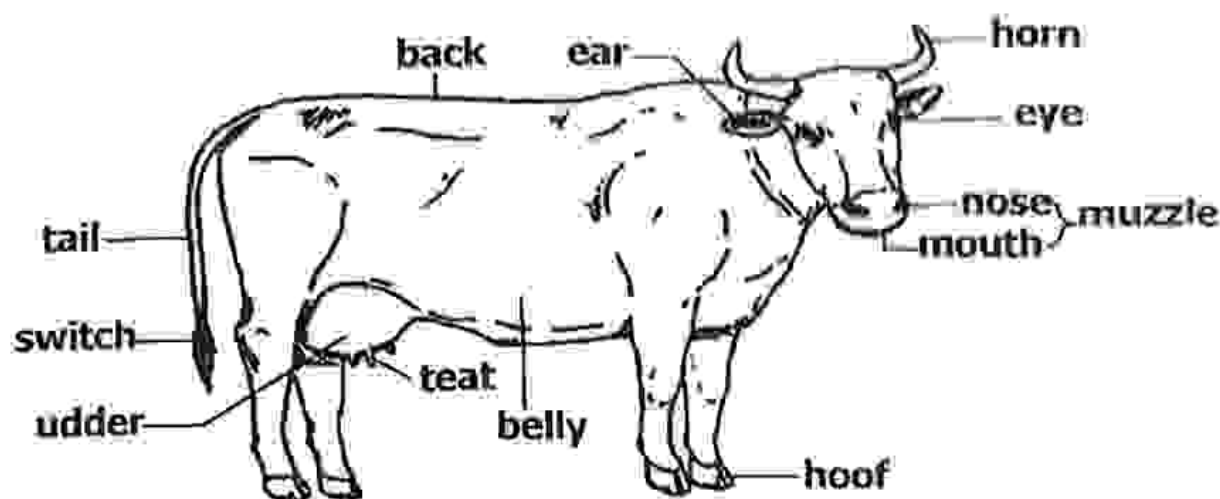
This is the mating of local breed with an exotic breed several times.

This helps to improve on the quality of local breeds and their products.

Evaluation Activity 5.2

1. Give any **one** disadvantage of inbreeding in cattle.
2. Name any **two** exotic breeds of cattle kept in Uganda.
3. State any **two** reasons why most farmers prefer keeping local breeds of cattle to exotic breeds.
4. Give any **two** ways in which a farmer can improve on the local breeds of cattle.
5. Mention any **two** methods of breeding in cattle.
6. State any **two** ways in which a cattle farmer can control inbreeding in cattle.

EXTERNAL FEATURES OF A COW.



Functions of each feature/ part of the cow.

Part/feature	Function to a cow
Nose	It is used for smelling. It also lets in air during breathing.
Muzzle	It prevents the cow from picking up foreign objects when eating.
Hooves	They protect the feet from mechanical injury.
Udders	They have mammary glands that produce milk. They store milk.
Switch	It enables the cow to chase away disturbing flies.
horns	They are used for protection.
Tail	It has the switch that enables a cow to chase flies.
Teats	They have pores that let out milk.
Mouth	It used for taking in food.

CARING FOR CATTLE

- Treating the sick cattle.
- Feeding cattle
- Building house for cattle.
- Deworming and vaccination

Grazing cattle

This is letting animals to eat grass.

Systems of grazing cattle are ; rotational grazing and zero grazing

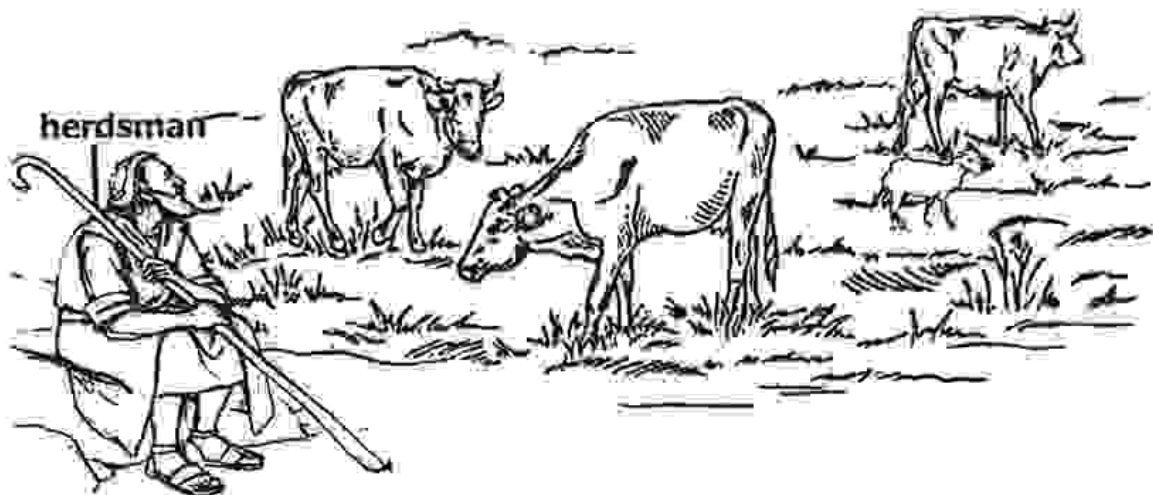
Methods of grazing cattle are; herding, paddocking, tethering and strip grazing.

Herding

This is a system where animals are left to move freely as they feed on different types of pasture and monitored by a herdsman.

This method of grazing is commonly used in rural areas.

Reason. There is enough land for grazing cattle.



Advantages of herding.

- ✓ Cattle feed on the variety of feeds.
- ✓ Cattle get enough physical exercises.
- ✓ It is cheap to use
- ✓ Less labour is required to look after cattle.

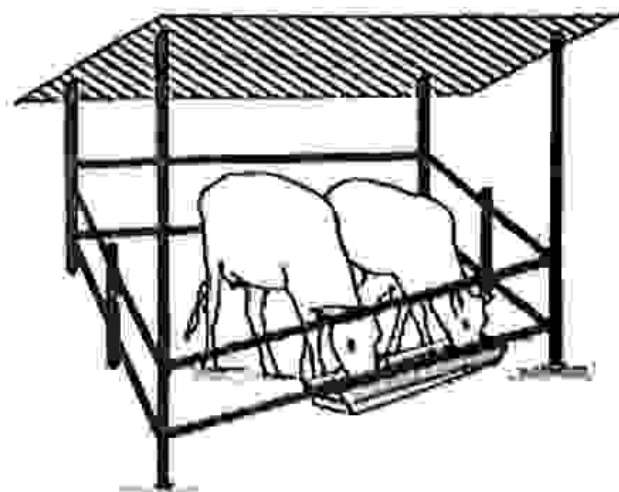
Disadvantages of herding

- ❖ The method requires a lot of land.
- ❖ Animals can stray and spoil people's crops.
- ❖ It is difficult to keep records.
- ❖ Cattle are exposed to predators and parasites
- ❖ Cattle can easily be stolen by thieves.
- ❖ Inbreeding is difficult to control.
- ❖ Diseases can easily be spread among the cattle.

Zero grazing

This is a system of grazing where cattle are fed on prepared feeds from indoors.

This system of grazing cattle is commonly used in towns due limited piece of land.



Advantages of zero grazing

- ↑ Feeds are not wasted.
- ↑ Farmyard manure is easily collected.
- ↑ Animals are protected from bad weather.

- ↑ Animals are protected from predators.
- ↑ There are less risks of getting diseases.

- ↑ It is easy to spot sick animals.
- ↑ There is no wastage of feeds.

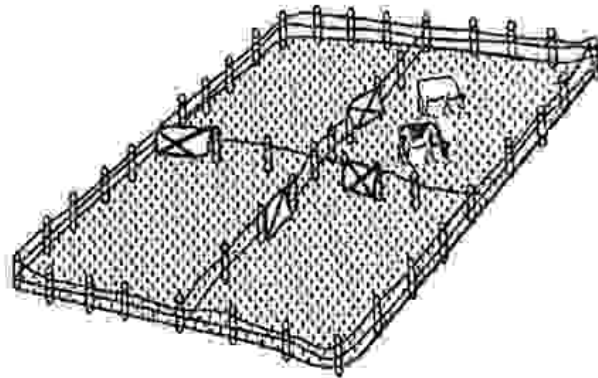
Disadvantages of zero grazing.

- ❖ The method is expensive to start and maintain.
- ❖ It requires a lot of labour.
- ❖ Animals do not get enough physical exercises.
- ❖ There is easy spread of diseases in case of an outbreak.

Paddock/paddock grazing

Paddock is a method of grazing where the grazing land is divided into units or plots called **paddocks**.

The paddocks are separated using a fence and the farmer keeps moving the animals from one paddock to another.



Advantages of paddocking

- The farmer gets time to do other activities.
- Pasture is properly used.
- Animal dung and urine are evenly distributed. *They decay to form humus for pasture to grow again.*
- It controls easy spread of tick-borne diseases in cattle. *The method breaks down the life cycle of ticks. It starves ticks to death.*
- Record keeping is easy.
- Paddock controls overgrazing.
- It can be used for commercial purpose.

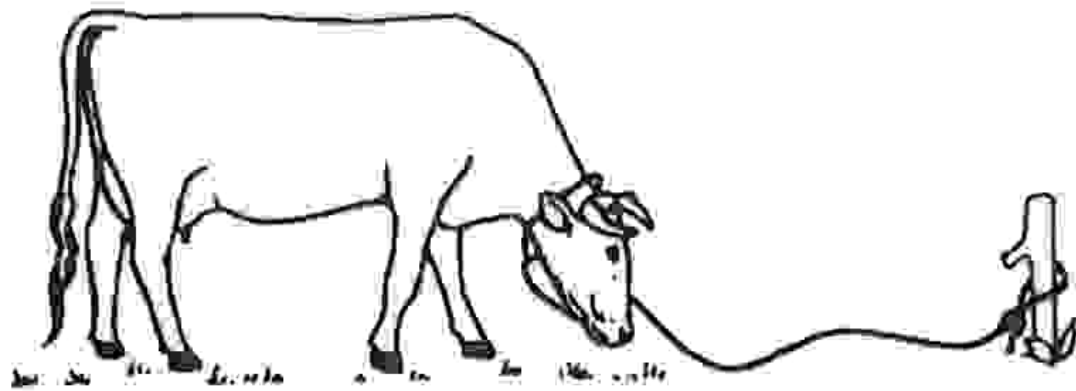
It manages a big number of animals.

Disadvantages of paddocking

- ✓ Paddock requires a large piece of land.
- ✓ Paddock is expensive to start.
- ✓ Animals do not get enough exercises.
- ✓ Animals have no choice on the type of grass to eat.

Tethering method

Tethering is a method of grazing where cattle are tied to pegs or trees using ropes. The farmer shifts the animal to another area when the pasture in the previous area is finished.



Advantages of tethering method.

- Tethering is cheap to use.
- Less labour is required.
- Cattle cannot stray and destroy crops.
- Animals do not move long distances.
- Small piece of land is used to graze on cattle.
- The farmer chooses the best pasture for the cattle.

Disadvantages of tethering method

- Animals can easily be stolen.
- The rope can easily strangle the animal.
- Animals have no choice for the grass to eat.
- Animals are exposed to predators.
- The animals do not get enough exercises.
- Only few animals can be kept.

Strip grazing

This is where the small sections called strips are created using temporary electric wires to keep animals in gazetted areas.

When pasture is fully grazed on by the animals, they are moved to a different area.

This method is similar to paddocking. However, in strip grazing the area is divided using electric wires instead of the other materials like poles and razor wires.

Advantages of strip grazing

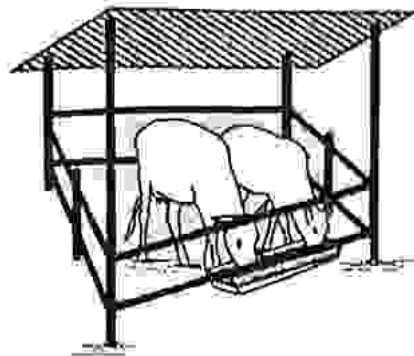
- ↑ Pasture is evenly used.
- ↑ Diseases and parasites are controlled.
- ↑ Less labour is used on the farm.

Disadvantages of strip grazing

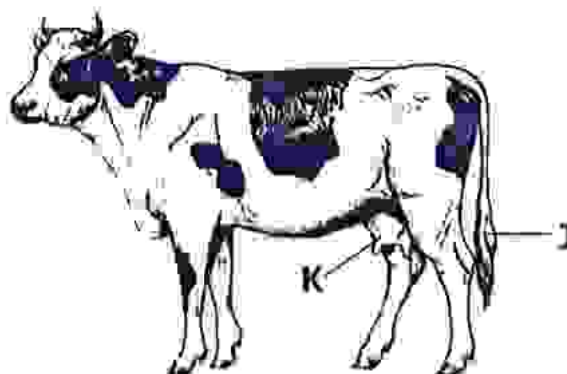
- o It requires electricity.
- o It is expensive to start.

Evaluation Activity 6.3

1. Name any **two** methods of grazing cattle.
2. Which method of grazing cattle involves the use of electricity?
3. The diagram below shows cattle grazing system. Study and use it to answer the questions that follow.



- a) Name the cattle grazing system shown above.
 - b) Which type of natural manure is got from the above system?
 - c) Apart from easy collection of manure, state any **two** advantages of the above cattle grazing system.
4. Give any **two** reasons why most cattle farmers use herding method of grazing.
 - 5.(a) How does paddocking help to control the spread of
 - i) Tick borne diseases among cattle?
 - ii) Soil erosion?(b) Apart from controlling tick borne diseases and soil erosion, give any **two** other advantages of paddock grazing.
 6. Name any **one** method of grazing cattle suitable for commercial purpose. The diagram below shows external parts of the cow. Use it to answer questions 7 and 8.



7. Name the part of the cow marked J.
8. How does a calf benefit from the part of the cow marked K?

FEEDING CATTLE

Advantages of proper feeding of cattle

- ✓ It improves the quality of milk and meat.
- ✓ It boosts the immunity against diseases.
- ✓ It promotes faster and proper growth of cattle.

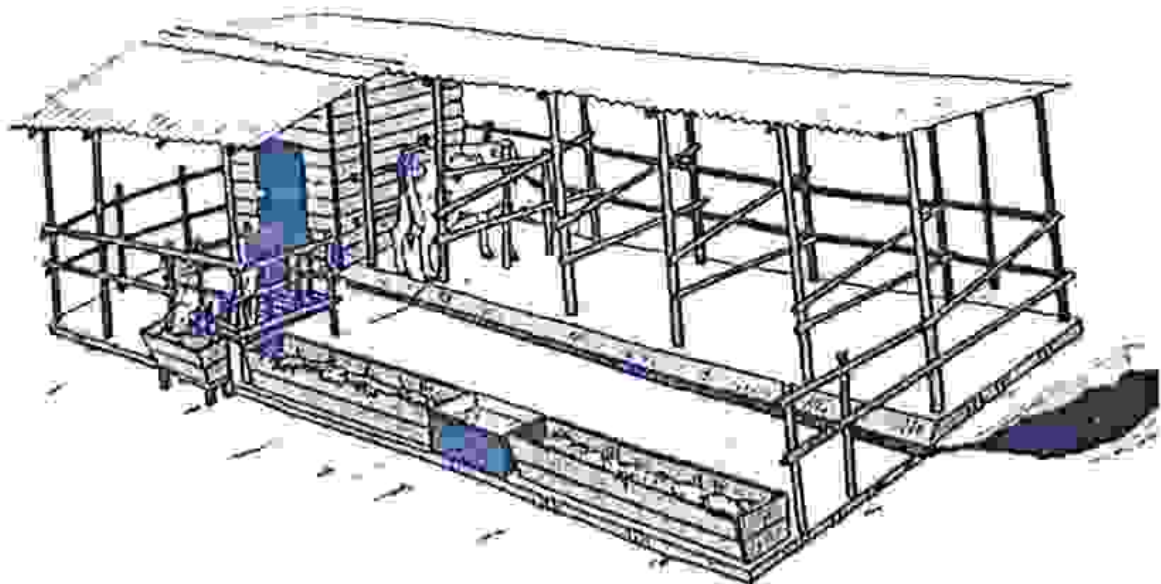
Types of feeds for cattle

Type of feeds	examples	Importance / characteristics
roughages	Young grass// fodder crops e.g. sorghum, millet and hay	They help to ease digestion of food.
concentrates	Bone meal Cotton seed cake Maize bran Blood meal Rock salt legumes	- source of calcium - source of proteins and fats. - source of carbohydrates. - source of iron. - source of sodium Concentrates enable an animal to fatten and to produce plenty of milk.
succulents	Sweet potato vines Maize stalks Banana peelings	They contain high moisture content and low Fibre content

HOUSING CATTLE

The house for cattle is called byre/kraal

The materials used to build cattle house include; wood, concrete, metals, bricks and stones.



Advantages of housing cattle

- ❖ It protects cattle from thieves and predators
- ❖ It protects cattle from harsh weather
- ❖ It is easier to manage animals in terms of feeding.
- ❖ It reduces easy spread of diseases.

Qualities of a good cattle house.

- ✓ It should have a slanting floor to allow easy flow of urine
- ✓ It should have a strong roof to protect cattle from harsh weather.
- ✓ It should be well ventilated to allow proper circulation of air.
- ✓ It should have ample space to accommodate animals well.
- ✓ It should be made of concrete for easy cleaning.

MATING, GESTATION PERIOD AND CALVING IN CATTLE

Heat period in cattle.

Heat period is the time when a cow is ready to mate with a bull.

Signs of a cow on heat

- ✓ The cow mounts others.
- ✓ Discharge of mucus from the vulva.
- ✓ The cow urinates frequently.
- ✓ The cow becomes restless.
- ✓ The cow loses appetite to graze.
- ✓ The cow makes a lot of noise.
- ✓ The vulva swells and changes its colour to red.

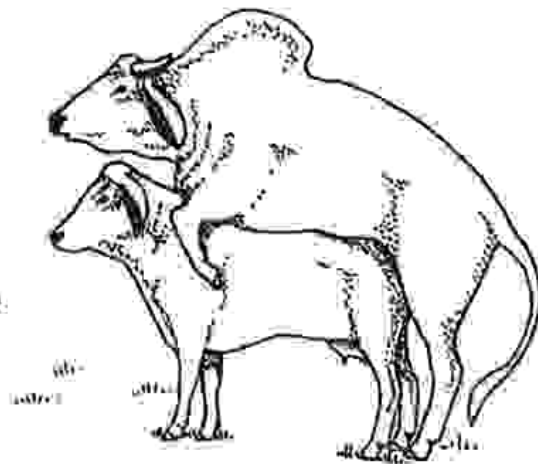
Mating in cattle

Mating in cattle is the sexual union between a bull and a cow.

Mating is also called **insemination**.

Insemination is the depositing of semen containing sperm cells into the female reproductive system.

Mating results into fertilization in the oviduct of a cow.



Types of insemination

Natural insemination

This involves the use of a bull in mating.

Advantages of natural insemination

- It saves time
- It is cheaper to use.
- It is easier for a bull to detect a cow on heat.
- There are less risks in storage of semen in the bull's body.

Disadvantages of natural insemination.

- It leads to easy spread of venereal diseases.
- Small cows can be injured by big bulls.
- It encourages inbreeding on the farm.

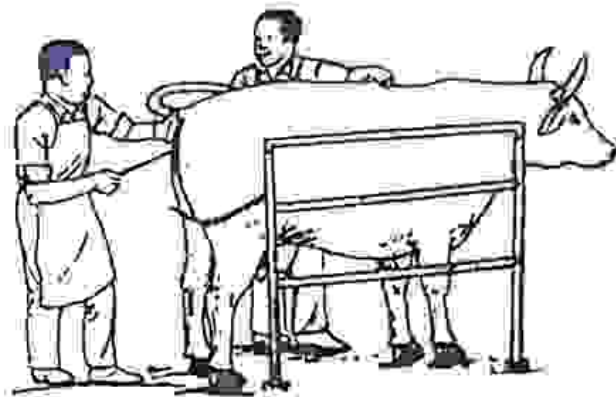
Artificial insemination (A.I)

This is where a farmer deposits semen into the cow's vagina using an inseminating syringe.



Advantages of artificial insemination.

- It controls inbreeding.
- It reduces the costs of buying bulls.
- Many cows are inseminated within a short time.
- It prevents easy spread of venereal diseases.



Disadvantages of artificial insemination

- It requires skilled labour.
- The farmer may fail to detect the cow on heat.
- If it is not done well, semen is wasted.
- Nitrogen used to preserve semen is expensive.

GESTATION PERIOD

Gestation period is the time from fertilization to birth.

A cow takes 9 months (270 days) to calve.

Signs of pregnancy in cows.

- The cow stops going on heat.
- The udder and teats enlarge.
- The belly enlarges due to the growth of the foetus.

Steaming up

Steaming up is the feeding of a pregnant animal on highly nutritious diet before delivery.

Proteins are mainly used during steaming up.

Advantages/importance of steaming up.

- It promotes proper growth of the foetus.
- It stimulates milk production and let down.
- It prepares the body of the pregnant animal for birth.

CALVING IN CATTLE

Calving is the giving birth to a calf by a cow.

It is done in a calving pen.

Signs of calving in a cow

- The udder becomes swollen.
- Discharge of mucus from the vulva.
- Movement of the foetus is seen.
- The cow lies down most of the time.
- The cow stops grazing.
- The cow isolates itself from others.

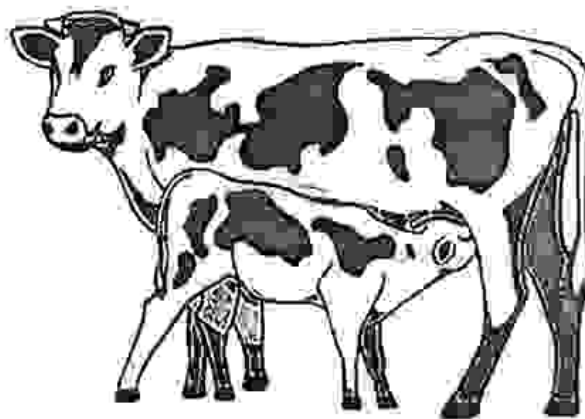


Colostrum.

Colostrum is the first thick nutritious milk from the udder of a lactating cow.

Importance of colostrum to a calf.

- It opens the digestive tract of the calf.
- It boosts the immunity of the calf.
- It provides protein to the calf for proper growth.



Evaluation Activity 6.4

1. Give any **one** reason why it is important to feed cattle.
2. State any **two** signs that a cow on heat shows.
3. What is meant by the term steaming up?
4. Give any **two** ways in which steaming up is important in cattle management.
5. State **one** way in which colostrum is important to a newly born calf.
6. State any **two** advantages of proper housing in cattle.
7. Why should a cow be given a block of salt to lick?
8. The diagram below shows an equipment found on a cattle farm. Study and use it to answer questions 9 and 10.



Evaluation Activity 6.4... continued

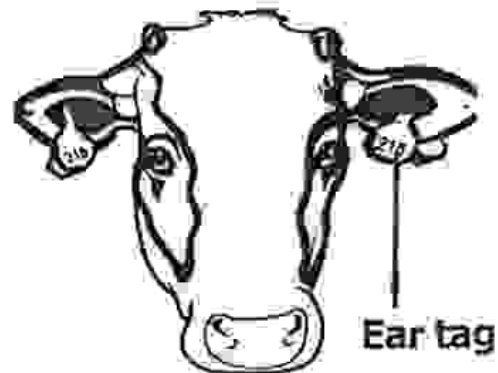
9. Name the equipment shown in the diagram above.
10. Give any **one** use of the above equipment to a cattle farmer.
11. Give any **two** advantages of artificial Insemination in keeping cattle.
12. Which gas is used to preserve semen used for artificial insemination?

CATTLE MANAGEMENT PRACTICES

1. Animal identification

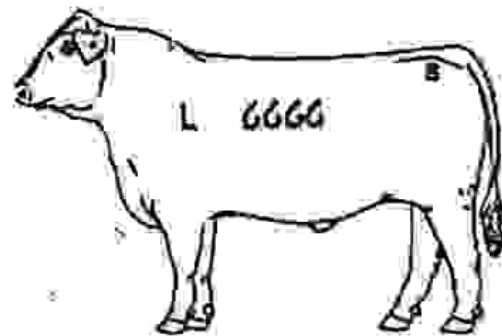
Ear tagging

Tags having numbers made of light metals or plastics are fixed onto the ear of an animal.



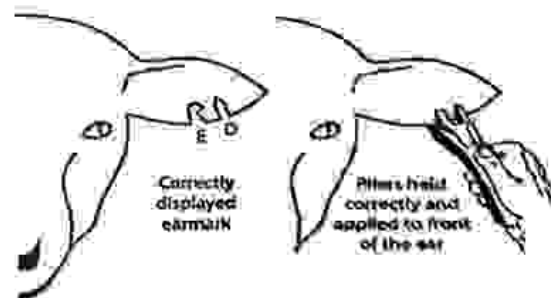
Branding

Some parts on the animal's skin are branded with symbols, designs or marks using a hot iron. It is done on the leg, shank, hump, ear, forehead or shoulder



Ear notching

This is when the animal's ear is cut with marks at its edges.



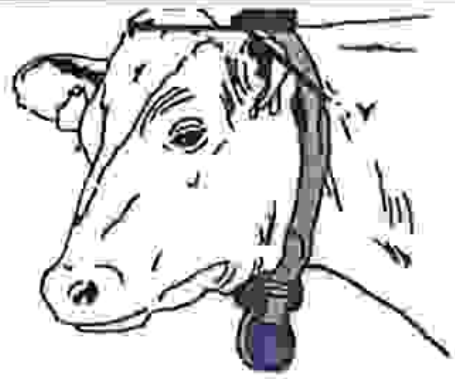
Tattooing

Permanent marks, patterns, a letter or number is put on the animal's body especially on the skin or ears.



Number lacing

A wooden piece or metallic plate is put around the neck of a cow or bull.



2. Castration

Castration is the removal or inactivation of testes of a male animal. It should be done at around 7 months of age. A castrated bull is called a bullock/steer.

Methods of castration

Open castration

This is where the scrotum is cut using a sharp knife or razorblade to remove the testes.

Closed castration

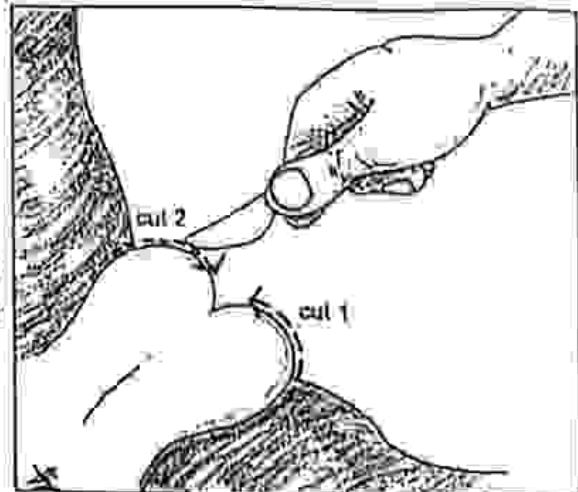
This is where a burdizzo is used to break the blood vessels leading to the testes.

The burdizzo has blunt edges.

Loop or rubber ring method

In this method, a rubber band is used to squeeze the testes.

This method is suitable for calves.



Advantages of castration.

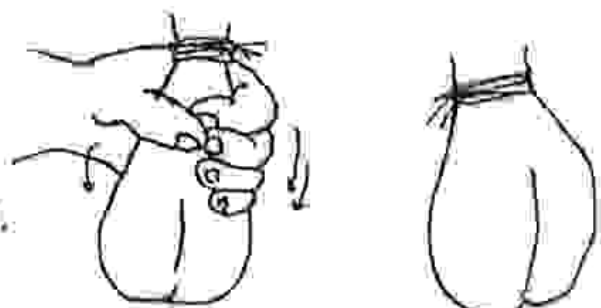
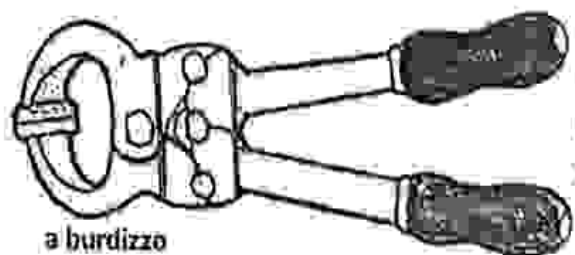
- Animals grow faster and fatten.
- It controls inbreeding in animals.
- It prevents unnecessary mating.

Testes that produce sperms are removed or deactivated.

- It makes animals docile.

Disadvantages of castration to animals

- It can cause too much loss of blood.
- It inflicts a lot of pain to an animal.



- The wounds may become septic to an animal.

Dehorning/disbudding

Dehorning is the removal of horn buds from an animal.

It is done using a dehorning iron or spoon dehorner

Dehorning iron



Spoon dehorner



Advantages of dehorning

- It makes animals easy to handle.
- It reduces the risk of injuries among animals.
- It makes transportation of cattle easy.
- It increases the space in the kraal.



Dehorning

Hoof trimming

This is the cutting of overgrown hooves from an animal

Hoof trimming controls foot rot in cattle.



Hoof trimming

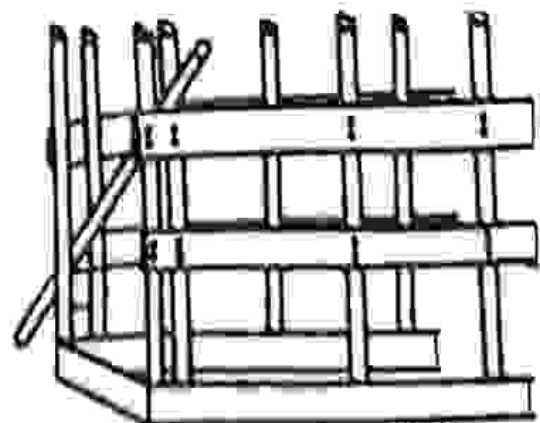
Cattle crush

- It restricts animal movement when being worked on.
- It protects people from being kicked by the animal.

Farm activities carried out in a

cattle crush

- artificial insemination
- deworming
- spraying
- drenching
- ear tagging
- branding
- deticking
- hoof trimming
- dusting
- Pregnancy test



Cattle crush

Don't speak for Quality, let Quality speak for itself

13/11

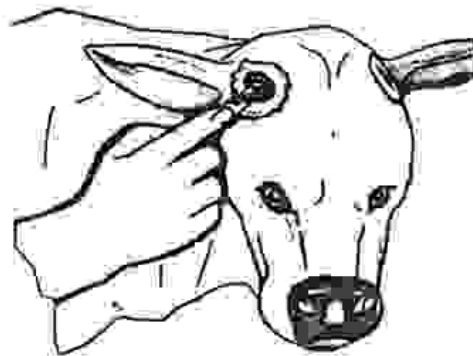


Learn, unlearn and relearn: Castration and milking are not farm activities carried out in a cattle crush. Castration is usually done on calves which do not need to be put in a cattle crush while milking is done in a milking parlour.

Evaluation Activity 6.5

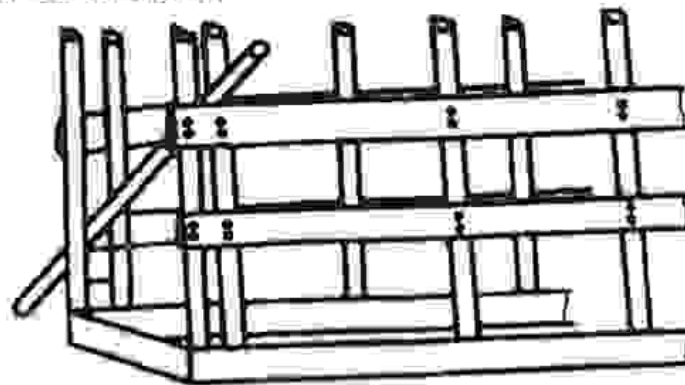
1. Name any **one** method of animal identification done on the ears.
2. Apart from animal identification, give any **two** other examples of cattle management practices.
3. Which cattle practice is carried out by farmers to control foot rot in cattle?
4. How does castration help to control inbreeding in cattle?
5. Apart from controlling inbreeding, mention any **two** other advantages of castrating bulls.
6. Give any **two** advantages of dehorning cattle.

The diagram below shows one of the management practices in cattle management. Use it to answer questions 7 and 8



7. Name the management practice shown above.
8. Give any **one** advantage of the above practice in cattle management.
9. State the advantage of hoof trimming in cattle.

The diagram below shows a structure in a cattle farm. Study and use it to answer questions 10 and 11.



10. Mention any **one** activity carried out in the above structure.
11. How is the above structure useful when carrying out the farm activity you have mentioned above?

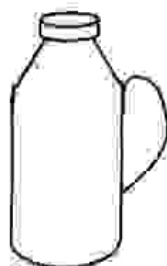
MILKING AND MILK PRODUCTS

Milking is the process of obtaining milk from the udder of a cow.

Milk let down is the flow of milk from the udder of a cow.

Equipment used in milking

Equipment	Importance
Strip cup	It is used to detect the presence of mastitis in milk. It has a black smooth surface that detects any abnormal change in milk. It sieves blood clots in milk.
lactometer	It used to determine the amount of water and fats in milk.
Milk pail	It used to keep milk to prevent it from contamination.



Milk can



Strip cup



Lactometer

Activities a cattle farmer can do to obtain clean milk during milking

- ❖ Clean the milking parlour.
- ❖ Wash hands with clean water and soap.
- ❖ Wash the udder and teats of the cow with warm water. Warm water removes dirt from the udder and teats. Warm water stimulates milk let down.
- ❖ Filter the milk before keeping it in the milk pail.
- ❖ Smear jelly on the teats to lubricate them and prevent them from cracking during milking.
- ❖ Use clean containers when milking.
- ❖ Put the milk in cool clean containers.

Methods of milking

Hand milking

In this method, the farmer uses his/her hands to extract milk from the udder of a cow.

Advantages of hand milking

- ✓ It is cheap to use.
- ✓ It does not require skilled labour.
- ✓ It can be used in places without electricity.



Don't speak for Quality, let Quality speak for itself

Disadvantages of hand milking

- ❖ Milk can easily get contaminated
- ❖ Few animals are milked.
- ❖ It is time consuming
- ❖ It is tiring to milk animals.

Machine milking

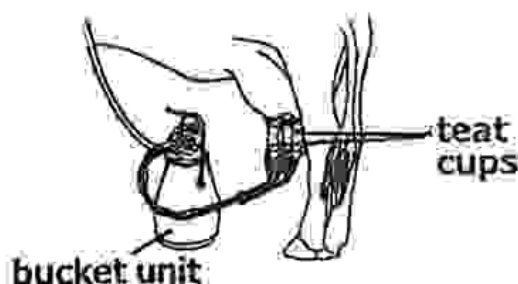
In this method, a machine is used to extract milk from the udder of the cow.

Advantages of machine milking

- Milk is not contaminated.
- Many cows can be milked within a short time.
- It is faster than hand milking.

Disadvantages of machine milking

- It requires skilled labour
- It requires electricity to use.
- Machines are expensive to buy and maintain.
- The udder may be injured by faulty machines.



Milk products

Milk product	How it is made	Its uses
cheese	By thickening sour milk and salting it.	It is eaten directly or mixed in dishes
fresh milk	By boiling and refrigeration.	It is drunk or added to other dishes.
butter	By churning cream.	It is used for frying food.
yoghurt	By churning skimmed milk.	It is mixed with rice and other dry food// It is drunk
ghee	By boiling butter.	It is used for frying food.
ice cream	By flavouring milk fats.	It is refrigerated and eaten or mixed with grain food like rice.
whey	By curdling and straining milk.	- It is a source of proteins. - It is used in baking to add proteins and moisture in bread and cakes. - It is used in the production of food
casein	It is precipitated from skimmed milk.	It is used in food products as an additive e.g. Processed foods.

skimmed milk	By removing fats from milk.	It is drunk by people who are conscious about their weight.
fortified milk	Milk enriched with vitamins and mineral salts.	- It boosts immunity - It provides vitamins and mineral salts e.g. calcium, vitamin A and vitamin D

Homogenisation is the process by which fat droplets from milk are emulsified and the cream does not separate.



Ice cream



Yoghurt

Methods of preserving milk

Milk preservation is the way of keeping milk safe for a long time without going bad.

Diseases like tuberculosis, brucellosis and typhoid can spread through drinking contaminated milk.

The methods of preserving milk are;

Boiling; Milk is heated in a container put on the heating place like charcoal stove.

Boiling kills all bacteria in milk.

Boiling is the best and cheapest method of boiling milk.

Refrigeration; Milk is put in a refrigerator to lower its temperature.

The low temperature in the refrigerator prevents the breeding of bacteria in milk.

Sterilisation; Milk is heated at high temperature, cooled and sealed in sterile containers.

Bacteria in milk are killed by maximum boiling.

Pasteurisation

Milk is heated at controlled temperature and allowed to cool.

Pasteurisation kills bacteria in milk.

Other products got from cattle are; beef, blood, hides, horns and hooves.

Uses of hides got from cattle

- They are used in leather industries. The process of turning hides into leather is called tanning.
- They are used as seats.
- They are used for making drums.
- They are used for dressing handles of pangas.

Methods of preserving hides

Wet salting

Salt is applied on the wet skin got from an animal.

Salt dehydrates the skin

Suspension drying

The skin is put in sunshine to dry by supporting it with frames of sticks.

Heat from the sun drains moisture from the hide.

Advantages of suspension drying

- It keeps the skin clean.
- It is cheap.
- It keeps the skin flat with no folds.



Examples of industries that use cattle products.

- Milk processing industry
- Beef processing industry
- Art and craft industry
- Leather industry
- Animal feeds industry

Practices that affect the quality of cattle products

- Poor feeding.
- Poor housing.
- Branding cattle.
- Beating cattle.

Ways in which cattle farmers can improve on the quality of cattle products.

- Crossbreeding.
- Upgrading.
- Castration
- Regular deworming of cattle.
- Avoid practices that damage cattle hides e.g. branding.
- Proper feeding.
- Proper housing.
- Selective breeding.

Evaluation Activity 6.6

1. Name the product obtained from the udder of a cow.
2. How is a lactometer useful to a cattle farmer?
3. Write down any **two** activities that a cattle farmer can do to obtain clean milk.
4. Apart from removing dirt, why should the udder and teats of a cow be washed with warm water before milking?
5. Which part of a cow is affected by mastitis disease?
6. Mention any **one** method used to preserve;
a) i) Hides from cattle.....ii) Milk from cattle.....

Evaluation Activity 6.6... Continued

The diagram below shows an equipment found on a cattle farm. Study and use it to answer the questions 7 and 8.



7. Name the cattle equipment shown above.
8. Which disease in cows requires the use of the equipment shown above?
9. List down any **two** products got from milk.
10. How does pasteurization help to preserve milk?
11. (a) State **one** way in which cattle farming contributes to the development of leather industry.
(b) Apart from leather industry, give any **one** other example of industry that uses cattle products.

CATTLE PESTS (PARASITES) AND DISEASES

A parasite is an organism that depends on another organism for food or shelter.

Types of parasites are ectoparasites and endoparasites.

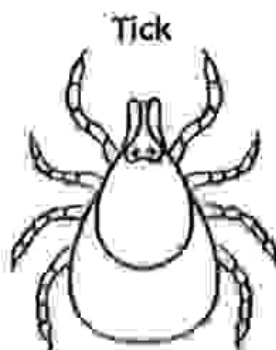
Ectoparasites are parasites that live on the skin of the host.

They are also called **external parasites**.

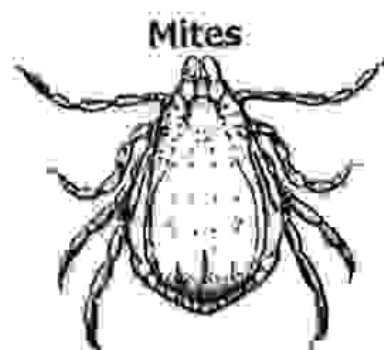
Examples of ectoparasites are; ticks, mites, tsetse flies, lice and stable flies.



Tsetse fly



Tick



Mites

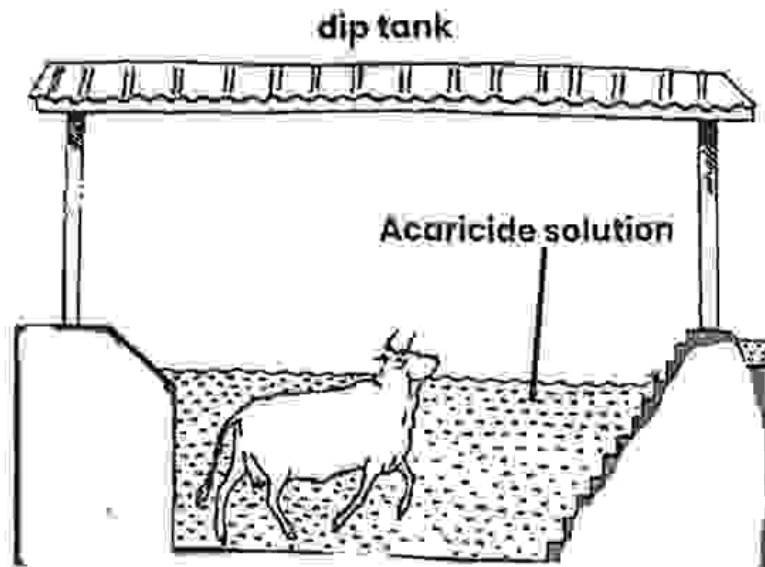
Ticks spread tick borne diseases like heartwater, east coast fever, anaplasmosis and redwater.

Remembered as; **HEAR**

Tsetse flies spread nagana.

Ways of controlling ectoparasites in cattle

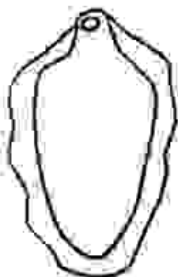
- ❖ Spraying using acaricides
- ❖ Dipping cattle in acaricide solution
- ❖ Dusting cattle with acaricides
- ❖ Practising paddock grazing
- ❖ Practising double fencing
- ❖ Creating tick buffers
- ❖ Hand picking



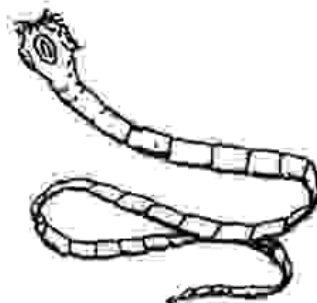
Endoparasites are parasites that live in the intestines or liver of the host.

They are also called **internal parasites**.

Examples of endoparasites are; tapeworms, roundworms and liver flukes.



Liver fluke



Tapeworm



hookworms

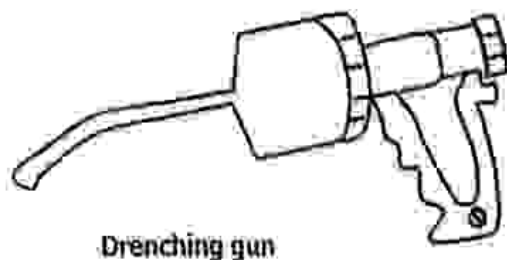
Endoparasites in cattle are controlled by deworming.

Methods of deworming are drenching and dosing.

Drenching is the giving of liquid medicine to cattle through the mouth.

Drenching is done using a drenching bottle or drenching gun.

A **drenching gun**. It is used for giving liquid medicine to cattle through the mouth



Drenching gun

Dosing is the giving of solid medicine to cattle through the mouth.

Effects of parasites to cattle

- ✓ They transmit diseases to cattle.
- ✓ They lower the quality of cattle products.
- ✓ They suck blood from cattle leading to anaemia.
- ✓ They cause damage to the skin of an animal.
- ✓ They cause irritation to an animal.

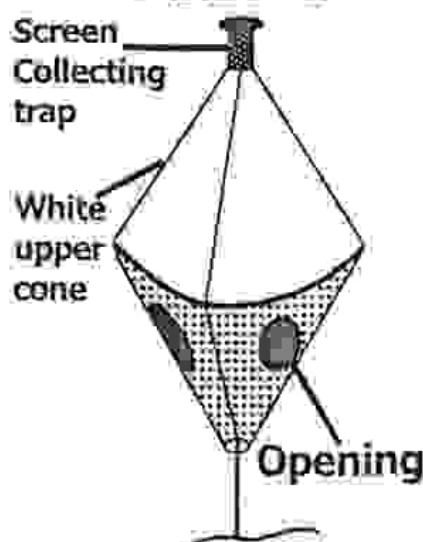
DISEASES THAT AFFECT/ATTACK CATTLE

DISEASE	HOW IT SPREADS	SIGNS/ SYMPTOMS	PREVENTION AND TREATMENT
Mastitis Germs: bacteria Attacks udder and teats	Through body contact with infected animals.	- Swollen udder and teats - Blood-stained milk. - Pus in milk - Painful udder and teats	- Early treatment using antibiotics. - Use a strip to detect the disease regularly. - disinfect the milk- ing parlour regularly.
Anthrax Germs: bacteria	Through body contact with infected animals.	- High fever - Loss of appetite - Blood-stained dung. - Bleeding from the body opening. - Sudden death	- Vaccinate animals regularly. - Impose quarantine. - Burn or bury the dead animals Deep in the soil.
Tuberculosis Germs: bacteria	- Breathing in air with tuberculosis bacteria - In humans, it is spread Through drinking milk from infected cows.	- Loss of appetite - Coughing - Loss of weight	- Treat infected ani- mals using antibiotics. - Isolate infected animals from healthy ones.
Nagana Germs: protozoa Pest: tsetse fly	Through bites of tsetse flies	- Loss of weight - Loss of appetite - Fever - Anaemia	- Control tsetse flies by clearing bushes or use traps - Treat with drugs as directed by the veterinary doctor.

DISEASE	HOW IT SPREADS	SIGNS/ SYMPTOMS	PREVENTION AND TREATMENT
Pneumonia Germs: bacteria Or virus	- Through breathing in air contaminated with bacteria or virus.	-Difficulty in breathing. -Runny nose -Coughing -Loss of appetite. -Loss of weight. -Nasal discharge	Keep the kraal dust free and clean. Treat using antibiotics.
Foot and mouth disease Germs: virus	Sharing food, containers and grazing land. Through body contact with infected cattle	-Swollen legs and lameness. -Blisters on the top of -Hooves and mouth. -Loss of appetite to graze -High temperature. -Reduced milk production	Isolate the infected animals. Vaccinate every after six months. Impose quarantine.
Rinderpest Germs. virus	Through body contact with infected cattle	-Sores in the mouth -Fast breathing -Nasal discharge -Tears from the eyes -Sunken eyes -Blood-stained dung -Fever	Separate infected animals from Healthy ones. Regular vaccination Slaughter the infected animals.
Heartwater Germs: bacteria	Through the bites of ticks (red ticks)	-Animals walk in circles -Animals place their heads on objects -Fever -When animals fall, the legs keep paddling in air.	- Dipping with acaricide solution. - Spraying cattle with acaricides - Practice paddocking - Establishing tick buffers - Treat animals using tetracycline or sulphadimidine

Redwater Germ: protozoa	Through the bites of Ticks.	- High temperature - Red coloured urine. - Animals lick soil.	- Dipping with acaricide solution. - Spraying cattle with acaricides. - Practice paddocking. - Establishing tick buffers. - Regular vaccination.
East Coast Fever Germ: protozoa	Through the bites of Ticks.	- Loss of weight - Loss of appetite - Continuous salivation. - Nasal discharge - Swollen legs.	- Dipping with acaricide solution. - Spraying cattle with acaricides. - Practice paddocking. - Establishing tick buffers. - Deticking.

A tsetse fly trap. It traps and kills tsetse flies



Points to note: Quarantine is the restriction of the movement of animals or their products to certain areas due to disease outbreak.

- It helps to reduce the spread of animal diseases in an area.
- Isolation is the separation of infected animals from others on the farm.
- Tick buffers are grass-free lanes around the paddocks.
- Ticks are unable to cross these lanes from one paddock to another.

i *Learn, unlearn and relearn. Not all tickborne diseases in cattle are caused by protozoa.*

Some are caused by bacteria e.g. heartwater and anaplasmosis (gall sickness) while East Coast Fever and redwater are caused by protozoa.

PRACTICES THAT HARM CATTLE AND OTHER DOMESTIC ANIMALS

Place	Practices
At home	➤ Overcrowding animals. ➤ Beating animals. ➤ Plucking feathers from poultry. ➤ Castrating animals ➤ Dehorning animals ➤ Pulling teats of the cow hard when milking.
In the field	↑ Beating animals ↑ Improper disposal of polythene bags. ↑ Making the rope too tight when tethering the animals. ↑ Overstocking ↑ Overloading animals like oxen during ploughing. ↑ Piercing animals ↑ Bestiality ↑ Making animals to fight. ↑ Keeping animals in unfavourable conditions during grazing.
In transit	-Making animals to walk long distances. -Overloading animals in the vehicles. -Beating animals. -Tying the legs and the neck of the animals to the side of the truck -Drugging animals -Transporting animals in the boot of the vehicle.
In abattoir	❖ Overcrowding of animals in the same room. ❖ Cruel methods of slaughtering animals. ❖ Starving animals before slaughtering. ❖ Slaughtering in-calf cows.

STARTING A LIVESTOCK FARM.

Livestock refers to the animals kept on the farm.

Examples of livestock are; cattle, goats, sheep, poultry, rabbits and pigs

Factors to consider when starting a livestock farm.

Factor	Importance of the factor
land	It used for setting up the farm. It is used for growing pasture for feeding livestock.

Factor	Importance of the factor
capital	- It is used to buy the farm tools and equipment. - It is used for paying workers. - It is used to buy livestock. - It is used to buy feeds.
labour	- Needed to look after animals and carry out farm activities like vaccination
management	- It is needed to carry out the day today activities on the farm.
market	- For selling livestock products like meat and milk. - For buying farm requirements like tools, drugs and feeds.

Mnemonic **LCM** can be used to recall these factors.

RECORD KEEPING IN A LIVESTOCK FARM.

Record keeping is the gathering and storing of information about various activities and transactions carried out on a farm.

Farm records are written information about different activities and transactions carried out on a farm.

Examples of information on farm records.

- ✓ Size of the farm
- ✓ Date of planning
- ✓ Size of paddocks
- ✓ Type of fertilisers used for planting
- ✓ Health status of animals and crops.
- ✓ Date when the animals will give birth.
- ✓ Amount spend on feeding and vaccination.

Examples of farm records.

Production records:

Provide information about milk and meat from cattle.

They help in planning schedules and related activities.

Feeding records:

Feeding records show the amount of feeds given to animals in a specific period of time.

Health records:

These show cases of diseases, their diagnosis and treatment. It also shows dates for vaccination and health conditions of different animals.

Importance of keeping health records.

- It allows farmers to carry out culling of animals easily.
- It helps to identify sick animals on the farm.
- It helps to know the expenditure for treating animals.
- It helps the farmer to plan for the health of animals.
- It helps to know common diseases and ways of preventing them.

Sales records:

These show the income, daily expenditure and total sales.

Inventory records:

These show detailed list of all the items used on the farm.

Breeding records,

These show the breeds of cattle kept on the farm.

Ways in which keeping farm records is important to a farmer.

- It helps a farmer to plan for the farm.
- It helps a farmer to know the income and expenditure on the farm.
- It helps a farmer to know the profits and losses made on a farm.
- It helps in decision making regarding the future of the project.
- Records can be used as reference when a farmer wants to get loans.
- It helps a farmer to be taxed fairly by the government.
- It helps a farm to identify areas development and investment

Example of a Farm record

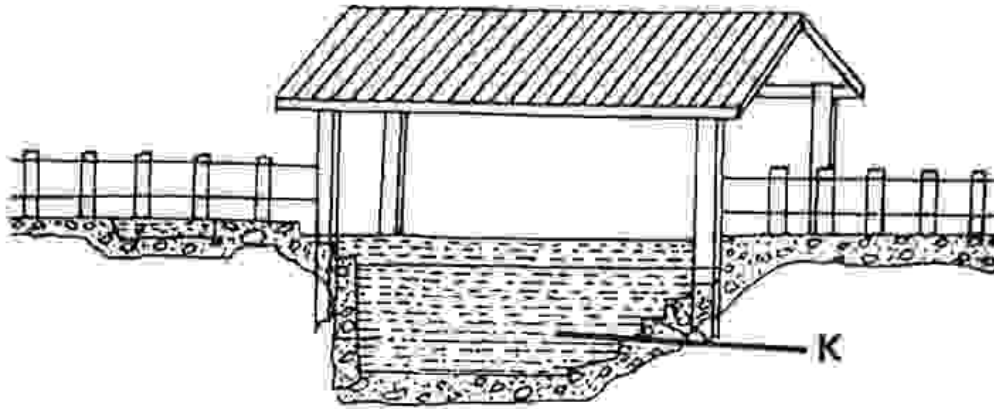
JAMIE FARM INC [®]				
FARM RECORD				
DATE	HEALTH PROBLEM	ANIMAL(S) AFFECTED	TREATMENT/ Control method	Cost of treatment
01/8/2023	Mastitis	Cows D and G	Antibiotics	17,000/=
12/10/2023	Worm Infestation	Calves (06)	Dewormers	25,000/=
03/11/2023	Anthrax	All Cows	Vaccination	50,000/=

Evaluation Activity 6.7

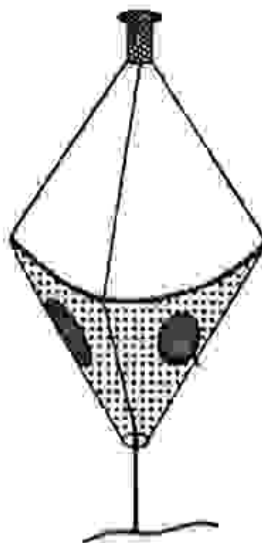
- 1.(a) What germ causes mastitis in cattle?
(b) Give any **two** signs of mastitis.
(c) State any **one** way of controlling mastitis in cattle.
- 2.(a) Apart from ticks, name any **one** other cattle pest.
(b) State any **two** diseases spread by ticks to cattle.
(c) Give any **one** way in which the tickborne diseases can be controlled.

Evaluation Activity 6.7... continued

3. Name the cattle product through which tuberculosis can spread to human beings.
4. Give the use of the drenching gun to a cattle farmer.
5. Which part of the cow is attacked by mastitis?
6. The diagram below shows a structure found in cattle management. Study and use it to answer the questions that follow.



- (a) Identify the farm structure shown above.
 - (b) How is the structure shown above useful in cattle management?
 - (c) Name the substance marked K
 - (d) Which cattle parasites are controlled using the structure above?
7. State any **two** factors to consider when starting a livestock farm.
 8. Give any **two** practices that harm cattle in the field when grazing them.
 9. Which disease in cattle makes animals to move in circles?
 10. Name any **one** disease that attacks both cattle and human beings.
- The diagram below shows a structure used to control cattle pests. Use it to answer questions 11 and 12.



11. Identify the cattle pests controlled using the structure above.
12. Which cattle disease is spread by the cattle pest controlled using the structure above?

Theme:

THE ENVIRONMENT

Topic

7:

RESOURCES IN THE ENVIRONMENT

Resources are things in the environment that people use to satisfy their needs.

NON-LIVING THINGS AS RESOURCES

These include; soil, minerals, sun, fossil fuels, water and air.

Soil as a resource

Soil as a renewable resource is replaced through decomposition of organic matter and weathering of rocks.

Ways in which people use soil as a resource

- ✓ Sand soil is used for building houses.
- ✓ Clay soil is used for making ceramics like clay pots.
- ✓ Loam soil is used for growing crops.
- ✓ Sand soil is used for making glasses.
- ✓ Sand soil is used for putting out fire. Sand soil smothers fire by cutting off its oxygen supply.

Ways of conserving soil as a resource.

- ❖ Planting trees and grass. This controls soil erosion
- ❖ Limiting the use of artificial fertilisers. Excessive use of artificial fertilisers destroy soil structure
- ❖ Avoid dumping polythene bags and plastics in the soil.
 - They prevent rainwater from entering the soil.
 - They kill soil organisms.
 - They block soil aeration.

Minerals as resources

Minerals are non-renewable resources because they cannot be replaced naturally once used up.

They are extracted from underground by mining.

Minerals	uses
Iron	For making garden tools, parts of vehicles, nails, iron bars and iron sheets
Copper	For making electric wires, lightning conductors, bullets, coins and freezers
Wolframite	It is a source of tungsten used to make the filament of electric bulbs.
Aluminium	For making electric wires, bodies of aeroplanes, roofing sheet, saucepans

Gold	For making earrings, necklaces, watches and trophies.
Limestone	For making cement used for building houses, bridges and roads.
Mercury	Used in thermometers as a temperature indicator. Used to make insecticides

Alloys. An alloy is the mixture of two or more metals or a metal and a non metal.

Examples of alloys

Alloy	Composition	Use
Brass	Copper and zinc	For decorating ornaments.
Bronze	Copper and tin	For making medals, swords and statues.
Solder	Lead and tin	Making flux for metals/for joining metals.
Dental amalgam	Copper and mercury	For filling teeth.
Steel	Iron, nickel, carbon, copper	For making magnets, cutting tools and drills.

Reasons why alloys are made.

- ✓ To lower the melting point of metals.
- ✓ To make metals resistant to rusting.
- ✓ To make metals strong and durable.

The sun as a resource

- The sun is regarded as the main and primary source of energy on earth.

Reason. All forms of energy originate directly or indirectly from the sun.

- The energy got from the sun is solar energy.
- Heat energy and light energy are forms of energy obtained from the sun.

Ways in which the sun is used as a resource.

- ✓ The sun provides light that enables us to see.
- ✓ Heat from the sun dries wet clothes and harvested crops.
- ✓ Light from the sun is used to produce solar electricity.
- ✓ Sunlight is used in photography
- ✓ Sun heat helps to preserve food by sun-drying.

Harvesting solar energy.

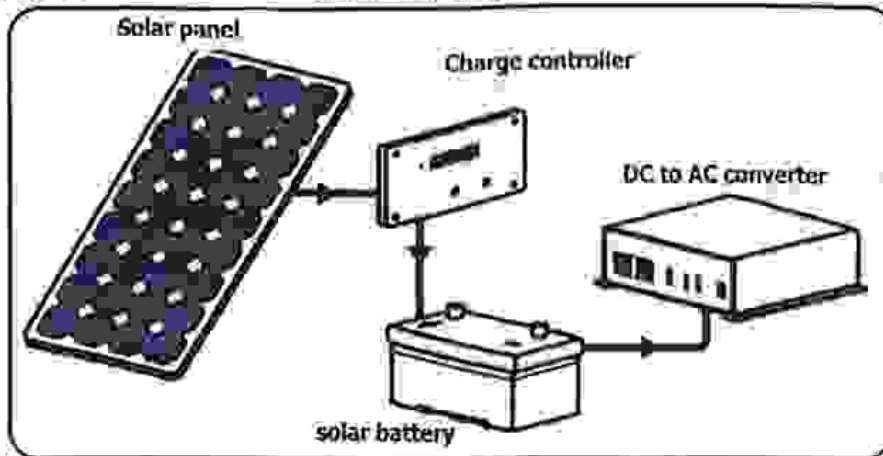
- The solar panels absorb the sunlight.
- Sunlight is changed to solar energy by the solar cells.
- Solar energy is carried and stored in solar batteries in form of chemical energy.

- An inverter changes solar electricity from direct current to alternating current for use.

A solar system

It consists of the;

- solar panel
- solar battery
- solar converter



A solar panel

- The solar panel absorbs sunlight.
- It is painted black in order to absorb sunlight.
- It is put on top of the building to access sunlight easily.

solar cells change sunlight energy into solar electricity.

The solar battery stores solar electricity

Devices that use solar energy are;

- solar lamps
- solar cookers
- solar dryers
- solar water pumps
- solar water heaters

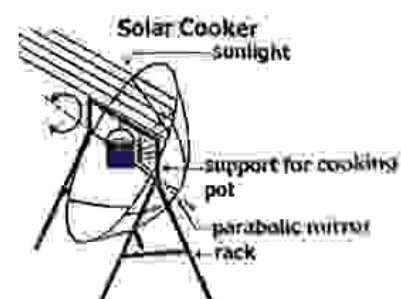
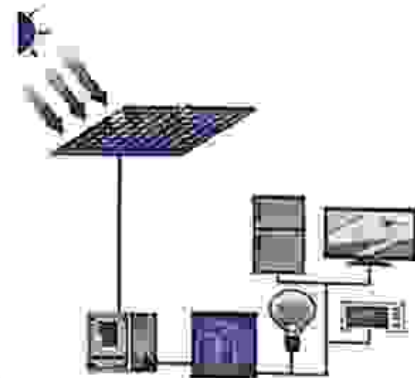
A solar water heater

It produces heat which helps to boil water.

A solar cooker.

It provides heat used for cooking food.

It has a parabolic mirror to converge sunlight rays into heat.



A solar drier

- It provides heat used to dry some items.

- Such items include;

- grains ➤ fish
- vegetables ➤ fruits ➤ meat

Explanation:

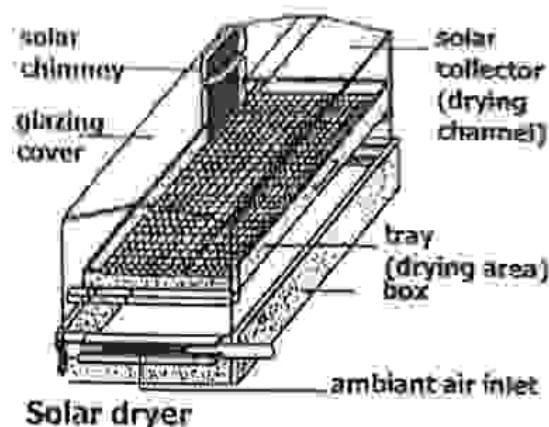
- The glazing cover increases the surface area exposed to sunlight.

- The solar drier is raised above the ground to prevent water and small animals from entering it.

Points to note:

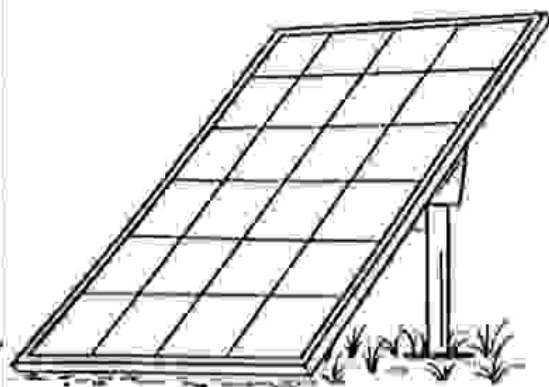
- ✍ • Solar energy is considered as the cleanest source of energy since it does not produce smoke that pollutes the environment.
- ✍ • A solar panel is painted black to easily absorb the sunlight.
- ✍ • A solar panel should be put in an open place to easily access sunlight.
- ✍ • The sun is regarded as a renewable resource.

Reason: The sun cannot get used up.



Evaluation Activity 7.1

1. Name the source of solar energy.
 2. Give any **two** ways in which soil is an important resource.
 3. State any **two** ways in which polythene bags affect the soil.
 4. Why is it not advisable to overuse artificial fertilisers in the gardens?
 5. Why is the sun regarded as the main source of energy on earth?
- The diagram below shows a device used in the production of solar energy. Use it to answer questions 6 and 7.



6. Name the device used to produce solar energy shown above.
7. Why is the device above usually put on top of the roof of the house?
- 8.a) Apart from brass, name any **one** other example of an alloy.
- b) Identify **two** metals used to make brass.
- c) Give any **one** reason why alloys are made.
9. Give any **one** reason why many people do not use solar energy.
10. Why is the sun regarded as a renewable resource?

Don't speak for Quality, let Quality speak for itself

Fossil fuels as resources

Fossil fuels are energy rich substances obtained from the remains of ancient buried organisms.

They are got from underground **by mining**.

A fuel is any substance burnt to produce heat and light energy.

Examples of fossil fuels /resources got from fossil fuels.

- petroleum/ crude oil
- coal
- natural gas
- peat

Petroleum/crude oil

Crude oil is a thick viscous liquid mixture formed from the remains of ancient buried animals.

It is a liquid fossil fuel.

Products obtained from petroleum are; Petrol, diesel, paint, dye, Vaseline, plastics, grease, ink, perfumes, jet fuel and polythene.

Fuels got from petroleum are; petrol, diesel, jet fuel, heating oil and paraffin/kerosene

Note well:

 **Petroleum products are separated by fractional distillation/fractionation.**

Reason: They have different boiling points.

Uses of petroleum products.

- ✓ Paraffin is burnt to produce heat and light.
- ✓ Petrol and diesel are used to run engines and generators.
- ✓ Jet fuel is used in aircraft engines.
- ✓ Vaseline is smeared to make the skin soft and shiny.

Dangers of using petroleum products.

- ✓ They are expensive to buy.
- ✓ They produce smoke that pollutes air.
- ✓ They lead to global warming.
- ✓ They lead to competition for oxygen when burnt.
- ✓ They can easily catch fire.

Natural gas

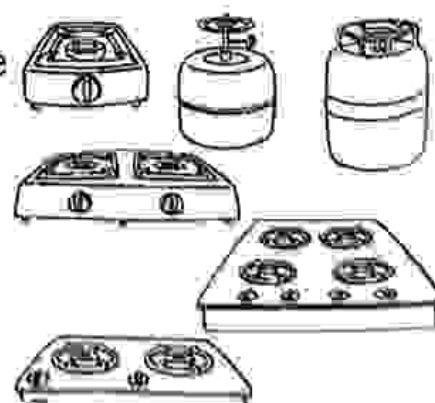
Natural gas is the gas that collects above the underground deposits of crude oil.

It usually exists on top of petroleum between rocks.

It is an example of gaseous fossil fuel.

Devices that use natural gas.

- gas cylinder
- bunsen burner
- gas vehicles
- gas lamps



Uses of natural gas.

- ✓ It is used as fuel in gas cookers.
- ✓ It is used to produce thermal electricity.
- ✓ It is burnt in gas lamps to produce light.
- ✓ It is used in air conditioning.

Advantages of using gas cookers for cooking.

- ✓ It reduces the cutting down of trees for wood fuel.
- ✓ It does not produce smoke that pollutes air.

Coal

It is an underground solid fossil fuel formed from the remains of ancient buried plants.

Importance of coal as a resource.

- ✓ Coal is burnt to produce heat energy.
- ✓ Coal is used to make tar for surfacing the road.
- ✓ Coal is used as fuel in steam engines.
- ✓ Coal is used to generate thermal energy.
- ✓ Coal is used in blast furnace to melt iron.

Points to note:



- Coal and peat are solid fossil fuels.
- Peat is got from papyrus remains in wetlands.
It can be burnt to produce heat for cooking.
- Crude oil is obtained from animal remains while coal is obtained from plant remains.
- Fossil fuels produce thermal energy.

Ways in which fossil fuels can be used as resources.

- ✓ They are used to produce thermal energy.
- ✓ Petrol and diesel from crude oil are used to run engines and generators.
- ✓ Paraffin from crude oil is burnt to produce heat and light.
- ✓ Natural gas is used as fuel in gas cookers.
- ✓ Coal is used in blast furnace to melt iron.
- ✓ Peat is burnt to produce heat for cooking.

Ways of conserving fossil fuels/non-renewable resources

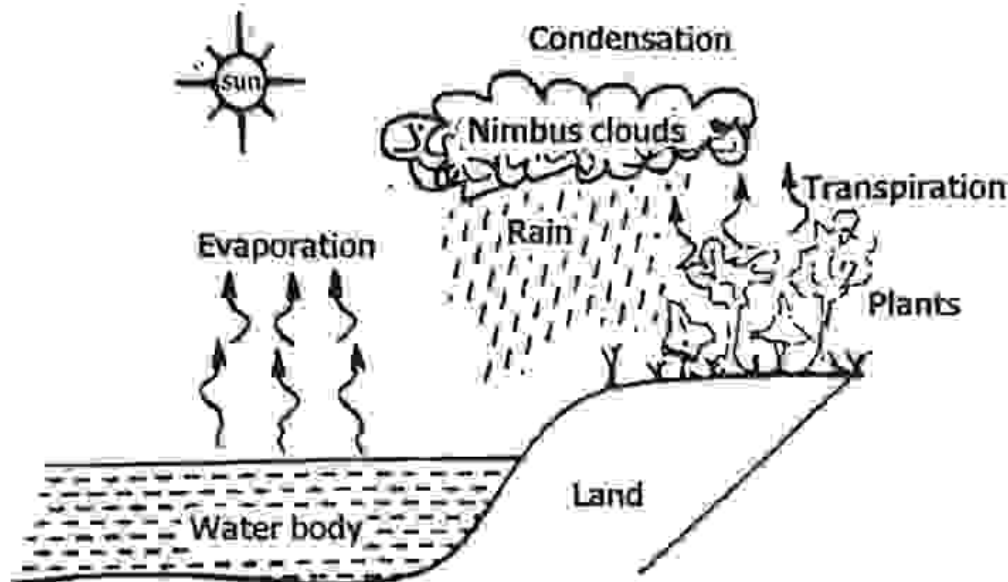
- ✓ Using fossil fuels sparingly and wisely.
- ✓ Using alternative sources of energy.
- ✓ Use of renewable energy resources.
- ✓ Walking short distances instead of driving.
- ✓ Switching off engines after use.

Water as a resource

Water is a renewable resource because it can be replaced naturally through water cycle.

How rain is formed/describing water cycle.

- The sunrays heat the water body and plants.
- Evapotranspiration takes place forming water vapour in the atmosphere.
- Water vapour rises and condenses to form nimbus clouds.
- When nimbus clouds become heavy, they fall as rain to the ground.



Roles of some components in rain formation

The sun. Heat from the sun causes evapotranspiration.

Heat from the sun causes evaporation in water bodies transpiration in plants.

Plants. They add vapour to the atmosphere through transpiration.

Water body. It adds water vapour to the atmosphere through evaporation.

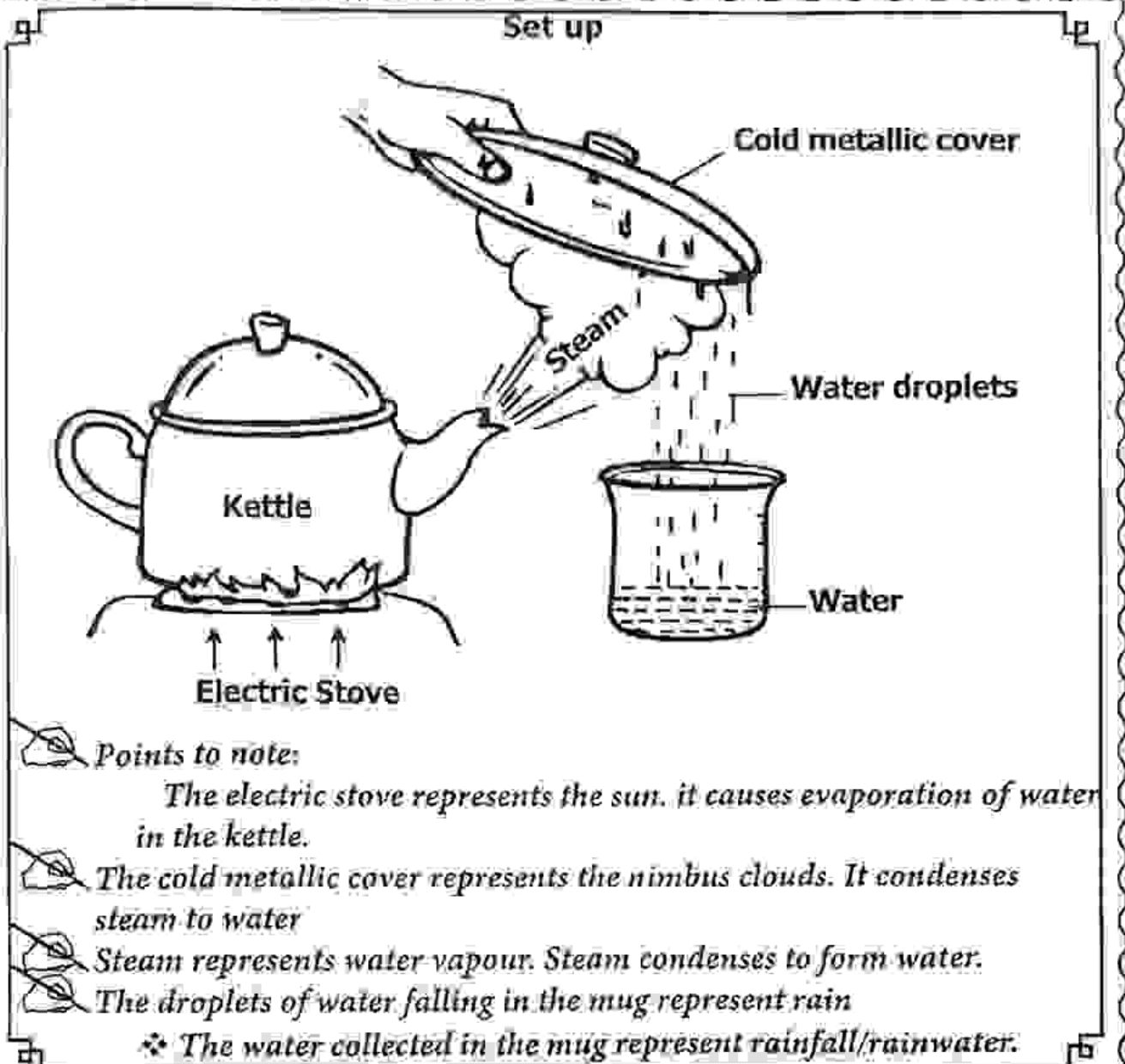
Nimbus clouds: They condense and form rain.

Experiment to show water cycle

Materials needed: electric stove, kettle, metallic cover and a mug

Steps taken

- Boil water in a kettle.
- Cover the top part until water evaporates.
- Hold a metallic cover at the opposite end where steam passes.
- Steam passes out to reach the cold metallic cover and condenses back to liquid in the mug.



Uses of water as a resources

- ✓ Water is used for drinking
- ✓ Water is used for bathing.
- ✓ Water is used for washing clothes and utensils.
- ✓ Water is used for irrigating crops

Ways of conserving water as a resource

- ❖ Planting trees
- ❖ Conserving swamps.
- ❖ Proper disposal of sewage.
- ❖ Fencing wells and water pumps

Air as a resource

Air is a renewable resource because it can be replaced naturally after use.

Photosynthesis, respiration, burning, fermentation and decomposition add air to the atmosphere.

Ways in which air is used as a resource

- ✓ Carbon dioxide is sued to preserve soft drinks.

- ✓ Oxygen supports burning
- ✓ Nitrogen is used to preserve semen for artificial insemination.
- ✓ Carbon dioxide is used in fire extinguishers.
- ✓ Oxygen is used to support the life of invalid patients in hospitals.
- ✓ Wind is used in winnowing crops.
- ✓ Wind energy is used to generate electricity in windmills.



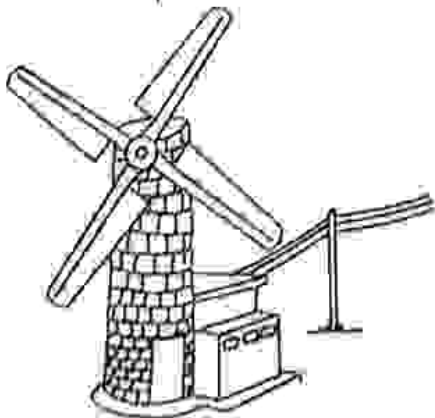
Fire extinguisher



Windmill

Evaluation Activity 7.2

1. Which gas in the atmosphere is used for;
 - i) Respiration
 - ii) Preserving soda
 - b) Give any **two** other ways in which air can be used as an energy resource.
 2. Why is water referred to as a renewable resource?
 3. How is fast flowing water used as a resource?
 4. Name the natural process that replaces water in the environment after use.
 5. Write down any **two** processes that help to replace air in the atmosphere.
- The diagram below is of a device that uses a resource to function. Use it to answer questions 6 and 7



6. Name the device shown above.
7. What source of energy is used to produce electricity in the device above?

LIVING THINGS AS RESOURCES

These include plants and animals.

Plants as resources in the environment

Plants are renewable resources because they can be replaced naturally through reproduction.

Ways in which plants can be used as resources

- ❖ Plants are source of food to people.
- ❖ Plants are source of herbal medicine e.g. aloe vera
- ❖ Plants are source of wood fuel.
- ❖ Plants provide timber for making furniture.
- ❖ Plants provide poles and timber for constructing houses.
- ❖ Plants provide fibres.

Ways in which plants support human life.

- Plants are source of food.
- Plants are source of herbal medicine.
- Plants provide oxygen for respiration.

Plant fibres

A fibre is a thin long thread-like structure used to make fabrics.

Examples of plant fibres are; sisal fibres, cotton wool, jute fibres, flax fibres, hemp fibres, raffia

Synthetic/man-made fibres are; nylon, polyester, spandex, rayon

Resources got from plants are; food, plant fibres, wood fuel, poles and timber.

Wood fuel got from plants are; firewood, charcoal, sawdust, wood briquettes

Ways of conserving plants.

- ✓ Using charcoal saving stoves.
- ✓ Planting more trees.
- ✓ Enforcing laws against deforestation.
- ✓ Using alternative sources of energy.
- ✓ Covering food when cooking.

This reduces heat loss from the saucepan

Animals as resources

Animals are renewable resources because they can be replaced naturally through reproduction.

Ways in which animals can be used as resources

- Animals are source of food to people.
- Some animals provide labour e.g. oxen
- Their hides/ skins are used in leather industries.
- Animals provide hooves and horns used for making buttons and glue.



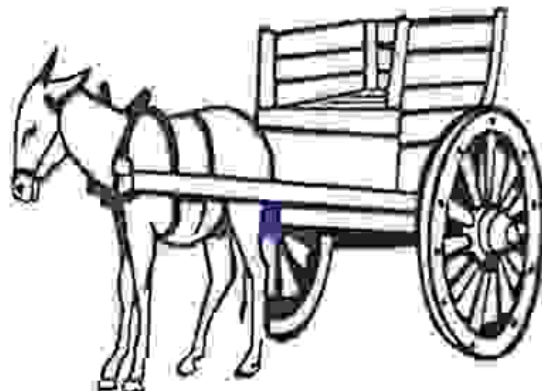
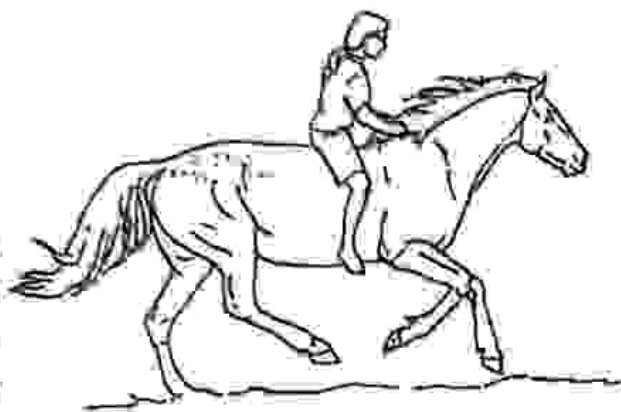
- Animals provide fibres for textile industries.

Examples of animal fibres are silk, mohair, wool/fleece and fur.

Silk is got from silkworms, mohair from goats, fur from rabbits and wool/fleece from sheep

How animal energy is useful

- Animals are used for transport
- Animals are used for pulling carts.
- Animals are used for ploughing.



Ways of conserving animals as resources

- Proper feeding of animals.
- Treating sick animals.
- Establishing zoos, game parks and game reserves.
- Enforcing strict laws against poaching
- Regular vaccination of animals.

Harvesting resources

A: Plants (trees)

Pollarding

Pollarding is the cutting off the top part of a tree.

Cutting should be done at an angle.

Reason. To drain off water from the remaining tree stump.



Pollarding

Advantages of pollarding.

- ✓ It encourages thicker growth of the lower branches.
- ✓ It helps to increase the yields of fruit trees.

Points to note:

Some fruit trees like mangoes, guavas, jackfruits and avocados tend to grow very tall. Pollarding keeps them shorter.

The cut off branches can be used as firewood.

Trees which grow near electricity power lines should be pollarded.

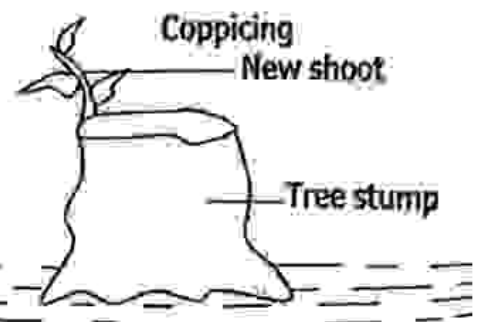
Reason. To prevent the trees from getting into contact with electric wires.

Coppicing

Coppicing is the cutting off a tree at the stump.

This method is suitable for harvesting trees for poles and timber.

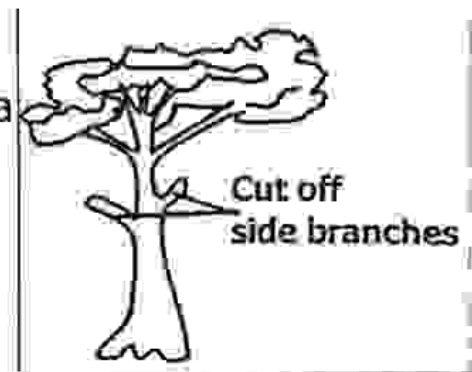
Coppicing encourages the growth of new shoot from the stump.



Lopping

Lopping is the cutting off the side branches of a tree.

The side branches can be used as firewood. This method does not cause damage to the trees.



Other methods of harvesting plants include;

Digging out, hand picking, cutting and uprooting

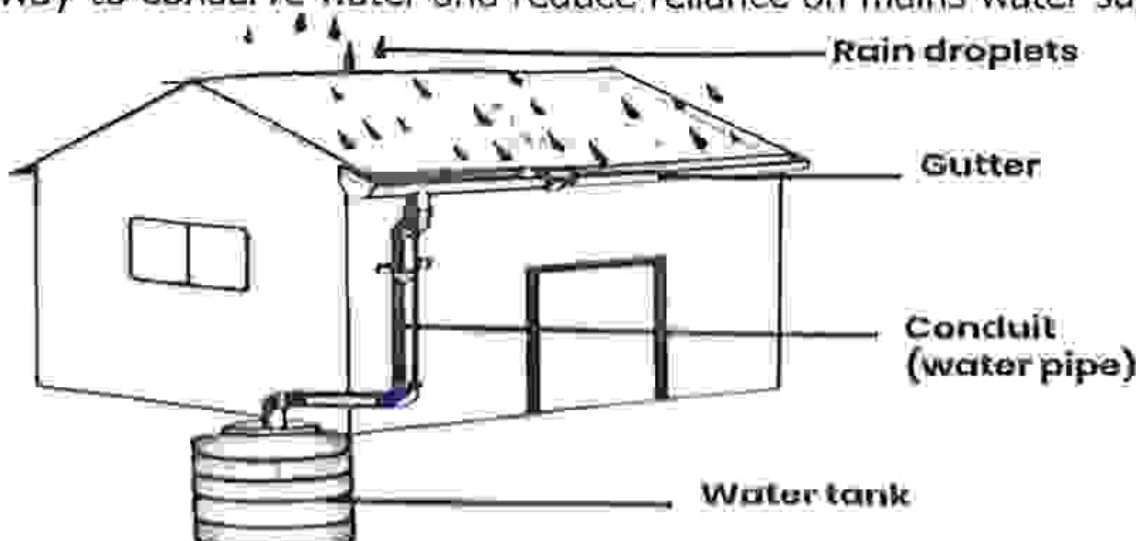
B: water

Rainwater can be harvested using;

- ❖ Rooftop rainwater harvesting
- ❖ Surface runoff harvesting.

Rooftop rainwater harvesting

This is a method of collecting rainwater that falls on the roof of a building and storing it for later use. This involves directing rainwater from the roof to a water storage tank via gutters and pipes (conduits), where it can be used for various purposes such as watering plants, flushing toilets and washing clothes, utensil etc. It is an effective way to conserve water and reduce reliance on mains water supply.



Note the following:

The gutter traps and directs rainwater to the water pipe.

The conduit directs rainwater to the water storage tank.

The water tank stores rainwater for future use.

Rain is the main source of water in the above method of harvesting water.

Rain is the main source of water because;

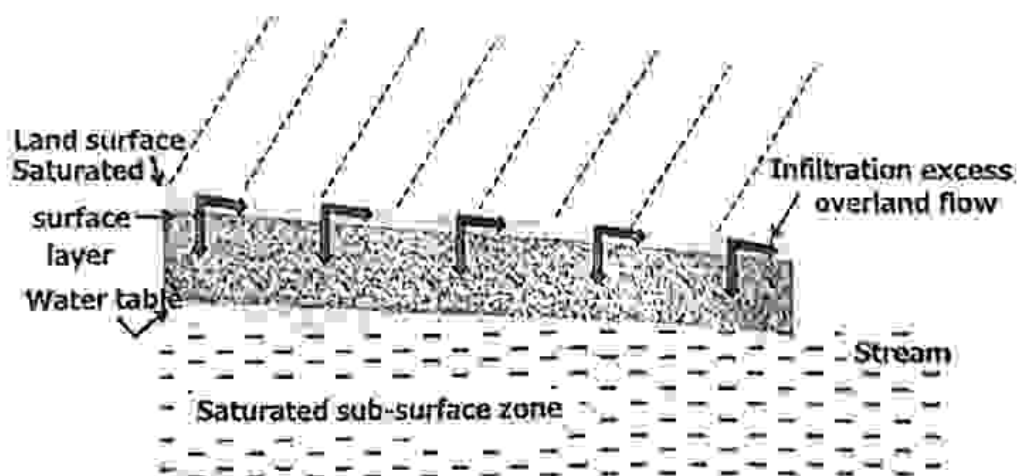
- ✓ Most of the water on earth comes directly or indirectly from rain.
- ✓ Rain supply water to other sources of water on earth.

Surface runoff rainwater harvesting

Surface runoff harvesting is a method of collecting rainwater that flows over the ground's surface during rainfall events. It involves capturing this runoff from rooftops, pavements or other impermeable surfaces and diverting it into storage tanks, reservoirs or other storage facilities for future use.

This helps to recharge ground water, reduce erosion, provide additional source of water for various purposes like irrigation, landscaping and non-potable uses.

Surface run-off rainwater harvesting



Other methods of harvesting water are;

- Fetching water from wells using containers like jerry cans.
- Pumping water from the ground using boreholes.
- Drawing water from deep wells using windlass.

C: Soil

- Sand mining
- Using machinery like bulldozers
- Excavation
- Shoveling

D: Animals

- Hunting
- Fishing
- Slaughtering
- Trapping
- Falconry

E: Minerals and rocks

- Quarrying ➤ Drilling ➤ Open cast mining ➤ Panning
- Shaft mining

F: Fossil fuels

- ❖ Drilling ❖ Open-pit mining ❖ Fracking
- ❖ Oil sands extraction ❖ Strip mining

G: Air/wind

- Using wind turbines. • Using propellers on windmills.
- Air compression. • Chemical absorption.

H: Sunshine

- Using solar panels. • Using solar thermal collectors.
- Using solar water heating system. • Using concentrated Solar Power.
- Using solar ponds. • Using solar cookers and dryers.
- Using solar chimneys. • Using solar greenhouses.

How people make use of available resources

- ✓ Using windmills to produce electricity from wind.
- ✓ Proper ventilation enables people to get fresh air.
- ✓ Fast flowing water is used to generate hydroelectricity.
- ✓ Heat from the sun is used to dry harvested crops and clothes.
- ✓ Solar electricity is used for lighting and operating devices like fridges.
- ✓ Sunshine enables the skin to manufacture vitamin D.
- ✓ Fossil fuels are used for lighting in kerosene lamps and candles.
- ✓ Fossil fuels are used in kerosene stoves to produce heat energy for cooking.
- ✓ Fossil fuels are used to run power-saw, mower, water pumps and vehicles.
- ✓ Biogas is used for cooking and lighting.

Types of resources

Renewables resources. These are resources that can be replaced naturally without getting used up.

Characteristics of renewable resources.

- They can be naturally replaced. - They exist in large amounts.
- They cannot get used up.

Examples of renewable resources are;

Plants, animals, water, soil, wind/air and the sun

Nonrenewable resources. These are resources that cannot be replaced naturally once used up.

Characteristics of non-renewable resources

- They cannot be replaced naturally once used up.
- They exist in limited amounts.

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159

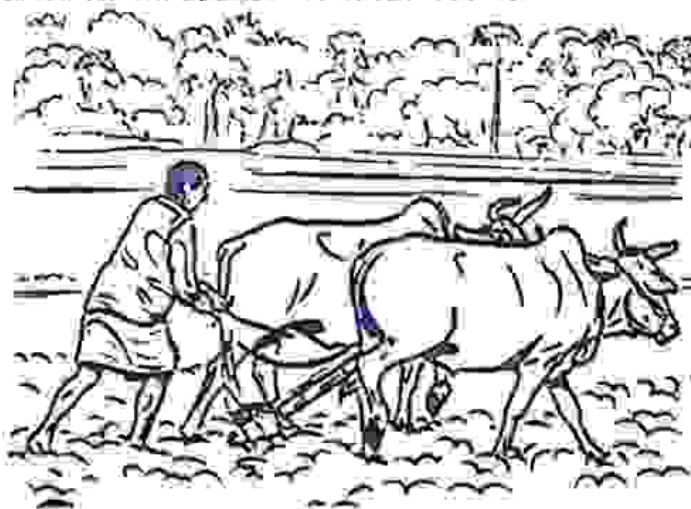
-They can get finished.

Examples of non-renewable resources

Minerals, fossil fuels and rocks

Evaluation Activity 7.3

1. Name the source of wood fuel in the environment.
2. a) Why are plants grouped under renewable resources?
b) Give any **two** ways in which plants can be used as resources.
3. Apart from plants, name any **one** other example of renewable resources.
3. State any **two** characteristics of non-renewable resources.
4. Write down any **two** examples of plant fibres.
5. Which animal resource is used in;
i) Leather industries?
ii) Textile industries?
6. The diagram below shows how animals can be used as resources. Study and use it to answer the questions that follow.



- a) Identify the activity done by the animals above.
 - b) What type of animals provide labour used to do the activity above?
 - c) How is the activity provided by animals above useful on the farm?
 - d) Name any **one** other animal that can provide labour to people.
7. How does covering food during cooking help to conserve wood fuel?
 8. Write down any two examples of;
i) Animal fibres
ii) Synthetic fibres
 9. Write down any **two** examples of non-renewable resources in the environment.
 10. State any **two** ways in which people use animals as resources.

ENVIRONMENTAL DEGRADATION

Environmental degradation is the lowering of the quality and productivity of the environment.

Natural causes of environmental degradation are; landslide, volcanic eruptions, floods, drought, earthquakes and hailstorms

Drought. Drought is the prolonged period of sunshine with little or no rain.

It is caused by deforestation, swamp drainage, bush burning and industrialization.

Effects of drought in the environment

- It leads to death of wildlife.
- It leads to drying of crops.
- It leads to dwindling in water bodies.

Ways of managing the effects of drought

- Planting drought resistant crops.
- Irrigating crops during drought.
- Conserving wetlands.
- Planting more trees

Floods. This refers to overflow of water bodies due to too much rainfall. Floods are caused by swamp drainage, deforestation, cultivating along water bodies, Uncontrolled mining and constructing in wetlands.

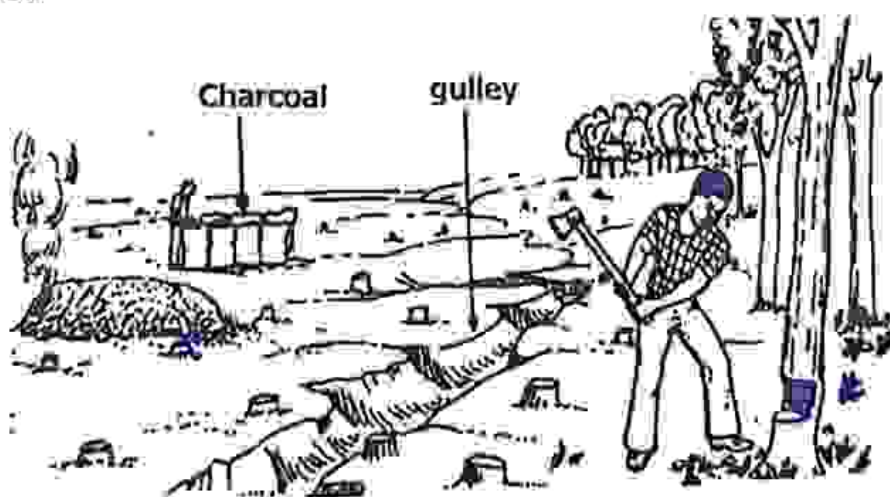
Effects of floods in the environment

- ↑ Floods cause displacement of people.
- ↑ Floods lead to death of aquatic animals.
- ↑ Floods lead to rotting of food crops.
- ↑ Floods lead to destruction of food crops.
- ↑ Floods cause leaching of plant nutrients.

Human activities that degrade the environment are;

deforestation, bush burning, swamp drainage, brick making, overgrazing, poaching and poor disposal of wastes like polythene bags.

Deforestation. Deforestation is the massive cutting down of trees in an area



Reasons why people cut down trees

- For making charcoal.
- To get firewood, timber and poles.
- To get land for farming.
- To get land for setting up industries

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161

- To get land for constructing houses.

Effects of deforestation and bush burning to the environment.

- ❖ It destroys habitats for wildlife.
- ❖ It leads to reduced rainfall.
- ❖ It leads to global warming.
- ❖ It leads to extinction of trees.
- ❖ It leads to soil erosion.

Ways of controlling deforestation

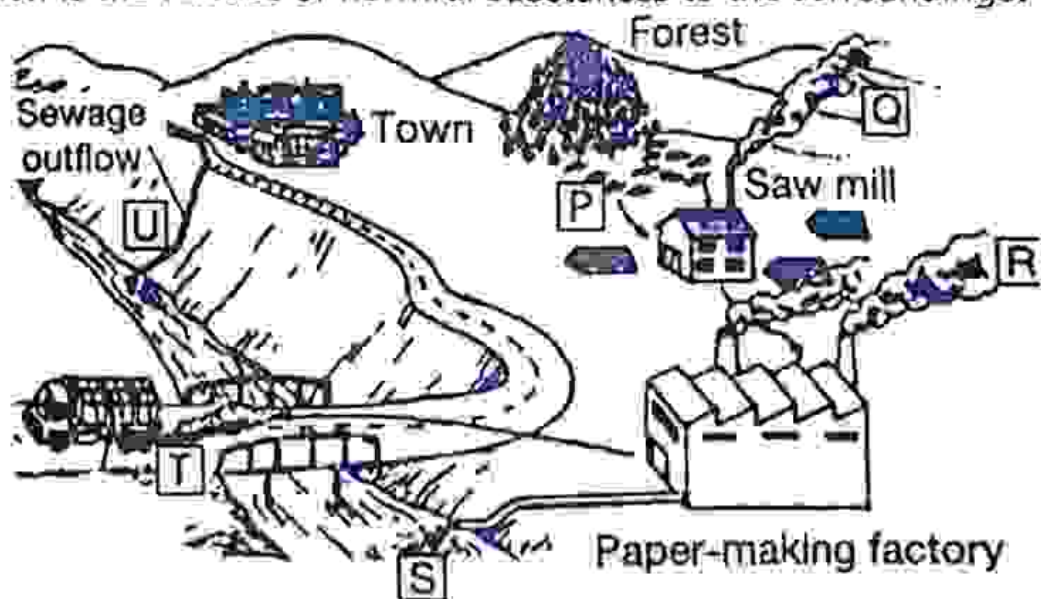
- ✓ Practising re-afforestation.
- ✓ Encourage people to use charcoal saving stoves.
- ✓ Use of alternative sources of heat e.g. gas cookers.
- ✓ Rural electrification.
- ✓ Gazetting forests
- ✓ Setting up forest reserves.

Evaluation Activity 7.4

1. Give the meaning of the term environmental degradation.
2. Give any **two** negative effects of floods in the environment.
3. Apart from floods, give any **two** other natural causes of environmental degradation.
4. Give any **two** reasons why people cut down trees.
5. Give any **one** way in which drought affects food production in an area.
6. Apart from deforestation, mention any **one** other human activity that can result into flooding in our environment.
7. In which **two** ways does bush burning degrade the environment?

Environmental pollution

Pollution is the release of harmful substances to the surroundings.



Forms of pollution

Water pollution. Water pollution is the addition of harmful substances that make water unfit to support life.

Agrochemicals, sediments, industrial forms, human wastes, animal dung, and urine are examples of water pollutants.

Causes of water pollution.

- Defecating and urinating in water sources.
- Discharge of industrial wastes in water bodies.
- Washing vehicles and clothes in water sources
- Fishing using poison.
- Silting of water bodies.

Effects of water pollution

- ❖ It leads to death of aquatic animals.
- ❖ It leads to the spread of water borne diseases.

Ways of controlling water pollution

- ⬇ Fencing water sources.
- ⬇ Avoid washing cars in/near water bodies.
- ⬇ Ensuring proper disposal of wastes.
- ⬇ Avoid discharging industrial wastes in water bodies.
- ⬇ Avoid using poison for fishing.

Air pollution.

Air pollution is the release of harmful substances in the atmosphere. Smoke, dust, industrial fumes, lead, smog and carbon dioxide are air pollutants.

Causes of air pollution

- | | |
|----------------------------------|-----------------------------------|
| • Industrialization | • Charcoal burning |
| • Quarrying | • Burning fossil fuels |
| • Tobacco smoking | • Volcanic eruption |
| • Excessive use of agrochemicals | • Decomposition of organic matter |
| • Poor disposal of human wastes | • Burning bushes and plastics |



Effects of air pollution

- ❖ It leads to acid rains.
- ❖ It leads to global warming.
- ❖ It leads to easy spread of airborne diseases

Soil pollution.

Soil pollution is the addition of harmful substances that degrade the soil. Polythene bags, plastics, metal like nails, chemicals like acid and broken glasses are soil pollutants.

Causes of soil pollution

- Dumping polythene bags and plastics in soil.
- Excessive use of artificial fertilisers.

- Pouring burnt oil and industrial wastes into the soil.

Effects of soil pollution

- It leads to loss of soil fertility.
- It leads to killing of soil organisms.
- It destroys soil structure and texture.

Ways of controlling soil pollution

- ✓ Proper disposal of plastics and polythene bags.
- ✓ Avoid excessive use of fertilisers.
- ✓ Treating industrial wastes before pouring them into soil.

Managing wastes in our environment

A waste is a material or product that is useless.

Some wastes are biodegradable (can decompose) e.g. human wastes, animal wastes, plant residues, paper and leftover food.

Some wastes are non-biodegradable (cannot decompose) e.g. plastics, polythene bag, glass and metallic wastes.

Dangers/ effects of poor waste disposal in the environment.

- It leads to easy spread of diarrhoeal diseases.
- It leads to air pollution.
- It leads to water pollution.
- It encourages the breeding of germs and vectors.
- It leads to blockages of sewage lines.
- It lowers soil fertility leading to poor crop yields.

5Rs in waste management

5Rs in waste management are; Reuse/Repair, Recycle, Reduce, Return, Refuse/Reject and Reinvent/Rethink

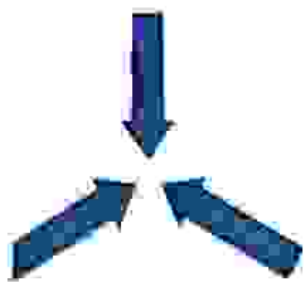
5RS	Materials	
Reuse This is putting wastes into other better alternative uses in their current state	- plastics - metals - old sacks - polythene bags	For growing trees, flowers and vegetables For carrying materials like food. For raising seedlings. For playing For fermenting alcohol They can be used as feeding troughs for animals. They can be used as laying nests. For making pieces of artwork. For drenching animals.

Recycle Recycling wastes is the making of re-usable materials from wastes	- old car tyres - plastics - glasses - paper - metals - yard wastes - strings - old radios - old woofers	For making manure from yard wastes. For making sandals from old car tyres. For making saucepans from old metals. For making knives and pangas from metals. For making water tanks, sandals and hand bags from plastics. For making kites from old papers and strings. For production of biogas from yard wastes. For making spare parts for radios, television sets and loud speakers.
Reject/Refuse Restricting the production of certain materials especially non-biodegradable materials and discarding wastes.	- polythene bags - rubbish	Banning the production and importation of Polythene bags. Burning rubbish Dumping rubbish in rubbish pits. Burying rubbish into the soil.
Return Return is the taking back waste objects to the manufacturers after use	- plastics - metals - old saucepans - old phones - old woofer - old radios	
Reduce This is the minimizing the production of harmful products.	plastics polythene bags	

Ways of managing wastes in the environment.

- Reusing wastes
- Recycling wastes
- Burning rubbish
- Dumping rubbish in rubbish pits.
- Proper disposal of sewage

Symbols used in waste management



Reduce



Reuse



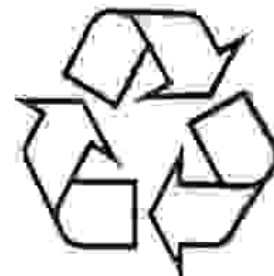
Recycle



Refuse



Proper disposal



Recycle

Importance of proper waste disposal in the environment.

- It prevents easy spread of diarrhoeal diseases
- It prevents water and air pollution.
- It reduces the breeding places for germs and vectors.
- It prevents soil exhaustion.

NEMA in conserving resources

NEMA stands for **N**ational **E**nvironment **M**anagement **A**uthority.

It conserves the environment.

Ways in which NEMA conserves the environment.

- It teaches people about the importance of forests and wetlands.
- It creates forest reserves.
- It enforces laws against wetland degradation.
- It evicts people settling in wetlands.

Evaluation Activity 7.5

1. a) Write down any **two** causes of water pollution.
b) How does water pollution affect aquatic animals?
c) State any **one** way of controlling water pollution.
 2. a) State any **two** ways in which plastics affect the soil.
b) Apart from plastics, name any **two** other examples of soil pollutants.
 3. a) Write down any **two** of the **5Rs** in waste management.
b) Give any **two** ways in which proper waste management is important in the environment.
 4. Give any **two** ways of recycling wastes in the environment.
 5. List down any **two** substances that degrade the quality of the atmosphere.
 6. a) Give any **two** ways in which you can reuse plastic objects at home.
b) Apart from plastic objects, mention any **two** other objects that can be reused at home.
 7. a) Give any **two** ways in which poor disposal of wastes is harmful to the environment.
b) Write down any **two** ways of managing wastes in the environment.
- The diagram below shows a method of managing wastes in the environment. Use it to answer questions 8 and 9.



8. State the method of managing wastes in the environment shown in the diagram above.
9. How is the method of managing resources shown above useful to the soil?
10. Give any **one** reason why the setting up of industries should be limited in an area.

Theme: **HUMAN BODY**
Topic **8: RESPIRATORY SYSTEM**

Respiratory system is the body system that helps the body to produce energy from food with the help of oxygen.

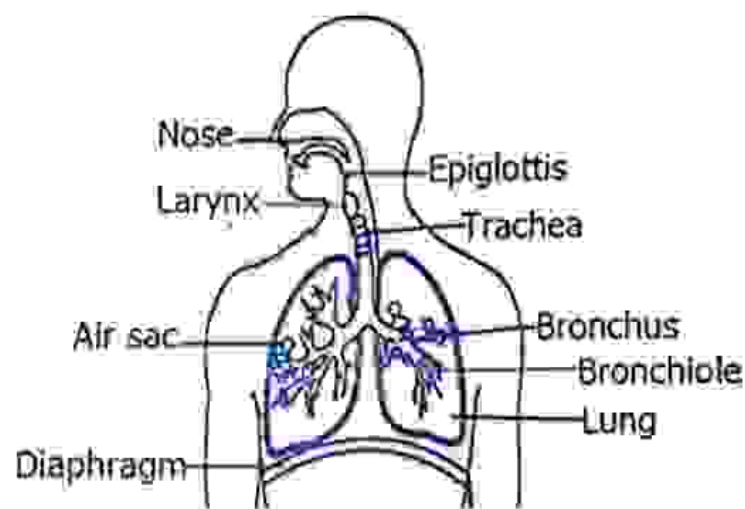
Breathing, gaseous exchange and oxidation are the body processes carried in the respiratory system.

Respiration

Respiration is the oxidation of glucose in the body cells to produce energy.

RESPIRATORY ORGANS

These include; lungs, nose and trachea



Functions of parts and organs of the respiratory system.

Nose

It lets in and out air. The nose has cilia (tiny hair) that trap dust, filter air and clean air.

The nose also contains mucus which traps dust and moistens air that passes through the nose.

Air in the nose is filtered, warmed and moistened.

It is not advisable to breathe through the mouth

Reason. The mouth lacks cilia to trap dust and filter air.

Epiglottis

It closes the trachea during swallowing. This prevents food from entering the trachea so as to avoid choking.

Trachea/windpipe

It is the passage of air to and from the lungs. It is made up of rings of cartilage to keep it open all the time.

It contains cilia and mucus which trap dust and filter air passing through it.

Lungs

They are used for breathing. Lungs are the major respiratory organs. They are protected by the rib cage.

Adaptations of lungs to their function.

- They are elastic to change their size during breathing.
- They have alveoli that help in gaseous exchange.
- They are soft and spongy for easy contraction and relaxation during gaseous exchange.
- They have a network of blood capillaries which supply oxygen.
- They have thin moist walls for easy diffusion of gases.

Alveoli/air sacs

They allow gaseous exchange to take place.

Oxygen joins bloodstream in the alveoli through diffusion.

Adaptations of the air sacs to gaseous exchange.

- ‡ They are many in number which increases the surface area.
- ‡ They have thin walls for easy diffusion of gases.
- ‡ They are surrounded by a dense network of blood capillaries to supply oxygen.

Bronchioles. They supply air to the air sacs.

Bronchi. They take air from the trachea to the lungs.

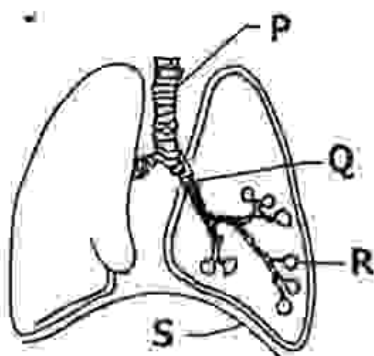
Diaphragm.

It flattens to create space for the lungs to expand.

It separates the chest cavity and abdominal cavity.

Evaluation Activity 8.1

1. The diagram below is of human lungs as excretory organs. Study the diagram and use it to answer the questions that follow.



- a) Name the parts marked **Q** and **S**.
- b) What substance is part **P** made of?
- c) Give the function of part marked **R**.

2. Why is the trachea made up of rings of cartilage?
3. By what process does oxygen join the bloodstream from alveoli?

LUNGS AND BREATHING

Breathing is the intake and outtake of air by an organism.

Breathing in human beings takes place in the lungs.

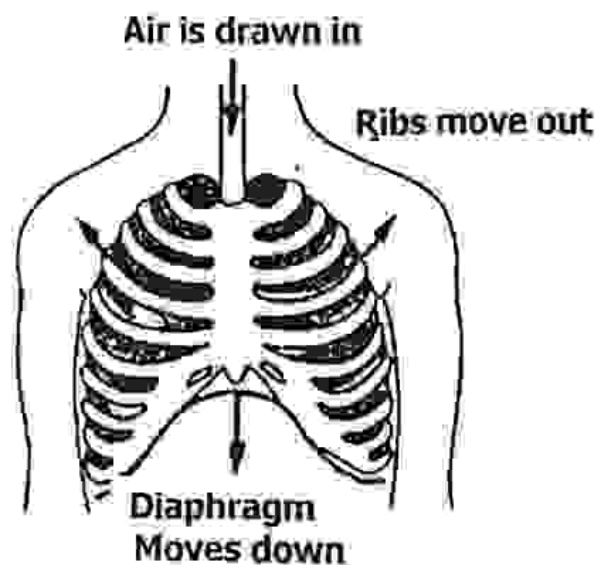
The types of breathing are breathing in and breathing out.

Breathing in/inhalation/inspiration

This is the taking in of air. We breathe in to get more oxygen for respiration.

Actions during breathing in

- The intercostal muscles contract.
- The ribs move upwards and outwards
- The lungs expand (to accommodate air inhaled)
- The volume of the chest cavity increases.
- The diaphragm contracts/flattens/moves downwards



Points to note.

We breathe faster when doing vigorous exercise to take in more air in order to get oxygen for respiration.

When an insect moves through flour, it dies.

Reason. Flour closes its spiracles.

We breathe faster after a race to replenish oxygen and remove carbon dioxide buildup from the body.

Breathing out/exhalation/expiration

Breathing out is the taking of air out of the body.

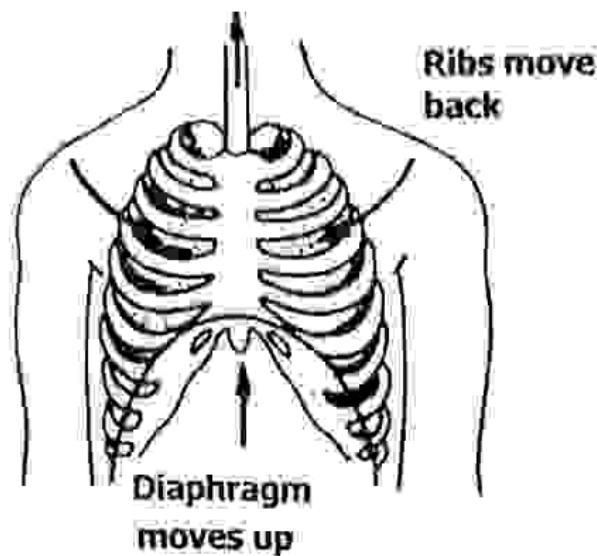
We breathe out to reduce carbon dioxide in the body.

Actions during breathing out

- ❖ The intercostal muscles relax.
- ❖ The ribs move downwards and inwards.
- ❖ The volume of the chest cavity decreases.
- ❖ The diaphragm moves upwards and forms a dome shape.

- ❖ The lungs contract. (to push out air)

Air forced out

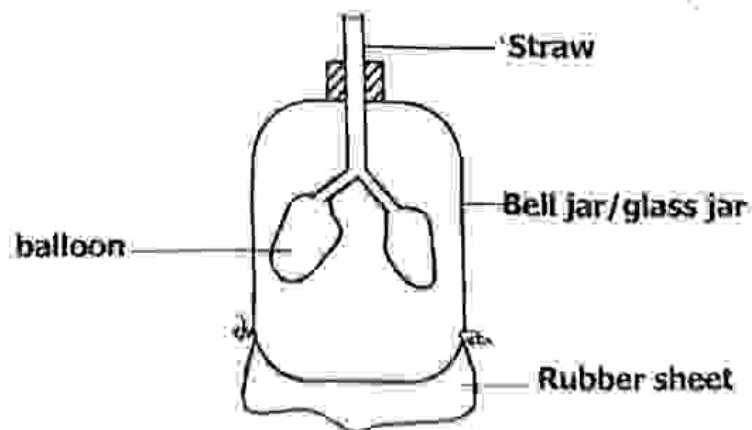


Experiment to show breathing using a model.

Materials needed: 2 balloons, a straw, rubber sheet and a plastic bottle/bell jar/glass jar

What to do:

- Cut the bottom of the plastic bottle/bell jar/glass jar to make one end open.
- Insert the two straws having inflated balloons on either side into the bottle through the bottom open end.
- Place the rubber sheet at the open end then tie it firmly onto the bottle.



Observation:

When the rubber sheet is pulled down, the balloons swell up with air/they are filled with air. The balloons expand when more air draws into them. When the air is sucked out of the straws, the balloons contract and air gets out. When the rubber sheet is released, air is pushed out of the balloons.

In the experiment above;

- ☛ The straws represent the trachea and bronchi
- ☛ The bell jar/plastic bottle represent the rib cage.
- ☛ The balloons represent the lungs.
- ☛ The rubber sheet represents the diaphragm.

Composition of air breathed in and air breathed out.

Component of air	Breathed in air	Breathed out air
Oxygen	21%	16%
Carbon dioxide	0.03%	4%
Nitrogen	78%	78%
Rare gases	0.97%	0.97%
Water vapour	water vapour	water vapour

Explanation

The percentage of oxygen breathed out is less than the one breathed in.

Reason. Some oxygen is used in the body for respiration.

There is an increase in the percentage of carbon dioxide breathed out.

Reason. More carbon dioxide is released from the body cells during respiration.

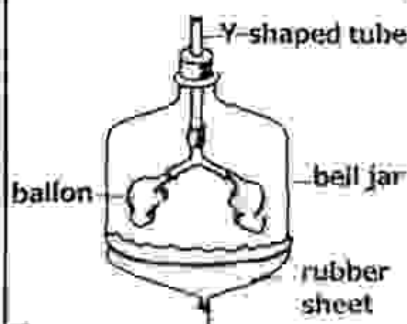
Nitrogen and rare gases remain unchanged because there is no human body process that uses them.

There is more water vapour breathed out than the one breathed in.

Reason. some water vapour is released from the body cells during respiration.

Evaluation Activity 8.2

1. (a) Name the respiratory organs which are protected by the rib cage.
(b) What happens to the rib cage during the following processes?
(i) Breathing in.....(ii) Breathing out.....
(c) Give the use of oxygen we breathe in.
2. The diagram below shows a model of a certain body system. Study and use it to answer the questions that follow.



- (a) Which body system is shown by the model above?
- (b) What do the Y-shaped tube and balloons represent on body system given in (a) above?
- (c) State what happens to the balloons when the rubber sheet is pulled downwards.

RESPIRATION IN RELATION TO FOOD AND OXYGEN Respiration is the oxidation of glucose in the body cells to produce energy.

Glucose + oxygen $\longrightarrow$ energy + carbon dioxide + water vapour

Respiration takes place in the **body cells**.

Oxygen and **glucose** are raw materials for respiration. Oxygen burns glucose to produce energy while glucose is oxidized to produce energy.

⇒ Carbohydrates are used during respiration to release energy in the body.

⇒ Energy is the main product produced during respiration. This is heat energy.

⇒ During respiration, **chemical energy changes to heat energy**.

 *Respiration is a chemical change because:*

✓ It forms new permanent substances.

✓ It is irreversible.

Carbon dioxide and **water vapour** are by-products of respiration.

Importance of respiration in the body.

- It enables the body to produce energy.

- It enables the body to get rid of waste products.

Importance of the circulatory system during respiration.

✓ Red blood cells supply oxygen to the body cells for respiration.

✓ Blood plasma supplies glucose to the body cells for respiration

Evaluation Activity 8.3

1. What is meant by the term respiration?
2. Where in the human body does respiration take place?
3. State the class of food used during respiration.
4. By what process does the body release energy from digested food?
5. The equation below shows the human body process. Study and use it to answer the questions that follow.



- a) Identify the human body process shown in the equation above.
 - b) Name the gaseous substances marked **W** and **Z**.
 - c) State the role of the gaseous substance marked **W** during the human body process above.
6. Which form of energy is produced during respiration?
 7. In which way are the red blood cells useful during respiration?
 8. Mention **two** raw materials used by the human body for respiration.
 9. In which way do plants benefit from the respiration in human beings?
 10. Which component of blood transports digested food to the body cells?

Diseases That Affect The Respiratory System

Disease and cause	Signs and Symptoms	Prevention and control
Asthma. - Hereditary disease	• Difficulty in breathing. • Persistent dry cough. • Wheezing sound.	Keep away from cold and dusty places.
Lung cancer - Tobacco smoking - It is non-contagious	❖ Coughing ❖ Chest pain ❖ Fever ❖ Shortness of breath.	Avoid smoking.
Emphysema Tobacco smoking - It is non-contagious	• Shortness of breath • Lung tissues peel off. • Enlargement of the chest. • Accumulation of air in the alveoli.	Avoid smoking.
Bronchitis Bacteria Spread - Through saliva and mucus droplets.	➤ Difficulty in breathing. ➤ Swollen bronchi ➤ Chronic cough.	Treating infected person using antibiotics.
Tuberculosis Bacteria Spread - Through saliva and mucus droplets - Through drinking unboiled milk from an infected cow.	• Mild fever • Chronic cough. • Blood-stained sputum. • Difficulty in breathing. • Chest pain. • Excessive sweating at night. • Loss of body weight.	- Immunising newly born babies using BCG vaccine. - Isolating infected people. - Wearing face masks. - Coughing in handkerchiefs. - Always boil milk from cows before drinking.
Whooping cough Bacteria Spread - Through saliva and mucus droplets - Through body contact with an infected person.	• Runny nose • Frequent coughing. • Shortness of breath. • Nausea (vomiting)	- Immunising children using DPT vaccine. - Isolating infected people.

Pneumonia Bacteria/virus Spread -Through saliva and mucus droplets	- Difficulty in breathing - Persistent dry cough - Pain in the chest. - Fever	- Treating infected person using antibiotics. - Immunising children using PCV.
Influenza Virus Spread -Through saliva and mucus droplets	• Frequent sneezing • Coughing • Nasal congestion. • Difficulty in breathing. • Runny nose and sneezing • fever	- Isolating infected people. - Early treatment of infected people. - Cough and sneeze from the handkerchief.
Diphtheria Bacteria Spread -Through saliva and mucus droplets	• Sore throat • Swollen neck • Severe fever • Difficulty in breathing.	- Immunising children using DPT vaccine. - Isolating infected people.

Disorders Of The Respiratory System

- 1. Choking:** Blockage of the airway by food or an object, preventing breathing.
- 2. Hiccups:** Involuntary spasms of the diaphragm causing sudden breaths and characteristic sounds.
- 3. Nasal congestion:** Swelling and mucus in nasal passages, making breathing through the nose difficult.
- 4. Sneezing:** Forceful expulsion of air from the lungs to clear irritants from the nose.
- 5. Wheezing:** High-pitched whistling sound during breathing, usually due to narrowed airways.
- 6. Cough:** Sudden expulsion of air to clear the throat or airways of irritants.
- 7. Yawning:** Involuntary deep breath often due to tiredness or lack of oxygen.

Practices that can damage the respiratory system.

- Regular smoking.
- Late treatment of lung diseases.
- Staying in areas where tear gas is exploded.
- Staying in dusty places.
- Failure to do regular exercises.
- Poor nutrition.

Ways of keeping the respiratory system in healthy working conditions

- ❖ Doing regular physical exercises.
- ❖ Feeding on a balanced diet.
- ❖ Avoid smoking.
- ❖ Immunizing children.

- ❖ Avoid staying in very cold areas. ❖ Going for medical check-ups.
- ❖ Avoid dusty environment.

Evaluation Activity 8.4

1. (a) Name the cattle product through which tuberculosis can spread to humans.
(b) Which body system is affected by tuberculosis?
(c) When is the vaccine against tuberculosis administered?
(d) Apart from immunizing, give **one** other way in which the spread of tuberculosis can be controlled in humans.
2. (a) Name the respiratory organ affected by tobacco smoking.
(b) Mention **two** respiratory illnesses caused by tobacco smoking.
(c) Apart from causing illnesses, in which **one** other way does smoking affect the human respiratory system?
3. (a) Write down any **two** disorders of the human respiratory system.
(b) Give any **two** ways of keeping the respiratory system in a healthy working condition.
4. (a) Name any **two** non-infectious diseases that affect the respiratory system.
(b) Mention any **two** practices that can damage the respiratory system.

Theme: SCIENCE IN HUMAN ACTIVITIES AND OCCUPATIONS

Topic 9: SCIENCE AT HOME AND IN OUR COMMUNITY

Preparation of Clean and Safe Water for Drinking and Washing.

Safe water is water that is free from germs.

Methods of preparing safe water for drinking are boiling, chlorination and fluoridation

Boiling

Boiling is the cheapest and commonest method of making water safe for drinking.

Water should be boiled to kill germs in it. It is the best method of making water safe for drinking.

Reason. Boiling kills all germs in water.

Steps taken when boiling water for drinking

- (i) Put clean water in a clean kettle or a saucepan.
- (ii) Put the kettle or saucepan on fire.
- (iii) Boil the water until it is ready. You will see steam coming out of the spout.



Remove the water from the fire. Keep it in a clean container and safe place to cool.

Cooled water should be put in a clean container and covered to prevent recontamination.

Water can also be put in small containers and kept in a refrigerator.

A refrigerator keeps water cold and keeps it free from germs.

Treating water using chemicals.

These chemicals include; chlorine, iodine, calcium chloride, fluorine and potassium permanganate.

The chemicals above kill germs in water.

Chlorination.

Chlorination is a method of purifying water by bubbling small amounts of chlorine through it to kill germs.

Chlorine kills germs in water.

Fluoridation

Fluoridation is the addition of small amounts of fluoride salts/fluorine to drinking water. Fluorine kills germs in water.

Disadvantages of using chemicals to treat water

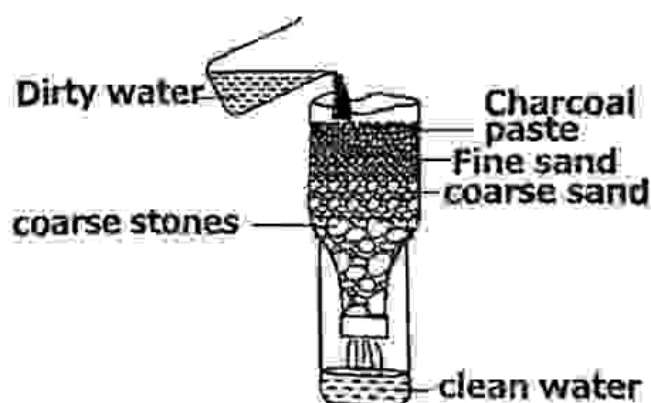
- ✓ Chemicals are expensive to buy.
- ✓ Chemicals add taste and smell in water.
- ✓ Chemicals do not make water clear.

Methods of Preparing Clean Water for Washing

Clean water is water that is free from impurities/dirt.

Filtration

Filtration is a method of separating liquid from solid particles. Filtration can be used to remove solid impurities from water.



Explanation:

- The solid particles that remain on the filter paper are called **residues** while the clear water that passes through the filter paper is called **filtrate**.
- The filter paper/cloth traps residues.
- The solid particles that remain behind when pouring water are called **sediments/precipitate**.
- The water filtered is not necessarily safe for drinking.
Reason. Filtered water still contains germs.
- Filtered water should be boiled before drinking to kill germs.

Local materials used for filtering water are; coarse sand, gravel, cotton wool, Charcoal paste, sisal fibre, banana fibre, pebble, grass and clean cloth.

Charcoal removes toxins and smell from water.

Fine sand, coarse sand and coarse stones trap impurities in water.

Other ways in which filtration is useful to people.

- It helps in preparing local salt from plant ash.
- It helps in separating fruit juice from seeds.
- It helps in separating tea leaves from tea.

Preparing local salt from plant ash using filtration.

Materials needed: water, sieve, banana leaf, dry bean plant leaves, 2 containers, flat metal sheet

Procedure:

- (i) Burn the dry bean plant leaves to get plant ash.
- (ii) Gently place the banana leaf on the sieve.
- (iii) Take the sieve with a banana leaf and place on the cup.
- (iv) Transfer the plant ash onto the sieve and add water to make a solution.
- (v) Leave the solution to collect in the cup for some time. The solution tastes salty.

Set up:



Note: the solution can also be heated/boiled to evaporation to dryness to obtain solid salt. Local salt is rich in sodium chloride. It tastes sour due to high concentration of sodium chloride.

Simple summary of steps followed when preparing local salt

- i. Burn the dry pods of legumes to ash.
- ii. Put ash into a perforated container.
- iii. Add little water into the perforated container of ash.
- iv. Leave the salt solution to collect in a clean container.

Ways in which local salt is important to people.

- It is used to soften food during cooking.
- It is used as a preservative.
- It is used as pesticide.
- It is used as local medicine to cure stomach upsets.
- It is added to some sauce to get ready faster.
- It is cooked with vegetables to add flavour.
- It is added to lemon to reduce its sour taste.

Foodstuffs that can be cooked with local salt

- tomatoes
- beans
- pigweeds
- cowpeas
- pumpkin leaves

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Decantation

Decantation is the process of pouring off clear water leaving settled dirt at the bottom of a Container.

In this method, impurities are first allowed to settle at the bottom of the container.

Clean water is then carefully poured out into another container leaving behind the impurities.



Note well:

The clean water obtained through decantation is not recommended for drinking.

Reason. Decanted water contains germs.

In order to make decanted water safe for drinking, it must be boiled or treated using chemicals so as to kill germs.

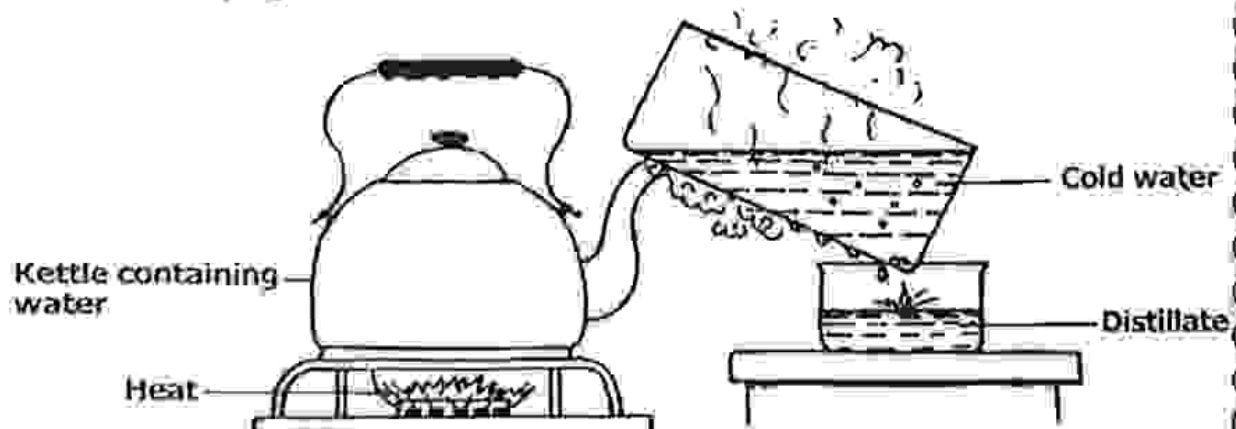
The glass rod is used to carefully stir the mixture during decantation.

It ensures clean decanting of water without disturbing the settled materials.

Distillation Method of Purifying Water.

In distillation method, the muddy water is heated until it evaporates.

The vapour is then Condensed to obtain pure water. Distillation is also used in purifying alcohol.



The liquid condensed is from water vapour during distillation is called **distillate**.

Distilled water is free from germs, dirt and other impurities but not recommended for drinking.

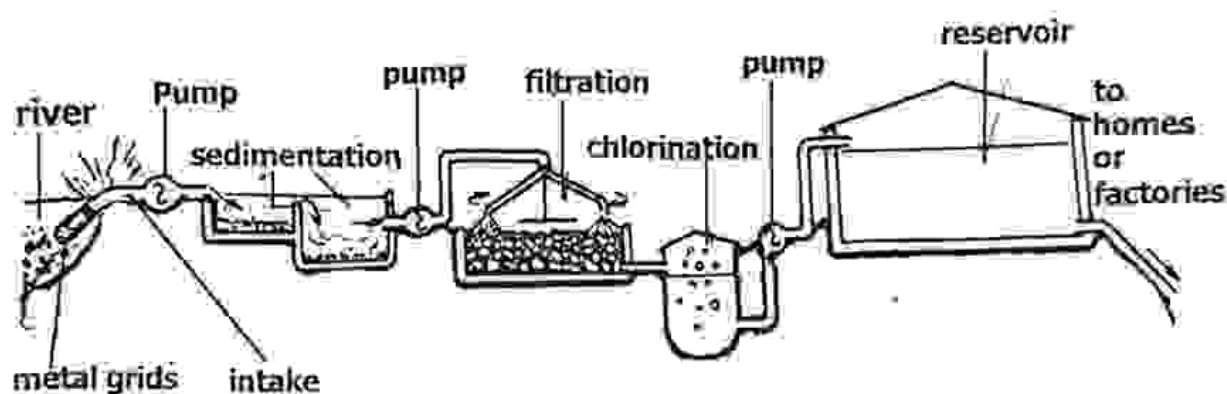
Reason. Distilled water lacks mineral salts.

Distilled water does not contain mineral salts because during distillation, mineral salts remain in the heating container as water evaporates.

Uses of Distilled Water

- It is used in laboratory experiments.
- It is used for cleaning wounds.
- It is used for mixing medicines in hospitals.
- It is used in automobile batteries to prevent mineral build up.
- It is used in steam irons to prevent mineral build up.
- It is used for cleaning medical equipment.

A water purification system.



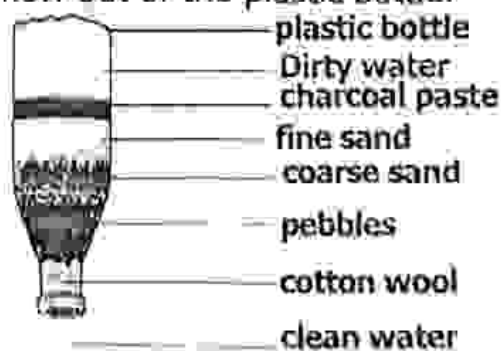
Stages of purifying water

i Learn, unlearn and relearn: filtering and decanting are not methods of obtaining clean water but activities used to make water clean from dirty water. Methods of making water clean are filtration and decantation.

How to make a simple water purifier.

- Cut off the bottom of a plastic bottle.
- Put a layer of cotton wool in the neck of the bottle.
- Add a layer of pebbles, coarse sand, fine sand and charcoal paste.
- The clear water will start to flow out of the plastic bottle.

Set up:



Evaluation activity 9.1

1. Name **two** methods of obtaining clean water for washing at home.
2. In which method of making water clean do you first allow the impurities to settle at the bottom of a container?
3. How does chlorine make water safe for drinking?
4. Why is water obtained by distillation method not good for drinking?
5. How does boiling make water safe for drinking?
6. The diagram below shows a simple method of making water clean. Study and use it to answer the questions that follow.



- a) Name the method used above.
 - b) Apart from coarse stones, name any **two** other local materials that can be placed in container T.
 - c) Why should the clean water obtained in the container S be boiled before drinking?
7. (a) Name **one** material from which local salt can be prepared at home.
- (b) State the method used to prepare local salt.
 - (c) Give any **two** ways in which the local salt prepared at home is useful to people
8. The diagram below shows a simple water purification system. Study and use it to answer the questions that follow:



- a) Name any **one** activity that can be done to water out at point W.
- b) Why is swampy water filtered before being taken at point W?
- c) State the importance of passing water through point W.
- d) Mention any **one** other activity that can be done to swampy water before being taken to point W.

Water Impurities

Human wastes, animal wastes, industrial wastes, silt from soil erosion, agrochemicals

Silting

Silting is the deposition of eroded soil particles into water bodies. Soil erosion is the major cause of silting in water bodies.

Human activities that lead to silting in water bodies.

- Cultivating along river banks.
- Grazing animals near water bodies.
- Spilling sediment near water bodies.
- Sand mining in and near lakes.
- Dam construction
- Road building near water sources.
- Deforestation near water bodies.

Note: The above activities lead to soil erosion which causes silting in water bodies.

Effects of silting of water bodies.

- ✓ Exposes aquatic animals to their predators
- ✓ Leads to death of aquatic animals.
- ✓ Blocks proper oxygen supply to the aquatic animals.
- ✓ Blocks food for the aquatic animals.
- ✓ Destroys habitats for aquatic life.
- ✓ It causes floods in an area.
- ✓ It causes the creation of shallow water bodies.

Ways of controlling silting in water bodies

- ⌚ Leaving river banks and lake shores uncultivated.
- ⌚ Dredging water sources.
- ⌚ Maintaining vegetation near river banks and lake shores.
- ⌚ Avoid grazing animals near river banks and lake shores.

Cleaning Clothes in a Home

Reasons why clothes should be cleaned regularly.

- ✓ To prevent the spread of skin diseases.
- ✓ To prevent the breeding of body lice.
- ✓ To prevent bad body smell.

Steps people should follow to clean clothes are sorting, soaking, washing, rinsing, wringing, drying and ironing

Sorting.

This is the separating of clothes to be cleaned.

Factors considered when sorting clothes

- Colour of clothes
- Intensity of dirt
- Texture of the clothes.

Importance of sorting clothes.

- It prevents bleaching clothes from staining other clothes.
- It prevents less dirty clothes from absorbing dirt from dirtier clothes.
- It helps to maintain the colour of clothes during washing.

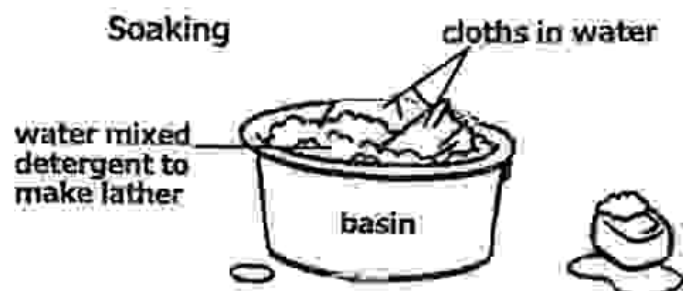


Sorting

Soaking.

This is the putting of clothes in water with detergent or jik for some time before washing.

This detergent can be Omo, Magic, Nomi, Aerial or Sunlight.



Soaking

Importance of soaking clothes

- It helps to soften and dissolve dirt in the clothes.
- It removes stains from clothes.

Washing.

It is done using soap.

Soap softens and dissolves stains or dirt on clothes.

Note well: Soap does not soften clothes during washing but softens dirt/stains



Washing

Importance of washing clothes

- It kills some parasites in clothes.
- It helps to keep clothes clean.
- It helps to remove bad smell from clothes.
- It helps to prevent skin diseases like ringworm.

Types of washing are;

hand washing and machine washing

Rinsing.

This is the putting of clothes in clean water without soap after washing.

Importance of rinsing clothes.

- It helps to remove lather from clothes
- It helps to remove dirty soapy water from clothes.

Wringing.

This is twisting and squeezing clothes in order to remove excess water.

- Wringing clothes helps to ease drying.
- It reduces water content in the clothes.

Note: Woollen clothes should be dried without wringing to maintain the quality of their fabric.



Drying.

This is the putting of clothes in sunshine to dry after washing.

Heat from the sun drains moisture from the clothes. Wind drives moisture from clothes by increasing the rate of evaporation.

Reasons for not over drying clothes

- To reserve moisture in clothes for easy ironing.
- To prevent clothes from fading or bleaching in sunshine.



Ironing

This is the last step in cleaning clothes.

Importance of ironing clothes

- It helps to kill some parasites like lice.
- It helps to kill germs in clothes.

The sun, water and wind help in cleaning clothes



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Evaluation activity 9.2

1. Mention **one** way in which you can prevent some clothes from staining others during washing.
2. Give any **two** examples of water impurities.
3. State the importance of the following steps in cleaning clothes at home.
 - (i) Sorting
 - (ii) Soaking
4. Mention any **one** factor considered when sorting clothes for washing.
5. State any **one** danger of hanging clothes under direct sunshine for a long time to dry.
6. Name any **one** natural resource that can be important when washing clothes.
7. State any **two** ways in which washing clothes is important to human health.
8. Name the step used in cleaning clothes that;
 - (a) Removes excess water from the clothes
 - (b) Removes soapy water
9. (a) What is meant by the term silting?
 (b) State the major cause of silting in water bodies.
 (c) Apart from killing fish, give any **two** other ways in which silting affects fish.
10. State the reason why woollen clothes should be dried without wringing.
11. In which way is the sun useful in the cleaning of clothes at home?
12. State any **one** way in which the ironing of clothes is useful when cleaning clothes.
13. In the table below, part **A** shows steps used in cleaning clothes and part **B** has the meanings of the terms.

Part A: Steps	Part B: Meanings
Sorting	Putting dirty clothes in soapy water.
Wringing	Putting clothes in water to remove soapy water.
Rinsing	Grouping clothes before washing or soaking.
Soaking	Squeezing clothes to remove excess water.

Use the meanings in part B of the table to match the terms below.

- (a) Sorting
- (b) Wringing
- (c) Rinsing
- (d) Soaking

Theme: **HUMAN HEALTH**
Topic **10: ACCIDENTS AND FIRST AID**

An accident is a sudden happening that can cause injury or death to a victim.

An accident victim is called a **casualty**.

Common forms of accidents

Burns and Scalds

A burn is a skin injury caused by dry hot substances.

Examples of dry hot substances are; hot electric iron, hot jikos, hot cooker, acid, hot saucepan and burning firewood

A scald is a skin injury caused by wet hot substances.

Examples of wet hot substances are; hot water, hot sauce, hot milk and hot porridge

Causes of burns

- Skin contact with hot metals.
- Skin contact with acids
- Skin contact with burning wood fuel
- Skin contact with bare electric wires carrying current

Causes of scalds

- Skin contact with steam.
- Skin contact with hot liquids.

Degrees of burns.

First degree burn. Blisters are not formed

Second degree burn. Blisters are formed.

Third degree burn. This is the severe burn in which the skin is burnt deeply and appears shiny white.

This degree of burns leads to dehydration because the casualty loses a lot of fluids in the body.

The casualty should be given a lot of fluids like fruit Juice.

First aid for burns and scalds.

Dip the burnt or scalded part in clean cold water.



scald from a hot liquid



burn from dry heat

Reasons why a scalded or burnt part is dipped in clean cold water.

- To cool temperature around the injured part.
- To reduce further damage on the skin tissues.
- To reduce swelling of the burnt area.
- To relieve pain
- To minimize the formation of blisters at the burnt part.



dipping hand in cold water

Precautions against burns and scalds

Never put cow dung or soil on the injured part.

Reason. Cow dung attracts germs to the wound.

Never apply cooking oil on the injured part.

Reason. It causes infections.

Never apply sugar. Sugar attracts bacteria and absorbs water from the skin.

Never apply salt on the part. Salt causes dehydration.

Never burst blisters. It causes wounds leading to germ infection

Prevention of burns and scalds.

- Use insulators to carry hot objects.
- Cook from raised places.
- Keep flammable liquids away from fire.
- Put out all fire after use.
- Avoid leaving burning candles in rooms where there are clothes.

Fever and convulsions

Fever is the condition in which the body temperature rises beyond the normal.

Excessive sweating is the main sign of fever.

Causes of fever are; diseases like malaria, food poisoning and putting on too tight clothes.

First aid for fever.

❖ Tepid sponging the casualty.

Tepid sponging is the use of a clean piece of cloth to apply lukewarm water on the body of a casualty with fever.

Tepid sponging cools the body temperature of the victim.

Lukewarm water opens the sweat pores to let out excess heat from the body.

- ❖ Loosen tight clothes on the victim to enable the victim get fresh air.
- ❖ Give the casualty plenty of drinks to prevent dehydration.

 **Note well.** Very cold water should not be used to tepid sponge the victim.
Reason. Cold water causes shivering to the victim which increases heat production in the body.

Convulsions.

Convulsion is the rapid uncontrolled shaking/jerking of the body.
 Fever is the main cause of convulsions.

Signs and symptoms of convulsions.

- ✓ High body temperature.
- ✓ Violent shaking of the body.
- ✓ Frequent urination.
- ✓ Excessive sweating.
- ✓ Bubbles of saliva in the mouth.

First aid for convulsions.

- Put a hard material in the mouth of convulsing victim.

Reason. To prevent the victim from biting the tongue.

- Place the victim in a safe place to avoid injury by surrounding objects.
- Make sure the victim gets plenty of fresh air.
- Loosen/remove most of the clothes to enable the victim get fresh air.
- Apply tepid sponging to cool the body temperature.

How to prevent fever and convulsions.

- ⚡ Controlling mosquitoes.
- ⚡ Early treatment of infections.
- ⚡ Immunizing children.
- ⚡ Maintain proper hygiene.

 **Notewell.** Do not give food or drink to the convulsing victim.

Reason. Food or drinks leads to choking and causes death.

Evaluation activity 10.1

1. How is a burn different from a scald?
2. A child suddenly removed the cover of a saucepan full of boiling water. One of the child's arms was burnt.
 - (a) Name the substance that burnt the child.
 - (b) What kind of accident did the child get?
 - (c) State the first aid you would give to the child.
 - (d) Give any **one** way in which the child would avoid the accident mentioned in (b) above.
3. Give any **one** reason why the burnt part on the skin is dipped in cold water.
4. Why is it not advisable to burst blisters on the burnt part of the skin?
5. How is tepid sponging useful to a victim with fever?
6. What kind of water is encouraged to use during tepid sponging?
7. Why is it not advisable to give a convulsing person food to eat?
8. State the first aid given to a casualty with burns.
9. Which health condition can result from severe burns and scalds?
10. Give a reason why a hard object should be put in the mouth of a convulsing victim.

Near Drowning

Near drowning is a condition in which one's lungs are filled with water and fails to breathe but not yet dead.

Drowning is dying as a result of having lungs filled with water.

Near drowning occurs in water bodies, fish ponds, swimming pools, bath tubs and basins.

Causes of near drowning.

- Flooding in an area.
- Capsizing of boats due to rain storms.
- Failure to swim.
- Inability to swim.
- Being drunk while in deep water.
- Attack by the disease during swimming.
- Muscle pull while in water.



First aid for near drowning

- I) Yell for help from people around.
- II) Pull the casualty from water in case you are able.
- III) Apply mouth to mouth breathing.

This opens up airways to restore breathing/Induces breathing



How to apply mouth to mouth breathing.

- (i) Lie the person on his/her back and open his/her mouth.
- (ii) Press the nostrils with your fingers to close them.
- (iii) Bring your mouth to that of the victim's mouth.
- (iv) Blow air strongly into his/her mouth and then pause to let the air come out. Blow in again. Repeat this several times.



IV) Press the belly of a casualty to expel water from the lungs.

Prevention of near drowning.

- Covering water tanks and deep pits.
- Fencing swimming pools.
- Wearing life jackets when sailing on water.
- Acquiring swimming skills.
- Fencing water sources.
- Remove the baby from water after bathing.
- Avoid playing near water bodies.
- Relocating to other areas in case of flood outbreak.



Fainting

Fainting is the sudden loss of consciousness for a short time.

The major cause of fainting is reduced supply of oxygen to the brain.

Signs of fainting are; loss of body balance, body weakness, loss of sense.

Conditions that can lead to fainting are;

prolonged hunger, vigorous exercise, shocking news, severe pain, anxiety and excitement, excessive bleeding, fear, being frightened, standing in sunshine for long, alcoholism, dehydration and heart problems like irregular heartbeat.

The above conditions lead to reduced supply of oxygen to the brain.

First Aid for fainting

- ❖ Lay the casualty on his/her back and raise the legs higher than the head.

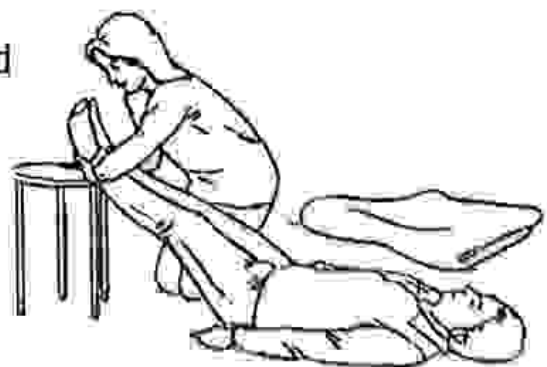
Reason. To allow the supply of oxygen to the brain.

- ❖ Loosen the tight clothes.

Reasons. To ensure adequate supply of fresh air//To allow free blood circulation.

- ❖ Fan the casualty in order for him/her get fresh air.

- ❖ Put the casualty in a cool and open place to enable him/her get fresh air.



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[19]

Evaluation activity 10.2

1. Name the kind accident that requires raising the legs higher than the head.
2. (a) State the difference between drowning and near drowning.
(b) Give any **two** causes of near drowning.
(c) State any **one** way in which you can help a near drowned casualty.
3. Why is it not possible to give first aid to a drowned casualty?
4. (a) What causes fainting?
(b) State any **two** conditions that can lead to fainting.
(c) Why are legs of a person who has fainted raised higher than the head when giving first aid?
5. State the purpose of pressing the belly of a casualty of near drowning.
6. Give **one** reason why a person who has fainted should be placed in a cool and open place.
7. How does prolonged hunger lead to fainting?

Foreign Bodies in Passages

A foreign body is something that is stuck inside the body part where it is not supposed to be.

You may inhale or swallow a foreign body, or you may get one from an injury to almost any part of the body.

They enter body parts like eyes, ears, skin, nose, anus, vagina.

Examples foreign bodies are; insects, dust, seeds, small stones, grass, iron filings and food particles.

Giving first aid for foreign bodies

Body opening	Foreign bodies	First aid given
In the eyes	small insects, sand, grass, sand particles, dust, tiny seeds and iron filings	• Wash the eyes with plenty of clean warm water. • Use a piece of soft clean cloth to gently wipe the eye.
In the ears	small insects, small seed and dust particles.	• Flush the ear with plenty of clean water.
In the nose	small insects, small seeds, small stones	• Press the nose without the foreign object and ask the the casualty to blow his/her nose. • Encourage him/her to breathe through the nose.

In the anus	sand, small stones, seeds, leaves and grass	• Use water and a clean finger to remove sand. • Use a finger and a piece of cloth to remove others
In the throat	food particles This causes choking	• If the casualty is a small child, press the belly with strong upward push. • If the casualty is an adult, give several blow on his/her back.

Choking is the blockage of the air passage to the lungs.

Choking mainly affects the respiratory system.

Choking can cause failure to breathe which leads to death.

This is because food in the throat squeezes and blocks the trachea used to let air in and out of the lungs.

Causes of choking

- Swallowing big chunks of food
- Talking with food in the mouth.
- Eating hurriedly.

Dangers of choking

- It leads to death.
- It damages the throat.

Giving first aid for choking/foreign body in the throat.



Nose bleeding

Nose bleeding is the condition in which blood flows from the nostrils.

Causes of nose bleeding are;

overblowing of the nose, fractured skull, high temperature, Inhaling very hot air, strong sneezing, high fever, punch on the nose, nose picking, allergies and standing in sunshine for long.



Giving first aid for nose bleeding.

- Lean the casualty forward to control the flow of blood through the nose.
- Pinch the soft part of the nose to promote clotting of blood.
- Apply cold compress to cool temperature in the nose.
- Encourage the casualty to breathe through the mouth to reduce the pressure of blood flowing through the nose.



Snake bite.

An injury caused when the snake bites an individual. Poisonous snakes have venom which they inject in the victim through their long fangs.

First aid for snake bite.

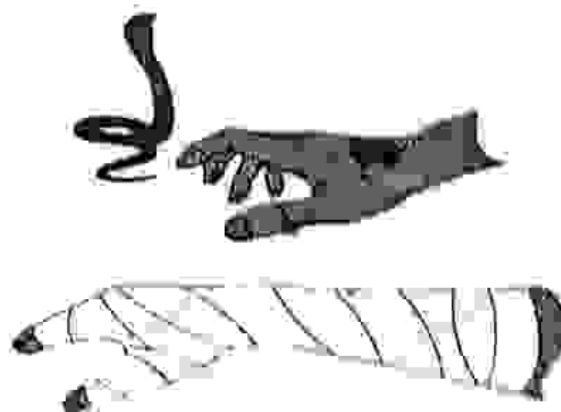
- (i) Calm the casualty
- (ii) Identify where the fang marks are on the victim's body. This is to identify whether the victim was bitten by a poisonous snake or non-poisonous snake.
- (iii) Tie a bandage slightly above the bitten part. This prevents venom from flowing to the heart and weaken the cardiac muscles.
- (iv) Take the casualty to the nearest health unit.



Note: if possible, kill the snake and go with it to the health centre to enable the health worker give the casualty the right antivenom/antidote/serum.



tourniquet



Poisoning

Poisoning is the act of taking toxic substances into the body.

Poison is any substance that is harmful to the body once taken.

Examples of poison are; jik, agrochemicals, rat poison, methanol, paraffin, drugs, expired food

Signs and symptoms of a poisoned victim;

rapid breathing, excessive vomiting, loss of body

Balance, fever and sweating, feeling a lot of thirsty, mental confusion

Causes of poisoning at home.

Eating expired food, taking expired drugs, taking overdose, ignorance, keeping unlabeled drugs, keeping drugs in reach of children and keeping chemicals in containers which are commonly used.

The major cause of child poisoning is keeping chemicals near the reach of children.

First Aid for poisoning

Give a casualty a lot of fluids like fruit juice and milk to dilute the concentration of poison.

In case of paraffin, diesel, petrol, jik and bleach, do not make the victim to vomit.

Reason. Vomiting may cause damage to the gullet, throat and stomach.

Prevention of poisoning

- ⚡ Label dangerous chemicals.
- ⚡ Keep chemicals out of reach of children.
- ⚡ Read and follow instructions on chemicals.
- ⚡ Provide proper storage of chemicals



Evaluation activity 10.3

1. (a) Name any **two** examples of foreign bodies in the eyes.
(b) State any **one** effect of foreign bodies in the eyes.
(c) Give any **one** way in which you can give first aid to a victim with a foreign body in the eye.
2. (a) Write down any **two** causes of nose bleeding.
(b) Why is a person whose nose is bleeding advised to;
(i) lean forward his/her head?
(ii) pinch the soft part of his/her nose?
3. Why is it advisable to tie the bandage slightly above the bitten when giving first aid to a victim of snake bite?
4. Give a reason why a poisoned victim is given a lot of fruit juice to drink.
5. Why is a person who has nose bleeding advised to breathe through mouth?
6. State the major cause of child poisoning at home.
7. Why should chemicals like rat poison be kept away from the reach of children?
8. Why is it not advisable to make a person who has taken jik to vomit?
9. The diagram below shows a form of accident. Study and use it to answer the questions that follow.



- (a) Identify the form of accident that happened to the casualty above.
- (b) Which body system is affected by the form of accident above?
- (c) State any **one** possible cause of the form of accident that happened to the casualty above.
- (d) What first aid would you give to the casualty shown in the diagram above?

Theme: HUMAN HEALTH

Topic 11: SANITATION

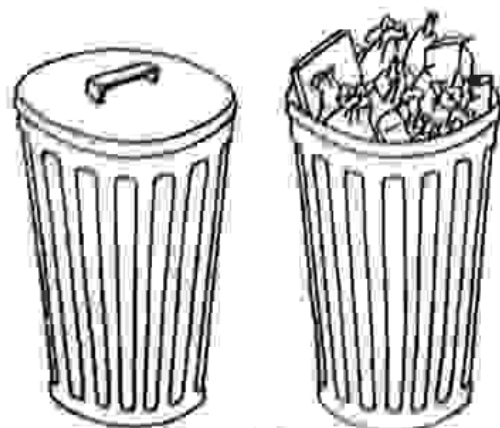
Sanitation is the general cleanliness of the environment.
It is the way of keeping an area clean.

Ways of promoting sanitation

- ✓ Mopping classrooms.
- ✓ Sweeping the compound.
- ✓ Draining stagnant water.
- ✓ Scrubbing verandas and floors.
- ✓ Constructing urinals.
- ✓ Flushing toilets after use.
- ✓ Proper disposal of rubbish.
- ✓ Dusting tables and chairs.
- ✓ Clearing bushes around the school.
- ✓ Smoking latrines.
- ✓ Washing toilets.
- ✓ Proper disposal of human wastes.

Elements of a clean home.

Element	Importance/use
Clean latrines/toilets	For proper disposal of human wastes.
Clean bathrooms	For bathing
Rubbish pit/dustbin	For proper disposal of rubbish.
Plate stand/rack	For proper drying of utensils and cutlery. It prevents utensils and cutlery from being dirtied by dust on the ground, animals and young children.
Tippy-tap/hand washing facility	For washing hands.



Dust bin

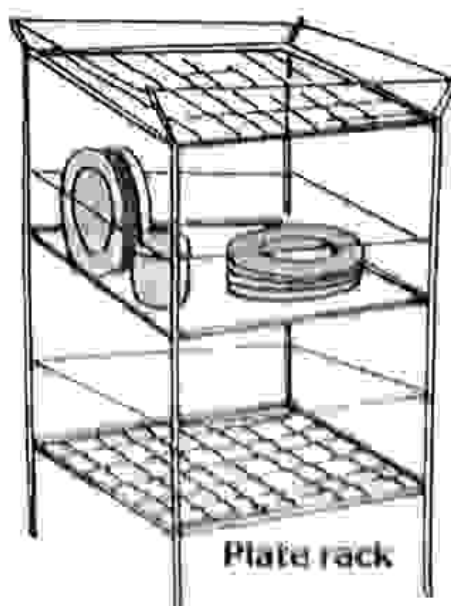


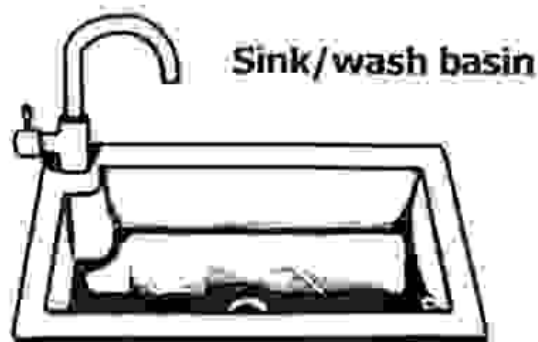
Plate rack

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197



tippy-tap



Sink/wash basin

Points to note.



The stand of the plate stand has to be raised at least 1 metre above the ground

Reasons

- To prevent animals and young children from dirtying washed utensils and cutlery.
- To prevent water splashes from dirtying utensils in case it is raining.
- To prevent dust from dirtying washed utensils.

A sink is used for washing utensils and cutlery.

Indicators of poor sanitation

Bushy compound, poor disposal of faeces, rubbish, domestic wastes, dusty classrooms, littered Compound, pools of water in the compound, high number of houseflies in the compound.

Diseases that spread due to poor sanitation are;

diarrhoea, dysentery, typhoid, and cholera.

Dangers of poor sanitation in an area.

- ⌘ It leads to easy spread of diarrhoeal diseases.
- ⌘ It leads to water and air pollution.
- ⌘ It leads to contamination of food by houseflies.
- ⌘ It encourages the breeding of vectors and germs.

Dangers/disadvantages of sharing the house with domestic animals.

- It leads to the spread of ectoparasites.
- It leads to competition for oxygen.
- It leads to bad smell in the house.
- It leads to spread of some diseases.
- Animals can cause noise pollution in the house
- Animals destroy household property.
- Animals can contaminate food or utensils.

Importance of promoting proper sanitation

- ❖ It prevents easy spread of diarrhoeal diseases.
- ❖ It prevents air and water pollution.
- ❖ It reduces the breeding of disease vectors and germs.
- ❖ It controls bad smell in the environment.

Evaluation activity 11.1

1. (a) Give the meaning of the term sanitation.
(b) Identify any **two** activities that school children can do to promote sanitation at school.
(c) Mention any **one** place at school that needs to be kept clean.
2. The diagram below shows a sanitary item. study and use it to answer the questions that follow.



- (a) Name the sanitary item shown above.
- (b) How is the above item useful in the promotion of sanitation at home?
- (c) State any **two** way of caring for the sanitary item shown above.
3. (a) State any **two** indicators of poor sanitation in a school.
(b) Give any **two** diseases that can break out due to poor sanitation in a school.
4. (a) Give **two** reasons for placing the utensils on a rack.
(b) Why should a rack be constructed with stands at least **1 metre** above the ground?
(c) Give any **one** danger of leaving utensils outside for a long time on the rack.
5. The diagram below shows a hand washing facility. Study and use it to answer the questions that follow.



- (a) Name the hand washing facility shown above.
- (b) How does the above hand washing facility help to prevent the spread of diarrhoeal diseases?
- (c) What is the importance of the stick on the above hand washing facility?
- (d) State any **one** way of keeping the hand washing facility above clean.

TOILETS AND LATRINES

A latrine is a structure built on a pit for disposing human wastes.

Examples/types of latrines

ventilated improved pit latrines, ecosan latrines, ordinary pit latrines and enviro-loo latrines.

Ventilated Improved Pit latrine (VIP latrine)

Characteristics of a Ventilated Improved Pit latrine

- ❖ A VIP latrine has a vent pipe.

A vent pipe lets out bad smell from the latrine. This is by convection currents. The smelly air from the VIP latrine moves through the vent because it is less dense than fresh air entering the squat hole.

A vent pipe directs houseflies to the screen.

The presence of sunlight in the vent pipe enables the houseflies to reach the screen.

- ❖ A VIP latrine has a screen which traps and kills houseflies.

A screen has network like structures with small holes which prevent houseflies from escaping.

Flies die due to excess heat which comes from the pit through the vent pipe and starvation due to lack of food.

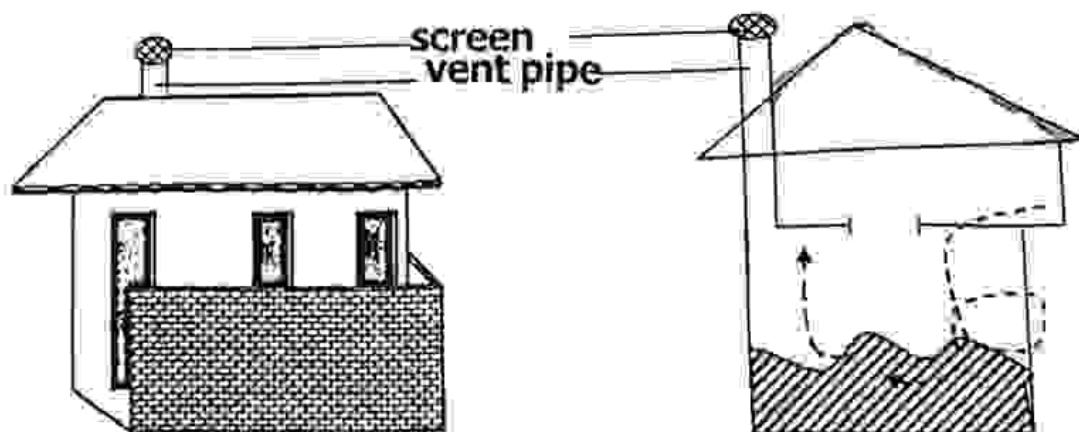
- ❖ A VIP latrine has no lid to the hole.

This allows free air circulation in the pit.

- ❖ A VIP latrine has a spiral shaped wall.

It provides privacy to the user.

It allows free air circulation.



Points to note.

A VIP latrine is said to be ventilated because it has a vent pipe to let out bad smell and improved because it has a screen that traps houseflies.

The arrows in the diagram show air circulation. Air moves through the vent pipe by convection currents.

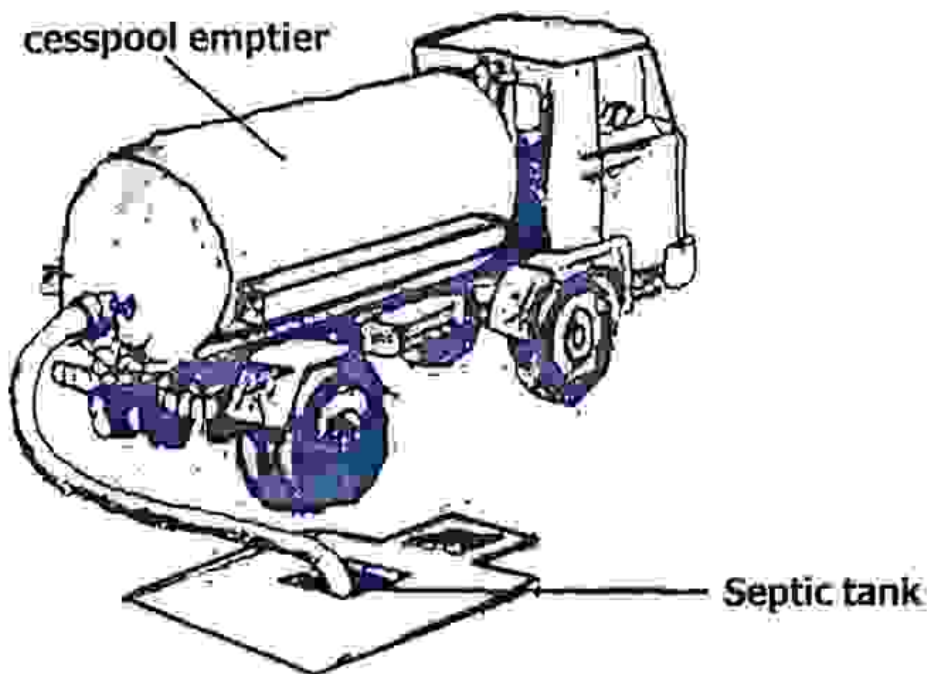
Maintenance of a VIP latrine.

- ❖ Dispose faeces in the pit properly.
- ❖ Sweep the floor of the latrine regularly.
- ❖ Repair the damaged floor of the and walls.
- ❖ Trim the grass around the latrine.
- ❖ Empty the latrine using a cesspool emptier.

A cesspool emptier.

It is commonly found at the sewage plants and water treatment plants.

- ✓ It helps in the proper disposal of sewage.
- ✓ It helps in transporting sewage safely to the treatment plants.
- ✓ It helps in emptying septic tanks.
- ✓ It helps in emptying latrines safely when full.
- ✓ It controls houseflies.



Ecosan latrines

Ecosan stands for Ecological Sanitation.

Ecosan refers to Ecosystem Sanitation Conveniences.

It separates faeces from urine and suitable for slum areas.

Characteristics of ecosan latrines

- It has a shallow pit.
- It has two separate chambers.
- It requires the use of ash to dry faeces.
- It requires regular emptying.

Types of ecosan latrines

- Urine Diversion Dry Toilet (UDDT)
- Arborloo toilet
- Fossa alterna

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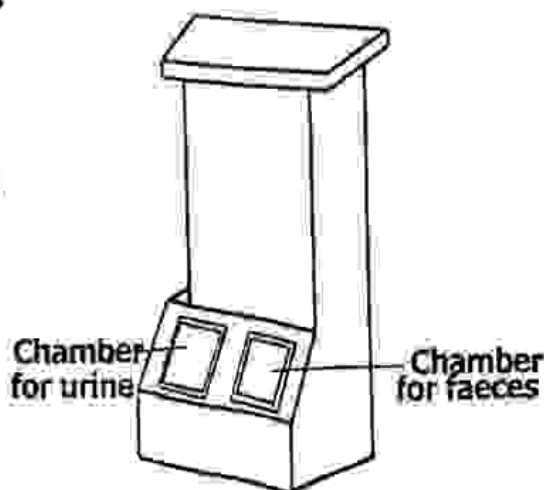
201

Advantages of using ecosan latrines

- They do not produce any bad smell.
- The dry faeces is used as manure in gardens for crop growth.
- Urine collected is used as pesticide and fertiliser.
- They requires a small space

Disadvantages of ecosan latrines

- ⚡ The urine diversion pipe often blocks.
- ⚡ There is need to have ash or lime to use daily.
- ⚡ They are expensive to construct.



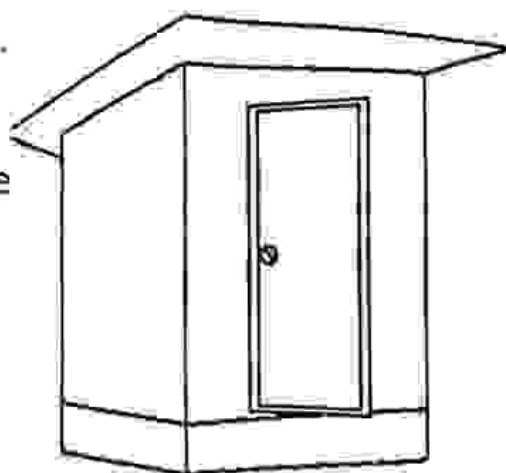
Conventional/Ordinary Pit Latrine

Ordinary pit latrine and potty

It is also called a conventional pit latrine.

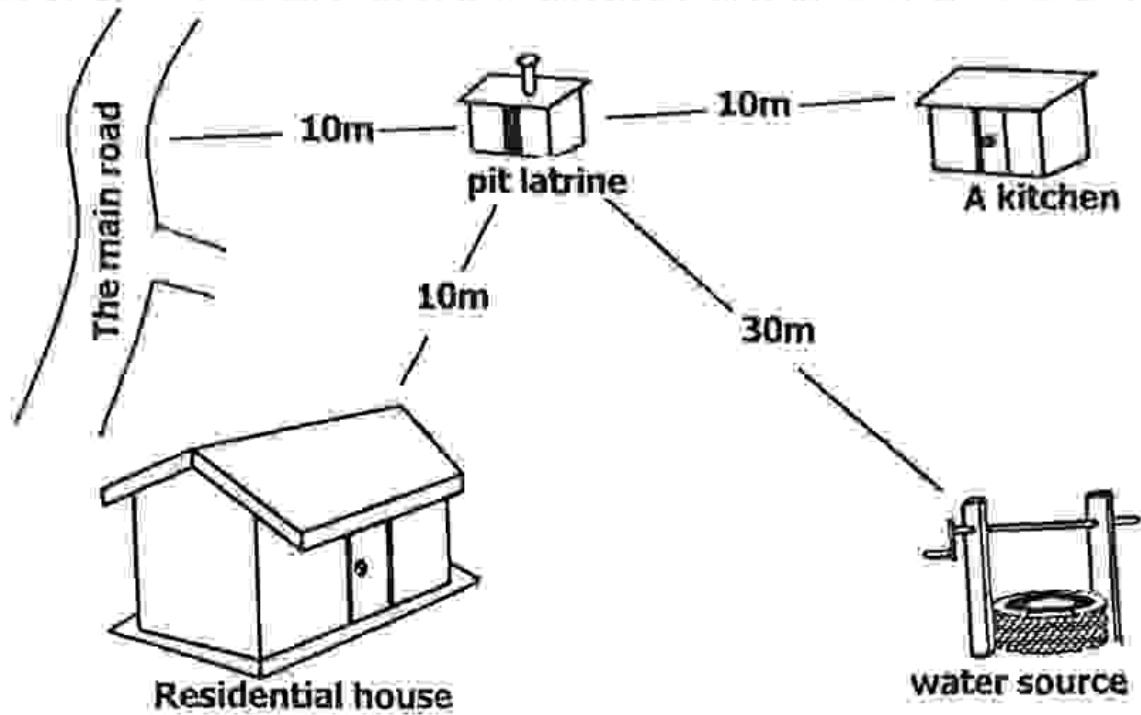
Characteristics of an ordinary pit latrine

- An ordinary pit latrine has the lid to the hole.
The lid reduces bad smell from coming out of the pit.
The lid reduces the free movement of houseflies.
- It does not have a vent pipe and a screen.



Recommended distance for pit latrine.

Building/item	Distance	Reason.
Water source	30 metres	To prevent water contamination by human wastes.
Residential house	10 metres	To prevent bad smell from reaching people.
Kitchen, hotel	10 metres	To prevent houseflies from reaching food.
Main road	10 metres	To keep away bad smell.



Factors to consider when choosing a site for digging a pit latrine.

Distance from the living house, distance from the water source, distance from the hotel and Kitchen, area downhill the water source and firm ground.

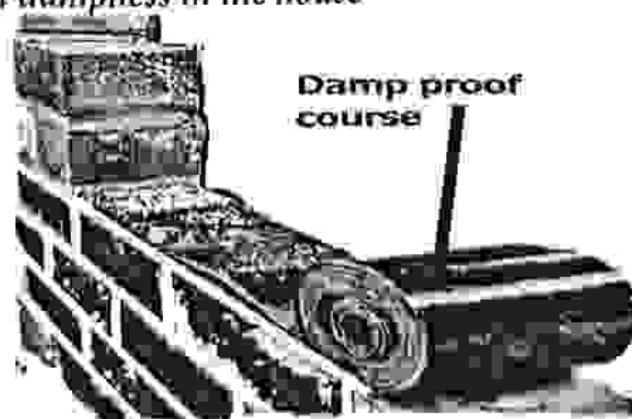
Factors to consider when choosing a site for building a house.

Distance from the water supply, Drainage of the area, Distance from the road, Accessibility to electricity supply, Distance from the health centres, schools and market, Security of the place, Distance from swampy areas.

Note: When building the foundation of the house, a damp proof course is used.

Reasons

- To prevent water from seeping into the wall.
- To prevent dampness in the house



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Proper maintenance of a pit latrine.

- Sweeping and mopping the floor regularly.
- Smoking pit latrines

Smoking reduces bad smell in the latrine.

Smoke drives smelly air out of the pit by increasing convection currents.

Smoking reduces the breeding of houseflies and cockroaches in the latrine.

- Slashing tall bushes near pit latrines.
- Repairing damaged floor and leaking roof.
- Scrubbing the floor
- Picking rubbish near the latrine,
- Applying ash around the pit hole of a latrine.

Ash reduces bad smell// Keeps the pit hole dry//Reduces houseflies in the latrine.

Differences between a VIP latrine and an ordinary pit latrine.

- A VIP latrine has a screen that traps and kills houseflies while an ordinary pit latrine does not have a screen.
- A VIP latrine has a vent pipe while an ordinary pit latrine does not have a vent pipe.
- A VIP latrine does not require a lid while an ordinary pit latrine requires a lid to the hole.
- A VIP latrine has spiral walls while an ordinary pit latrine does not have spiral walls.

Potty. It is a container used to dispose excreta from small children.

A potty is used by small children who are unable to use latrines/toilets.

Water should be poured in it before disposing wastes to prevent faeces from sticking to the sides of the potty.

Potties should not be kept closed away from food. Keeping them near food may lead to food contamination and transmission of diarrhoeal diseases.



Potty



Child using potty

Ways of caring for a potty

- Regular emptying and rinsing.
- Regular washing.
- Regular disinfecting.
- Storing in a clean dry place.
- Sprinkle small amounts of baking soda to control odour.
- Keeping the potty dry.

Evaluation activity 11.2

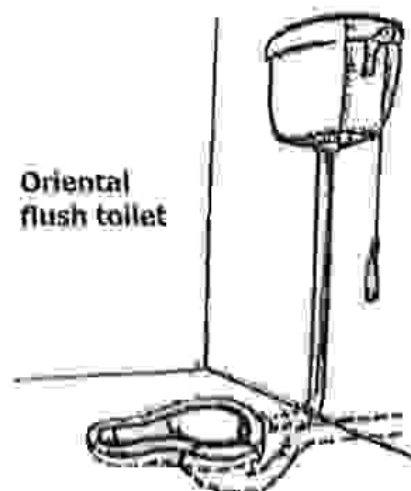
1. (a) State any **one** factor to consider when choosing site to dig a pit latrine.
(b) Mention any **one** characteristic of a VIP latrine.
(c) How are latrines important at home?
2. Why should a ventilated Improved Pit (VIP) latrine be left without a lid to the hole?
3. Which type of latrines are used in the maintenance of soil fertility?
4. Give a reason why a pit latrines should be built at least 30 metres away from the water source.
5. State the importance of each of the following parts of a Ventilated Improved Pit (VIP) latrine.
 - a) Vent pipe
 - b) screen
6. Give any **one** reason why ash is usually put around the pit hole of a latrine.
7. Give any **one** way in which a VIP latrine is different from an ordinary pit latrine.
8. State **one** way in which the screen on the vent pipe of a VIP latrine helps in the control of diarrhoeal diseases
9. How does smoking help to prevent bad smell in an ordinary pit latrine?
10. State any **one** way in which a cesspool emptier is important in the promotion of sanitation.
11. State any **one** way of controlling bad smell in ordinary pit latrines.
12. Apart from pit latrines, name **one** other type of latrine that requires the use of ash.

The water closet/water borne toilets

They are also called flush toilets because water is used to wash away faeces and urine from the bowl.

A **toilet** is a waterborne facility for disposing human wastes.

Examples of water closet toilets



Parts of the flush/water closet toilet.



Functions of each part of the flush toilet

Toilet seat. It is where the user sits when defecating or urinating. A lid is attached to the seat for covering/closing the bowl.

Toilet bowl. For depositing faeces and urine. Some water remains in the bowl after flushing. This water is called toilet bowl trap seal.

Uses of some water that remains in the bowl after flushing.

- It prevents smelly air from coming out of the septic tank.
- It keeps the toilet bowl hygienically clean.
- It prevents houseflies from reaching faeces.
- It prevents the backflow of sewage in the toilet room.

Cistern tank. It holds water for flushing the toilet after use. It is usually raised above the bowl to increase the pressure of water flow when flushed.

Handle. It is pulled or pushed when flushing the toilet.

Water pipe. It brings water to the toilet bowl for flushing faeces and urine.

Septic pipe. It takes faeces and urine to the septic tank.

Septic tank. It stores faeces and urine.

There are bacteria that decompose faeces .

It is not advisable to pour chemicals like acids in the septic tank.

Reason. They kill bacteria that help to decompose faeces.

Proper maintenance of the water closet toilet.

- ✓ Flush the toilet with water after use.
- ✓ Scrubbing the floor with a toilet brush.
- ✓ Use a toilet paper to clean yourself after defecating.
Using hard materials like papers may block sewage pipes.
- ✓ Do not squat or stand on the bowl. This may make the bowl dirty.
It may cause the bowl to break
- ✓ Disinfect the toilet regularly to kill germs.



Things needed to clean toilets and latrine



Cleaning a western toilet



Cleaning Asian or Oriental toilet

Steps taken when using a water closet toilet.

- (i) Lift the lid.
- (ii) Sit on the seat.
- (iii) Flush the toilet after use.
- (iv) Place back the lid and wash your hands.



lift the lid



Sit on the seat
Do not stand on it



Flush the toilet
after use



wash your hands

Advantages of water closet toilet.

- ❖ It can be put in the house and vehicle.
- ❖ It is easy to clean.
- ❖ It is user friendly.

Disadvantages of water closet toilet.

- It is very expensive to install.
- It requires a lot of water to function.
- Improper use can lead to bad smell in the house.
- Sewage pipes can easily block when hard materials are used.

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Problems faced by urban toilets.

- ‡ Inadequate water for flushing.
- ‡ Blockage of pipes by hard materials.
- ‡ Bursting of pipes when not treated.
- ‡ Spilling of dirty water due to blockage of sewage system.

Importance of using a latrine/toilet correctly.

- It prevents easy spread of diarrhoeal diseases.
- It prevents pollution of air and water.
- It prevents houseflies from contaminating food

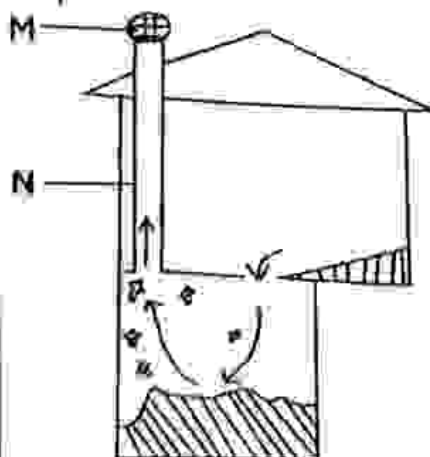
Evaluation activity 11.3

1. The diagram below shows a flush toilet. Study carefully and use it to answer the questions that follow.



- (a) Name the parts labelled **D** and **E**.
 - (b) State the function of pipe marked **F** in the above toilet.
 - (c) Give any **one** way of maintaining the above toilet in good working conditions.
2. Why is the cistern tank of a flush toilet raised above the toilet bowl?

3. Give any **one** way in which water that remains in the toilet bowl after flushing is useful in the flush toilet.
4. Why is it not advisable to pour acid in the septic tank?
5. Which organisms prevent the septic tank from filling up quickly with faeces?
6. Give any **one** way in which the use of latrines is important to human health.
7. The diagram below of a VIP latrine. Study and use it to answer the questions that follow.



- (a) Name the parts marked **M** and **N**.
- (b) Give the function of the part marked **M**.
- (c) What do the arrows in the diagram above show?

Theme: **HUMAN BODY**

Topic **12: THE REPRODUCTIVE SYSTEM**

The reproductive system is a group of organs in the human body that helps human beings to reproduce.

Reproduction is the process by which living things multiply by giving rise to offspring.

Types of reproduction

Sexual reproduction. This is the type of reproduction which involves the union of male and female gametes (reproductive cells)

Asexual reproduction. This is the type of reproduction which does not involve the union of male and female gametes.

Examples of asexual reproduction are; binary fission, sporulation and vegetative propagation.

Growth and Development in Human Beings

Growth is the permanent increase in size of a person.

It is caused by increase in the number of cells or increase in size of cells.

Development is the increase in complexity of a person.

Puberty and adolescence

Puberty is the period of sexual maturity of a person.

Adolescence is the transitional stage between childhood and adulthood.

Primary and Secondary sex Changes

Primary Sex Changes

Primary sex changes are the changes that occur to the sexual organs.

Primary sex organs in males are testes while in females are ovaries.

Primary sex changes in adolescent boys

- ✓ Testes and penis enlarge.
- ✓ Production of sperms.
- ✓ Wet dreams begin.

Primary sex changes in adolescent girls

- Ovulation begins.
Ovulation is the release of mature ova from ovaries.
- Menstruation periods begin.
Menstruation is the discharge of blood from the womb with unfertilized ovum.
- The uterus walls mature and thicken.

Primary sex changes that occur both in adolescent boys and girls.

- Production of gametes.
- Development of reproductive organs.

Secondary Sex Characteristics/changes.

Secondary sex changes are changes which give a person male or female appearance.

Secondary sex changes in adolescent boys/males

These changes are controlled by testosterone hormone.

- ❖ Deepening of the voice.
- ❖ Growth of hair in the armpits and pubes.
- ❖ Sweat glands become more active.
- ❖ Growth of beard.
- ❖ The body becomes more muscular.

Secondary sex changes in adolescent girls/females

These changes are controlled by oestrogen hormone.

- The hips enlarge.
- The breasts enlarge.
- Growth of hair in the armpit and pubes.
- The voice becomes soft.
- Sweat glands become more active.
- Pimples may appear on the face.

Secondary sex changes that occur in both adolescent boys and girls.

- Growth of hair in the armpits and pubes.
- Sweat glands become more active

Social and Emotional Changes.

- ‡ Adolescents become interested to people of opposite sex.
- ‡ Adolescents rebel against rules of their parents and teachers.
- ‡ They move in peer groups.
- ‡ They become confused about body changes.
- ‡ They develop desire for freedom.

Ways in which adolescents can manage their body changes.

- ✓ Bathing regularly.
- ✓ Shaving hair around the pubes and in armpits.
- ✓ Washing underpants regularly.
- ✓ Use of deodorants.
- ✓ Using antiperspirants.
- ✓ Regular changing of sanitary pads during menses.

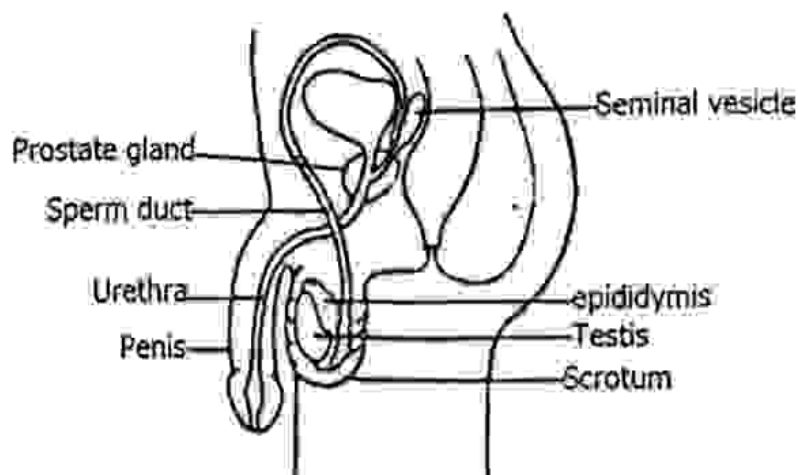
 *Note well: the practices above help to control bad odour from adolescents.*

Evaluation activity 12.1

1. (a) Give the meaning of the following terms;
(i) Puberty..... (ii) Adolescence.....
(b) State any **two** secondary sex changes that occur in adolescent girls.
2. What type of reproduction do human beings undergo?
3. Name the human body system that enables human beings to give rise to new offspring.
4. Write down any **two** primary sex changes that occur in adolescent boys.
5. State any **two** secondary sex characteristics that occur in both adolescent girls and boys.
6. Name the reproductive organ where menstruation period takes place.
7. What is meant by the term ovulation in reproduction?
8. Give any **two** examples of social and emotional changes that adolescent girls and boys experience in life.
9. State the difference between growth and development.
10. State the characteristic of living things that enable humans to increase in number.
11. State any **one** way in which an adolescent can control bad odour that results from body changes.

Reproductive Organs and Cells.

Structure of the male reproductive system



Functions of each part of the male reproductive system

Testes. They produce sperms and testosterone hormone.

Testes in males hang outside the body.

Reason. The production of sperms requires temperature slightly lower than the normal temperature. This temperature should be between 34°C to 36°C or 93.2°F to 96.8°F

Epididymis. It stores sperms.

Scrotum. It regulates temperature of the testes.

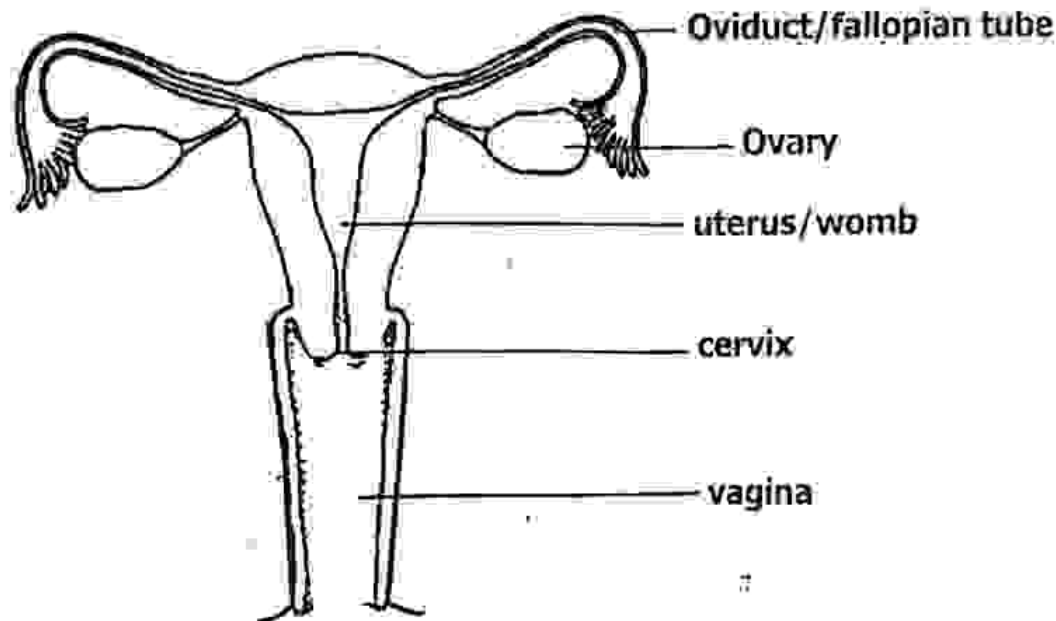
Sperm duct. It transports sperms from the epididymis to the urethra.

Seminal vesicle. It produces semen.

Urethra. It is a passage of semen containing sperms.

Penis. It is used for depositing semen into the vagina.

The structure of the female reproductive system.



Functions of each part of the female reproductive system.

Ovaries.

They produce ova.

They produce oestrogen and progesterone hormones.

Oestrogen prepares the uterus for implantation.

Oviduct (fallopian tube). It is where fertilization and conception take place.

Uterus/womb. It is where implantation and development of the foetus take place.

Cervix. It closes the uterus during pregnancy. This prevents the foetus from coming out before its time.

It also separates the vagina from the uterus.

Vagina. It is where copulation takes place.

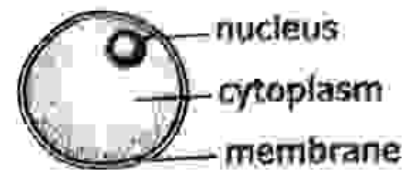
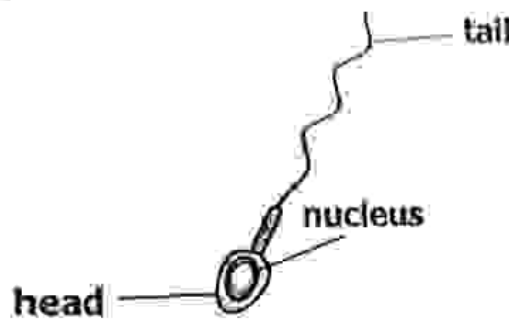
It is the passage for the baby during childbirth.

Reproductive cells.

Reproductive cells are called gametes.

Gametes are produced by gonads.

Sperms are male gametes produced by testes while ova are female gametes produced by ovaries



The tail helps the sperm cell to swim easily through the semen.
The nucleus controls all the sex activities of the gamete.

Evaluation activity 12.2

1. What name is given to male gametes in human beings?
2. How are ova in humans similar to ovules in plants?
3. In the human reproductive, state where each of the following processes take place.
 - (i) Production of sperms
 - (ii) Implantation
 - (iii) Production of female hormones
 - (iv) Conception
4. In the table below, part A shows organs of the reproductive system and part B has their functions.

Part A: Organs	Part B: Functions
Ovary	Passage for semen during ejaculation.
Epididymis	Protects the foetus from external damage.
Cervix	Produces oestrogen hormone
Urethra	Stores sperms.

Use the functions in part B of the table to match the organs below.

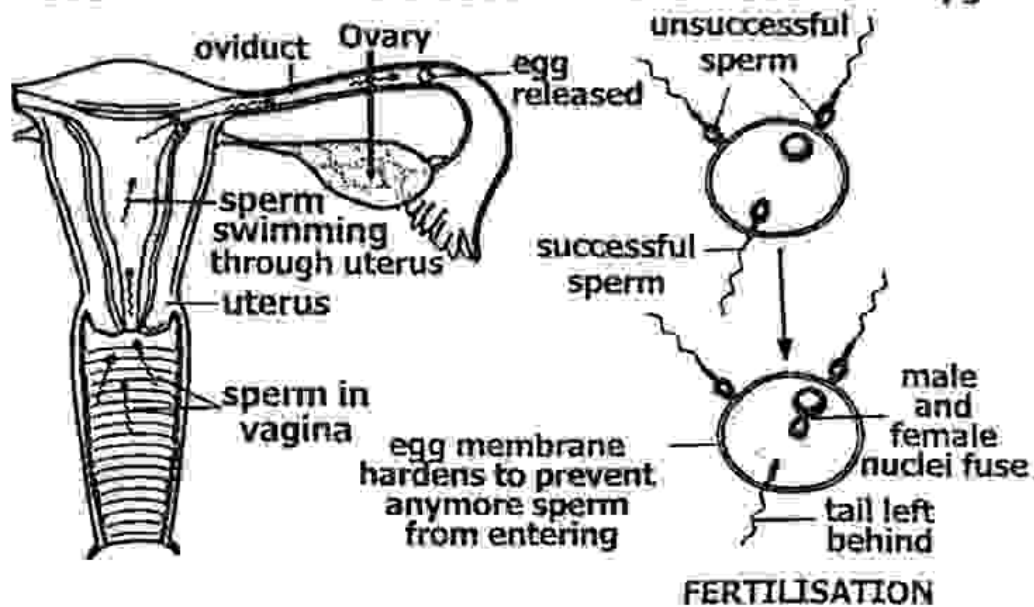
- (i) Ovary
- (ii) Epididymis
- (iii) Cervix
- (iv) Urethra
5. How are the following organs useful in the process of reproduction?
 - (a) Testes
 - (b) ovaries

Fertilization, Conception and Pregnancy

Fertilisation is the fusion of male and female gametes to form a zygote. A zygote is a fertilized egg produced by the fusion of the male and female gametes.

In humans, fertilization is internal and it takes place in the oviduct.

The immediate result after fertilization is the formation of a **zygote**.



Conception is successful fertilization in the oviduct.

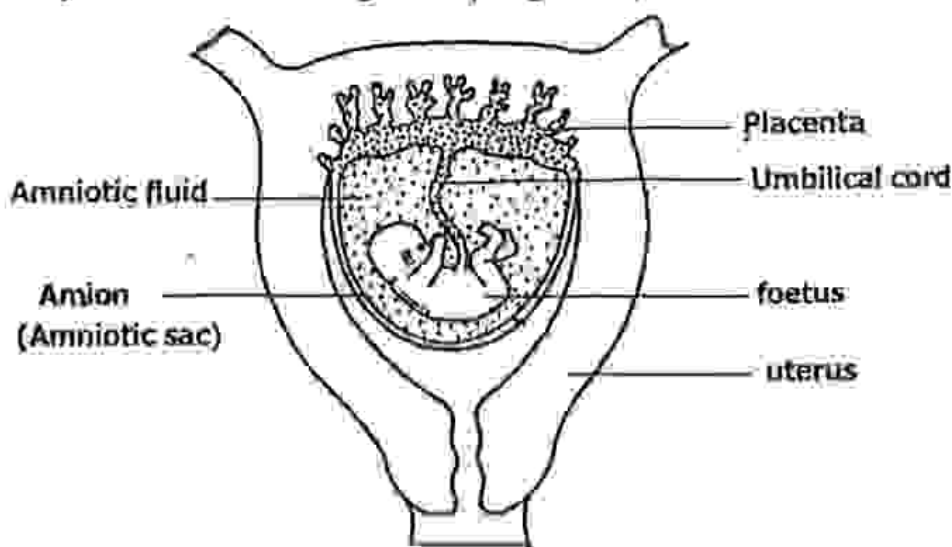
Implantation is the attachment of the zygote to the uterus.

These processes occur as follow; ovulation, fertilisation, conception and implantation.

Pregnancy. Pregnancy is the period from fertilisation to birth.

The gestation period of a woman is 9 months (270 days)

Gestation period is the length of pregnancy.



Functions of each part.

Placenta

- It prevents the mother's blood from mixing with that of the foetus.
- It supplies food and oxygen to the foetus.
- It attaches the foetus to the uterus.
- It receives and stores waste materials from the foetus.
- It prevents toxic substances from reaching the foetus.

Umbilical cord

- It transports oxygen and food to the embryo.
- It attaches the foetus to the placenta.
- It transports wastes from the foetus to the placenta.

Amniotic sac (amnion)

It produces amniotic fluid.

Amniotic fluid. It reduces friction and absorbs shock from the foetus.

Signs of pregnancy

- Menstruation period stops.
- Breasts grow bigger and become tender.
- The belly expands due to increasing size of the foetus.
- Frequent urination.



Care and requirements of a pregnant woman

Requirement	Importance/uses
Balanced diet	❖ It enables the pregnant mother to remain healthy. ❖ Carbohydrates provide energy to the pregnant woman. ❖ Proteins repair worn out body tissues. ❖ Proteins promote proper growth of the foetus in the womb. ❖ Proteins replace the worn-out body cells. ❖ Proteins make antibodies to boost immunity of the mother and unborn child. ❖ Vitamins and mineral salts boost the immunity of the mother.
Enough rest and sleep	- It helps to refresh the brain and relieve stress.

Requirement	Importance/uses
Physical exercises	• They make the muscles of the womb flexible. • They enable the baby to move and turn easily in the womb. • They prevent complications during birth. • They improve blood circulation. • They reduce overweight of the body.
Proper hygiene	➤ It helps to prevent skin diseases. ➤ It helps to prevent bad odour from the body.
Appropriate dressing	🍏 It helps to support the body's changes. 🍏 It boosts self-esteem during pregnancy. 🍏 It ensures comfort to the pregnant woman.
Ante-natal care	⚡ It helps to promote the health of the mother and foetus. ⚡ It helps to prepare the mother for safe childbirth. ⚡ It helps to prevent complications during childbirth. ⚡ It helps to detect and treat conditions arising during pregnancy. ⚡ It enables HIV positive pregnant women to protect their unborn baby from contracting HIV during childbirth.

Ante-natal care

Ante-natal care is the care given to the pregnant women before giving birth.

Reasons why pregnant women go for ante-natal care.

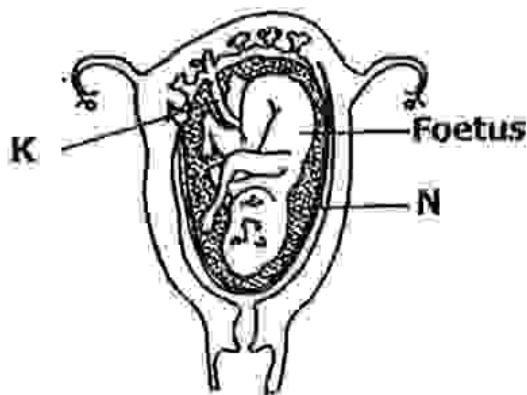
- To check for the foetal growth and position.
- To get Tetanus Toxoid vaccine.
T.T vaccine protects the mother and unborn baby against tetanus during childbirth.
- To get advice on personal hygiene.
- To test for HIV status. This helps to prevent unborn baby from contracting HIV during and after childbirth.

Services provided to pregnant mothers during ante-natal care

Guidance and counselling, testing for HIV/AIDS and other STDs and immunization against tetanus

Evaluation activity 12.3

1. The diagram below is of a developing foetus in a mother's womb. Study it to answer the questions that follow.



- (a) Name the substance marked **N**.
- (b) Give **two** uses of the part marked **K** to the foetus.
- (c) On the diagram above, label the placenta.
2. Name the part that attaches the foetus to the womb.
3. What type of fertilisation do human beings undergo?
4. What is meant by the term implantation?
5. Write down any **two** signs of pregnancy in humans.
6. Which part of the reproductive system prevents the mother's blood from mixing with that of the foetus?
7. Mention any **one** diseases that the developing baby in the uterus can contract when its blood mixes with that of the mother.
8. Give any **two** functions of the placenta to the developing foetus in the womb.
9. Write down any **two** services provided to pregnant mothers during ante-natal care.
10. Why should pregnant mothers be given T.T vaccine before giving birth?
11. Give any **one** reason why pregnant mothers are advised to feed on food rich in proteins.
12. List down any **two** requirements of a pregnant woman.

Reproductive Health

Reproductive health is the state of being free from diseases and disorders of the reproductive system.

It is the state of being free from problems related to reproduction.

Problems in pregnancy.

- Anaemia
- Excessive and frequent vomiting.
- Backache
- Heartburns.

Teenage pregnancy.

Teenage pregnancy is the pregnancy acquired by a girl below the age of twenty years.

Factors that may lead to teenage pregnancy.

- ⬆ Peer influence into early sex.
- ⬆ Failure to provide basic needs to girls by parents.
- ⬆ Rape or defilement.
- ⬆ Lack of proper guidance and counselling to adolescents.

Consequences/dangers of teenage pregnancy

- ❖ It leads to school dropouts.
- ❖ It leads to stigmatization.
- ❖ It causes birth complications.
- ❖ It leads to forced marriage.
- ❖ It may lead to death due to attempted abortion.

Ways of avoiding/preventing teenage pregnancy.

- Abstaining from pre-marital sex.
- Educating adolescents about dangers of teenage pregnancy.
- Keep away from bad peer groups.
- Reporting cases of rape to concerned people.
- Educating adolescents about body growth and body changes.

Life skills to safeguard adolescents against teenage pregnancy.

- Decision making.
- Self-control
- Assertiveness
- Peer resistance
- Self esteem
- Self-awareness

Ways of caring for reproductive system/organs in humans.

- ✓ Washing the reproductive organs regularly.
- ✓ Abstaining from pre-marital sex.
- ✓ Early treatment of diseases of the reproductive system.
- ✓ Shaving overgrown pubic hair.
- ✓ Wearing clean and dry underwears.

Common Diseases and Disorders of the Reproductive System (STIs)

Diseases that Affect the Reproductive System

- gonorrhoea ➤ syphilis ➤ chancroids ➤ candidiasis
- genital herpes ➤ genital warts ➤ trichomoniasis ➤ cervical cancer



Learn, unlearn and relearn:

-AIDS is not among the diseases that affect/attack the reproductive system because it attacks the white blood cells which is part of the circulatory system.

-Conception does not take place in the womb/uterus. It takes place in the oviduct.

Sexually Transmitted Diseases (STDs)

Sexually transmitted diseases are diseases that spread from one person to another through having unprotected sexual intercourse with an infected person.

They are also called sexually transmitted infections (STIs)/venereal diseases.

Ways through which sexually transmitted diseases spread.

- ✓ Through having unprotected sexual intercourse with an infected person.
- ✓ Through sharing sharp objects with an infected person.
- ✓ Through the placenta during pregnancy.
- ✓ Through blood during childbirth.
- ✓ Through taking breast milk.
- ✓ Through traditional circumcision.

Examples of STDs are; AIDS, gonorrhoea, syphilis, trichomoniasis, candidiasis, genital warts

Gonorrhoea

It is caused by bacteria.

Signs and symptoms of gonorrhoea

- Pain in the lower belly.
- Discharge of pus from the vagina or penis.
- Blockage of the oviducts.
- Painful urination.
- Blindness in newly born babies.

Effects of gonorrhoea

- It causes bareness/infertility.
- It blocks the oviducts.
- It causes blindness in newly born babies.
- It makes the baby to contract bacteria in the birth canal which causes a severe eye infection.
- It increases the chances of contracting HIV/AIDS.

It causes sores on sexual organs through which HIV passes.

- It causes ectopic pregnancy.

It creates the scar on the oviducts which makes it rough preventing the movement of a fertilised ovum.

 *Point to note: it is advisable for both married couples to go for gonorrhoea treatment together.*

Reason. To prevent reinfection.

Syphilis

It is caused by bacteria.

Signs and symptoms of syphilis.

- Painless sores on the penis or in the vagina.

- Itching body rash.
- Mouth and throat sores.
- Swollen joints and lymph nodes.
- Madness if left untreated.
- Mild fever.
- Ring patches on the skin.
- Loss of hair on the head.

Candidiasis/thrush

It is caused by fungus. This fungus is called candida.

It is spread through poor personal hygiene.

Signs and symptoms of candidiasis.

- Itching of the private parts.
- Burning pain when urinating.
- Sore genital organs.
- Smelly discharge from the vagina.

Prevention of candidiasis.

- ❖ Keeping private parts clean and dry.
- ❖ Washing the reproductive parts regularly.
- ❖ Keep latrines clean regularly.
- ❖ Treat infected people using fungicides.

Trichomoniasis

It is caused by protozoa.

Signs and symptoms of trichomoniasis.

- Smelly discharge from the vagina.
- Itching in the vagina.
- Sometimes the discharge contains blood.

HIV/AIDS

- AIDS stands for acquired immune deficiency syndrome.
- Acquired means got from, immune means protected against, deficiency means lack of and syndrome means many signs and symptoms.
- AIDS is caused by a virus called HIV.
- HIV stands for human immunodeficiency virus.
- HIV destroys the white blood cells making them unable to fight against germs in the body.

Ways in which HIV/AIDS spreads.

- ✓ Through having unprotected sexual intercourse with an infected person.
- ✓ Through sharing unsterilized sharp objects like syringes with an infected person.
- ✓ Mother to child transmission.
 - Through placenta during pregnancy.
 - Through blood during breastfeeding
 - Through blood during birth.
- ✓ Through cultural circumcision and genital mutilation



*Learn, unlearn and relearn: blood is screened before transfusion.
Therefore, reject through blood transfusion.*

Signs and symptoms of HIV/AIDS

- ✓ Prolonged dry cough
- ✓ Widespread itching and skin rash.
- ✓ Loss of body weight.
- ✓ Blood-stained sputum.
- ✓ Prolonged diarrhoea
- ✓ Persistent fever
- ✓ Excessive sweating at night.

Diseases mistaken as AIDS are;

Tuberculosis, cholera, measles and typhoid.

Such diseases are called opportunistic diseases/secondary diseases.

They attack the body when its immunity becomes weak.

A person suffering from HIV/AIDS is easily attacked by other diseases.

Reasons. HIV/AIDS weakens the immunity of a person.

- HIV/AIDS destroys white blood cells which fight against germs in the body.

Effects of HIV/AIDS to an individual.

- ❖ It weakens the immunity of a person.
- ❖ It increases the chances of getting other diseases.
- ❖ It leads to stigmatization.
- ❖ It leads to loss of one's job.
- ❖ It leads to death of an individual.

Effects of HIV/AIDS to the family

- ⚡ It leads to poverty due to high treatment costs.
- ⚡ It leads to loss of important people in a family.
- ⚡ It causes loss of trust and respect among family members.

Effects of HIV/AIDS to the community.

- It leads to loss of skilled labour in the family.
- It leads to loss of friends in the community.
- It leads to shortage of productive population.
- It causes child headed families.
- It increases the number of street children.

Ways of preventing the spread of HIV/AIDS among people.

- ✓ Abstaining from pre-marital sex. This is the best way for school children.
- ✓ Being faithful to one's sexual partner. This is best for married couples.
- ✓ Avoid sharing sharp objects like nail clippers and razorblades.
- ✓ Using condoms when having sex with untrusted sex partner.

Ways in which HIV positive mother can protect unborn baby from getting HIV/AIDS.

- Taking ARVs.
- Giving birth in hospitals under the supervision of health workers.

 Note well.

- It is easier to prevent AIDS than malaria.

Reason. It is easier to abstain from sex than avoiding mosquito bites.

- A person suffering from malaria can die faster than one suffering from AIDS.

Reason. Malaria destroys red blood cells while AIDS makes the white blood cells weak.

- A person who has HIV is one who is infected with the virus that causes AIDS while an AIDS patient is one who suffers from AIDS and shows signs like chronic cough, herpes zoster and skin rash.

- Female anopheles mosquito cannot spread HIV/AIDS.

Reasons.

- HIV is destroyed in its abdomen.

- HIV cannot survive for long outside human blood.

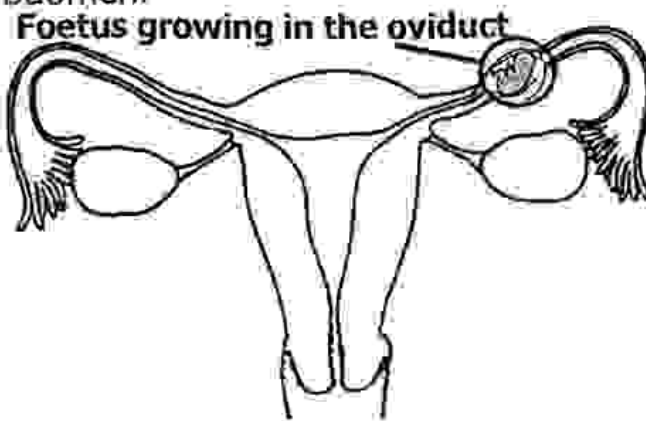
Ways of preventing the spread of Sexually Transmitted Diseases (STDs)

- Abstaining from pre-marital sex.
- Using condoms properly during sexual intercourse.
- Being faithful to one's sexual partner.
- Avoid sharing sharp objects with other people.
- Sterilising sharp medical equipment like syringes and pair of scissors.

Disorders of the Reproductive System in females are;

- Miscarriage;** The spontaneous loss of a pregnancy before the foetus can survive outside the womb, typically before 26 weeks.
- Fibroids;** Noncancerous growths of the uterus that often appear during childbearing years, potentially causing pain and heavy menstrual bleeding.
- Sterility;** The inability to conceive children after a year of regular unprotected sexual intercourse, due to various medical conditions.
- Stillbirth;** The death of a foetus at 26 weeks or more of pregnancy, resulting in the birth of a baby who shows no signs of life.
- Blocked oviduct;** a condition where one or both fallopian tubes are obstructed, preventing the sperm from reaching the egg, leading to infertility.

- f. **Misplaced uterus;** also known as uterine prolapse, it occurs when the uterus descends into or outside of the vaginal canal due to weakened pelvic muscles.
- g. **Ectopic pregnancy.** This is a condition in which the foetus grows in the oviduct. It can be caused by gonorrhoea. It causes severe pain in the lower abdomen.



Disorders of the Reproductive System in males are;

- a. **Impotence;** Inability to achieve or maintain an erection for sexual intercourse.
- b. **Premature ejaculation;** Ejaculation that occurs sooner than desired, often within one minute of penetration.
- c. **Low sperm count;** Having fewer than 15 million sperm per milliliter of semen.
- d. **Malformed sperms;** Sperm with abnormal shape or structure affecting fertility.
- e. **Deformed penis;** this refers to an abnormal shape or curvature of the penis, which may be congenital or result from conditions such as Peyronie's disease.

Family planning

Family planning is the use of the birth control methods to decide when to have or not to have a child in a family.

Birth control means preventing conception or pregnancy in women.

Family planning services are provided by **The Reproductive Health Uganda.**

Child spacing is the reasonable time lapse between the births of children in a family.

Importance of child spacing to the mother.

- It reduces maternal anaemia.
- It reduces maternal mortality rate.
- It gives the mother's body time to repair.

Advantages of family planning.

Don't speak for Quality, let Quality speak for itself

227

- ✓ It reduces infant mortality rate.
- ✓ It enables children to get enough care from their parents.
- ✓ It reduces the risks of maternal anaemia in women.
- ✓ It controls high population growth in a country.
- ✓ It prevents the mother from wearing off quickly due to frequent childbirths.

Reasons why some parents produce many children.

- Ignorance of family planning.
- Desire to have a certain sex.
- Prestige
- High infant mortality rate.
- Cultural affairs
- Religious affairs.

Problems associated with producing many children.

Social problems

- ⇒ Constant quarrels within the family.
- ⇒ High immorality rate among children.
- ⇒ Inadequate basic needs.
- ⇒ Lack of parental care.

Health problems.

- ⇓ Inadequate food.
- ⇓ Premature birth
- ⇓ Proneness to diseases
- ⇓ Lack of proper health care.
- ⇓ Maternal anaemia

Natural Methods of Family Planning

Method	How it works.
Abstinence from sex	It avoids sexual intercourse till the right time. It is best for school children.
Prolonged breastfeeding	It delays monthly ovulation. /It delays the next pregnancy.
Withdrawal method	It prevents sperms from flowing to the oviduct.
Rhythm method	It prevents sexual intercourse during un-safe days.
Basal body temperature	It enables the couple to decide when to have sex. The body temperature of the woman slightly changes during menses These changes relate with the days of ovulation.

Cervical mucus method

Thick mucus in the cervix prevents sperms from reaching the oviduct during sexual intercourse.

Advantages of natural methods

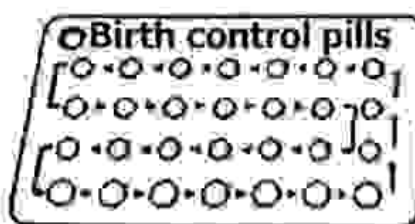
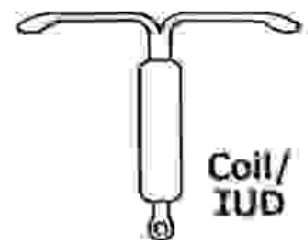
- They are easy and cheap to use.
- They have no side effects to the body.

Disadvantages of natural methods

- They are not effective to use.
- Illiterate women may not be able to use the thermometer.

Artificial methods of family planning.

Method	How it works,
Using condoms	Condoms block sperms from reaching the vagina during ejaculation.
Using birth control pills	The pills stop or reduce chances of ovulation.
Using intra uterine device	The coil prevents a fertilised ovum from implanting in the uterus.
Using diaphragm	The diaphragm prevents semen from entering the uterus.
Using foams and jellies	Foams and jellies kill sperms to prevent fertilisation.
Vasectomy	It prevents sperms from reaching the vagina. It is the permanent method of family planning in men.
Tubal ligation	It prevents the released ovum from meeting with the sperms. It is the permanent method of family planning in women.
Using birth control injections	Depro-provera prevents pregnancy.
Using implants	The implants prevent the woman from conceiving.

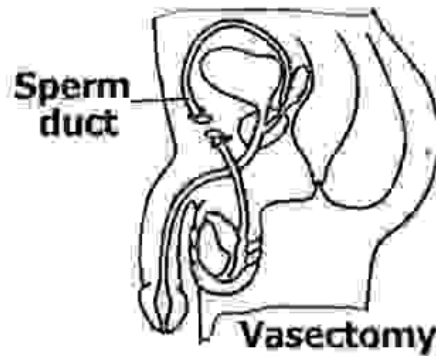
Condom**Birth control pills****Coil/
IUD**



diaphragm



intra-uterine device



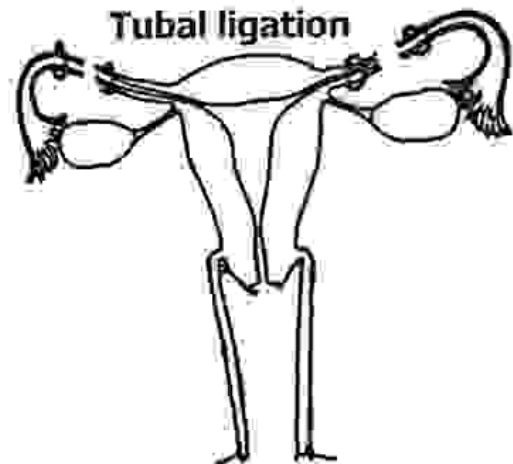
Vasectomy



condom

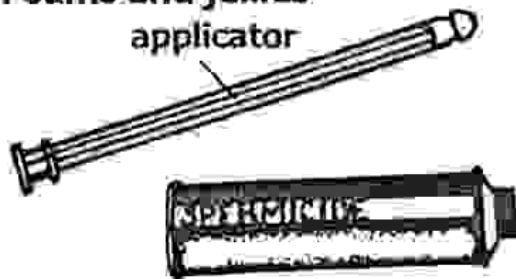


pill
(oral contraceptive)



Tubal ligation

Foams and jellies
applicator



Advantage of artificial methods. They are effective if correctly used.	Disadvantages of artificial methods They are expensive to use. They have side effects to the user's body.
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Myths and misconceptions about family planning

- ⚡ Using birth control pills or injections can cause cancer.
- ⚡ Family planning increases fornication and adultery among people.
- ⚡ Vasectomy is castration of men.
- ⚡ Birth control methods should be used by women not men.
- ⚡ If a woman uses birth control methods, she may produce little or no breast milk.
- ⚡ The whites want to reduce the population of Africans.

PIASCY Messages about Adolescence and Reproductive Health

PIASCY stands for Presidential Initiative on AIDS Strategy for Communication to Youth.

Examples of PIASCY Messages

- AIDS kills and has no cure
- Abstinence is health.
- Avoid early marriage.
- Abstain from sex.
- Avoid bad touches.
- Do not accept gifts for sex.

Importance of PIASCY messages to school children.

- ✓ They prevent school children from contracting HIV/AIDS.
- ✓ They prevent school dropouts.
- ✓ They help to prevent teenage pregnancies among school girls.
- ✓ They prevent early marriages among school children.

Roles of school children in promoting PIASCY.

- Designing posters with PIASCY messages.
- Educating fellow children about body changes at puberty.
- Reciting poems about PIASCY messages.
- Acting drama about PIASCY messages.

Evaluation activity 12.4

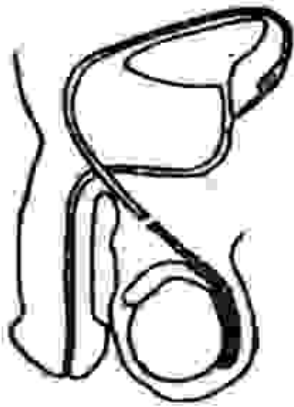
1. (a) Give the meaning of the following terms:
 - (i) Family planning
 - (ii) Child spacing
 - (b) State any **two** advantages of family planning.
 2. Write down any **two** diseases that affect the reproductive system.
 3. Which sexually transmitted disease causes blindness in babies?
 4. How does untreated syphilis affect an individual?
 5. (a) Apart from prolonged breastfeeding, give any **two** other natural methods of family planning.
 - (b) How does prolonged breastfeeding help to promote family planning?
 - (c) State any **one** reason why some parents produce many children.
 6. (a) Name the germ that causes gonorrhoea.
 - (b) Give any **two** signs of gonorrhoea infection in people.
 - (c) How does gonorrhoea infection increase the chances of a person getting HIV/AIDS?
 7. Mention any **two** disorders of the female reproductive system.
- The diagram below shows a message in the signpost commonly displayed in the school compound. Use it to answer question 8.



8. In which way is the message in the signpost above helpful to school children?
9. In which way is a person who is HIV positive different from the one who has developed AIDS?

Evaluation activity 12.4 ...continued

10. Why would you advise school children to abstain from sex?
11. Which method of family planning is suitable for school children?
12. Why should the married couple who are infected with gonorrhoea go for treatment together?
13. Name the fungal caused sexually transmitted disease, which is spread through poor hygiene.
14. Mention **one** permanent method of birth control in males.
15. (a) Give any **one** example of PIASCY message.
(b) State any **one** role played by school children in promoting PIASCY at school.
(c) Give any **two** ways in which PIASCY messages are important to school children.
16. The diagram below shows a permanent method of family planning. Study and use it to answer the questions that follow.



- (a) Identify the permanent method of family planning shown above.
- (b) Give any **one** disadvantage of using the above method of family planning.
- (c) Apart from the method above, mention any **two** other examples of artificial methods of family planning.

ABOUT THE AUTHOR



Jamie HUMPHREY is a professional teacher of Integrated Science and Mathematics. He has taught in prominent schools within Kampala and Wakiso districts in Uganda. He has sufficient knowledge and experience in teaching Integrated Science at primary level as well as Biology and Chemistry at secondary level.

In addition, Jamie doubles as a facilitator of Integrated Science, Item Writing and Customer Care. He is also an item writer of Integrated Science in the firm of **SUREKEY EXAMINATION BOARD (SKEB)**. He is a moderator of Pass PLE items in newspapers as well as a senior examiner.

ABOUT THE BOOK

The SureKey Integrated Science Pupil's Book 6 has been developed basing on the current Primary Six Integrated Science standard curriculum prescribed by the NCDC of Uganda. The book covers all the topics of Primary Six Integrated Science syllabus.

The book contains some highlighted areas of common misconceptions on certain concepts and irrelevant content in Primary Six Integrated Science to enable learners to get the right content at their level.

Similarly, the book has been written in simple and precise English, thus making it easy for the learner to use. At the end of each lesson, a learner's evaluation activity has been provided, catering for all the three levels of assessment—Recalling, Understanding and Application (RUA) as prescribed by the UNEB PLE syllabus.

It is my sincere hope that this reference book will be of unwavering value to Primary Six learners, teachers and researchers.

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